

Every supplied page. The full source trail.

One combined edition covering all 95 supplied URLs: 9 fundamentals, 41 crawling/indexing, and 45 ranking/search-appearance pages. All 669 existing question-and-answer cards are retained, including the 10 cross-topic scenarios and the 76 additional appearance questions. A separate glossary contains 127 source-backed definitions. Every answer has supporting Google excerpts and section links. The source library preserves all main-article text, tables and code from the 94 licensed supplied pages; the supplied Google Support article has a labeled complete-topic brief and original link. Seven official supplements clarify current guidance. Linked subguides are not recursively included.

95

REQUESTED ARTICLES

7

OFFICIAL SUPPLEMENTS

699

SOURCE HEADINGS

Reviewed 6 October 2026 · Combined fundamentals, crawling, ranking and appearance edition

Scope and attribution

One combined edition covering all 95 supplied URLs: 9 fundamentals, 41 crawling/indexing, and 45 ranking/search-appearance pages. All 669 existing question-and-answer cards are retained, including the 10 cross-topic scenarios and the 76 additional appearance questions. A separate glossary contains 127 source-backed definitions. Every answer has supporting Google excerpts and section links. The source library preserves all main-article text, tables and code from the 94 licensed supplied pages; the supplied Google Support article has a labeled complete-topic brief and original link. Seven official supplements clarify current guidance. Linked subguides are not recursively included.

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All supplied links

1. [D01 · Seo starter guide](#)
2. [D02 · How search works](#)
3. [D03 · Creating helpful content](#)
4. [D04 · Ai optimization guide](#)
5. [D05 · Using gen ai content](#)
6. [D06 · Get started](#)
7. [D07 · Get started developers](#)
8. [D08 · Do i need seo](#)
9. [D09 · Third party seo](#)
10. [D10 · Crawling overview](#)
11. [D11 · Indexable file types](#)
12. [D12 · URL structure](#)
13. [D13 · Crawlable links](#)
14. [D14 · Sitemap overview](#)
15. [D15 · Build a sitemap](#)
16. [D16 · Sitemap indexes](#)
17. [D17 · Image sitemaps](#)
18. [D18 · News sitemaps](#)
19. [D19 · Video sitemaps](#)

20. [D20 · Combine extensions](#)
21. [D21 · Request recrawling](#)
22. [D22 · Crawling errors](#)
23. [D23 · Crawler overview](#)
24. [D24 · Googlebot](#)
25. [D25 · Reduce crawl rate](#)
26. [D26 · Verify Google requests](#)
27. [D27 · Robots.txt introduction](#)
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29. [D29 · Canonicalization](#)
30. [D30 · Choose canonical URLs](#)
31. [D31 · Canonical issues](#)
32. [D32 · Mobile-first indexing](#)
33. [D33 · JavaScript basics](#)
34. [D34 · Debug JavaScript](#)
35. [D35 · Lazy loading](#)
36. [D36 · Dynamic rendering](#)
37. [D37 · Valid metadata HTML](#)
38. [D38 · Supported meta tags](#)
39. [D39 · Robots meta & headers](#)
40. [D40 · Noindex](#)
41. [D41 · Outbound link qualifiers](#)
42. [D42 · Control sharing](#)
43. [D43 · Remove pages](#)
44. [D44 · Remove images](#)
45. [D45 · Safe redaction](#)
46. [D46 · Redirects](#)
47. [D47 · Hosting migration](#)
48. [D48 · URL migration](#)
49. [D49 · A/B testing](#)
50. [D50 · Temporary closure](#)
51. [D51 · Appearance overview](#)
52. [D52 · AI features](#)
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- 54. [D54 · Favicons](#)
- 55. [D55 · Featured snippets](#)
- 56. [D56 · Flexible sampling](#)
- 57. [D57 · Discover](#)
- 58. [D58 · Google Images](#)
- 59. [D59 · Business details](#)
- 60. [D60 · Top places lists](#)
- 61. [D61 · Property display opt-out](#)
- 62. [D62 · Page experience](#)
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- 85. [D85 · JavaScript structured data](#)
- 86. [D86 · Search gallery](#)
- 87. [D87 · Title links](#)

- 88. [D88 · Translated results](#)
- 89. [D89 · Ads & translation](#)
- 90. [D90 · Video SEO](#)
- 91. [D91 · Visual elements](#)
- 92. [D92 · Enable Web Stories](#)
- 93. [D93 · Story creation practices](#)
- 94. [D94 · Story content policies](#)
- 95. [D95 · Package tracking](#)

<https://developers.google.com/search/docs/fundamentals/seo-starter-guide>

Search Engine Optimization (SEO) Starter Guide

When you built your website, you likely created it with your users in mind, trying to make it easy for them to find and explore your content. One of those users is a search engine, which helps people discover your content. SEO—short for search engine optimization—is about helping search engines understand your content, and helping users find your site and make a decision about whether they should visit your site through a search engine.

The [Search Essentials](#) outline the most important elements of what makes your website eligible to appear on Google Search. While there's no guarantee that any particular site will be added to Google's index, sites that follow the **Search Essentials are more likely to show up in Google's search results**. SEO is about taking the next step and **working on improving your site's presence in Search**. This guide will walk you through some of the most common and effective improvements you can do on your site.

There are no secrets here that'll automatically rank your site first in Google (sorry!). In fact some of the suggestions might not even apply to your business, but following the best practices will hopefully make it easier for search engines (not just Google) to crawl, index, and understand your content.

How does Google Search work?

Google is a fully automated search engine that uses programs called crawlers to explore the web constantly, looking for pages to add to our index. You usually don't need to do anything except publish your site on the web. In fact, the vast majority of sites listed in our results are found and added automatically as we crawl the web. If you're hungry for more, we have documentation about how [Google discovers, crawls, and serves web pages](#).

Short on time or not feeling adventurous? You might consider hiring a professional. [Here's what to consider](#).

How long until I see impact in search results?

Every change you make will take some time to be reflected on Google's end. Some changes might take effect in a few hours, others could take several months. In general, you likely want to wait a few weeks to assess whether your work had beneficial effects in Google Search results. Keep in mind that not all changes you make to your website will result in noticeable impact in search results; if you're not satisfied with your results and your business strategies allow it, try iterating with the changes and see if they make a difference.

Help Google find your content

Before you actually do anything mentioned in this section, check if Google has already found your content (maybe you don't need to do anything!). Try searching on Google for your site with the [site: search operator](#). If you see results pointing to your site, you're in the index. For example, a search for `site:wikipedia.org` returns [these results](#). If you don't see your site, check out the [technical requirements](#) to make sure there's nothing technically preventing your site from showing in Google Search, and then come back here.

Google primarily finds pages through links from other pages it already crawled. In many cases, these are other websites that are linking to your pages. Other sites linking to you is something that happens naturally over time, and you can also encourage people to discover your content by [promoting your site](#).

If you're open to a little technical challenge, you could also [submit a sitemap](#)—which is a file that contains all the URLs on your site that you care about. Some content management systems (CMS) may even do this automatically for you. However this isn't required, and you should first focus on making sure [people know about your site](#).

Check if Google can see your page the same way a user does

When Google crawls a page, it should ideally [see the page the same way an average user does](#). For this, Google needs to be able to access the same resources as the user's browser. If your site is hiding important components that make up your website (like [CSS](#) and [JavaScript](#)), Google might not be able to understand your pages, which means they might not show up in search results or rank well for the terms you're targeting.

If your pages have different information depending on the user's physical location, make sure you're satisfied with the information that Google sees from its crawler's location, which is generally the US.

To check how Google sees your page, use the [URL Inspection Tool in Search Console](#).

Don't want a page in Google's search results?

It might be important for you to opt out your site as a whole or sections of it from appearing in search results. For example, you might not want your posts about your new embarrassing haircut to show up in search results. Google supports various ways that lets you opt out of crawling and indexing of your URLs. If you need to block some files, directories, or even your whole site from Google Search, check out our guide about [ways to prevent content from appearing in search results](#).

Organize your site

When you're setting up or redoing your site, it can be good to organize it in a logical way because it can help search engines and users understand how your pages relate to the rest of your site. Don't drop everything and start reorganizing your site right now though: while these suggestions can be helpful long term (especially if you're working on a larger website), search engines will likely understand your pages as they are right now, regardless of how your site is organized.

Use descriptive URLs

Parts of the URL can be displayed in search results as breadcrumbs, so users can also use the URLs to understand whether a result will be useful for them.

Google learns breadcrumbs automatically based on the words in the URL, but you can also influence them with [structured data](#) if you like a technical challenge. Try to include words in the URL that may be useful for users; for

example:

`https://www.example.com/pets/cats.html`

A URL that only contains random identifiers is less helpful for users; for example:

`https://www.example.com/2/6772756D707920636174`

Group topically similar pages in directories

[Original illustration: An illustration of how to group pages in directories]

If you have more than a few thousand URLs on your site, how you organize your content may have effects on how Google crawls and indexes your site. Specifically, using directories (or folders) to group similar topics can help Google learn how often the URLs in individual directories change.

For example, consider the following URLs:

`https://www.example.com/policies/return-policy.html`

`https://www.example.com/promotions/new-promos.html`

The content in the policies directory seldomly changes, however the content in the promotions directory likely changes very often. Google can learn this information and crawl the different directories at different frequencies. To learn more about search-friendly site structures, check out our [guide for ecommerce sites](#), for which a good URL structure is more important as they tend to be larger.

Reduce duplicate content

Some websites show the same content under different URLs, which is called *duplicate content*. Search engines choose a single URL (the *canonical* URL) to show users, per piece of content.

Having duplicate content on your site is not a violation of our spam policies, but it can be a bad user experience and search engines might waste crawling resources on URLs that you don't even care about. If you're feeling adventurous, it's worth figuring out if you can [specify a canonical version](#) for your pages. But if you don't canonicalize your URLs yourself, Google will try to automatically do it for you.

When working on canonicalization, try to ensure that each piece of content on your site is only accessible through one individual URL; having two pages that contain the same information about your promotions can be a confusing user experience (for example, people might wonder which is the right page, and whether there's a difference between the two).

If you have multiple pages that have the same information, try setting up a [redirect](#) from non-preferred URLs to a URL that best represents that information. If you can't redirect, use the `rel="canonical"` link element instead. But again, don't worry too much about this; search engines can generally figure this out for you on their own most of the time.

Make your site interesting and useful

Creating content that people find compelling and useful will likely influence your website's presence in search results more than any of the other suggestions in this guide. While "compelling and useful content" can mean different things to different people, content like this generally shares some common attributes, such as:

- **The text is easy-to-read and well organized:** Write content naturally and make sure the content is well written, easy to follow, and free of spelling and grammatical mistakes. Break up long content into paragraphs and sections, and provide headings to help users navigate your pages.
- **The content is unique:** When you're writing new content, don't copy others' content in part or in its entirety: create the content yourself based on what you know about the topic. Don't just rehash what others already published.
- **The content is up-to-date:** Check in on previously published content and update it as needed, or even delete it if it's not relevant anymore.
- **The content is helpful, reliable, and people-first:** Be sure that you're writing content that your readers will find helpful and reliable. For example, providing expert or experienced sources can help people understand your articles' expertise.

Expect your readers' search terms

Think about the words that a user might search for to find a piece of your content. Users who know a lot about the topic might use different keywords in their search queries than someone who is new to the topic. For example, some users might search for "charcuterie", while others might search for "cheese board". Anticipating these differences in search behavior and writing with your readers in mind could produce positive effects on how your site performs in search results.

However, don't worry if you don't anticipate every variation of how someone might seek your content. Google's language matching systems are sophisticated and can understand how your page relates to many queries, even if you don't explicitly use the exact terms in them.

Avoid distracting advertisements

While ads are a part of the internet and are meant to be seen by users, don't let them become overly distracting or prevent your users from reading your content. For example, advertisements, or [interstitial pages](#) (pages displayed before or after the content you're expecting) that make it difficult to use the website.

Link to relevant resources

Links are a great way to connect your users and search engines to other parts of your site, or relevant pages on other sites. In fact, the vast majority of the new pages Google finds every day are through links, making links a crucial resource you need to consider to help your pages be discovered by Google and potentially shown in search results. Additionally, links can also add value by connecting users (and Google) to another resource that corroborates what you're writing about.

[Original illustration: An illustration that shows how one web page is linking to other relevant resources]

Write good link text

Link text (also known as *anchor text*) is the text part of a link that you can see. This text tells users and Google something about the page you're linking to. With [appropriate anchor text](#), users and search engines can easily understand what your linked pages contain before they visit.

[Original illustration: An illustration that shows text part of a link]

Link when you need to

Links can provide more context on a topic, both for users and search engines, which may help demonstrate your knowledge on a topic. However when you're linking to pages outside of your control, for example content on other sites, make sure you trust the resource you're linking to. If you can't trust the content and you still want to link to them, add a [nofollow or similar annotation](#) to the link to avoid search engines associating your site with the site you're linking to. This helps avoid potential negative consequences in your rankings in Google Search.

If you're accepting user-generated content on your site, such as forum posts or comments, make sure every link that's posted by users has a `nofollow` or similar annotation automatically added by your CMS. Since you're not creating the content in this case, you likely don't want your site to be blindly associated with the sites users are linking to. This can also help discourage spammers from abusing your website.

Influence how your site looks in Google Search

[Original embedded video]

A typical Google Search results page consists of a [few different visual elements](#) that you can influence to help users decide whether they should visit your site through those search results. In this section, we're focusing on the *title link* and the *snippet* because these are the more visually significant elements.

Influence your title links

The *title link* is the headline part of the search result and it can help people decide which search result to click. There are a few sources that Google uses to generate this title link, including the words inside the `<title>` element (also called the title text) and other headings on the page. This title text can also be used for the title that's shown in browsers and bookmarks.

If you use a CMS, you might not need to do anything technical to your titles, beyond just focusing on writing good titles. Most CMSes can automatically turn the titles you write into a `<title>` element in the HTML.

[Original illustration: An illustration of how title text looks on a web page, and then how it looks in the HTML]

You can influence the title links in Search by writing good titles: a good title is unique to the page, clear and concise, and accurately describes the contents of the page. For example, your title could include the name of your website or business, other bits of important information like the physical location of the business, and maybe some information about what the particular page has to offer for users. Our [documentation about title links](#) has more tips about how to create good titles and how to influence your site's search results' title links.

Control your snippets

Below the title link, a search result typically has a description of the target page to help users decide whether they should click the search result. This is called a *snippet*.

The snippet is sourced from the actual content of the page the search result is linking to, thus you have complete control over the words that can be used to generate the snippet. Occasionally the snippet may be sourced from the contents of the meta description tag, which is typically a succinct, one- or two-sentence summary of the page. A good meta description is short, unique to one particular page, and includes the most relevant points of the page. Check out our tips for [writing good meta descriptions](#) for more inspiration.

Add images to your site, and optimize them

Many people search visually, and images can be how people find your website for the first time. For example, if you have a recipe blog, people might find your content by searching for "fruit tart recipes" and browsing photos of various types of fruit tarts.

As you add images to your site, make sure that people and search engines can find and understand them.

Add high-quality images near relevant text

When you use high quality images, you give users enough context and detail to decide which image best matches what they were looking for. For example, if people are looking for "daisies" and come across a rogue edelweiss in search results, a higher quality image would help them distinguish the type of flower.

Use images that are sharp and clear, and place them near text that's relevant to the image. The text that's near images can help Google better understand what the image is about and what it means in context to your page.

For example, if the page is reviewing yarn shops in London, then it would make sense to embed one of your photos of the yarn shop in the section that details the location, description, and review information for that yarn shop. This helps Google and users associate the image with text that provides more context to what the page is about.

Add descriptive alt text to the image

Alt text is a short, but descriptive piece of text that explains the relationship between the image and your content. It helps search engines understand what your image is about and the context of how your image relates to your page, so writing [good alt text](#) is quite important. You can add this to your HTML with the alt attribute of the `img` element, or your CMS may have an easy way to specify a description for an image when you're uploading it to your site. Learn more about [how to write good alt text](#), and how to add it to your images.

Optimize your videos

If your website includes pages that are primarily about individual videos, people may also be able to discover your site through video results in Google Search. Many of the best practices for images and text also apply to videos:

- Create high-quality video content, and embed the video on a standalone page, near text that's relevant to that video.
- Write descriptive text in the titles and description fields of a video (the title of a video is still a title, and so you can apply the best practices for writing titles here too).

If your site is particularly video-focused, then continue reading about more things you can do to [optimize your videos for search engines](#).

Promote your website

Effectively promoting your new content will lead to faster discovery by those who are interested in the same subject, and also by search engines. You can do this in many ways:

- Social media promotion
- Community engagement
- Advertisement, both offline and online
- Word of mouth, and many other methods

One of the [most effective and lasting ways](#) is word of mouth: that is, people familiar with your site tell their friends about it, who in turn visit your site. This can take time, and usually you need to invest some time and effort in other practices first, such as community engagement. Our friends over at Google for Creators have excellent resources about [building and engaging your audience](#).

Putting effort into the offline promotion of your company or site can also be rewarding. For example, if you have a business site, make sure its URL is listed on your business cards, letterhead, posters, and other materials. With their permission, you could also send out recurring newsletters to your audience letting them know about new content on your website.

As with everything in life, you can overdo promoting your site and actually harm it: people may get fatigued of your promotions, and search engines may perceive some of the practices as [manipulation of search results](#).

Things we believe you shouldn't focus on

As SEO has evolved, so have the ideas and practices (and at times, misconceptions) related to it. What was considered best practice or top priority in the past may no longer be relevant or effective due to the way search engines (and the internet) have developed over time.

To help you focus on the things that are actually important when it comes to SEO, we collected some of the most common and prominent topics we've seen circulating the internet. In general, our message on these topics is that you should do what's best for your business area; we will elaborate on a few specific points here:

	Meta keywords Google Search doesn't use the keywords meta tag . Keyword stuffing	Subdomains versus subdirectories From a business point of view, do whatever makes sense for your business. For
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Excessively repeating the same words over and over (even in variations) is tiring for users, and [keyword stuffing is against Google's spam policies](#).

Keywords in the domain name or URL path

When picking the name of your site, do what's best for your business. Users will use this name to find you, so we recommend following general marketing best practices. From a ranking perspective, the keywords in the name of the domain (or URL path) alone have hardly any effect beyond appearing in [breadcrumbs](#).

And while still on the topic of domain names: the TLD (the domain name ending like ".com" or ".guru") only matters if you're targeting a specific country's users, and even then it's usually a low impact signal. For example, if you're trying to sell Dutch cheese to people searching from Switzerland, it makes some sense (both from business and [SEO point of view](#)) to use a .ch domain name. Otherwise Google Search doesn't care which TLD you're using (whether it's a .com or .org or .asia).

Minimum or maximum content length

The length of the content alone doesn't matter for ranking purposes (there's no magical word count target, minimum or maximum, though you probably want to have at least one word). If you are varying the words (writing naturally to not be repetitive), you have more chances to show up in Search

example, it might be easier to manage the site if it's segmented by subdirectories, but other times it might make sense to partition topics into subdomains, depending on your site's topic or industry.

PageRank

While [PageRank](#) uses links and is one of the fundamental algorithms at Google, there's much more to Google Search than just links. We have many ranking signals, and PageRank is just one of those.

Duplicate content "penalty"

If you have some content that's accessible under multiple URLs, it's fine; don't fret about it. It's inefficient, but it's not something that will cause a manual action. [Copying others' content, however, is a different story](#).

Number and order of headings

Having your headings in semantic order is fantastic for screen readers, but from Google Search perspective, it doesn't matter if you're using them out of order. The web in general is not valid HTML, so Google Search can rarely depend on semantic meanings hidden in the HTML specification.

There's also no magical, ideal amount of headings a given page should have. However, if you think it's too much, then it probably is.

Thinking E-E-A-T is a ranking factor

[No, it's not](#).

	simply because you are using more keywords.	
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Next steps

- **Get started with Search Console:** Setting up a Search Console account helps you monitor and optimize how your website performs on Google Search. Learn how to [set up your account and what reports to check out first](#).
- **Maintain your website's SEO over time:** Learn more about [managing your site's presence in the long term](#), including more in-depth SEO tasks and scenarios, such as preparing for a site move, or managing a multi-lingual site.
- **Enhance how your site looks in Google Search results:** Valid [structured data](#) on your pages also makes your pages eligible for many special features in Google Search results, including review stars, carousels, and more. Explore the [gallery of search result types](#) that your page can be eligible for.

Stay informed and ask questions

As you embark on your SEO journey, here are some resources that can help you stay on top of changes and new resources we publish:

	• Google Search Central blog: Get the latest information from our Google Search Central blog. You can find information about updates to Google Search, new Search Console features, and much more. • Google Search Central on LinkedIn and X (Twitter): Follow us for updates on Google Search and resources to help you make a great site.	• Google Search Central Help Forum: Post questions about your site's SEO issues and find tips to create high quality sites from the product forum for website owners. There are many experienced contributors in the forum, including Product Experts and occasionally Googlers. • Google Search Central YouTube Channel: Watch hundreds of helpful videos created for website owners.
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<https://developers.google.com/search/docs/fundamentals/how-search-works>

In-depth guide to how Google Search works

Google Search is a fully-automated search engine that uses software known as web crawlers that explore the web regularly to find pages to add to our index. In fact, the vast majority of pages listed in our results aren't manually submitted for inclusion, but are found and added automatically when our web crawlers explore the web. This document explains the stages of how Search works in the context of your website. Having this base knowledge can help you fix crawling issues, get your pages indexed, and learn how to optimize how your site appears in Google Search.

Looking for something less technical? Check out our [How Search Works site](#), which explains how Search works from a searcher's perspective.

A few notes before we get started

Before we get into the details of how Search works, it's important to note that Google doesn't accept payment to crawl a site more frequently, or rank it higher. If anyone tells you otherwise, they're wrong.

Google doesn't guarantee that it will crawl, index, or serve your page, even if your page follows the [Google Search Essentials](#).

Introducing the three stages of Google Search

Google Search works in three stages, and not all pages make it through each stage:

1. **Crawling:** Google downloads text, images, and videos from pages it found on the internet with automated programs called crawlers.
2. **Indexing:** Google analyzes the text, images, and video files on the page, and stores the information in the Google index, which is a large database.
3. **Serving search results:** When a user searches on Google, Google returns information that's relevant to the user's query.

Crawling

The first stage is finding out what pages exist on the web. There isn't a central registry of all web pages, so Google must constantly look for new and updated pages and add them to its list of known pages. This process is called "URL discovery". Some pages are known because Google has already visited them. Other pages are discovered when Google extracts a link from a known page to a new page: for example, a hub page, such as a category page, links to a new blog post. Still other pages are discovered when you submit a list of pages (a [sitemap](#)) for Google to crawl.

[Original embedded video]

Once Google discovers a page's URL, it may visit (or "crawl") the page to find out what's on it. We use a huge set of computers to crawl billions of pages on the web. The program that does the fetching is called [Googlebot](#) (also

known as a crawler, robot, bot, or spider). Googlebot uses an algorithmic process to determine which sites to crawl, how often, and how many pages to fetch from each site. [Google's crawlers](#) are also programmed such that they try not to crawl the site too fast to avoid overloading it. This mechanism is based on the responses of the site (for example, [HTTP 500 errors mean "slow down"](#)).

However, Googlebot doesn't crawl all the pages it discovered. Some pages may be [disallowed for crawling](#) by the site owner, other pages may not be accessible without logging in to the site.

During the crawl, Google renders the page and [runs any JavaScript it finds](#) using a recent version of [Chrome](#), similar to how your browser renders pages you visit. Rendering is important because websites often rely on JavaScript to bring content to the page, and without rendering Google might not see that content.

Crawling depends on whether Google's crawlers can access the site. Some common issues with Googlebot accessing sites include:

- [Problems with the server handling the site](#)
- [Network issues](#)
- [robots.txt rules preventing Googlebot's access to the page](#)

Indexing

After a page is crawled, Google tries to understand what the page is about. This stage is called indexing and it includes processing and analyzing the textual content and key content tags and attributes, such as [<title> elements](#) and alt attributes, [images](#), [videos](#), and more.

[\[Original embedded video\]](#)

During the indexing process, Google determines if a page is a [duplicate of another page on the internet or canonical](#). The canonical is the page that may be shown in search results. To select the canonical, we first group together (also known as clustering) the pages that we found on the internet that have similar content, and then we select the one that's most representative of the group. The other pages in the group are alternate versions that may be served in different contexts, like if the user is searching from a mobile device or they're looking for a very specific page from that cluster.

Google also collects signals about the canonical page and its contents, which may be used in the next stage, where we serve the page in search results. Some signals include the language of the page, the country the content is local to, and the usability of the page.

The collected information about the canonical page and its cluster may be stored in the Google index, a large database hosted on thousands of computers. Indexing isn't guaranteed; not every page that Google processes will be indexed.

Indexing also depends on the content of the page and its metadata. Some common indexing issues can include:

- [The quality of the content on page is low](#)
- [Robots meta rules disallow indexing](#)
- [The design of the website might make indexing difficult](#)

Serving search results

Google doesn't accept payment to rank pages higher, and ranking is done programmatically. [Learn more about ads on Google Search.](#)

When a user enters a query, our machines search the index for matching pages and return the results we believe are the highest quality and most relevant to the user's query. Relevancy is determined by hundreds of factors, which could include information such as the user's location, language, and device (desktop or phone). For example, searching for "bicycle repair shops" would show different results to a user in Paris than it would to a user in Hong Kong.

[Original embedded video]

Based on the user's query the search features that appear on the search results page also change. For example, searching for "bicycle repair shops" will likely show local results and no [image results](#), however searching for "modern bicycle" is more likely to show image results, but not local results. You can explore the most common UI elements of Google web search in our [Visual Element gallery](#).

Search Console might tell you that a page is indexed, but you don't see it in search results. This might be because:

- [The content on the page is irrelevant to users' queries](#)
- [The quality of the content is low](#)
- [Robots meta rules prevent serving](#)

While this guide explains how Search works, we are always working on improving our algorithms. You can keep track of these changes by following the [Google Search Central blog](#).

<https://developers.google.com/search/docs/fundamentals/creating-helpful-content>

Creating helpful, reliable, people-first content

Google's [automated ranking systems](#) are designed to prioritize helpful, reliable information that's created to benefit people, and not content that's created to manipulate search engine rankings. This page is designed to help creators evaluate if they're producing such content.

Self-assess your content

Evaluating your own content against these questions can help you gauge if the content you're making is helpful and reliable. Beyond asking yourself these questions, consider having others you trust but who are unaffiliated with your site provide an honest assessment.

Also consider an audit of the drops you may have experienced. What pages were most impacted and for what types of searches? Look closely at these to understand how they're assessed against some of the questions outlined here.

Content and quality questions

- Does the content provide original information, reporting, research, or analysis?
- Does the content provide a substantial, complete, or comprehensive description of the topic?
- Does the content provide insightful analysis or interesting information that is beyond the obvious?
- If the content draws on other sources, does it avoid simply copying or rewriting those sources, and instead provide substantial additional value and originality?
- Does the main heading or page title provide a descriptive, helpful summary of the content?
- Does the main heading or page title avoid exaggerating or being shocking in nature?
- Is this the sort of page you'd want to bookmark, share with a friend, or recommend?
- Would you expect to see this content in or referenced by a printed magazine, encyclopedia, or book?
- Does the content provide substantial value when compared to other pages in search results?
- Does the content have any spelling or stylistic issues?
- Is the content produced well, or does it appear sloppy or hastily produced?
- Is the content mass-produced by or outsourced to a large number of creators, or spread across a large network of sites, so that individual pages or sites don't get as much attention or care?

Expertise questions

- Does the content present information in a way that makes you want to trust it, such as clear sourcing, evidence of the expertise involved, background about the author or the site that publishes it, such as through links to an author page or a site's About page?
- If someone researched the site producing the content, would they come away with an impression that it is well-trusted or widely-recognized as an authority on its topic?
- Is this content written or reviewed by an expert or enthusiast who demonstrably knows the topic well?
- Does the content have any easily-verified factual errors?

Provide a great page experience

Google's core ranking systems look to reward content that provides a good page experience. Site owners seeking to be successful with our systems should not focus on only one or two aspects of page experience. Instead, check if you're providing an overall great page experience across many aspects. For more advice, see our page, [Understanding page experience in Google Search results](#).

Focus on people-first content

People-first content means content that's created primarily for people, and not to manipulate search engine rankings. How can you evaluate if you're creating people-first content? Answering yes to the questions below means you're probably on the right track with a people-first approach:

- Do you have an existing or intended audience for your business or site that would find the content useful if they came directly to you?
- Does your content clearly demonstrate first-hand expertise and a depth of knowledge (for example, expertise that comes from having actually used a product or service, or visiting a place)?
- Does your site have a primary purpose or focus?
- After reading your content, will someone leave feeling they've learned enough about a topic to help achieve their goal?
- Will someone reading your content leave feeling like they've had a satisfying experience?

Avoid creating search engine-first content

We recommend that you focus on creating people-first content to be successful with Google Search, rather than search engine-first content made primarily to gain search engine rankings. Answering yes to some or all of the questions below is a warning sign that you should reevaluate how you're creating content:

- Is the content primarily made to attract visits from search engines?
- Are you producing lots of content on many different topics in hopes that some of it might perform well in search results?
- Are you using extensive automation to produce content on many topics?
- Are you mainly summarizing what others have to say without adding much value?
- Are you writing about things simply because they seem trending and not because you'd write about them otherwise for your existing audience?
- Does your content leave readers feeling like they need to search again to get better information from other sources?
- Are you writing to a particular word count because you've heard or read that Google has a preferred word count? (No, we don't.)
- Did you decide to enter some niche topic area without any real expertise, but instead mainly because you thought you'd get search traffic?
- Does your content promise to answer a question that actually has no answer, such as suggesting there's a release date for a product, movie, or TV show when one isn't confirmed?
- Are you changing the date of pages to make them seem fresh when the content has not substantially changed?
- Are you adding a lot of new content or removing a lot of older content primarily because you believe it will help your search rankings overall by somehow making your site seem "fresh?" (No, it won't)

What about SEO? Isn't that search engine-first?

There are some things you could do that are specifically meant to help search engines better discover and understand your content. Collectively, this is called "search engine optimization" or SEO, for short. [Google's own SEO guide](#) covers best practices to consider. SEO can be a helpful activity when it is applied to people-first content, rather than search engine-first content.

Get to know E-E-A-T and the quality rater guidelines

Google's automated systems are designed to use [many different factors](#) to rank great content. After identifying relevant content, our systems aim to prioritize those that seem most helpful. To do this, they identify a mix of factors that can help determine which content demonstrates aspects of experience, expertise, authoritativeness, and trustworthiness, or what we call E-E-A-T.

Of these aspects, trust is most important. The others contribute to trust, but content doesn't necessarily have to demonstrate all of them. For example, some content might be helpful based on the experience it demonstrates, while other content might be helpful because of the expertise it shares.

While E-E-A-T itself isn't a specific ranking factor, using a mix of factors that can identify content with good E-E-A-T is useful. For example, our systems give even more weight to content that aligns with strong E-E-A-T for topics that could significantly impact the health, financial stability, or safety of people, or the welfare or well-being of society. We call these "Your Money or Your Life" topics, or YMYL for short.

[Search quality raters](#) are people who give us insights on if our algorithms seem to be providing good results, a way to help confirm our changes are working well. In particular, raters are trained to understand if content has strong E-E-A-T. The criteria they use to do this is outlined in our [search quality rater guidelines](#).

Search raters have no control over how pages rank. Rater data is not used directly in our ranking algorithms. Rather, we use them as a restaurant might get feedback cards from diners. The feedback helps us know if our systems seem to be working.

Reading the guidelines may help you self-assess how your content is doing from an E-E-A-T perspective, improvements to consider, and help align it conceptually with the different signals that our automated systems use to rank content.

Beyond E-E-A-T, the [Search Quality Evaluator Guidelines](#) emphasize that one of the most critical factors for assessing page quality is the quality of the main content, specifically how effectively it allows the page to achieve its purpose and provide a satisfying user experience.

Main content

Google Search defines "main content" as any part of the web page that directly helps the page achieve its purpose. Depending on the primary intent of the page, the main content directly fulfills the page's purpose and could include:

- **Primary text and media:** The main text, articles, images, audio, or video files on the page.
- **Interactive features:** Built-in functionality such as calculators, online tools, games, or search functionality.
- **User-generated contributions:** Forum discussions, customer reviews, comments, or user-uploaded media when they directly fulfill the page's purpose (for example, when the page is about discussing a particular user question).

- **Tabbed or expanded sections:** Information housed behind interactive tabs (such as product specifications, safety notes, or user reviews).
- **Page titles and headings:** The visible title and headings throughout a page that summarize its topic and helps users make informed decisions about visiting.

When reviewing content quality, Search Quality raters are trained to evaluate these fundamental attributes of the main content:

- **Effort:** The extent to which human work went into creating the content or the systems powering it. For example, writing original analysis, translating a poem manually, or building a custom interactive map for a game represents high effort. Conversely, automatically generating pages from feeds or using generative AI to produce large amounts of text without manual oversight or curation represents little to no effort. Attribution or giving credit to other sources doesn't replace the need for original effort.
- **Originality:** The extent to which the content offers unique, original information or perspectives that aren't already available on other websites.
- **Talent or skill:** Whether the content shows the talent, skill, or expertise necessary to provide a satisfying experience for visitors (such as clear writing, well-produced video, or functional page tools). Keep in mind that not all content requires specialized expertise. For example, a person sharing their experience about finding a new way to clear their driveway of snow generally doesn't necessarily require specialized expertise.
- **Accuracy:** For informational pages, the content should be factually accurate. For topics that could significantly impact people's lives or well-being (YMYL topics), the content must be highly accurate and consistent with established expert consensus.

Ask "Who, How, and Why" about your content

Consider evaluating your content in terms of "Who, How, and Why" as a way to stay on course with what our systems seek to reward.

Who (created the content)

Something that helps people intuitively understand the E-E-A-T of content is when it's clear who created it. That's the "Who" to consider. When creating content, here are some who-related questions to ask yourself:

- Is it self-evident to your visitors who authored your content?
- Do pages carry a byline, where one might be expected?
- Do bylines lead to further information about the author or authors involved, giving background about them and the areas they write about?

If you're clearly indicating who created the content, you're likely aligned with the concepts of E-E-A-T and on a path to success. We strongly encourage adding accurate authorship information, such as bylines to content where readers might expect it.

However, avoid using deceptive authorship information. Fabricating creator profiles (such as by using AI-generated headshots, made-up names, or false credentials to make content appear as if it was written by human experts) is a form of deception. Any form of deception makes a page untrustworthy to both users and our automated quality systems, and is a signal of a low-quality page.

How (the content was created)

It's helpful to readers to know how a piece of content was produced: this is the "How" to consider including in your content.

For example, with product reviews, it can build trust with readers when they understand the number of products that were tested, what the test results were, and how the tests were conducted, all accompanied by evidence of the work involved, such as photographs. It's advice we share more about in our [Write high quality product reviews help page](#).

Many types of content may have a "How" component to them. That can include automated, AI-generated, and AI-assisted content. Sharing details about the processes involved can help readers and visitors better understand any unique and useful role automation may have served.

If automation is used to substantially generate content, here are some questions to ask yourself:

- Is the use of automation, including AI-generation, self-evident to visitors through disclosures or in other ways?
- Are you providing background about how automation or AI-generation was used to create content?
- Are you explaining why automation or AI was seen as useful to produce content?

Overall, AI or automation disclosures are useful for content where someone might think "How was this created?" Consider adding these when it would be reasonably expected. For more, see our blog post and FAQ: [How Google Search views AI-generated content](#).

Why (was the content created)

"Why" is perhaps the most important question to answer about your content. Why is it being created in the first place?

The "why" should be that you're creating content primarily to help people, content that is useful to visitors if they come to your site directly. If you're doing this, you're aligning with E-E-A-T generally and what our [core ranking systems](#) seek to reward.

If the "why" is that you're primarily making content to attract search engine visits, that's not aligned with what our systems seek to reward. If you use automation, including AI-generation, to produce content for the primary purpose of manipulating search rankings, that's a [violation of our spam policies](#).

<https://developers.google.com/search/docs/fundamentals/ai-optimization-guide>

Optimizing your website for generative AI features on Google Search

User preferences are rapidly evolving and people are increasingly gravitating to generative AI experiences to help them find information. As we upgrade Search to meet these changing expectations, this transformation offers new opportunities to reach people who may be more inclined to engage with your site, spend more time with your content, or even convert by becoming a subscriber or making a purchase. This guide is for website owners looking for official best practices from Google Search on how to succeed in generative AI features in Google Search (such as AI Overviews and AI Mode).

Is SEO still relevant for generative AI search?

In short, yes! The [best practices for SEO](#) continue to be relevant because our generative AI features on Google Search are rooted in our core Search ranking and quality systems. These features rely on AI techniques to highlight content from our Search index, such as:

- **Retrieval-augmented generation (RAG):** A technique (also known as grounding) used to improve the quality, accuracy, and freshness of AI responses by relying on our core Search ranking systems to retrieve relevant, up-to-date web pages from our Search index. Our systems then review the specific information from those retrieved pages to generate a more reliable and helpful response, showing prominent, clickable links to relevant web pages that support the information in the response.
- **Query fan-out:** A set of concurrent, related queries generated by the model to request more information and fetch additional relevant search results to address the user's query. For example, if the original user's query is "how to fix a lawn that's full of weeds", fanout queries might include "best herbicides for lawns", "remove weeds without chemicals", and "how to prevent weeds in lawn".

What about "AEO" and "GEO"? "AEO" stands for "answer engine optimization" and "GEO" for "generative engine optimization". These are both terms you may see used to describe work specifically focused on improving visibility in AI search experiences. From Google Search's perspective, optimizing for generative AI search is optimizing for the search experience, and thus still SEO. If you're considering third-party "AEO" or "GEO" advice or services, review our [guidance on evaluating third-party SEO advice](#).

Apply foundational SEO best practices to generative AI search

This section focuses on reframing SEO best practices to understand what matters most to AI systems today and how you can implement them in the context of generative AI search, and ultimately with the goal of improving your website's visibility in both generative AI search experiences and Google Search overall.

Create valuable, non-commodity content for your audience

Creating content that people find unique, compelling, and useful will likely influence your website's presence in generative AI search in the long run more than any of the other suggestions in this guide. While "unique, valuable,

good content" can mean different things to different people, content like this generally shares some common attributes, such as:

- **Providing a unique point of view:** Our AI systems take a look at a variety of sources, so it can be helpful to have a unique viewpoint that stands out. For example, a first-hand review provides a unique perspective based on personal experience, whereas a summary of existing content simply restates information already available elsewhere. Create the content yourself based on what you know about the topic, and consider what in-depth experience you can bring to your content. Don't just recycle what others on the internet have already said, or could easily be produced by a generative AI model.
- **Creating non-commodity content that's helpful, reliable, and people-first:** Be sure that you're writing non-commodity content that your readers will find helpful and reliable. Commodity content (for example, something like "7 Tips for First-Time Homebuyers") is often based on common knowledge, which could originate from anyone, and typically adds little unique insight for readers. In contrast, non-commodity content (such as "Why We Waived the Inspection & Saved Money: A Look Inside the Sewer Line") provides unique expert or experienced takes that go beyond common knowledge and the ordinary.
- **Organizing content in a way that helps your readers:** Write content for your human audience and make sure the content is well written and easy to follow. People generally appreciate it when web pages are organized by paragraphs and sections, along with headings that provide a clear structure to navigate content.
- **Add high-quality images and video:** Many people appreciate finding images and videos as they search for things online. As with Google Search overall, our generative AI search features can bring in relevant images and video, which means more opportunities for your website to appear beyond web page links. When it makes sense, look for ways to support your textual content with high-quality, relevant images and videos on your pages. If you're already following our [image SEO best practices](#) and [video SEO documentation](#), you're already optimizing for generative AI search.
- **Focus on what your users want, and avoid overdoing it.** While it might be tempting to create separate content for every possible variation of how people might search (for example, by focusing on other queries that people have asked, or fan-out queries), doing so primarily to manipulate rankings or generative AI responses in Google Search violates Google's [scaled content abuse spam policy](#). This is also an ineffective long-term strategy, as a high quantity of pages doesn't make a website higher quality or more relevant to users. Google's AI systems have advanced even further and improved upon our ability to [understand the relevance of pages](#), even when there is no exact match between the query and the page's primary content.
- **If you're using generative AI tools to assist in content creation,** be sure that your work meets the standards of the [Search Essentials](#) and our [spam policies](#). For more details on our approach, see our [guidance on AI-generated content](#).

You can simplify your approach by focusing on one core principle: focus on what your visitors would enjoy, find helpful, and feel satisfied with after visiting your website. If you're ever unsure about a decision for your site, ask yourself: "Is this content that my visitors would find satisfying?" If the answer is yes, then you're on the right track, as our systems are designed to connect people with exactly that kind of useful information. For more, check out our guide to [creating helpful, reliable, people-first content](#).

Build and maintain a clear technical structure

The way Google Search finds and processes your pages remains the core of how our AI systems access your data. Technical clarity ensures your content is ready for discovery and indexing, and all existing technical [SEO best practices](#) continue to be worthwhile, for example:

- **Meet the Search technical requirements:** To be eligible to be shown in generative AI features on Google Search, a page must be indexed and eligible to be shown in Google Search with a snippet, fulfilling the [Search technical requirements](#).

In addition to the technical requirements for Search, a site must be [included in Search generative AI features in Search Console](#) to be eligible for display in generative AI features on Google Search.

Just because a page meets all requirements, best practices, and complies with the policies, doesn't mean that Google will crawl, index, or serve its content. Indexing and serving aren't guaranteed. Learn more about [How Search Works](#).

- **Follow crawling best practices.** To maximize your site's visibility in generative AI search features, ensure your content is crawlable, as Google Search generative AI models use publicly accessible, crawlable content to learn patterns and provide relevant, grounded responses. For very large and frequently updated sites, review our guide to [optimizing your crawl budget](#).
- **When it comes to semantic HTML, focus on human readability and don't worry about perfect code:** While it's not required to have perfectly semantic HTML (the web in general is not valid HTML, and Google can understand it), it's generally a good idea to try to use semantic HTML when possible, as it helps other types of users, such as screen readers, [parse and navigate your web page more easily](#).
- **If you're using JavaScript, be sure to follow JavaScript SEO best practices.** Google is able to process content within JavaScript as long as it isn't blocked. That said, working on SEO with a website that uses JavaScript frameworks is generally more complex than when working with other kinds of websites. Make sure to follow the usual [SEO best practices for JavaScript](#).
- **Provide a good page experience** for those who arrive at your site. This includes ensuring your site displays well across all devices, reducing latency, and making it easy for people to distinguish your main content from other elements on the page.
- **Reduce duplicate content:** Having duplicate content can be a bad user experience and search engines might waste crawling resources on URLs that you don't even care about. If you have time, [try to reduce it](#).

To discover and diagnose potential technical issues quickly, [verify your site in Search Console](#). For more, check out our [technical guide to SEO](#) and [maintaining your website's SEO](#).

Optimize your local business and ecommerce details

Where appropriate, generative AI responses can include product listings, product information, and information about local businesses. Using products like [Merchant Center](#) (such as [Merchant Center feeds](#)) and [Google Business Profiles](#) can help your products and services to be visible in both AI responses and other Google Search results. Learn more about how to [add and manage your business details on Google Search](#).

Depending on your business type and goals, consider other merchant experiences like [Business Agent](#), which is a conversational experience on Google Search that helps customers chat with your brand.

Mythbusting generative AI search: what you don't need to do

As generative AI search evolves, so have the theories and practices—and sometimes, the misconceptions—surrounding it. While terms like Answer Engine Optimization (AEO) or Generative Engine Optimization (GEO) are common online, many suggested "hacks" aren't effective or supported by how Google Search actually works.

To help you focus on what matters for your website's visibility, we've collected some of the most prominent topics circulating the internet around generative AI and Google Search. Here are a few things you can ignore for Google Search:

- **LLMS.txt files and other "special" markup:** You don't need to create new machine readable files, AI text files, markup, or Markdown to appear in Google Search (including its generative AI capabilities), as Google Search itself doesn't use them. Note that Google may discover, crawl, and index [many kinds of files](#) in addition to HTML on a website: this doesn't mean that the file is treated in a special way.

It's completely fine if you decide to create and maintain LLMS.txt files (or other similar files) for other services or systems that use these files. Doing so will neither harm nor help your site's visibility or rankings in Google Search, as Google Search ignores them.

- **"Chunking" content:** There's no requirement to break your content into tiny pieces for AI to better understand it. Google systems are able to understand the nuance of multiple topics on a page and show the relevant piece to users. However, sometimes shorter (or longer!) pages can work well depending on your audience and subject matter. There's no ideal page length, and in the end, make pages for your audience, not just for generative AI search.
- **Rewriting content just for AI systems:** You don't need to write in a specific way just for generative AI search. AI systems can understand synonyms and general meanings of what someone is seeking, in order to connect them with content that might not use the same precise words. This means you don't have to worry that you don't have enough "long-tail" keywords or haven't captured every variation of how someone might seek content like yours.
- **Seeking inauthentic "mentions":** Just like the rest of Google Search, our generative AI features can show what's being said about products and services across the web, including in blogs, videos, and forum discussions. However, seeking inauthentic "mentions" across the web isn't as helpful as it might seem. Our core ranking systems focus on [high-quality content](#) while other systems [block spam](#); our generative AI features depend on both.
- **Overfocusing on structured data:** Structured data isn't required for generative AI search, and there's no special schema.org markup you need to add. However, it's a good idea to continue using it as part of your overall SEO strategy, as it helps with being eligible for rich results on Google Search.

Measure visibility in generative AI features with Search Console

To measure how your content is performing in generative AI features on Google Search and Discover, use the [Generative AI performance report](#) in Search Console. This can help you get an idea of how people are discovering your content through generative AI features on Google Search.

Be wary of third-party tools that promise ranking success or claim to use "internal" Google metrics. No third-party tool has access to our internal ranking or AI systems. Use these tools if they help your workflow, but be sure to [evaluate their advice against our official guidance](#).

Explore agentic experiences

AI agents are autonomous systems that can perform tasks on behalf of people, such as booking a reservation or comparing product specifications. These agents can take many forms; for example, browser agents may access your website to gather the data they need to complete these tasks, such as analyzing visual renderings (like screenshots), inspecting the DOM structure, and interpreting the accessibility tree.

If this is something that's relevant to your business and you have extra time, check out the available agentic experiences and review the guide to [agent-friendly website best practices](#), which gives some insights into how a

website can generally prepare for current browser agents. Protocols like [Universal Commerce Protocol \(UCP\)](#) are emerging that will allow Search agents to do more.

Next steps: what to focus on

As you continue working on your website, remember that plenty of content thrives in Google Search (including generative AI experiences) without any overt SEO at all, and you don't need to accomplish everything in this guide in order to succeed on Google Search. To recap, here are the key takeaways from this guide:

- **Apply SEO best practices to generative AI search:** Continue prioritizing foundational SEO best practices, such as [building a clear technical structure](#) and [creating unique, valuable content](#); these are the foundation for visibility in generative AI search experiences (and Google Search overall).
- **Create non-commodity content that's helpful, reliable, and people-first:** Focus on developing unique, expert-led content that provides value beyond common knowledge.
- **Prioritize effective SEO strategies over "AEO/GEO hacks":** For Google Search, you can ignore tactics like "chunking" content, creating unnecessary AI text files (like llms.txt), or pursuing inauthentic mentions.
- **Monitor visibility in Search Console:** Use the [Generative AI performance report](#) to see how your content is performing in generative AI features on Google Search.
- **Explore agentic experiences:** Stay informed about emerging technologies that allow AI agents to interact with your site, such as browser agents and new protocols.

Stay informed and ask questions

If you want to learn more about SEO, here are some resources that can help you stay on top of changes and new resources we publish:

	• Google Search Central blog: Get the latest information from our Google Search Central blog. You can find information about generative AI in Search, new Search Console features, and much more. • Google Search Central on LinkedIn and X (Twitter): Follow us for updates on Google Search and resources to help you make a great site.	• Google Search Central Help Forum: Post questions about your site's SEO issues and find tips to create high quality sites from the product forum for website owners. There are many experienced contributors in the forum, including Product Experts and occasionally Googlers. • Google Search Central YouTube Channel: Watch hundreds of helpful videos created for website owners.
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<https://developers.google.com/search/docs/fundamentals/using-gen-ai-content>

Google Search's guidance on using generative AI content on your website

Generative AI can be particularly useful when researching a topic, and to add structure to original content. However, using generative AI tools or other similar tools to generate many pages without adding value for users may violate [Google's spam policy on scaled content abuse](#). If you're using generative AI content on your website, **make sure your work meets the standards of the [Search Essentials](#) and our [spam policies](#).**

You might find value in looking at the [Search Quality Raters guidelines](#) on how to evaluate both scaled content abuse (section 4.6.5) and main content created with little to no effort, little to no originality, and little to no added value (section 4.6.6). These guidelines are not a guide to ranking first in Google; they're used by our [search raters](#) to help evaluate the performance of our [various search ranking systems](#), and their ratings don't directly influence ranking.

Focus on accuracy, quality, and relevance

When creating content for the web, focus on accuracy, quality, and relevance, especially when automatically generating the content. Keep in mind that generative models don't retrieve facts, but predict a likely sequence of words based on their training data. Because of this, generative AI outputs may contain inaccuracies (also known as hallucinations). It is critical to manually factcheck and review all AI-generated content for accuracy and trustworthiness before publishing. This review also applies to metadata like [<title> elements](#), [meta description elements](#), [structured data](#), and [alternate texts for images](#), which can appear in Search results.

For structured data, also ensure compliance with the [general guidelines](#), the specific policies for the individual search features, and [validate the markup](#) to ensure eligibility for [Search features](#).

Give users context

Sharing [information about how a piece of content was created](#) can help give your readers more context. If you're automatically generating content, consider adding information on how your content was created in a way that makes sense for your audience, such as by providing more background information on how automation was used and adding [image metadata](#).

For ecommerce sites, Google Merchant Center has [policies for AI-generated content](#). In particular, AI-generated images must contain metadata using the IPTC DigitalSourceType [TrainedAlgorithmicMedia](#) metadata. AI-generated product data such as title and description attributes must be specified separately and labeled as AI-generated.

For more, see our [FAQs in our blog post on AI-generated content](#).

<https://developers.google.com/search/docs/fundamentals/get-started>

Maintaining your website's SEO

If your site is on Google and you're familiar with the [fundamentals of SEO](#), there are more things you can do to improve how your site appears on Google. As you manage and maintain your website, you may come across more unique scenarios that affect Google Search. This guide covers more in-depth SEO tasks, such as preparing for a site move or managing a multi-lingual site.

Control how Google crawls and indexes your site

Read our guide to understand [how Google Search works](#); if you don't understand the crawl/index/serving pipeline well, it will be difficult to debug issues or anticipate Search behavior on your site.

Duplicate content

Be sure that you understand what [canonical pages](#) are, and how they affect crawling and indexing of your site.

Resources

Be sure that any resources (images, CSS files, and so on) or pages that Google is meant to crawl are accessible to Google; that is, they are not blocked by any robots.txt rules and are accessible to an anonymous user. Inaccessible pages will not appear in the [Page Indexing report](#), and the [URL Inspection tool](#) will show them as not crawled. Blocked resources are shown only at the individual URL level, in the URL Inspection tool. If important resources on a page are blocked, this can prevent Google from crawling your page properly. Use the URL Inspection tool to render the live page to verify whether Google sees the page as you expect.

Robots.txt

Use robots.txt rules to prevent crawling, and sitemaps to encourage crawling. Block crawling of duplicate content on your site, or unimportant resources (such as small, frequently used graphics such as icons or logos) that might overload your server with requests. Don't use robots.txt as a mechanism to prevent indexing; use the noindex tag or login requirements for that. [Read more about blocking access to your content](#).

Sitemaps

Sitemaps are a very important way to tell Google which pages are important to your site, and also provide additional information (such as update frequency), and are very important for crawling non-textual content (such as images or video). Although Google won't limit crawling to pages listed in your sitemaps, it will prioritize crawling these pages. This is especially important for sites with rapidly changing content, or with pages that might not be discovered through links. Using sitemaps helps Google discover and prioritize which pages to crawl on your site. [Read all about sitemaps here](#).

Internationalized or multi-lingual sites

If your site includes multiple languages, or is targeted at users in specific locales:

- [Read about multi-regional and multi-lingual sites](#) for high-level advice on how to manage sites that have localized content for different languages or regions.
- [Use hreflang](#) to tell Google about different language variations of pages on your site.
- If your site adapts the content of its pages based on the locale of the request, read [how this can affect Google's crawl of your site](#).

Migrating a page or a site

On the occasion that you might need to move a single URL or even a whole site, follow these guidelines:

Migrating a single URL

If you move a page permanently to another location, don't forget [to implement 301 redirects for your page](#). If the move is only temporary for some reason, return 302 instead to tell Google to continue to crawl your page.

When a user requests a page that has been removed, you can create a custom 404 page to provide a better experience. Just be sure that when a user requests a page that is no longer there, you return a true 404 error, not a [soft 404](#).

Migrating a site

If you're migrating an entire site, implement all the 301 and sitemap changes you need, then tell Google about the move so that we can start crawling the new site and forwarding your signals to the new site. [Learn how to migrate your site](#).

Follow crawling and indexing best practices

- [Make your links crawlable](#).
- [Use rel=nofollow](#) for paid links, links that require login, or untrusted content (such as user-submitted content) to avoid passing your quality signals on to them, or having their bad quality reflect on you.
- [Managing your crawl budget](#): If your site is particularly large (hundreds of millions of pages that change periodically, or perhaps tens of millions of pages that change frequently), Google might not be able to crawl your entire site as often as you'd like, so you might need to point Google to the most important pages on your site. The best mechanism for doing so at present is to list your most recently updated or most important pages in your sitemaps, and hiding your less important pages using robots.txt rules.
- [JavaScript usage](#): Follow [Google's recommendations for JavaScript on websites](#).
- [Multi-page articles](#): If you have an article broken into several pages, be sure that there are prominent next and previous links for users to click (and that these are crawlable links). That's all you need for the page set to be crawled by Google.
- [Infinite scroll pages](#): Google can have trouble scrolling through infinite scroll pages; provide a paginated version if you want the page to be crawled. [Learn more about search-friendly infinite scroll pages](#).
- [Block access to URLs that change state](#), such as posting comments, creating accounts, adding items to a cart, and so on. Use [robots.txt](#) to block these URLs.
- Review the [list of which file types are indexable by Google](#).
- In the unlikely situation that [Google seems to be crawling your site too much](#), you can [reduce the crawl rate](#) for your site. However, this should be a rare occurrence.
- If your site is still HTTP, we recommend [migrating to HTTPS](#), for your users' security, as well as your own.

Help Google understand your site

Put key information in text, not graphics, on the site. Although Google can parse and index [many file types](#), text is still the safest bet to help us understand the content of the page. If you use non-text content, or if you want to provide additional guidance about the content of the site, add [structured data](#) to your pages to help us understand your content (and in some cases, provide special search features such as [rich results](#)).

If you feel comfortable with HTML and basic coding, you can add structured data by hand following the [developer guidelines](#). If you want a little help, you can use the WYSIWYG [Structured Data Markup helper](#) to help you generate basic structured data for you.

If you don't have the ability to add structured data to your pages, you might use the [Data Highlighter tool](#), which lets you highlight portions of a page and tell Google what each section represents (an event, a date, a price, and so on). This is simple, but it can break if you change the layout of your page.

[Read more about helping Google understand your site content.](#)

Follow our guidelines

Caution: Be sure to follow our [Search Essentials](#). Some of these are recommendations and best practices; others can lead to a site being removed from the Google index if you do not follow them.

Content-specific guidelines

If you have specific types of content on your site, here are some recommendations for getting them on Google in the best way:

- **Video:** Be sure to follow our [video best practices](#) to enable Google to find, crawl, and show results for videos hosted on your site.
- **Images:** Follow our [image best practices](#) to get your images to appear in Search. You can show additional information about your image in Google Images by [providing image metadata](#) on the image host page. To block an image from being indexed, [use a robots.txt Disallow rule](#).
- **For children:** If your content is specifically for children, [tag your pages or site as child-directed](#) in order to comply with the Children's Online Privacy Protection Act (COPPA).
- **Adult sites:** If your site (or specific pages) contain adult-only content, you might consider [tagging it as adult content](#), which will filter it in SafeSearch results.
- **News:** If you run a news site, here are some important considerations:
 - If you have news content, be sure to read the [Google Publisher Center help documentation](#).
 - In addition, create a [News sitemap](#) to help Google discover content more quickly.
 - Be sure to [prevent abuse](#) on your site.
 - If you want to provide a limited number of views to visitors without a subscription or login, read about [flexible sampling](#) to learn some best practices about providing limited access to your content.
 - See how to [indicate subscription and paywalled content](#) on your site to Google while still enabling crawling.
 - See how to use meta tags to [limit text or image use when generating search result snippets](#).
 - Consider using [AMP](#) or [Web Stories](#) for fast-loading content.
- **Other sites** (for example, sites about businesses, books, apps, scholarly works): See [other Google services](#) where you can post your information.

- See if [Google supports a Search feature specific for your content type](#). Google supports specialized Search features for recipes, events, job posting sites, and more.

Manage the user experience

Providing a good user experience should be your site's top goal, and a good user experience is a ranking factor. There are many elements to providing a good user experience; here are a few of them.

Google recommends that websites use [HTTPS](#), rather than HTTP, to improve user and site security. Sites that use HTTP can be marked as "not secure" in the Chrome browser. [Learn how to secure your site with HTTPS](#).

A fast page usually beats a slow page in user satisfaction. You can use the [Core Web Vitals report](#) to see your site-wide performance numbers, or [PageSpeed Insights](#) to test performance for individual pages. You can learn more about building fast pages on the [web.dev site](#). Also consider using [AMP](#) for fast pages.

Mobile considerations

With over 60 percent of the global internet population using a mobile device to go online, it's important that your site be mobile-friendly. Google now uses a mobile crawler as the default crawler for websites. [Read about how to make your site mobile friendly](#).

Control your search appearance

Google provides many kinds of search result features and experiences in Google Search, including review stars and special result types for specific types of information such as events or recipes. See which ones are appropriate for your site and consider implementing them. You can [provide a favicon](#) to show in search results for your site. You can also [provide an article date](#) to appear in search results.

Be sure to read the articles on how to help Google provide good [titles links](#) and [snippets](#). You can also restrict the snippet length, or omit it entirely if you wish. See how to use meta tags to [limit text or image use when generating search result snippets](#).

Using Search Console

Search Console offers a broad range of reports to help you monitor and optimize your site performance on Google Search. Learn more about [what reports to use](#).

<https://developers.google.com/search/docs/fundamentals/get-started-developers>

Get started with Search: a developer's guide

Making your content search-friendly matters because it's how you get more relevant users viewing your content. This is called search engine optimization (SEO), which can result in more interested users coming to your site. If Google Search has trouble understanding your page, you're possibly missing out on an important source of traffic.

This guide covers what developers can do to make sure that their sites work well with Google Search. In addition to the items in this guide, make sure that your site is [secure](#), [fast](#), [accessible to all](#), and [works on all devices](#).

For help that's not so technical, visit the [SEO starter guide](#). The SEO starter guide covers other aspects of SEO, like authoring content.

Find out how Google sees your site

To get started, test your site in the [URL inspection tool](#) or [Rich Results Test](#) to see how Google sees your site. [Googlebot is Google's web crawling bot](#) that discovers new and updated pages for the Google index. For more information about the process, go to [How Google Search Works](#).

You may be surprised to find that Google doesn't always see everything that you see in the browser. In the following example, Google doesn't know there are images on this page because the page uses a JavaScript feature that isn't supported by Google.

User view

Here's how a user views the page. Users can view the images and text in the browser.

[Original illustration: A website that shows 6 different images of cats. The title of the website is Cute cat content chronicles.]

Google view

Here's how Google views the page. Google doesn't know there are images on this page because the page uses a JavaScript feature that isn't supported by Google.

[Original illustration: A website that shows the title of the website. There are supposed to be images of cats on the page, but it just shows loading icons.]

Check your links

Googlebot navigates from URL to URL by fetching and parsing links, sitemaps, and redirects. Googlebot treats every URL as if it's the first and only URL it has seen from your site. To make sure that Googlebot can find all the URLs on your site:

- Use [<a> elements that Google can crawl](#). Ensure that all pages on the site can be reached by a link from another findable page. Make sure the referring link includes either text or, for images, an alt attribute, that

is relevant to the target page.

- [Build and submit a sitemap](#) to help Googlebot more intelligently crawl your site. A sitemap is a file where you provide information about the pages, videos, and other files on your site, and the relationships between them.
- For JavaScript apps that have only one HTML page, make sure that each screen or piece of individual content has a URL.

Check how you're using JavaScript

While Google does run JavaScript, there are some differences and limitations that you need to account for when designing your pages and applications to accommodate how crawlers access and render your content. Learn more about the [basics of JavaScript SEO](#) or how to [fix Search-related JavaScript problems](#).

To learn more about how Google handles JavaScript when crawling, rendering, and indexing, see the following video.

[\[Original embedded video\]](#)

Keep Google updated when content changes

To make sure that Google finds your new or updated pages quickly:

- [Submit sitemaps](#).
- [Ask Google to recrawl your URLs](#).

If you're still having trouble getting your page indexed, check your server logs for errors.

Don't forget about the words on the page

Googlebot can only find content that is textually visible. For example, text in videos is invisible to Googlebot. To make sure that Google Search understands what your page is about:

- **Make sure that your visual content is expressed in text form.** For example, a product category page that contains a list of images of shirts with no textual context about each image is suboptimal. The product category page should include some textual explanation for each image.
- **Make sure that every page has a [descriptive title](#) and [meta description](#).** Unique titles and meta descriptions help Google show how your pages are relevant to users, which in turn can increase your search traffic.
- **Use semantic HTML.** While Google indexes HTML, PDF content, images, and videos, it doesn't index content that requires plugins (for example, Java or Silverlight) or content that is rendered in a canvas. Instead of using a plugin, use semantic HTML markup for your content whenever possible.
- **Make sure your text content is accessible in the [DOM](#).** For example, content that is added via the [CSS content property](#) is not part of the DOM and Google Search ignores it at the moment. It's fine to use the content property for decorative content; Google Search may not index this content.

Tell Google about other versions of your content

Google doesn't automatically know that there are multiple versions of your site or content. For example, a mobile and desktop version, or international versions of your site. To make sure that Google serves the right version to

users, you can:

- [Consolidate duplicate URLs](#).
- [Tell Google about localized versions of your site](#).
- [Make your AMP pages discoverable](#).

Control what content Google sees

There are several ways to block Googlebot:

- To block Google from finding your page, restrict access to your content to logged in users (for example, use a login page or [password-protect your page](#)).
- To block Googlebot from crawling your page, [create a robots.txt](#).

A robots.txt is not a mechanism for keeping a web page out of Google. To keep a web page out of Google, use the noindex robots rules, or [password-protect your page](#).

- To block Google from indexing your page but still allow crawling, [add a noindex tag](#).

Combining multiple crawling and indexing rules might cause some rules to counteract other rules. Learn how to configure these rules properly by reading [Combining crawling with indexing / serving rules](#).

If your content isn't showing up in Google Search and you want it to show up, follow these steps:

- Check if Googlebot can access the page with the [URL inspection tool](#).
- [Test your robot.txt](#) file to see if you're unintentionally blocking Googlebot from crawling your site.
- Check your HTML for noindex rules in meta tags.

Enable rich results for your site

A rich result can include styling, images, or other interactive features that can help your site stand out more in Search results. You can help Google understand your page better and show rich results for it in Search by providing explicit clues about the meaning of a page [with structured data on the page](#). If you're not sure where to start, [explore our gallery of available features](#).

[Original illustration: A carousel of recipes in search results]

<https://developers.google.com/search/docs/fundamentals/do-i-need-seo>

Do you need an SEO?

[Original embedded video]

As with many business activities, sometimes hiring a professional can be helpful. Deciding to hire a company or person to do [search engine optimization \(SEO\)](#) for your website in Google Search is a big decision that can potentially improve your site's visibility and save you time, but you can also risk damage to your site and reputation. Make sure to research the potential advantages as well as the damage that an irresponsible SEO can do to your site. Many SEOs and other agencies and consultants provide useful services for website owners, including:

- Review of your site content or structure
- Technical advice on website development: for example, hosting, redirects, error pages, use of JavaScript
- Content development
- Management of online business development campaigns
- Keyword research
- SEO training
- Expertise in specific markets and geographies
- Optimizing for generative AI

Advertising with Google won't have any effect on your site's presence in our search results. Google never accepts money to include or rank sites in our search results, and it costs nothing to appear in our organic search results. Resources such as [Search Console](#), the official [Google Search Central blog](#), and our [discussion forum](#) can provide you with a great deal of information about how to optimize your site for Google's non-advertising search results (often called "organic search").

Deciding on whether you need an SEO

If you run a small local business, you can probably do much of the work yourself. We recommend starting with the [SEO starter guide](#), as it's a practical guide to improving your site's SEO and it describes much of what your SEO will do for you. If you decide to hire a professional to do the work for you in the end, it's also useful to be familiar with these techniques so that you can be aware if an SEO wants to use a technique that's not recommended or, worse, strongly discouraged.

If you think that you still need extra help from a professional, continue reading about hiring an SEO.

Things to consider when hiring an SEO

As with many business activities, sometimes hiring a professional can be helpful. For example, a great time to hire is when you're considering a site redesign (the earlier, the better), or planning to launch a new site. That way, you and your SEO can ensure that your site is designed to be search engine-friendly from the bottom up. However, a good SEO can also help improve an existing site.

Here are some things to consider if you're considering hiring someone to do SEO for your site:

- **Interview your potential SEO.** Some useful questions to ask an SEO include:

- Can you show me examples of your previous work and share some success stories?
- Do you follow the [Google Search Essentials](#) (previously known as Webmaster Guidelines)?
- What kind of results do you expect to see, and in what timeframe? How do you measure your success?
- What's your experience in my industry?
- What's your experience in my country/city?
- What's your experience developing international sites?
- How long have you been in business?
- How can I expect to communicate with you? Will you share with me all the changes you make to my site, and provide detailed information about your recommendations and the reasoning behind them?
- See if the SEO is interested in you and your business. If they're not interested, find someone who is. Your SEO should ask questions about what makes your business unique, who your competitors are, how search results can help, and how your customers find you.
- **Check your SEO's business references.** Ask past clients if they felt that this SEO provided useful service, was easy to work with, and produced positive results.
- **If your SEO offers to do an SEO audit for you, be sure to carefully consider what's involved and only grant read access to Search Console (at this stage, don't grant them write access).** An SEO audit should be about giving you realistic estimates of improvement, and an estimate of the work involved. If they guarantee you that their changes will give you first place in search results, find someone else.

If your SEO uses a third-party tool, keep in mind that Google doesn't evaluate or endorse third-party SEO tools, and these tools don't have access to Google's internal ranking data. Be wary of tools that claim to be "acceptable" or "approved" by Google Search.

- **Evaluate your SEO's recommendations and tools they use.** Before making significant changes to your site based on a third-party tool's audit, be sure to check their recommendations against [official guidance from Google Search](#), think critically about any claims or recommendations you hear, and make your own informed decisions.
 - Do they cite official Google documentation as supporting evidence for their recommendations?
 - If they have advice on optimizing for AI experiences (also known as "AEO" "GEO" services), is their advice aligned with [Google Search's official guidance on optimizing for generative AI features](#)?
 - Do they use tools that are aligned with Google's guidance?
- **Be wary of SEO firms and web consultants or agencies that email you out of the blue.**

Amazingly, we get these spam emails too:

*"Dear google.com,
I visited your website and noticed that you are not listed in most of the major search engines and directories..."*

Reserve the same skepticism for unsolicited email about search engines as you do for "burn fat at night" diet pills or requests to help transfer funds from deposed dictators.

- **No one can guarantee a #1 ranking on Google.** Beware of SEOs that claim to guarantee rankings, allege a "special relationship" with Google, or advertise a "priority submit" to Google.
- **Be careful if a company is secretive or won't clearly explain what they intend to do.**

Ask for explanations if something is unclear. If an SEO creates deceptive or misleading content on your behalf, your site could be removed entirely from Google's index. Ultimately, you are responsible for the actions of any companies you hire, so it's best to be sure you know exactly how they intend to "help" you. If

an SEO has FTP access to your server, they should be willing to explain all the changes they are making to your site.

- **You should never have to link to an SEO.**

Avoid SEOs that talk about link popularity schemes or submitting your site to thousands of search engines. These are typically useless exercises that don't affect your ranking in the results of the major search engines—at least, not in a way you would likely consider to be positive.

Important: While SEOs can provide clients with valuable services, some unethical SEOs have given the industry a black eye by using overly aggressive marketing efforts or using techniques that violate [our spam policies](#), which may result in a negative adjustment of your site's presence in Google, or even the removal of your site from our index.

Reporting issues

If you feel that you were deceived by an SEO in some way, you may want to report it.

In the United States, the Federal Trade Commission (FTC) handles complaints about deceptive or unfair business practices. To file a complaint, visit the FTC website to [file a complaint online](#) or call 1-877-FTC-HELP.

If your complaint is against a company in a country other than the United States, file it at <https://www.econsumer.gov/>.

<https://developers.google.com/search/docs/fundamentals/third-party-seo>

Google Search's guidance on using third-party SEO tools, services, and advice

If you're thinking about improving the SEO for your website, you might also be considering advice from third parties and using third-party SEO services and tools. This guide provides our recommendations for evaluating external SEO resources and tools to make sure they align with official Google Search guidance.

If you're considering hiring a company to perform SEO services, check out our guide that includes more tips on [how to choose and evaluate their work](#).

Evaluate and verify external SEO advice against official Google guidelines

There's plenty of third-party SEO advice on the internet related to SEO, search listings, and AI experiences (sometimes called AEO for "answer engine optimization" or GEO for "generative engine optimization"). While some advice is helpful, others may misinterpret or make claims about what "Google says" or how Google ranking systems work. In general, good advice either qualifies their claims as opinion based on data or experience, or backs up their claims by citing official Google Search guidance.

We recommend carefully evaluating any advice you might be considering implementing against our [official SEO guidance](#), including our [guidance on optimizing for generative AI](#), and making your own informed decisions.

Think critically about using third-party SEO tools and services

If you're planning to do SEO yourself, you may be considering third-party SEO services and tools, such as:

- Assisting in sitemap generation
- Establishing indexing directives
- Offering to generate "SEO-optimized" content for you
- Providing advice they claim will improve the ranking of your existing content
- Promising improvements for AI experiences and search formats (also known as "AEO" or "GEO" tools)

Some of these services may be helpful in your work, while others may make claims or imply that what they do is somehow "acceptable" or "approved" by Google Search. Google doesn't evaluate third-party services, so be wary of such claims and those making them. Keep in mind that using a service or tool doesn't guarantee ranking success.

Some third-party services provide data that some users of those tools misinterpret as somehow being from Google. Third-party tools don't have access to our internal ranking data. They can't guarantee performance. Any predictions are their own and like predictions generally, may not happen.

Whether you use a third-party tool or not, we strongly encourage using our first-party tool, [Google Search Console](#), which provides you with key information and data directly from Google Search itself.

<https://developers.google.com/search/docs/crawling-indexing>

Overview of crawling and indexing topics

The topics in this section describe how you can control Google's ability to find and parse your content in order to show it in Search and other Google properties, as well as how to prevent Google from crawling specific content on your site.

Here's a brief description of each page. To get an overview of crawling and indexing, read our [How Search works](#) guide.

Topics	
File types indexable by Google	Google can index the content of most types of pages and files. Explore a list of the most common file types that Google Search can index.
URL structure	Consider organizing your content so that URLs are constructed logically and in a manner that is most intelligible to humans.
Sitemaps	Tell Google about pages on your site that are new or updated.
Crawler management	• Ask Google to recrawl your URLs • Managing crawling of faceted navigation URLs • Large site owner's guide to managing your crawl budget • How HTTP status codes, and network and DNS errors affect Google Search • Google crawlers
robots.txt	A robots.txt file tells search engine crawlers which pages or files the crawler can or can't request from your site.
Canonicalization	Learn what URL canonicalization is and how to tell Google about any duplicate pages on your site in order to avoid excessive crawling. Learn how Google auto-detects duplicate content, how it treats duplicate content, and how it assigns a <i>canonical URL</i> to any duplicate page groups found.
Mobile sites	Learn how you can optimize your site for mobile devices and ensure that it's crawled and indexed properly.
AMP	If you have AMP pages, learn how AMP works in Google Search.
JavaScript	There are some differences and limitations that you need to account for when designing your pages and applications to accommodate how crawlers access and render your content.
Page and content metadata	• Use valid HTML to specify page metadata • All meta tags that Google understands

Topics	
	• Robots meta tag, data-nosnippet, and X-Robots-Tag specifications • Block indexing with the noindex meta tag • Make your links crawlable • Qualify your outbound links to Google with rel attributes
Removals	• Control what you share with Google • Remove a page hosted on your site from Google • Remove images hosted on your page from search results • Keep redacted information out of Google Search
Site moves and changes	• Redirects and Google Search • Site moves • Minimize A/B testing impact in Google Search • Temporarily pause or disable a website

<https://developers.google.com/search/docs/crawling-indexing/indexable-file-types>

File types indexable by Google

Google can index the content of most text-based files and certain encoded document formats. The file type is determined by the Content-Type HTTP header returned when Google crawls the file, though in some cases Google may use the file extension or re-parse the file using a different parser if the Content-Type header is missing or incorrect.

Supported flat file types

The following flat file types are supported. These are files where the content is stored in plain, unencoded text (though they may use markup tags).

- Comma-Separated Values (.csv)
- Google Earth (.kml, .kmz)
- GPS eXchange Format (.gpx)
- HTML (.htm, .html, other file extensions)
- Scalable Vector Graphics (.svg)
- TeX/LaTeX (.tex)
- Text (.txt, .text, other file extensions), including source code in common programming languages, such as:
 - Basic source code (.bas)
 - C/C++ source code (.c, .cc, .cpp, .cxx, .h, .hpp)
 - C# source code (.cs)
 - Java source code (.java)
 - Perl source code (.pl)
 - Python source code (.py)
- Wireless Markup Language (.wml, .wap)
- XML (.xml)

Supported encoded file types

The following encoded file types are supported. These are binary files or complex containers that require a specific parser to extract the human-readable text.

- Adobe Portable Document Format (.pdf)
- Adobe PostScript (.ps)
- Electronic Publication (.epub)
- Hancom Hanword (.hwp)
- Microsoft Excel (.xls, .xlsx)
- Microsoft PowerPoint (.ppt, .pptx)
- Microsoft Word (.doc, .docx)
- OpenOffice presentation (.odp)
- OpenOffice spreadsheet (.ods)
- OpenOffice text (.odt)
- Rich Text Format (.rtf)

Supported media formats

Google can also index the following media formats:

- Image formats: BMP, GIF, JPEG, PNG, WebP, SVG, and AVIF
- Video formats: 3GP, 3G2, ASF, AVI, DivX, M2V, M3U, M3U8, M4V, MKV, MOV, MP4, MPEG, OGV, QVT, RAM, RM, VOB, WebM, WMV, and XAP

Search by file type

You can use the `filetype:` operator in Google Search to limit results to a specific file type or file extension. For example, `filetype:rtf galway` will search for RTF files and URLs ending in `.rtf` whose content contains the term "galway".

<https://developers.google.com/search/docs/crawling-indexing/url-structure>

URL structure best practices for Google Search

To make sure Google Search can crawl your site effectively, use a crawlable URL structure that meets the following requirements. If your URLs don't meet the following criteria, Google Search will likely crawl your site inefficiently – including but not limited to extremely high crawl rates, or not at all.

Requirements for a crawlable URL structure							
Follow IETF STD 66	Google Search supports URLs as defined by IETF STD 66. Characters defined by the standard as reserved must be percent encoded.						
Don't use URL fragments to change content	Don't use fragments to change the content of a page, as Google Search generally doesn't support URL fragments. Here's an example of a URL fragment: https://example.com/#/potatoes If you're using JavaScript to change content, use the History API instead.						
Use a common encoding for URL parameters	When specifying URL parameters, use the following common encoding: an equal sign (=) to separate key-value pairs and add additional parameters with an ampersand (&). To list multiple values for the same key within a key-value pair, you can use any character that doesn't conflict with IETF STD 66, such as a comma (,).						
	<table><tr><th>Recommended</th><th>Not recommended</th></tr><tr><td> Using an equal sign (=) to separate key-value pairs and an ampersand (&) to add additional parameters: https://example.com/category?category=dresses&sort=low-to-high&sid=789 </td><td> Using a colon (:) to separate key-value pairs and brackets ([]) to add additional parameters: https://example.com/category?[category:dresses][sort:price-low-to-high][sid:789] </td></tr><tr><td> Using a comma (,) to list multiple values for the same key, an equal </td><td> Using a single comma (,) to separate key-value pairs and </td></tr></table>	Recommended	Not recommended	Using an equal sign (=) to separate key-value pairs and an ampersand (&) to add additional parameters: https://example.com/category?category=dresses&sort=low-to-high&sid=789	Using a colon (:) to separate key-value pairs and brackets ([]) to add additional parameters: https://example.com/category?[category:dresses][sort:price-low-to-high][sid:789]	Using a comma (,) to list multiple values for the same key, an equal	Using a single comma (,) to separate key-value pairs and
	Recommended	Not recommended					
Using an equal sign (=) to separate key-value pairs and an ampersand (&) to add additional parameters: https://example.com/category?category=dresses&sort=low-to-high&sid=789	Using a colon (:) to separate key-value pairs and brackets ([]) to add additional parameters: https://example.com/category?[category:dresses][sort:price-low-to-high][sid:789]						
Using a comma (,) to list multiple values for the same key, an equal	Using a single comma (,) to separate key-value pairs and						

sign (=) to separate key-value pairs, and an ampersand (&) to add additional parameters:

```
https://example.com/category?
category=dresses&color=purple,
pink,salmon&sort=low-to-high&sid=789
```

double commas (,,) to add additional parameters:

```
https://example.com/category?
category,dresses,,sort,lowtohigh,,sid,789
```

Make it easy to understand your URL structure

To help Google Search (and your users) better understand your site, we recommend creating a simple URL structure, applying the following best practices when possible.

Consider organizing your content so that URLs are constructed logically and in a manner that is most intelligible to humans. For information on structuring your site as a whole, check out [this section of the SEO Starter Guide](#).

Best practices

Use descriptive URLs	When possible, use readable words rather than long ID numbers in your URLs. <table data-bbox="373 232 1222 620"> <tr> <th data-bbox="373 232 798 344">Recommended (simple, descriptive words)</th><th data-bbox="798 232 1222 344">Not recommended (unreadable, long ID numbers)</th></tr> <tr> <td data-bbox="373 344 798 620"> https://example.com/wiki/Aviation </td><td data-bbox="798 344 1222 620"> https://example.com/index.php?topic=42&area=3a5ebc944f41d aa6f849f730f1 </td></tr> </table>	Recommended (simple, descriptive words)	Not recommended (unreadable, long ID numbers)	https://example.com/wiki/Aviation	https://example.com/index.php?topic=42&area=3a5ebc944f41d aa6f849f730f1		
Recommended (simple, descriptive words)	Not recommended (unreadable, long ID numbers)						
https://example.com/wiki/Aviation	https://example.com/index.php?topic=42&area=3a5ebc944f41d aa6f849f730f1						
Use your audience's language	Use words in your audience's language in the URL (and, if applicable, transliterated words). For example, if your audience is searching in German, use German words in the URL: https://example.com/lebensmittel/pfefferminz Or if your audience is searching in Japanese, use Japanese words in the URL: https://example.com/ペパーミント						
Use percent encoding as necessary	When linking to pages on your site, use percent encoding in your links's href attributes as necessary. Unreserved ASCII characters may be left in the non-encoded form. Additionally, characters in the non-ASCII range should be percent encoded. For example: <table data-bbox="373 1346 1222 1939"> <tr> <th data-bbox="373 1346 798 1458">Recommended (percent encoding)</th><th data-bbox="798 1346 1222 1458">Not recommended (non-ASCII characters)</th></tr> <tr> <td data-bbox="373 1458 798 1697"> https://example.com/%D9%86%D8%B9%D9%86%D8%A7%D8%B9/%D8%A8%D9%82%D8%A7%D9%84%D8%A9 </td><td data-bbox="798 1458 1222 1697"> https://example.com/نعناع </td></tr> <tr> <td data-bbox="373 1697 798 1939"> https://example.com/%E6%9D%82%E8%B4%A7/%E8%96%84%E8%8D%B7 </td><td data-bbox="798 1697 1222 1939"> https://example.com/杂货/薄荷 </td></tr> </table>	Recommended (percent encoding)	Not recommended (non-ASCII characters)	https://example.com/%D9%86%D8%B9%D9%86%D8%A7%D8%B9/%D8%A8%D9%82%D8%A7%D9%84%D8%A9	https://example.com/نعناع	https://example.com/%E6%9D%82%E8%B4%A7/%E8%96%84%E8%8D%B7	https://example.com/杂货/薄荷
Recommended (percent encoding)	Not recommended (non-ASCII characters)						
https://example.com/%D9%86%D8%B9%D9%86%D8%A7%D8%B9/%D8%A8%D9%82%D8%A7%D9%84%D8%A9	https://example.com/نعناع						
https://example.com/%E6%9D%82%E8%B4%A7/%E8%96%84%E8%8D%B7	https://example.com/杂货/薄荷						

https://example.com/gem%C3%BCse	https://example.com/gemüse
https://example.com/%F0%9F%A6%99%E2%9C%A8	https://example.com/🦊✨

Use hyphens to separate words	We recommend separating words in your URLs, when possible. Specifically, we recommend using hyphens (-) instead of underscores (_) to separate words in your URLs, as it helps users and search engines better identify concepts in the URL. For historical reasons, we don't recommend using underscores, as this style is already commonly used for denoting concepts that should be kept together, for example, by various programming languages to name functions (such as <code>format_date</code>). <table data-bbox="373 456 1224 1064"> <tr> <th data-bbox="373 456 798 533">Recommended</th><th data-bbox="798 456 1224 533">Not recommended</th></tr> <tr> <td data-bbox="373 533 798 1064"> Using hyphens (-) to separate words: <pre>https://example.com/summer-clothing/filter?color-profile=dark-grey</pre> </td><td data-bbox="798 533 1224 1064"> Using underscores (_) to separate words: <pre>https://example.com/summer_clothing/filter?color_profile=dark_grey</pre> Joining words together in the URL: <pre>https://example.com/greendress</pre> </td></tr> </table>	Recommended	Not recommended	Using hyphens (-) to separate words: <pre>https://example.com/summer-clothing/filter?color-profile=dark-grey</pre>	Using underscores (_) to separate words: <pre>https://example.com/summer_clothing/filter?color_profile=dark_grey</pre> Joining words together in the URL: <pre>https://example.com/greendress</pre>
Recommended	Not recommended				
Using hyphens (-) to separate words: <pre>https://example.com/summer-clothing/filter?color-profile=dark-grey</pre>	Using underscores (_) to separate words: <pre>https://example.com/summer_clothing/filter?color_profile=dark_grey</pre> Joining words together in the URL: <pre>https://example.com/greendress</pre>				
Use as few parameters as you can	Whenever possible, shorten URLs by trimming unnecessary parameters (meaning, parameters that don't change the content).				
Be aware that URLs are case sensitive	Like any other HTTP client following IETF STD 66, Google Search's URL handling is case sensitive (for example, Google treats both <code>/APPLE</code> and <code>/apple</code> as distinct URLs with their own content). If upper and lower case text in a URL is treated the same by your web server, convert all text to the same case so it's easier for Google to determine that URLs reference the same page.				

For multi-regional sites	If your site is multi-regional, consider using a URL structure that makes it easy to geotarget your site. For more examples of how you can structure your URLs, refer to using locale-specific URLs. Recommended (using a country-specific domain): <code>https://example.de</code> Recommended (using a country-specific subdirectory with gTLD): <code>https://example.com/de/</code>
---------------------------------	--

Avoid common issues related to URLs

Overly complex URLs, especially those containing multiple parameters, can cause problems for crawlers by creating unnecessarily high numbers of URLs that point to identical or similar content on your site. As a result, Googlebot may consume much more bandwidth than necessary, or Google Search may be unable to completely index all the content on your site.

Unnecessarily high numbers of URLs can be caused by a number of issues. These include:

Common issues

Additive filtering of a set of items

Many sites provide different views of the same set of items or search results, often allowing the user to filter this set using defined criteria (for example: show me hotels on the beach). When filters can be combined in an additive manner (for example: hotels on the beach and with a fitness center), the number of URLs (views of data) in the sites explodes. Creating a large number of slightly different lists of hotels is redundant, as Googlebot only needs to see a small number of lists from which it can reach the page for each hotel. For example:

- Hotel properties at "value rates":

```
https://example.com/hotel-search-results.jsp?Ne=292&N=461
```

- Hotel properties at "value rates" on the beach:

```
https://example.com/hotel-search-results.jsp?
Ne=292&N=461+4294967240
```

- Hotel properties at "value rates" on the beach and with a fitness center:

```
https://example.com/hotel-search-results.jsp?
Ne=292&N=461+4294967240+4294967270
```

Irrelevant parameters

Irrelevant parameters in the URL can cause a large number of URLs, such as:

- Referral parameters:

```
https://example.com/search/noheaders?
click=6EE2BF1AF6A3D705D5561B7C3564D9C2&clickPage=OPD+Product
+Page&cat=79
```

```
https://example.com/discuss/showthread.php?
referrerid=249406&threadid=535913
```

```
https://example.com/products/products.asp?
N=200063&Ne=500955&ref=foo%2Cbar&Cn=Accessories
```

- Shopping sorting parameters:

```
https://example.com/results?
search_type=search_videos&search_query=tpb&search_sort=relev
ance&search_category=25
```

- Session IDs:

	https://example.com/search/noheaders?sessionId=6EE2BF1AF6A3D705D5561B7C3564D9C2 Wherever possible, avoid the use of session IDs in URLs and consider using cookies instead. Consider using a robots.txt file to block Googlebot's access to these problematic URLs.
Calendar issues	A dynamically generated calendar might generate links to future and previous dates with no restrictions on start or end dates. For example: https://example.com/calendar.php?d=13&m=8&y=2011 If your site has an infinite calendar, add a nofollow attribute to links to dynamically created future calendar pages.
Broken relative links	Placing a parent-relative link on the wrong page may create infinite spaces if your server doesn't respond with the right HTTP status code for nonexistent pages. For example, a parent-relative link such as <code>...</code> on <code>https://example.com/category/community/070413/html/FAQ.htm</code> may lead to bogus URLs such as <code>https://example.com/category/community/category/stuff</code>. To fix, use root-relative URLs in your links (instead of parent-relative).

Fixing crawling-related URL structure problems

If you notice that Google Search is crawling these problematic URLs, we recommend the following:

- Consider using a [robots.txt file](#) to block Googlebot's access to [problematic URLs](#). Typically, consider blocking dynamic URLs, such as URLs that generate search results, or URLs that can create infinite spaces, such as calendars, and ordering and filtering functions.
- If your site has faceted navigation, learn how to [manage crawling of those faceted navigation URLs](#).

<https://developers.google.com/search/docs/crawling-indexing/links-crawlable>

Link best practices for Google

Google uses links as a signal when determining the relevancy of pages and to find new pages to crawl. Learn how to make your links crawlable so that Google can find other pages on your site via the links on your page, and how to improve your anchor text so that it's easier for people and Google to make sense of your content.

Make your links crawlable

Generally, Google can only crawl your link if it's an `<a>` HTML element (also known as *anchor element*) with an `href` attribute. Most links in other formats won't be parsed and extracted by Google's crawlers. Google can't reliably extract URLs from `<a>` elements that don't have an `href` attribute or other tags that perform as links because of script events. Here are examples of links that Google can and can't parse:

Recommended (Google can parse)

```
<a href="https://example.com">
```

```
<a href="/products/category/shoes">
```

```
<a href="./products/category/shoes">
```

```
<a href="/products/category/shoes" onclick="javascript:goto('shoes')">
```

```
<a href="/products/category/shoes" class="pretty">
```

Links are also crawlable when you use JavaScript to insert them into a page dynamically as long as it uses the HTML markup shown above.

Not recommended (but Google may still attempt to parse this):

```
<a routerLink="products/category">
```

```
<span href="https://example.com">
```

```
<a onclick="goto('https://example.com')">
```

Make sure that the URL in your `<a>` element resolves into an actual web address (meaning, it resembles a URI) that Google crawlers can send requests to, for example:

Recommended (Google can resolve):

```
<a href="https://example.com/stuff">
```

```
<a href="/products">
```

```
<a href="/products.php?id=123">
```

Not recommended (but Google may still attempt to resolve this):

```
<a href="javascript:goto('products')">
```

```
<a href="javascript:window.location.href='/products'">
```

Anchor text placement

Anchor text (also known as *link text*) is the visible text of a link. This text tells people and Google something about the page you're linking to. Place anchor text between `<a>` [elements that Google can crawl](#).

Good:

```
<a href="https://example.com/ghost-peppers">ghost peppers</a>
```

Bad (empty link text):

```
<a href="https://example.com"></a>
```

As a fallback, Google can use the `title` attribute as anchor text if the `<a>` element is for some reason empty.

```
<a href="https://example.com/ghost-pepper-recipe" title="how to pickle ghost peppers"></a>
```

For images as links, Google uses the `alt` attribute of the `img` element as anchor text, so be sure to [add descriptive alt text to your images](#):

Good:

```
<a href="/add-to-cart.html">
</a>
```

Bad (empty alt text and empty link text):

```
<a href="/add-to-cart.html"></a>
```

If you are using JavaScript to insert anchor text, use the [URL Inspection Tool](#) to make sure it's present in the rendered HTML.

Write good anchor text

Good anchor text is descriptive, reasonably concise, and relevant to the page that it's on and to the page it links to. It provides context for the link, and sets the expectation for your readers. The better your anchor text, the easier it is for people to navigate your site and for Google to understand what the page you're linking to is about.

Bad (too generic):

```
<a href="https://example.com">Click here</a> to learn more.
```

```
<a href="https://example.com">Read more</a>.
```

Learn more about our cheese on our [website](https://example.com).

We have an [article](https://example.com) that provides more background on how the cheese is made.

Tip: Try reading only the anchor text (out of context) and check if it's specific enough to make sense by itself. If you don't know what the page could be about, you need more descriptive anchor text.

Better (more descriptive):

For a full list of cheese available for purchase, see the [list of cheese types](https://example.com).

Bad (weirdly long):

Starting next Tuesday, the [Knitted Cow invites local residents of Wisconsin to their grand re-opening by also offering complimentary cow-shaped ice sculptures](https://example.com) to the first 20 customers.

Better (more concise):

Starting next Tuesday, the [Knitted Cow invites local residents of Wisconsin](https://example.com) to their grand re-opening by also offering complimentary cow-shaped ice sculptures to the

first 20 customers.

Write as naturally as possible, and resist the urge to cram every keyword that's related to the page that you're linking to (remember, [keyword stuffing](#) is a violation of our spam policies). Ask yourself, does the reader need these keywords to understand the next page? If it feels like you're forcing keywords into the anchor text, then it's probably too much.

Remember to give context to your links: the words before and after links matter, so pay attention to the sentence as a whole. Don't chain up links next to each other; it's harder for your readers to distinguish between links, and you lose surrounding text for each link.

Bad (too many links next to each other):

I've written about cheese [so](https://example.com/page1) [many](https://example.com/page2) [times](https://example.com/page3) [this](https://example.com/page4) [year](https://example.com/page5).

Better (links are spaced out with context):

I've written about cheese so many times this year: who can forget the [controversy over blue cheese and gorgonzola](https://example.com/blue-cheese-vs-gorgonzola), the [world's oldest brie](https://example.com/worlds-oldest-brie) piece that won the Cheesiest Research Medal, the epic retelling of [The Lost Cheese](https://example.com/the-lost-cheese), and my personal favorite, [A Boy and His Cheese: a story of two unlikely friends](https://example.com/boy-and-his-cheese).

Internal links: cross-reference your own content

[Original embedded video]

You may usually think about linking in terms of pointing to external websites, but paying more attention to the anchor text used for internal links can help both people and Google make sense of your site more easily and find other pages on your site. Every page you care about should have a link from at least one other page on your site. Think about what other resources on your site could help your readers understand a given page on your site, and link to those pages in context.

There's no magical ideal number of links a given page should contain. However, if you think it's too much, then it probably is.

External links: link to other sites

Linking to other sites isn't something to be scared of; in fact, using external links can help establish trustworthiness (for example, citing your sources). Link out to external sites when it makes sense, and provide context to your readers about what they can expect.

Good (citing your sources):

According to a recent study from Swiss researchers, Emmental cheese wheels that were exposed to music had a milder flavor when compared to the control cheese wheels (which experienced no such musical treatment), with the full findings available in <https://example.com> **Cheese in Surround Sound—a culinary art experiment**.

Use `nofollow` only when you don't trust the source, and not for every external link on your site. For example, you're a cheese enthusiast and someone published a story badmouthing your favorite cheese, so you want to write an article in response; however, you don't want to give the site some of your reputation from your link. This would be a good time to use `nofollow`.

If you were paid in some way for the link, qualify these links with `sponsored` or `nofollow`. If users can insert links on your site (for example, you have a forum section or Q&A site), add `ugc` or `nofollow` to these links too.

<https://developers.google.com/search/docs/crawling-indexing/sitemaps/overview>

Learn about sitemaps

[Original embedded video]

A *sitemap* is a file where you provide information about the pages, videos, and other files on your site, and the relationships between them. Search engines like Google read this file to crawl your site more efficiently. A sitemap tells search engines which pages and files you think are important in your site, and also provides valuable information about these files. For example, when the page was last updated and any alternate language versions of the page.

You can use a sitemap to provide information about specific types of content on your pages, including [video](#), [image](#), and [news](#) content. For example:

- A sitemap *video entry* can specify the video running time, rating, and age-appropriateness rating.
- A sitemap *image entry* can include the location of the images included in a page.
- A sitemap *news entry* can include the article title and publication date.

If you're using a CMS such as WordPress, Wix, or Blogger, it's likely that your CMS has already [made a sitemap available to search engines](#) and you don't have to do anything.

Do I need a sitemap?

If your site's pages are properly linked, Google can usually discover most of your site. Proper linking means that all pages that you deem important can be reached through some form of navigation, be that your site's menu or links that you placed on pages. Even so, a sitemap can improve the crawling of larger or more complex sites, or more specialized files.

A sitemap helps search engines discover URLs on your site, but it doesn't guarantee that all the items in your sitemap will be crawled and indexed. However, in most cases, your site will benefit from having a sitemap.

You might need a sitemap if:

- **Your site is large.** Generally, on large sites it's more difficult to make sure that every page is linked by at least one other page on the site. As a result, it's more likely [Googlebot](#) might not discover some of your new pages.
- **Your site is new and has few external links to it.** Googlebot and other web crawlers crawl the web by accessing URLs found in previously crawled pages. As a result, Googlebot might not discover your pages if no other sites link to them.
- **Your site has a lot of rich media content (video, images) or is shown in Google News.** Google can take additional information from sitemaps into account for Search.

You might not need a sitemap if:

- **Your site is "small".** By small, we mean about 500 pages or fewer on your site. Only pages that you think need to be in search results count toward this total.
- **Your site is comprehensively linked internally.** This means that Googlebot can find all the important pages on your site by following links starting from the home page.
- **You don't have many media files (video, image) or news pages** that you want to show in search results. Sitemaps can help Google find and understand video and image files, or news articles, on your site. If you don't need these results to appear in Search you might not need a sitemap.

Build a sitemap

If you decided that you need a sitemap, [learn more about how to create one](#).

<https://developers.google.com/search/docs/crawling-indexing/sitemaps/build-sitemap>

Build and submit a sitemap

This page describes how to build a sitemap and make it available to Google. If you're new to sitemaps, [read our introduction first](#).

Google supports the sitemap formats defined by the [sitemaps protocol](#). Each format has its own benefits and shortcomings; choose the one that is the most appropriate for your site and setup (Google doesn't have a preference). The following table compares the different sitemap formats:

Sitemaps comparison	
XML sitemap	XML sitemaps are the most versatile of the sitemaps formats. It's extensible and can be used to supply additional data about images, video, and news content, as well as the localized versions of your pages. Pros: • Extensible and versatile. • It can provide the most information about your URLs. • Most content management systems (CMS) automatically generate sitemaps or CMS users can find plenty of sitemap plugins. Cons: • Can be cumbersome to work with. • Can be complex to maintain the mapping on larger sites, or sites where the URLs change often.
RSS, mRSS, and Atom 1.0	RSS, mRSS, and Atom 1.0 sitemaps are similar in structure to XML sitemaps, however they are often the easiest to provide because CMSes automatically create them. Pros: • Most CMSes automatically generate RSS and Atom feeds. • Can be used to provide Google information about your videos. Cons: • Besides HTML and other indexable textual content, it can only provide information about videos, not images or news. • Can be cumbersome to work with.
Text sitemap	The simplest of sitemap formats, it can only list URLs to HTML and other indexable pages.

Pros:

- Simple to do and maintain, especially on large sites.

Cons:

- Limited to HTML and other indexable textual content.

Sitemap best practices

The best practices for sitemaps are defined by the [sitemaps protocol](#). The most overlooked best practices are related to the size limits, sitemap location, and the URLs included in the sitemaps.

Sitemap size limits: All formats limit a single sitemap to 50MB (uncompressed) or 50,000 URLs. If you have a larger file or more URLs, you must break your sitemap into multiple sitemaps. You can optionally create a [sitemap index](#) file and submit that single index file to Google. You can submit multiple sitemaps and sitemap index files to Google. This may be useful if you want to track the search performance of each individual sitemap in Search Console.

Sitemap file encoding and location: The sitemap file must be UTF-8 encoded. You can host your sitemaps anywhere on your site, but unless you submit your sitemap through [Search Console](#), a sitemap affects only descendants of the parent directory. Therefore, a sitemap posted at the site root can affect all files on the site, which is where we recommend posting your sitemaps.

Referenced URLs' properties: Use fully-qualified, absolute URLs in your sitemaps. Google will attempt to crawl your URLs exactly as listed. For example, if your site is at <https://www.example.com/>, don't specify a URL such as </mypage.html> (a relative URL), use the complete, absolute URL: <https://www.example.com/mypage.html>.

Include the URLs in your sitemap that you want to see in Google's search results. Google generally shows the [canonical URLs](#) in its search results, which you can influence with sitemaps. If you have different URLs for mobile and desktop versions of a page, we recommend pointing to only one version in a sitemap. However, if you want to point to both URLs, [annotate](#) your URLs to indicate the desktop and mobile versions.

For a complete list of best practices, check out the [sitemaps protocol](#).

XML sitemap

The XML sitemap format is the most versatile of the supported formats. Using the Google supported sitemap extensions, you can also provide additional information about your [images](#), [video](#), and [news](#) content, as well as the [localized versions](#) of your pages.

Here is a very basic XML sitemap that includes the location of a single URL:

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
  <url>
    <loc>https://www.example.com/foo.html</loc>
    <lastmod>2022-06-04</lastmod>
  </url>
</urlset>
```

You can find more complex examples and full documentation at sitemaps.org.

Additional notes about XML sitemaps

- As with all XML files, all tag values must be [entity escaped](#).
- Google ignores <priority> and <changefreq> values.
- Google uses the <lastmod> value if it's consistently and verifiably (for example by comparing to the last modification of the page) accurate.

The <lastmod> value should reflect the date and time of the last significant update to the page. For example, an update to the main content, the structured data, or links on the page is generally considered significant, however an update to the copyright date is not.

RSS, mRSS, and Atom 1.0

If your CMS generates an RSS or Atom feed, you can submit the feed's URL as a sitemap. Most CMSes create a feed for you, however keep in mind that this feed only provides information on recent URLs.

Additional notes about RSS, mRSS, and Atom 1.0

- Google accepts RSS 2.0 and Atom 1.0 feeds.
- You can use an [mRSS \(media RSS\) feed](#) to provide Google details about video content on your site.
- As with all XML files, all tag values must be [entity escaped](#).

Text sitemap

If you only want to provide web page URLs, you can create a common text file that contains one URL per line and submit that to Google. For example, if you have two pages on your site, you could add them to your text sitemap located at <https://www.example.com/sitemap.txt> as follows:

```
https://www.example.com/file1.html  
https://www.example.com/file2.html
```

Additional notes for text file sitemaps

- Don't put anything other than URLs in the sitemap file.
- You can name the text file anything you want, provided it has a .txt extension (for instance, sitemap.txt).

How to create a sitemap

[\[Original embedded video\]](#)

When creating a sitemap, you're telling search engines about which URLs you prefer to show in search results. These are the [canonical URLs](#). If you have the same content accessible under different URLs, choose the URL you prefer and include that in the sitemap instead of all URLs that lead to the same content.

Once you've decided which URLs to include in the sitemap, pick one of the following ways to create a sitemap, depending on your site architecture and size:

- [Let your CMS generate a sitemap for you.](#)

- For sitemaps with less than a few dozen URLs, you can [manually create a sitemap](#).
- For sitemaps with more than a few dozen URLs, [automatically generate a sitemap](#).

Let your CMS generate a sitemap for you

If you're using a CMS such as WordPress, Wix, or Blogger, it's likely that your CMS has already made a sitemap available to search engines. Try searching for information about how your CMS generates sitemaps, or how to create a sitemap if your CMS doesn't generate a sitemap automatically. For example, in case of Wix, search for "wix sitemap", or in case of Blogger, search for "Blogger RSS".

Manually create a sitemap

For sitemaps with less than a few dozen URLs, you may be able to manually create a sitemap. For this, open a text editor such as [Windows Notepad](#) or [Nano \(Linux, MacOS\)](#), and follow a syntax described in the [Sitemap Formats](#) section. You can name the file anything you like as long as [the characters are allowed in a URL](#).

You can manually create larger sitemaps, but it's a tedious process and hard to maintain long term.

Automatically generate a sitemap with tools

For sitemaps with more than a few dozen URLs, you will need to generate the sitemap. There are various tools that can [generate a sitemap](#). However, the best way is to have your website software generate it for you. For example, you can extract your site's URLs from your website's database and then export the URLs to either the screen or actual file on your web server. Talk to your developers or server manager about this solution. If you need inspiration for the code, check out our old, unmaintained collection of [third-party sitemap generators](#).

You don't have to worry about the order of the URLs in your sitemap, it doesn't matter to Google. Note the [size requirements for sitemaps](#); if the sitemap becomes too large, you must split it into smaller sitemaps. Learn more about [managing large sitemaps](#).

Submit your sitemap to Google

Keep in mind that submitting a sitemap is merely a hint: it doesn't guarantee that Google will download the sitemap or use the sitemap for crawling URLs on the site. There are a few different ways to make your sitemap available to Google.

- **Submit a sitemap in Search Console** using the [Sitemaps report](#). This will allow you to see when Googlebot accessed the sitemap and also potential processing errors.
- **Use the Search Console API** to programmatically [submit a sitemap](#).
- **Insert the following line anywhere in your robots.txt file**, specifying the path to your sitemap. We will find it the next time we crawl your robots.txt file:

```
Sitemap: https://example.com/my_sitemap.xml
```

You can specify multiple sitemap lines, and there's no limit to the number of sitemaps you can include in your robots.txt file.

- If you use Atom or RSS, you can use [WebSub](#) to broadcast your changes to search engines, including Google.

How to cross-submit sitemaps for multiple sites

If you have multiple websites, you can simplify the submission process by creating one or more sitemaps that include URLs for all your verified sites, and saving the sitemaps to a single location. You can choose to use:

- A single sitemap that includes URLs for multiple websites, including sites from different domains. For example, the sitemap located at `https://host1.example.com/sitemap.xml` can include the following URLs.
 - `https://host1.example.com`
 - `https://host2.example.com`
 - `https://host3.example.com`
 - `https://host1.example1.com`
 - `https://host1.example.ch`
- Individual sitemaps (one for each site) that all reside in a single location. For example, `https://host1.example.com` may host all of these sitemaps:
 - `https://host1.example.com/host1-example-sitemap.xml`
 - `https://host1.example.com/host2-example-sitemap.xml`
 - `https://host1.example.com/host3-example-sitemap.xml`
 - `https://host1.example.com/host1-example1-sitemap.xml`
 - `https://host1.example.com/host1-example-ch-sitemap.xml`

To submit cross-site sitemaps that are hosted in a single location, you can either use Search Console or robots.txt.

Sitemap cross-submission with Search Console

1. Make sure that you have [verified ownership](#) of all the sites that you will add in the sitemap.
2. [Create a sitemap](#) (or more if you prefer) that includes URLs from all the sites that you want to cover. You can include the sitemaps in a [sitemap index](#) file if you prefer and work with that sitemap index from here on.
3. Using Google Search Console, [submit your sitemaps or sitemap index file](#).

Sitemap cross-submission with robots.txt

1. [Create one or more sitemaps](#) for each individual site. For each individual sitemap file, make sure you include only URLs from that particular site.
2. Upload all sitemaps to a single site you have control over, for example `https://sitemaps.example.com`.
3. For each individual site, make sure that the robots.txt file references the sitemap for that individual site. For example, if you created a sitemap for `https://example.com/` and you're hosting the sitemap at `https://sitemaps.example.com/sitemap-example-com.xml`, reference the sitemap in the robots.txt file at `https://example.com/robots.txt`.

```
# robots.txt file of https://example.com/  
sitemap: https://sitemaps.example.com/sitemap-example-com.xml
```

Troubleshooting sitemaps

If you're having trouble with your sitemap, you can investigate the errors with Google Search Console. See Search Console's [sitemaps troubleshooting guide](#) for help.

<https://developers.google.com/search/docs/crawling-indexing/sitemaps/large-sitemaps>

Manage your sitemaps with a sitemap index file

If you have a sitemap that exceeds the [size limits](#), you'll need to split up your large sitemap into multiple sitemaps such that each new sitemap is below the size limit. Once you've split up your sitemap, you can use a sitemap index file as a way to submit many sitemaps at once.

Sitemap index best practices

The XML format of a sitemap index file is very similar to the XML format of a sitemap file, and it's defined by the [Sitemap Protocol](#). This means that all the sitemap requirements apply to sitemap index files also.

The referenced sitemaps must be hosted on the same site as your sitemap index file. This requirement is waived if you set up [cross-site submission](#).

Sitemaps that are referenced in the sitemap index file must be in the same directory as the sitemap index file, or lower in the site hierarchy. For example, if the sitemap index file is at https://example.com/public/sitemap_index.xml, it can only contain sitemaps that are in the same or deeper directory, like <https://example.com/public/shared/...>

You can submit up to 500 sitemap index files for each site in your Search Console account.

Example sitemap index

The following example shows a sitemap index in XML format that lists two sitemaps:

```
<?xml version="1.0" encoding="UTF-8"?>
<sitemapindex xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
  <sitemap>
    <loc>https://www.example.com/sitemap1.xml.gz</loc>
    <lastmod>2024-08-15</lastmod>
  </sitemap>
  <sitemap>
    <loc>https://www.example.com/sitemap2.xml.gz</loc>
    <lastmod>2022-06-05</lastmod>
  </sitemap>
</sitemapindex>
```

Sitemap index reference

The sitemap index tags are defined by the same namespace as generic sitemaps:

<http://www.sitemaps.org/schemas/sitemap/0.9>

To make sure Google can use your sitemap index, you must use the following required tags:

Required tags	
sitemapindex	The root tag of the XML tree. It contains all the other tags.
sitemap	The parent tag for each sitemap listed in the file. It's the only direct child of the sitemapindex tag.
loc	The location (URL) of the sitemap. It's a child of the sitemap tag. A sitemap index file may have up to 50,000 loc tags.

Additionally, the following optional tags may help Google schedule your sitemaps for crawling:

Optional tags	
lastmod	Identifies the time that the corresponding sitemap file was modified. It can be a child of a sitemap tag. The value for the lastmod tag must be in W3C Datetime format .

Troubleshooting sitemaps

If you're having trouble with your sitemap, you can investigate the errors with Google Search Console. See Search Console's [sitemaps troubleshooting guide](#) for help.

Additional resources

Want to learn more? Check out the following resources:

- [Submit your sitemap to Google](#)
- [Learn how to combine sitemap extensions](#)

<https://developers.google.com/search/docs/crawling-indexing/sitemaps/image-sitemaps>

Image sitemaps

Image sitemaps are a way of telling Google about other images on your site, especially those that we might not otherwise find (such as images your site reaches with JavaScript code). You can create a separate image sitemap or add image sitemap tags to your existing sitemap; either approach is equally fine for Google.

Image sitemaps are based on generic sitemaps so the [general sitemap best practices](#) also apply to image sitemaps. We also recommend that you follow the [general best practices for publishing images](#).

Example image sitemap

The following example shows a regular sitemap with image sitemap extension, with two `<url>` elements:

- <https://example.com/sample1.html>, which contains two images
- <https://example.com/sample2.html>, which contains one image

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"
  xmlns:image="http://www.google.com/schemas/sitemap-image/1.1">
  <url>
    <loc>https://example.com/sample1.html</loc>
    <image:image>
      <image:loc>https://example.com/image.jpg</image:loc>
    </image:image>
    <image:image>
      <image:loc>https://example.com/photo.jpg</image:loc>
    </image:image>
  </url>
  <url>
    <loc>https://example.com/sample2.html</loc>
    <image:image>
      <image:loc>https://example.com/picture.jpg</image:loc>
    </image:image>
  </url>
</urlset>
```

Image sitemap reference

The image tags are defined in the Image Sitemaps namespace: <http://www.google.com/schemas/sitemap-image/1.1>

To make sure Google can use your image sitemap, you must use the following required tags:

Required tags

<code><image:image></code>	Encloses all information about a single image. Each <code><url></code> tag can contain up to 1,000 <code><image:image></code> tags.
<code><image:loc></code>	The URL of the image. In some cases, the image URL may not be on the same domain as your main site. This is fine, as long as you verify both domains in Search Console. If, for example, you use a content delivery network such as Google Sites to host your images, make sure that the hosting site is verified in Search Console. In addition, make sure that your robots.txt file doesn't disallow the crawling of any content you want indexed.

Deprecated tags and attributes

We removed the following tags and attributes from our documentation: `<image:caption>`, `<image:geo_location>`, `<image:title>`, `<image:license>`. See the [deprecation announcement](#) for more information.

Troubleshooting sitemaps

If you're having trouble with your sitemap, you can investigate the errors with Google Search Console. See Search Console's [sitemaps troubleshooting guide](#) for help.

Additional resources

Want to learn more? Check out the following resources:

- [Submit your sitemap to Google](#)
- [Learn how to combine sitemap extensions](#)

<https://developers.google.com/search/docs/crawling-indexing/sitemaps/news-sitemap>

News sitemaps

If you are a news publisher, use news sitemaps to tell Google about your news articles and additional information about them. You can either extend your existing sitemap with news specific tags, or create a separate news sitemap that's reserved just for your news articles. Either option is fine with Google, however creating a separate sitemap just for your news articles may enable better tracking of your content in Search in Search Console.

News sitemap best practices

News sitemaps are based on generic sitemaps, so the [general sitemap best practices](#) also apply to news sitemaps.

Update your news sitemap with fresh articles as they're published. Don't create a new sitemap with each update. Google News crawls news sitemaps as often as it crawls the rest of your site.

Only include recent URLs for articles that were created in the last two days. Once the articles are older than two days, either remove those URLs from the news sitemap or remove the `<news:news>` metadata in your sitemap from the older URLs.

If you choose the method of removing old URLs from your news sitemap, this could mean that your sitemap becomes empty for a period of time (for example, if you haven't published articles in the last few days). You may see an Empty Sitemap warning in Search Console, but this is just to make sure it was intentional on your behalf. It won't cause any problems with Google Search if the file is empty.

Example news sitemap

The following example shows a regular sitemap with news extension. It contains one `<url>` tag and a single `<news:news>` tag with its required child tags:

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"
  xmlns:news="http://www.google.com/schemas/sitemap-news/0.9">
  <url>
    <loc>http://www.example.org/business/article55.html</loc>
    <news:news>
      <news:publication>
        <news:name>The Example Times</news:name>
        <news:language>en</news:language>
      </news:publication>
      <news:publication_date>2008-12-23</news:publication_date>
      <news:title>Companies A, B in Merger Talks</news:title>
    </news:news>
  </url>
</urlset>
```

News sitemap reference

The news tags are defined in the news sitemap namespace: <http://www.google.com/schemas/sitemap-news/0.9>

To make sure Google can use your news sitemap, you must use the following required tags:

Required tags	
<news:news>	The parent tag of other tags in the news: namespace. Each url sitemap tag can have only one news:news tag (plus the respective closing tag) and a sitemap may have up to 1,000 news:news tags. If there are more than 1,000 <news:news> tags in a news sitemap, split your sitemap into several smaller sitemaps .
<news:publication>	The parent tag for the <news:name> and <news:language> tags. Each <news:news> parent tag may only have one <news:publication> tag.
<news:name>	The <news:name> tag is the name of the news publication. It must exactly match the name as it appears on your articles on news.google.com , omitting anything in parentheses.
<news:language>	The <news:language> tag is the language of your publication. Use an ISO 639 language code (two or three letters). Exception: For Simplified Chinese, use zh-cn and for Traditional Chinese, use zh-tw.
<news:publication_date>	The article publication date in W3C format. Use either the "complete date" format (YYYY-MM-DD) or the "complete date plus hours, minutes, and seconds" format with time zone designator format (YYYY-MM-DDThh:mm:ssTZD). Specify the original date and time when the article was first published on your site. Don't specify the time when you added the article to your sitemap. Google accepts any of the following formats: • Complete date: YYYY-MM-DD (1997-07-16) • Complete date plus hours and minutes: YYYY-MM-DDThh:mmTZD (1997-07-16T19:20+01:00) • Complete date plus hours, minutes, and seconds: YYYY-MM-DDThh:mm:ssTZD (1997-07-16T19:20:30+01:00) • Complete date plus hours, minutes, seconds, and a decimal fraction of a second: YYYY-MM-DDThh:mm:ss.sTZD (1997-07-16T19:20:30.45+01:00)
<news:title>	The title of the news article. Tip: Google may shorten the title of the news article for space reasons when displaying the article on various devices. Include the title of the article as it appears on your site. Don't include the author name, publication name, or publication date in the <news:title> tag. Learn more about creating better titles.

Troubleshooting sitemaps

If you're having trouble with your sitemap, you can investigate the errors with Google Search Console. See Search Console's [sitemaps troubleshooting guide](#) for help.

Additional resources

Want to learn more? Check out the following resources:

- [Submit your sitemap to Google](#)
- [Learn how to combine sitemap extensions](#)

<https://developers.google.com/search/docs/crawling-indexing/sitemaps/video-sitemaps>

Video sitemaps and alternatives

A video sitemap is a [sitemap](#) with additional information about videos hosted on your pages. Creating a video sitemap is a good way to help Google find and understand the video content on your site, especially content that was recently added or that we might not otherwise discover with our usual crawling mechanisms.

Google recommends using video sitemaps, however we also support [mRSS feeds](#).

Video sitemap best practices

Video sitemaps are based on generic sitemaps, so the [general sitemap best practices](#) also apply to video sitemaps. You can create a separate sitemap or mRSS feed just for video, or you can add video sitemap tags within an existing [sitemap](#), whichever is more convenient for you.

Additionally, the following requirements apply to video sitemaps specifically:

- Don't list videos that are unrelated to the content of the host page. For example, a video that is a small addendum to the page, or unrelated to the main text content.
- All files referenced in the video sitemap must be accessible to Googlebot. This means that all URLs in the video sitemap:
 - must not be disallowed for crawling by [robots.txt](#) rules,
 - must be accessible without metafiles and without logging in,
 - must not be blocked by firewalls or similar mechanism,
 - and must be accessible on a supported protocol: HTTP and FTP (streaming protocols are not supported).

If you want to prevent spammers from accessing your video content at the `<player_loc>` or `<content_loc>` URLs, [verify that any bots accessing your server are really Googlebot](#).

For more tips about videos in Google Search, see our [video best practices](#).

Example video sitemap

The following example shows a regular sitemap with video extension. It includes two video entries nested in the single `<url>` tag. The first `<video>` entry includes all the tags that Google can use while the second only the required tags.


```

<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"
  xmlns:video="http://www.google.com/schemas/sitemap-video/1.1">
  <url>
    <loc>https://www.example.com/videos/some_video_landing_page.html</loc>
    <video:video>
      <video:thumbnail_loc>https://www.example.com/thumbs/123.jpg</video:thumbnail_loc>
      <video:title>Grilling steaks for summer</video:title>
      <video:description>
        Alkis shows you how to get perfectly done steaks every time
      </video:description>
      <video:content_loc>
        http://streamserver.example.com/video123.mp4
      </video:content_loc>
      <video:player_loc>
        https://www.example.com/videoplayer.php?video=123
      </video:player_loc>
      <video:duration>600</video:duration>
      <video:expiration_date>2021-11-05T19:20:30+08:00</video:expiration_date>
      <video:rating>4.2</video:rating>
      <video:view_count>12345</video:view_count>
      <video:publication_date>2007-11-05T19:20:30+08:00</video:publication_date>
      <video:family_friendly>yes</video:family_friendly>
      <video:restriction relationship="allow">IE GB US CA</video:restriction>
      <video:platform relationship="allow">web tv</video:platform>
      <video:requires_subscription>yes</video:requires_subscription>
      <video:uploader
        info="https://www.example.com/users/grillymcgrillerson">GrillyMcGrillerson
      </video:uploader>
      <video:live>no</video:live>
      <video:tag>steak</video:tag>
      <video:tag>meat</video:tag>
      <video:tag>summer</video:tag>
    </video:video>
    <video:video>
      <video:thumbnail_loc>https://www.example.com/thumbs/345.jpg</video:thumbnail_loc>
      <video:title>Grilling steaks for winter</video:title>
      <video:description>
        In the freezing cold, Roman shows you how to get perfectly done steaks every time.
      </video:description>
      <video:content_loc>
        http://streamserver.example.com/video345.mp4
      </video:content_loc>
      <video:player_loc>
        https://www.example.com/videoplayer.php?video=345
      </video:player_loc>
    </video:video>
  </url>
</urlset>

```

More examples

The following example demonstrates how to add a Vimeo video embed to a video sitemap:

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"
  xmlns:video="http://www.google.com/schemas/sitemap-video/1.1">
  <url>
    <loc>https://www.example.com/videos/some_video_landing_page.html</loc>
    <video:video>
      <video:thumbnail_loc>https://www.example.com/thumbs/123.jpg</video:thumbnail_loc>
      <video:title>Lizzi is painting the wall</video:title>
      <video:description>
        Gary is watching the paint dry on the wall Lizzi painted.
      </video:description>
      <video:player_loc>
        https://player.vimeo.com/video/987654321
      </video:player_loc>
    </video:video>
  </url>
</urlset>
```

The following example demonstrates how to add a YouTube video embed to a video sitemap:

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"
  xmlns:video="http://www.google.com/schemas/sitemap-video/1.1">
  <url>
    <loc>https://www.example.com/videos/some_video_landing_page.html</loc>
    <video:video>
      <video:thumbnail_loc>https://www.example.com/thumbs/345.jpg</video:thumbnail_loc>
      <video:title>John teaches cheese</video:title>
      <video:description>
        John explains the differences between a banana and cheese.
      </video:description>
      <video:player_loc>
        https://www.youtube.com/embed/1a2b3c4d
      </video:player_loc>
    </video:video>
  </url>
</urlset>
```

Video sitemap reference

The video tags are defined in the video sitemaps namespace: <http://www.google.com/schemas/sitemap-video/1.1>. Each tag can only be added one time per video, unless otherwise specified.

To make sure Google can use your video sitemap, you must use the following required tags:

Required tags	
<video:video>	The parent element for all information about a single video on the page specified by the <loc> tag. You can include multiple <video:video> tags nested in the <loc> tag, one for each video on the hosting page.

<code><video:thumbnail_loc></code>	A URL pointing to the video thumbnail image file. Follow the video thumbnail requirements .
<code><video:title></code>	The title of the video. All HTML entities must be escaped or wrapped in a CDATA block . We recommend that this match the video title displayed on the web page where the video is embedded.
<code><video:description></code>	A description of the video. Maximum 2048 characters. All HTML entities must be escaped or wrapped in a CDATA block . It must match the description displayed on the web page where the video is embedded, but it doesn't need to be a word-for-word match.
<code><video:content_loc></code>	A URL pointing to the actual video media file. The file must be one of the supported formats. It's required to provide either a <code><video:content_loc></code> or <code><video:player_loc></code> tag. We recommend that you provide the <code><video:content_loc></code> tag, if possible. This is the most effective way for Google to fetch your video content files. If <code><video:content_loc></code> isn't available, provide <code><video:player_loc></code> as an alternative. Additional guidelines • HTML and Flash aren't supported formats. • Must not be the same as the URL in the parent <code><loc></code> tag. • This is the equivalent of <code>VideoObject.contentUrl</code> in structured data. • Best practice: If you want to restrict access to your content but still have it crawled, ensure that Googlebot can access your content by using verifying Googlebot.
<code><video:player_loc></code>	A URL pointing to a player for a specific video. Usually this is the information in the <code>src</code> attribute of an <code><embed></code> tag. It's required to provide either a <code><video:content_loc></code> or <code><video:player_loc></code> tag. We recommend that you provide the <code><video:content_loc></code> tag, if possible. This is the most effective way for Google to fetch your video content files. If <code><video:content_loc></code> isn't available, provide <code><video:player_loc></code> as an alternative. Additional guidelines • Must not be the same as the <code><loc></code> URL. • For Vimeo, YouTube, and other video hosting platforms that allow embedding videos through <code>iframe</code> videos, this value is used rather than <code>video:content_loc</code>. This is the equivalent of <code>VideoObject.embedUrl</code> in structured data.

- **Best practice:** If you want to restrict access to your content but still have it crawled, ensure that Googlebot can access your content by using [verifying Googlebot](#).

Additionally, the following optional tags may help Google better understand your videos and its properties:

Optional tags	
<code><video:duration></code>	The duration of the video, in seconds. Value must be from 1 to 28800 (8 hours).
<code><video:expiration_date></code>	The date after which the video is no longer be available, in W3C format. Omit this tag if your video doesn't expire. If present, Google Search won't show your video after this date. For recurring videos at the same URL, update the expiration date to the new expiration date. Supported values are complete date (YYYY-MM-DD), or complete date plus hours, minutes and seconds, and timezone (YYYY-MM-DDThh:mm:ss+TZD). Example: 2012-07-16T19:20:30+08:00.
<code><video:rating></code>	The rating of the video. Supported values are float numbers in the range 0.0 (low) to 5.0 (high).
<code><video:view_count></code>	The number of times the video has been viewed.
<code><video:publication_date></code>	The date the video was first published, in W3C format. Supported values are complete date (YYYY-MM-DD) or complete date plus hours, minutes and seconds, and timezone (YYYY-MM-DDThh:mm:ss+TZD). Example: 2007-07-16T19:20:30+08:00
<code><video:family_friendly></code>	Whether the video is available with SafeSearch. If you omit this tag, the video is available when SafeSearch is turned on. Supported values: • yes: The video is available when SafeSearch is turned on. • no: The video is only available when SafeSearch is turned off.
<code><video:restriction></code>	Whether to show or hide your video in search results from specific countries. Specify a space-delimited list of country codes in ISO 3166 format. If there's no <code><video:restriction></code> tag, Google assumes that the video can be shown in all locations. <i>Note that this tag only affects search results; it doesn't prevent a user from finding or playing your video in a restricted location through other means.</i> Learn more about applying country restrictions. Attributes: If the parent tag <code><video:restriction></code> is used, the following attribute is required:

	- relationship: Whether the video is allowed or denied in search results in the specified countries. Supported values are: - allow: The listed countries are allowed and unlisted countries are denied. - deny: The listed countries are denied and unlisted countries are allowed. Example: This example allows the video search result to be shown only in Canada and Mexico: <pre><video:restriction relationship="allow">CA MX</video:restriction></pre>
<video:platform>	Whether to show or hide your video in search results on specified platform types. This is a list of space-delimited platform types. <i>Note that this only affects search results on the specified device types; it doesn't prevent a user from playing your video on a restricted platform.</i> If there's no <video:platform> tag, Google assumes that the video can be played on all platforms. Learn more about applying platform restrictions. Supported values: - web: Computer browsers on desktops and laptops. - mobile: Mobile browsers, such as those on cellular phones or tablets. - tv: TV browsers, such as those available through Google TV devices and game consoles. Attributes: If the parent tag <video:platform> is used, the following attributes are required: - relationship: Specifies whether the video is restricted or permitted for the specified platforms. Supported values are: - allow: Any omitted platforms will be denied. - deny: Any omitted platforms will be allowed. Example: The following example allows users on web or TV, but not mobile devices: <pre><video:platform relationship="allow">web tv</video:platform></pre>
<video:requires_subscription>	Indicates whether a subscription (either paid or free) is required to view the video. Supported values are: - yes: Subscription is required. - no: Subscription is not required.
<video:uploader>	The video uploader's name. The string value can be a maximum of 255 characters. Attributes: info [<i>Optional</i>]: Specifies the URL of a web page with additional information about this uploader. This URL must be in the same domain as the <loc> tag.
<video:live>	Indicates whether the video is a livestream. Supported values are: yes: The video is a livestream.

	• no: The video is not a livestream.
<code><video:tag></code>	An arbitrary string tag describing the video. Tags are generally very short descriptions of key concepts associated with a video or piece of content. A single video could have several tags, although it might belong to only one category. For example, a video about grilling food may belong in the "grilling" category, but could be tagged "steak", "meat", "summer", and "outdoor". Create a new <code><video:tag></code> element for each tag associated with a video. A maximum of 32 tags is permitted per video.

Deprecated tags and attributes

We removed the following tags and attributes from our documentation: `<video:category>`, `<video:gallery_loc>`, the `autoplay` and `allow_embed` attributes of the `<video:player_loc>` tag, the `<video:price>` tag and its attributes, and the `<video:tvshow>` tag and its attributes. See the [deprecation announcement](#) for more information.

Sitemap alternative: mRSS

While Google recommends using video sitemaps, we also support mRSS feeds.

Google supports [mRSS](#), an RSS module that supplements the element capabilities of [RSS 2.0](#). mRSS feeds are very similar to video sitemaps and can be tested, submitted, and updated just like sitemaps.

For more information about media feeds, see the [official media RSS documentation](#).

RSS vs mRSS: mRSS is a RSS extension used for syndicating multimedia files. It allows for a much more detailed description of the content than the RSS standard.

mRSS Example

Here's an example of an mRSS entry that provides all the tags that Google uses.

```

<?xml version="1.0" encoding="UTF-8"?>
<rss version="2.0" xmlns:media="http://search.yahoo.com/mrss/" xmlns:dcterms="http://purl.org/dc/terms/">
  <channel>
    <title>Example MRSS</title>
    <link>https://www.example.com/examples/mrss/</link>
    <description>MRSS Example</description>
    <item xmlns:media="http://search.yahoo.com/mrss/" xmlns:dcterms="http://purl.org/dc/terms/">
      <link>https://www.example.com/examples/mrss/example.html</link>
      <media:content url="https://www.example.com/examples/mrss/example.flv" fileSize="405321"
        type="video/x-flv" height="240" width="320" duration="120" medium="video"
isDefault="true">
        <media:player url="https://www.example.com/shows/example/video.swf?flash_params" />
        <media:title>Grilling Steaks for Summer</media:title>
        <media:description>Get perfectly done steaks every time</media:description>
        <media:thumbnail url="https://www.example.com/examples/mrss/example.png" height="120" width="160"/>
        <media:price price="19.99" currency="EUR" />
        <media:price type="subscription" />
      </media:content>
      <media:restriction relationship="allow" type="country">us ca</media:restriction>
      <dcterms:valid xmlns:dcterms="http://purl.org/dc/terms/">end=2020-10-15T00:00+01:00; scheme=W3C-
DTF</dcterms:valid>
      <dcterms:type>live-video</dcterms:type>
    </item>
  </channel>
</rss>

```

mRSS reference

The [full mRSS specification](#) contains more optional tags, best practices, and examples.

To make sure Google can use your mRSS feed, you must use the following required tags:

Required tags	
<media:content>	Encloses information about the video. Attributes: • medium: Type of content. Set to video. • url: The direct URL to the raw video content. If this isn't specified, you must specify the <media:player> tag. • duration [<i>Optional but recommended</i>]: Length of the video in seconds. For all of the other optional attributes and child fields of the <media:content> tag, see the mRSS specification.
<media:player>	You must specify at least one of <media:player> or the url attribute in <media:content>. A URL pointing to a player for a specific video. Usually this is the information in the src attribute of an <embed> tag and must not be the same as the content of the <loc> tag. It can't be the same URL as the <link> tag. The <link> tag points to the URL of the page hosting the video, while this tag points to a player.

<code><media:title></code>	The title of the video. Maximum 100 characters. All HTML entities must be escaped or wrapped in a CDATA block .
<code><media:description></code>	The description of the video. Maximum 2048 characters. All HTML entities must be escaped or wrapped in a CDATA block .
<code><media:thumbnail></code>	A URL pointing to a preview thumbnail. Follow the Video thumbnail requirements .

Additionally, the following optional tags may help Google better understand your videos and its properties:

Optional tags	
<code><dcterms:valid></code>	The publication and expiration date of the video. Here's the Full specification of the dcterms:valid tag. Example: <pre><dcterms:valid> start=2002-10-13T09:00+01:00; end=2002-10-17T17:00+01:00; scheme=W3C-DTF <dcterms:valid></pre>
<code><media:restriction></code>	A space-delimited list of countries where the video may or may not be played, in ISO 3166 format. If there's no <code><media:restriction></code> tag, Google assumes that the video can be played in all countries. Attributes: If the parent tag <code><media:restriction></code> is used, the following attributes are required: • type: Set the type attribute to country. Only country restrictions are supported. • relationship: Specifies whether the video may or may not be played in the specified list of countries. Supported values: ◦ allow: The listed countries are allowed and unlisted countries are denied. ◦ deny: The listed countries are denied and unlisted countries are allowed. Learn more about using country restrictions. Example: <pre><media:restriction relationship="allow" type="country">us ca</media:restriction></pre>
<code><media:price></code>	The price to download or view the video. Don't use this tag for videos that are available without payment. More than one <code><media:price></code> element can be listed (for example, in order to specify various currencies or purchasing options).

Attributes:

If the parent tag `<media:price>` is used, the following attributes are required:

- **currency:** The currency in [ISO 4217 format](#).
- **type:** The purchase option. Supported values are:
 - **rent:** The video is available for rent.
 - **purchase:** The video is available for purchase.
 - **package:** The video is part of a package deal.
 - **subscription:** The video is available with a subscription.

Troubleshooting sitemaps

If you're having trouble with your sitemap, you can investigate the errors with Google Search Console. See Search Console's [sitemaps troubleshooting guide](#) for help.

Additional resources

Want to learn more? Check out the following resources:

- [Submit your sitemap to Google](#)
- [Learn how to combine sitemap extensions](#)

<https://developers.google.com/search/docs/crawling-indexing/sitemaps/combine-sitemap-extensions>

How to combine sitemap extensions

Sitemap extensions are a great way to tell Google about the different kinds of content and their metadata that you're using on your site. Often the content on your pages may fit into multiple kinds of extensions; for example, you might be publishing news articles that embed images and videos. Additionally, your pages may be localized as well, which might mean that you could add hreflang annotations for your localized pages.

Namespaces

For each sitemap extension that you want to use in a sitemap you need to specify the respective namespace that declares the tags the extension supports. This is done with the xmlns attribute of the urlset tag. The namespaces for the sitemap extensions Google supports are:

Extension tags and their namespace definitions	
image:	http://www.google.com/schemas/sitemap-image/1.1
news:	http://www.google.com/schemas/sitemap-news/0.9
video:	http://www.google.com/schemas/sitemap-video/1.1
xhtml: (for hreflang)	http://www.w3.org/1999/xhtml

Declaring multiple namespaces

To declare multiple namespaces, add the respective namespace references to your sitemap as described in the documentation of the respective extensions. Here's an example that shows how to add the news, video, and xhtml (for hreflang) extensions to a sitemap:

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"
        xmlns:news="http://www.google.com/schemas/sitemap-news/0.9"
        xmlns:video="http://www.google.com/schemas/sitemap-video/1.1"
        xmlns:xhtml="http://www.w3.org/1999/xhtml">

  <url>
    <!-- rest of the sitemap -->
```

Combining sitemap extensions

Once you declared the namespaces, follow the implementation details from the respective sitemap extension documentation you're going to use.

- [Image sitemap](#)

- [News sitemap](#)
- [Video sitemap](#)
- [hreflang](#)

To combine extensions, add the tags from whatever sitemap extensions you're using, one after the other, to the appropriate `<url>` tag, as described in each sitemap extension's documentation.

For example, to add news, video, and xhtml (hreflang) extensions to a sitemap:

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"
  xmlns:news="http://www.google.com/schemas/sitemap-news/0.9"
  xmlns:video="http://www.google.com/schemas/sitemap-video/1.1"
  xmlns:xhtml="http://www.w3.org/1999/xhtml">
  <url>
    <loc>https://www.example.com/english/page.html</loc>
    <!-- Starting with the news extension tags -->
    <news:news>
      <news:publication>
        <news:name>The Example Times</news:name>
        <news:language>en</news:language>
      </news:publication>
      <news:publication_date>2008-12-23</news:publication_date>
      <news:title>Companies A, B in Merger Talks</news:title>
    </news:news>
    <!-- Next we add video extension tags -->
    <video:video>
      <video:thumbnail_loc>https://www.example.com/thumbs/123.jpg</video:thumbnail_loc>
      <video:title>Lizzi is painting the wall</video:title>
      <video:description>
        Gary is watching the paint dry on the wall Lizzi painted.
      </video:description>
      <video:player_loc>
        https://player.example.com/video/987654321
      </video:player_loc>
    </video:video>
    <!-- And finally the xhtml tags for hreflang -->
    <xhtml:link
      rel="alternate"
      hreflang="de"
      href="https://www.example.de/deutsch/page.html"/>
    <xhtml:link
      rel="alternate"
      hreflang="de-ch"
      href="https://www.example.de/schweiz-deutsch/page.html"/>
    <xhtml:link
      rel="alternate"
      hreflang="en"
      href="https://www.example.com/english/page.html"/>
  </url>
  <!-- Add more <url> tags -->
```

The order of the extension in the sitemap is irrelevant after the `<loc>` tag. Keep in mind the [general sitemap best practices](#), especially the file size limits. Combining sitemap extensions increases the file size of your sitemap significantly.

Troubleshooting sitemaps

If you're having trouble with your sitemap, you can investigate the errors with Google Search Console. See Search Console's [sitemaps troubleshooting guide](#) for help.

<https://developers.google.com/search/docs/crawling-indexing/ask-google-to-recrawl>

Ask Google to recrawl your URLs

Are you using a hosted content management platform (for example, Blogger or WordPress)? In most cases, the platform submits your new content to search engines automatically. Check your platform's support articles.

If you've recently added or made changes to a page on your site, you can request that Google re-index your page using any of the methods listed here. You can't request indexing for URLs that you don't manage.

Crawling can take anywhere from a few days to a few weeks. Be patient and monitor progress using either the [Index Status report](#) or the [URL Inspection tool](#).

Requesting a crawl does not guarantee that inclusion in search results will happen instantly or even at all. Our systems prioritize the fast inclusion of high quality, useful content.

Use the URL Inspection tool (just a few URLs)

To request a crawl of individual URLs, use the [URL Inspection tool](#). You must be an [owner or full user of the Search Console property](#) to be able to request indexing in the URL Inspection tool.

Keep in mind that there's a quota for submitting individual URLs and requesting a recrawl multiple times for the same URL won't get it crawled any faster.

Submit a sitemap (many URLs at once)

If you have large numbers of URLs, submit a sitemap. A sitemap is an important way for Google to discover URLs on your site. It can be very helpful if you just launched your site or recently performed a [site move](#). A sitemap can also include additional metadata about alternate language versions, video, image, or news-specific pages. [Learn how to create and submit a sitemap](#).

<https://developers.google.com/search/docs/crawling-indexing/troubleshoot-crawling-errors>

Troubleshoot Google Search crawling errors

Here are the key steps to troubleshooting and fixing Google Search crawling issues for your site:

1. See if Googlebot is encountering availability issues on your site.
2. See whether you have pages that aren't being crawled, but should be.
3. See whether any parts of your site need to be crawled more quickly than they already are.
4. Improve your site's crawl efficiency.
5. Handle overcrawling of your site.

See if Googlebot is encountering availability issues on your site

Improving your site availability won't necessarily increase your crawl budget; Google determines the best crawl rate based on the crawl demand, as described previously. However, availability issues do prevent Google from crawling your site as much as it might want to.

Diagnosing:

Use the [Crawl Stats report](#) to see Googlebot's crawling history for your site. The report shows when Google encountered availability issues on your site. If availability errors or warnings are reported for your site, look for instances in the **Host availability** graphs where Googlebot requests exceeded the red limit line, click into the graph to see which URLs were failing, and try to correlate those with issues on your site.

Additionally, you can also use the [URL Inspection Tool](#) to test a few URLs on your site. If the tool returns **Hostload exceeded** warnings, that means that Googlebot can't crawl as many URLs from your site as it discovered.

Treating:

- Read the [documentation for the Crawl Stats report](#) to learn how to find and handle some availability issues.
- **Block pages from crawling if you don't want them to be crawled.** (See [manage your inventory](#))
- **Increase page loading and rendering speed.** (See [Improve your site's crawl efficiency](#))
- **Increase your server capacity.** If Google consistently seems to be crawling your site at its serving capacity limit, but you still have important URLs that aren't being crawled or updated as much as they need, having more serving resources might enable Google to request more pages on your site. Check your host availability history in the [Crawl Stats report](#) to see if Google's crawl rate seems to be crossing the limit line often. If so, increase your serving resources for a month and see whether crawling requests increased during that same period.

See if any parts of your site are not crawled, but should be

Google spends as much time as necessary on your site in order to index all the high-quality, user-valuable content that it can find. If you think that Googlebot is missing important content, either it doesn't know about the content, the content is blocked from Google, or your site availability is throttling Google's access (or Google is trying not to overload your site).

Remember the difference between *crawling* and *indexing*. This page is about helping Google *crawl* your site efficiently, not whether the pages found make it into the index.

Diagnosing:

Search Console doesn't provide a crawl history for your site that can be filtered by URL or path, but you can inspect your site logs to see whether specific URLs have been crawled by [Googlebot](#). Whether or not those crawled URLs have been indexed is another story.

Remember that for most sites, new pages will take several days minimum to be noticed; most sites shouldn't expect same-day crawling for URLs, with the exception of time-sensitive sites such as news sites.

Treating:

If you are adding pages to your site and they are not being crawled in a reasonable amount of time, either Google doesn't know about them, the content is blocked, your site has reached its maximum serving capacity, or you are [out of crawl budget](#).

1. Tell Google about your new pages: update your sitemaps to reflect new URLs.
2. Examine your robots.txt rules to confirm that you're not accidentally blocking pages.
3. Review your crawling priorities (a.k.a. use your crawl budget wisely). [Manage your inventory](#) and [improve your site's crawling efficiency](#).
4. [Check that you're not running out of serving capacity](#). Googlebot will scale back its crawling if it detects that your servers are having trouble responding to crawl requests.

Note that pages might not be shown in search results, even if crawled, if there isn't sufficient value or user demand for the content.

See if updates are crawled quickly enough

If we're missing new or updated pages on your site, perhaps it's because we haven't seen them, or haven't noticed that they are updated. Here is how you can help us be aware of page updates.

Note that Google strives to check and index pages in a reasonably timely manner. For most sites, this is three days or more. Don't expect Google to index pages the same day that you publish them unless you are a news site or have other high-value, extremely time-sensitive content.

Diagnosing:

Examine your site logs to see when specific URLs were crawled by [Googlebot](#).

To learn the indexing date, use the URL Inspection tool or do a search for URLs that you updated.

Treating:

Do:

- Use a [news sitemap](#) if your site has news content.
- Use the `<lastmod>` tag in sitemaps to indicate when an indexed URL has been updated.
- Use a [crawlable URL structure](#) to help Google find your pages.
- Provide standard, [crawlable <a> links](#) to help Google find your pages.
- If your site uses separate HTML for mobile and desktop versions, provide the same set of links on the mobile version as you have on the desktop version. If it's not possible to provide the same set of links on the mobile

version, ensure that they're included in a [sitemap](#) file. Google only [indexes the mobile version](#) of pages, and limiting the links shown there can slow down discovery of new pages.

Avoid:

- Submitting the same, unchanged sitemap multiple times per day.
- Expecting that Googlebot will crawl everything in a sitemap, or crawl them immediately. Sitemaps are useful suggestions to Googlebot, not absolute requirements.
- Including URLs in your sitemaps that [you don't want to appear in Search](#). This can waste your crawl budget on pages that you don't want indexed.

Improve your site's crawl efficiency

Increase your page loading speed

Google's crawling is limited by bandwidth, time, and availability of Googlebot instances. If your server responds to requests quicker, we might be able to crawl more pages on your site. That said, Google only wants to crawl high quality content, so just making low quality pages faster won't encourage Googlebot to crawl more of your site; conversely, if we think that we're missing high-quality content on your site, we'll probably increase your budget to crawl that content.

Here's how you can optimize your pages and resources for crawling:

- Prevent large but unimportant resources from being loaded by Googlebot using robots.txt. Be sure to block only non-critical resources—that is, resources that aren't important to understanding the meaning of the page (such as decorative images).
- Make sure that your pages are fast to load.
- Watch out for long redirect chains, which have a negative effect on crawling.
- Both the time to respond to server requests, as well as the time needed to render pages, matters, including load and run time for embedded resources such as images and scripts. Be aware of large or slow resources required for indexing.

Specify content changes with HTTP status codes

Google generally supports the [If-Modified-Since](#) and [If-None-Match](#) HTTP request headers for crawling. Google's crawlers don't send the headers with all crawl attempts; it depends on the use case of the request (for example, [AdsBot](#) is more likely to set the If-Modified-Since and If-None-Match HTTP request headers). If our crawlers send the If-Modified-Since header, the header's value is the [date and time](#) the content was last crawled. Based on that value, the server may choose to return a 304 (Not Modified) HTTP status code with no response body, in which case Google will reuse the content version it crawled the last time. If the content is newer than the date specified by the crawler in the If-Modified-Since header, the server can return a 200 (OK) HTTP status code with the response body.

Independently of the request headers, you can send a 304 (Not Modified) HTTP status code and no response body for any Googlebot request if the content hasn't changed since Googlebot last visited the URL. This will save your server processing time and resources, which may indirectly improve crawl efficiency.

Hide URLs that you don't want in search results

Wasting server resources on unnecessary pages can reduce crawl activity from pages that are important to you, which may cause a significant delay in discovering great new or updated content on a site.

Blocking or hiding already crawled pages from recrawls won't shift your crawl budget to another part of your site unless Google is already hitting your site's serving limits.

Exposing many URLs on your site that you don't want crawled by Search can negatively affect a site's crawling and indexing. Typically these URLs fall into the following categories:

- **Faceted navigation** and **session identifiers**: Faceted navigation is typically duplicate content from the site; session identifiers and other URL parameters that simply sort or filter the page don't provide new content. Learn how to [manage crawling of faceted navigation pages](#).
- **Duplicate content**: Help Google identify duplicate content to avoid unnecessary crawling.
- **soft 404 pages**: Return a 404 code when a page no longer exists.
- **Hacked pages**: Be sure to check the [Security Issues report](#) and fix or remove any hacked pages you find.
- **Infinite spaces** and proxies: Block these from crawling with robots.txt.
- **Low quality and spam content**: Good to avoid, obviously.
- Shopping cart pages, infinite scrolling pages, and pages that perform an action (such as "sign up" or "buy now" pages).

Do:

- Use robots.txt if you don't want Google to crawl a resource or page at all.
- If a common resource is reused on multiple pages (such as a shared image or JavaScript file), reference the resource from the same URL in each page, so that Google can cache and reuse the same resource without needing to request the same resource multiple times.

Avoid:

- Don't add or remove pages or directories from robots.txt regularly as a way of reallocating crawl budget for your site. Use robots.txt only for pages or resources that you don't want to appear on Google for the long run.
- Don't rotate sitemaps or use other temporary hiding mechanisms to reallocate budget.

soft 404 errors

A soft 404 error is when a URL that returns a page telling the user that the page does not exist **and also a 200 (success) status code**. In some cases, it might be a page with no main content or empty page.

Such pages may be generated for various reasons by your website's web server or content management system, or the user's browser. For example:

- A missing server-side include file.
- A broken connection to the database.
- An empty internal search result page.
- An unloaded or otherwise missing JavaScript file.

It's a bad user experience to return a 200 (success) status code, but then display or suggest an error message or some kind of error on the page. Users may think the page is a live working page, but then are presented with some kind of error. Such pages are excluded from Search.

When Google's algorithms detect that the page is actually an error page based on its content, Search Console will show a soft 404 error in the site's [Page Indexing report](#).

Fix soft 404 errors

Depending on the state of the page and the outcome you want, you can solve soft 404 errors in multiple ways:

- The page and content are no longer available.
- The page or content is now somewhere else.
- The page and content still exist.

Try to determine which solution would be the best for your users.

The page and content are no longer available

If you removed the page and there's no replacement page on your site with similar content, return a [404 \(not found\)](#) or [410 \(gone\)](#) response (status) code for the page. These status codes indicate to search engines that the page doesn't exist and you don't want search engines to index the page.

If you have access to your server's configuration files, you can make these error pages useful to users by customizing them. A good custom 404 page helps people find the information they're looking for, and also provides other helpful content that encourages people to explore your site further. Here are some tips for designing a useful custom 404 page:

- Tell visitors clearly that the page they're looking for can't be found. Use language that is friendly and inviting.
- Make sure your 404 page has the same look and feel (including navigation) as the rest of your site.
- Consider adding links to your most popular articles or posts, as well as a link to your site's home page.
- Think about providing a way for users to report a broken link.

Custom 404 pages are created solely for users. Since these pages are useless from a search engine's perspective, make sure the server returns a 404 HTTP status code to prevent having the pages indexed.

The page or content is now somewhere else

If your page has moved or has a clear replacement on your site, return a [301 \(permanent redirect\)](#) to redirect the user. This won't interrupt their browsing experience and it's also a great way to tell search engines about the new location of the page. Use the [URL Inspection tool](#) to verify whether your URL is actually returning the correct code.

The page and content still exist

If an otherwise good page was flagged with a soft 404 error, it's likely it didn't load properly for Googlebot, it was missing critical resources, or it displayed a prominent error message during rendering. Use the [URL Inspection tool](#) to examine the rendered content and the returned HTTP code. If the rendered page is blank, nearly blank, or the content has an error message, it could be that your page references many resources that can't be loaded (images, scripts, and other non-textual elements), which can be interpreted as a soft 404. Reasons that resources can't be loaded include blocked resources (blocked by [robots.txt](#)), having too many resources on a page, various server errors, or slow loading or very large resources.

Handle overcrawling of your site (emergencies)

Googlebot has algorithms to prevent it from overwhelming your site with crawl requests. However, if you find that Googlebot is overwhelming your site, there are a few things you can do.

Diagnosing:

Monitor your server for excessive Googlebot requests to your site.

Treating:

In an emergency, we recommend the following steps to slow down an overwhelming crawl from Googlebot:

1. Return 503 or 429 HTTP response status codes *temporarily* for Googlebot requests when your server is overloaded. Googlebot will retry these URLs for about 2 days. Note that returning "no availability" codes for more than a few days will cause Google to permanently slow or stop crawling URLs on your site, so follow the additional next steps.
2. When the crawl rate goes down, stop returning 503 or 429 HTTP response status codes for crawl requests; returning 503 or 429 for more than 2 days will cause Google to drop those URLs from the index.
3. Monitor your crawling and your host capacity over time.
4. If the problematic crawler is one of the [AdsBot crawlers](#), the problem is likely that you have created [Dynamic Search Ad targets](#) for your site that Google is trying to crawl. This crawl will reoccur every 3 weeks. If you don't have the server capacity to handle these crawls, either limit your ad targets or get increased serving capacity.

<https://developers.google.com/crawling/docs/crawlers-fetchers/overview-google-crawlers>

Overview of Google crawlers and fetchers (user agents)

Google uses crawlers and fetchers to perform actions for its products, either automatically or triggered by user request. Crawler (sometimes also called a "robot" or "spider") is a generic term for any program that is used to [automatically discover and scan websites](#). Fetchers act as a program like [wget](#) that typically make a single request on behalf of a user. Google's clients fall into three categories:

Looking for the latest updates to this page? See our [updates to our documentation](#).

Common crawlers	The common crawlers used for Google's products (such as Googlebot). They always respect robots.txt rules for automatic crawls.
Special-case crawlers	Special-case crawlers are similar to common crawlers, however are used by specific products where there's an agreement between the crawled site and the Google product about the crawl process. For example, AdsBot ignores the global robots.txt user agent (*) with the ad publisher's permission.
User-triggered fetchers	User-triggered fetchers are part of tools and product functions where the end user triggers a fetch. For example, Google Site Verifier acts on the request of a user.

Technical properties of Google's crawlers and fetchers

Google's crawlers and fetchers are designed to be run simultaneously by thousands of machines to improve performance and scale as the web grows. To optimize bandwidth usage, these clients are distributed across many datacenters across the world so they're located near the sites that they might access. Therefore, your logs may show visits from several IP addresses. Google egresses primarily from IP addresses in the United States. In case Google detects that a site is blocking requests from the United States, it may attempt to crawl from IP addresses located in other countries.

Supported transfer protocols

Google's crawlers and fetchers support HTTP/1.1 and [HTTP/2](#). The crawlers will use the protocol version that provides the best crawling performance and may switch protocols between crawling sessions depending on previous crawling statistics. The default protocol version used by Google's crawlers is HTTP/1.1; crawling over HTTP/2 may save computing resources (for example, CPU, RAM) for your site and Googlebot, but otherwise there's no Google-product specific benefit to the site (for example, no ranking boost in Google Search). To opt out from crawling over HTTP/2, instruct the server that's hosting your site to respond with a 421 HTTP status code when Google attempts to access your site over HTTP/2. If that's not feasible, you [can send a message to the Crawling team](#) (however this solution is temporary).

Google's crawler infrastructure also supports crawling through FTP (as defined by [RFC959](#) and its updates) and FTPS (as defined by [RFC4217](#) and its updates), however crawling through these protocols is rare.

Supported content encodings

Google's crawlers and fetchers support the following content encodings (compressions): [gzip](#), [deflate](#), and [Brotli \(br\)](#). The content encodings supported by each Google user agent is advertised in the Accept-Encoding header of each request they make. For example, Accept-Encoding: gzip, deflate, br.

File size limits

By default, Google's crawlers and fetchers only crawl the first 15MB of a file, and any content beyond this limit is ignored. However, individual projects may set different limits for their crawlers and fetchers, and also for different file types. For example, a Google crawler [like Googlebot](#) may have a smaller size limit (for example, 2MB), or specify a larger file size limit for a PDF than for HTML.

Crawl rate and host load

Our goal is to crawl as many pages from your site as we can on each visit without overwhelming your server. If your site is having trouble keeping up with Google's crawling requests, you can [reduce the crawl rate](#). Note that sending the inappropriate [HTTP response code](#) to Google's crawlers may affect how your site appears in Google products.

HTTP Caching

Google's crawling infrastructure supports heuristic HTTP caching as defined by the [HTTP caching standard](#), specifically through the ETag response- and If-None-Match request header, and the Last-Modified response- and If-Modified-Since request header.

Note: Consider setting both the Etag and Last-Modified values regardless of the preference of Google's crawlers. These headers are also used by other applications such as CMSes.

If both ETag and Last-Modified response header fields are present in the HTTP response, Google's crawlers use the ETag value as [required by the HTTP standard](#). For Google's crawlers specifically, we recommend using [ETag](#) instead of the Last-Modified header to indicate caching preference as ETag doesn't have date formatting issues.

Other HTTP caching directives aren't supported.

Individual Google crawlers and fetchers may or may not make use of caching, depending on the needs of the product they're associated with. For example, Googlebot supports caching when re-crawling URLs for Google Search, and Storebot-Google only supports caching in certain conditions.

To implement HTTP caching for your site, get in touch with your hosting or content management system provider.

ETag and If-None-Match

Google's crawling infrastructure supports ETag and If-None-Match as defined by the [HTTP Caching standard](#). Learn more about the [ETag](#) response header and its request header counterpart, [If-None-Match](#).

Last-Modified and If-Modified-Since

Google's crawling infrastructure supports Last-Modified and If-Modified-Since as defined by the [HTTP Caching standard](#) with the following caveats:

- The date in the `Last-Modified` header must be formatted according to the [HTTP standard](#). To avoid parsing issues, we recommend using the following date format: "Weekday, DD Mon YYYY HH:MM:SS Timezone". For example, "Fri, 4 Sep 1998 19:15:56 GMT".
- While not required, consider also setting the `max-age` field of the `Cache-Control` response header to help crawlers determine when to recrawl the specific URL. Set the value of the `max-age` field to the expected number of seconds the content will be unchanged. For example, `Cache-Control: max-age=94043`.

Learn more about the [Last-Modified](#) response header and its request header counterpart, [If-Modified-Since](#).

Verify Google's crawlers and fetchers

Google's crawlers identify themselves in three ways:

1. The HTTP user-agent request header.
2. The source IP address of the request.
3. The reverse DNS hostname of the source IP.

Learn how to use these details to [verify Google's crawlers and fetchers](#).

<https://developers.google.com/search/docs/crawling-indexing/googlebot>

Googlebot

Googlebot is the generic name for two types of [web crawlers](#) used by Google Search:

- **Googlebot Smartphone:** a mobile crawler that simulates a user on a mobile device.
- **Googlebot Desktop:** a desktop crawler that simulates a user on desktop.

You can identify the subtype of Googlebot by looking at the [HTTP user-agent request header](#) in the request. However, both crawler types obey the same product token (user agent token) in robots.txt, and so you cannot selectively target either Googlebot Smartphone or Googlebot Desktop using robots.txt.

For most sites Google Search primarily [indexes the mobile version](#) of the content. As such the majority of Googlebot crawl requests will be made using the mobile crawler, and a minority using the desktop crawler.

How Googlebot accesses your site

For most sites, Googlebot shouldn't access your site more than once every few seconds on average. However, due to delays it's possible that the rate will appear to be slightly higher over short periods. If your site is having trouble keeping up with Google's crawling requests, you can [reduce the crawl rate](#).

When crawling for Google Search, Googlebot crawls the first 2MB of a [supported file type](#), and the first 64MB of a PDF file. From a rendering perspective, each resource referenced in the HTML (such as CSS and JavaScript) is fetched separately, and each resource fetch is bound by the same file size limit that applies to other files (except PDF files).

Once the cutoff limit is reached, Googlebot stops the fetch and only sends the already downloaded part of the file for indexing consideration. The file size limit is applied on the uncompressed data. Other Google crawlers, for example Googlebot Video and Googlebot Image, may have [different limits](#).

When crawling from IP addresses in the US, the timezone of Googlebot is [Pacific Time](#).

Other [technical properties of Googlebot](#) are described in the overview of Google's crawlers.

Blocking Googlebot from visiting your site

Googlebot discovers new URLs to crawl primarily from links embedded in previously crawled pages. It's almost impossible to keep a site secret by not publishing links to it. For example, as soon as someone clicks a link from your "secret" site to another site, your "secret" site URL may appear in the referrer tag and can be stored and published by the other site in its referrer log.

If you want to prevent Googlebot from crawling content on your site, you have a [number of options](#). Remember there's a difference between *crawling* and *indexing*; blocking Googlebot from crawling a page doesn't prevent the URL of the page from appearing in search results:

- **Prevent Googlebot from crawling a page?** Use a [robots.txt file](#).
- **Don't want Google to index a page?** Use [noindex](#).

- Prevent a page from being accessible at all by both crawlers or users? Use [another method](#), such as [password protection](#).

Blocking Googlebot affects Google Search (including Discover and all Google Search features), as well as other products such as Google Images, Google Video, and Google News.

Verifying Googlebot

Before you decide to block Googlebot, be aware that the HTTP user-agent request header used by Googlebot is often spoofed by other crawlers. It's important to verify that a problematic request actually comes from Google. The best way to verify that a request actually comes from Googlebot is to [use a reverse DNS lookup](#) on the source IP of the request, or to match the source IP against the [Googlebot IP ranges](#).

<https://developers.google.com/crawling/docs/crawlers-fetchers/reduce-crawl-rate>

Reduce the Google crawl rate

Google's crawler infrastructure has sophisticated algorithms to determine the optimal crawl rate for a site. Our goal is to crawl as many pages from your site as we can on each visit without overwhelming your server. In some cases, Google's crawling of your site might be causing a critical load on your infrastructure, or cause unwanted costs during an outage. To alleviate this, you may decide to reduce the number of requests made by Google's crawlers.

Understand the cause of the sharp increase in crawling

Sharp increase in crawling may be caused by inefficiencies in your site's structure or issues with your site otherwise. Based on the reports we've received in the past, the most common causes are:

- Inefficient configuration of URLs on the site, which is typically caused by a specific functionality of the site:
 - Faceted navigation or other sorting and filtering functionality of the site
 - A calendar with a lot of URLs for specific dates
- [A Dynamic Search Ad target](#)

We strongly recommend that you check with your hosting company and look at recent access logs of your server to understand the source of the traffic, and see if it fits in the aforementioned common causes of the sharp increase in crawling. Then, check our guides about [managing crawling of faceted navigation URLs](#) and [optimizing crawling efficiency](#).

Urgently reduce crawler traffic (for emergencies)

Warning: When considering reducing the Google's crawl rate, keep in mind that this will have broad effects. For Search, Googlebot will discover fewer new pages, and existing pages will be refreshed less frequently (for example, prices and product availability may take longer to be reflected in Search), and removed pages may stay in the index longer. For Google Ads, your campaigns may be cancelled or paused, and your ads may not serve.

If you need to urgently reduce the crawl rate for short period of time (for example, a couple of hours, or 1-2 days), then return 500, 503, or 429 HTTP response status code instead of 200 to the crawl requests. Google's crawling infrastructure reduces your site's crawling rate when it encounters a significant number of URLs with 500, 503, or 429 HTTP response status codes (for example, if you [disabled your website](#)). The reduced crawl rate affects the whole hostname of your site (for example, subdomain.example.com), both the crawling of the URLs that return errors, as well as the URLs that return content. Once the number of these errors is reduced, the crawl rate will automatically start increasing again.

Warning: We don't recommend that you do this for a long period of time (meaning, longer than 1-2 days) as it may have a negative effect on how your site appears in Google products. For example, in case of Search, if Googlebot observes these status codes on the same URL for multiple days, the URL may be dropped from Google's index.

Exceptional requests to reduce crawl rate

If serving errors to Google's crawlers is not feasible on your infrastructure, [file a special request](#) to report a problem with unusually high crawl rate, mentioning the optimal rate for your site in your request. You cannot request an increase in crawl rate, and it may take several days for the request to be evaluated and fulfilled.

<https://developers.google.com/crawling/docs/crawlers-fetchers/verify-google-requests>

Verify requests from Google crawlers and fetchers

You can verify if a request to your server really is [from Google](#). Verification is possible for crawlers such as Googlebot, as well as other requests. This is useful if you're concerned that spammers or other troublemakers are accessing your site while claiming to be from Google.

Google's crawlers and fetchers fall into three categories:

Type	Description	Reverse DNS mask	IP ranges
Common crawlers	The common crawlers used for Google's products (such as Googlebot). They always respect robots.txt rules for automatic crawls.	crawl-***-***-***-***.googlebot.com or geo-crawl-***-***-***-***.geo.googlebot.com	common-crawlers.json
Special-case crawlers	Crawlers or fetchers that perform specific functions for Google products (such as AdsBot) where there's an agreement between the crawled site and the product about the access or for abuse-specific crawling or fetching. These crawlers or fetchers may or may not respect robots.txt rules.	rate-limited-proxy-***-***-***-***.google.com	special-crawlers.json
User-triggered fetchers	Tools and product functions where the end user triggers a fetch. For example, Google Site Verifier acts on the request of a user. Because the fetch was requested by a user, these fetchers ignore robots.txt rules. Fetchers controlled by Google originate from IPs in the user-triggered-fetchers-google.json object and resolve to a google.com hostname. IPs in the user-triggered-fetchers.json object	***-***-***-***.gae.googleusercontent.com or google-proxy-***-***-***-***.google.com	user-triggered-fetchers.json, user-triggered-fetchers-google.json, and user-triggered-agents.json

	resolve to gae.googleusercontent.com hostnames. These IPs are used, for example, if a site running on Google Cloud (GCP) has a feature that requires fetching external RSS feeds on the request of the user of that site.		
--	---	--	--

There are two methods for verifying requests from Google:

- **Manually:** For one-off lookups, use command line tools. This method is sufficient for most use cases.
- **Automatically:** For large scale lookups, use an automatic solution to match a crawler's IP address against the list of published Google IP addresses.

Use command line tools

1. Run a reverse DNS lookup on the accessing IP address from your logs, using the host command.
2. Verify that the domain name is either googlebot.com, google.com, or googleusercontent.com.
3. Run a forward DNS lookup on the domain name retrieved in step 1 using the host command on the retrieved domain name.
4. Verify that it's the same as the original accessing IP address from your logs.

Example 1:

```
host 66.249.66.1
1.66.249.66.in-addr.arpa domain name pointer crawl-66-249-66-1.googlebot.com.

host crawl-66-249-66-1.googlebot.com
crawl-66-249-66-1.googlebot.com has address 66.249.66.1
```

Example 2:

```
host 35.247.243.240
240.243.247.35.in-addr.arpa domain name pointer geo-crawl-35-247-243-240.geo.googlebot.com.

host geo-crawl-35-247-243-240.geo.googlebot.com
geo-crawl-35-247-243-240.geo.googlebot.com has address 35.247.243.240
```

Example 3:

```
host 66.249.90.77
77.90.249.66.in-addr.arpa domain name pointer rate-limited-proxy-66-249-90-77.google.com.

host rate-limited-proxy-66-249-90-77.google.com
rate-limited-proxy-66-249-90-77.google.com has address 66.249.90.77
```

Use automatic solutions

Alternatively, you can identify Googlebot by IP address by matching the crawler's IP address to the lists of Google crawlers' and fetchers' IP ranges:

- [Common crawlers like Googlebot](#)
- [Special crawlers like AdsBot](#)
- [User-triggered fetchers \(users\)](#)
- [User-triggered fetchers \(Google\)](#)
- [User-triggered agents](#)

For other Google IP addresses from where your site may be accessed (for example, [Apps Scripts](#)), match the accessing IP address against the general [list of Google IP addresses](#). Note that the IP addresses in the JSON files are represented in [CIDR format](#).

<https://developers.google.com/search/docs/crawling-indexing/robots/intro>

Introduction to robots.txt

[Original embedded video]

A robots.txt file tells search engine crawlers which URLs the crawler can access on your site. This is used mainly to avoid overloading your site with requests; **it is not a mechanism for keeping a web page out of Google**. To keep a web page out of Google, [block indexing with noindex](#) or password-protect the page.

If you use a CMS, such as Wix or Blogger, you might not need to (or be able to) edit your robots.txt file directly. Instead, your CMS might expose a search settings page or some other mechanism to tell search engines whether or not to crawl your page.

If you want to hide or unhide one of your pages from search engines, search for instructions about modifying your page visibility in search engines on your CMS (for example, search for "wix hide page from search engines").

What is a robots.txt file used for?

A robots.txt file is used primarily to manage crawler traffic to your site, and *usually* to keep a file off Google, depending on the file type:

robots.txt effect on different file types	
Web page	You can use a robots.txt file for web pages (HTML, PDF, or other non-media formats that Google can read), to manage crawling traffic if you think your server will be overwhelmed by requests from Google's crawler, or to avoid crawling unimportant or similar pages on your site. Warning: Don't use a robots.txt file as a means to hide your web pages (including PDFs and other text-based formats supported by Google) from Google Search results. If other pages point to your page with descriptive text, Google could still index the URL without visiting the page. If you want to block your page from search results, use another method such as password protection or noindex. If your web page is blocked with a robots.txt file, its URL can still appear in search results, but the search result won't have a description. Image files, video files, PDFs, and other non-HTML files embedded in the blocked page will be excluded from crawling, too, unless they're referenced by other pages that are allowed for crawling. If you see this search result for your page and want to fix it, remove the robots.txt

robots.txt effect on different file types	
	entry blocking the page. If you want to hide the page completely from Search, use another method .
Media file	Use a robots.txt file to manage crawl traffic, and also to prevent image, video, and audio files from appearing in Google Search results. This won't prevent other pages or users from linking to your image, video, or audio file. • Read more about preventing images from appearing on Google. • Read more about how to remove or restrict your video files from appearing on Google.
Resource file	You can use a robots.txt file to block resource files such as unimportant image, script, or style files, if you think that pages loaded without these resources won't be significantly affected by the loss. However, if the absence of these resources make the page harder for Google's crawler to understand the page, don't block them, or else Google won't do a good job of analyzing pages that depend on those resources.

Understand the limitations of a robots.txt file

Before you create or edit a robots.txt file, you should know the limits of this URL blocking method. Depending on your goals and situation, you might want to consider other mechanisms to ensure your URLs are not findable on the web.

- **robots.txt rules may not be supported by all search engines.**
The instructions in robots.txt files cannot enforce crawler behavior to your site; it's up to the crawler to obey them. While Googlebot and other respectable web crawlers obey the instructions in a robots.txt file, other crawlers might not. Therefore, if you want to keep information secure from web crawlers, it's better to use other blocking methods, such as [password-protecting private files on your server](#).
- **Different crawlers interpret syntax differently.**
Although respectable web crawlers follow the rules in a robots.txt file, each crawler might interpret the rules differently. You should know the [proper syntax](#) for addressing different web crawlers as some might not understand certain instructions.
- **A page that's disallowed in robots.txt can still be indexed if linked to from other sites.**
While Google won't crawl or index the content blocked by a robots.txt file, we might still find and index a disallowed URL if it is linked from other places on the web. As a result, the URL address and, potentially, other publicly available information such as anchor text in links to the page can still appear in Google Search results. To properly prevent your URL from appearing in Google Search results, [password-protect the files on your server, use the noindex meta tag or response header](#), or remove the page entirely.

Caution: Combining multiple crawling and indexing rules might cause some rules to counteract other rules. Learn how to [combine crawling with indexing and serving rules](#).

Create or update a robots.txt file

If you decided that you need one, learn how to [create a robots.txt file](#). Or if you already have one, learn how to [update it](#).

Want to learn more? Check out the following resources:

- [How to write and submit a robots.txt file](#)
- [Update your robots.txt file](#)
- [How Google interprets the robots.txt specification](#)

<https://developers.google.com/crawling/docs/robots-txt/robots-txt-spec>

How Google interprets the robots.txt specification

Google's automated [crawlers](#) support the [Robots Exclusion Protocol \(REP\)](#). This means that before crawling a site, Google's crawlers download and parse the site's robots.txt file to extract information about which parts of the site may be crawled. The REP isn't applicable to Google's crawlers that are controlled by users (for example, feed subscriptions), or crawlers that are used to increase user safety (for example, malware analysis).

This page describes Google's interpretation of the REP. For the original standard, check [RFC 9309](#).

What is a robots.txt file

If you don't want crawlers to access sections of your site, you can create a robots.txt file with appropriate rules. A robots.txt file is a text file containing rules about which crawlers may access which parts of a site. For example, the robots.txt file for example.com may look like this:

```
# This robots.txt file controls crawling of URLs under https://example.com.
# All crawlers are disallowed to crawl files in the "includes" directory, such
# as .css, .js, but Google needs them for rendering, so Googlebot is allowed
# to crawl them.
User-agent: *
Disallow: /includes/

User-agent: Googlebot
Allow: /includes/

Sitemap: https://example.com/sitemap.xml
```

If you're new to robots.txt, start with our [intro to robots.txt](#). You can also find [tips for creating a robots.txt file](#).

File location and range of validity

You must place the robots.txt file in the top-level directory of a site, on a supported protocol. The URL for the robots.txt file is (like other URLs) case-sensitive. In case of Google Search, the supported protocols are HTTP, HTTPS, and FTP. On HTTP and HTTPS, crawlers fetch the robots.txt file with an HTTP non-conditional GET request; on FTP, crawlers use a standard RETR (RETRIEVE) command, using anonymous login.

The rules listed in the robots.txt file apply only to the host, protocol, and port number where the robots.txt file is hosted.

Examples of valid robots.txt URLs

The following table contains examples of robots.txt URLs and what URL paths they're valid for. Column one contains the URL of a robots.txt file, and column two contains domains that that robots.txt file would and wouldn't apply to.

Robots.txt URL examples	
https://example.com/robots.txt	This is the general case. It's not valid for other subdomains, protocols, or port numbers. It's valid for all files in all subdirectories on the same host, protocol, and port number. Valid for: • https://example.com/ • https://example.com/folder/file Not valid for: • https://other.example.com/ • http://example.com/ • https://example.com:8181/
https://www.example.com/robots.txt	A robots.txt on a subdomain is only valid for that subdomain. Valid for: https://www.example.com/ Not valid for: • https://example.com/ • https://shop.www.example.com/ • https://www.shop.example.com/
https://example.com/folder/robots.txt	Not a valid robots.txt file. Crawlers don't check for robots.txt files in subdirectories.
https://www.exäample.com/robots.txt	IDNs are equivalent to their punycode versions. See also RFC 3492. Valid for: • https://www.exäample.com/ • https://xn--exmple-cua.com/ Not valid for: https://www.example.com/
ftp://example.com/robots.txt	Valid for: ftp://example.com/ Not valid for: https://example.com/
https://212.96.82.21/robots.txt	A robots.txt with an IP-address as the hostname is only valid for crawling of that IP address as the hostname. It isn't automatically valid for all websites hosted on that IP address (though it's possible that the robots.txt file is shared, in which case it would also be available under the shared hostname). Valid for: https://212.96.82.21/ Not valid for: https://example.com/ (even if hosted on 212.96.82.21)
https://example.com:443/robots.txt	Standard port numbers (80 for HTTP, 443 for HTTPS, 21 for FTP) are equivalent to their default hostnames. Valid for:

Robots.txt URL examples	
	• https://example.com:443/ • https://example.com/ Not valid for: https://example.com:444/
https://example.com:8181/robots.txt	Robots.txt files on non-standard port numbers are only valid for content made available through those port numbers. Valid for: https://example.com:8181/ Not valid for: https://example.com/

Handling of errors and HTTP status codes

When requesting a robots.txt file, the HTTP status code of the server's response affects how the robots.txt file will be used by Google's crawlers. The following table summarizes how Googlebot treats robots.txt files for different HTTP status codes.

Handling of errors and HTTP status codes	
2xx (success)	HTTP status codes that signal success prompt Google's crawlers to process the robots.txt file as provided by the server.
3xx (redirection)	Google follows at least five redirect hops as defined by RFC 1945 and then stops and treats it as a 404 for the robots.txt file. This also applies to any disallowed URLs in the redirect chain, since the crawler couldn't fetch rules due to the redirects. Google doesn't follow logical redirects in robots.txt files (frames, JavaScript, or meta refresh-type redirects).
4xx (client errors)	Google's crawlers treat all 4xx errors, except 429, as if a valid robots.txt file didn't exist. This means that Google assumes that there are no crawl restrictions. Don't use 401 and 403 status codes for limiting the crawl rate. The 4xx status codes, except 429, have no effect on crawl rate. Learn how to limit your crawl rate.
5xx (server errors)	If Google finds a robots.txt file but can't fetch it, Google follows this behavior: 1. For the first 12 hours, Google stops crawling the site but keeps trying to fetch the robots.txt file. 2. If Google can't fetch a new version, for the next 30 days Google will use the last good version, while still trying to fetch a new version. A 503 (service unavailable) error results in fairly frequent retrying. If there's no cached version available, Google assumes there's no crawl restrictions. 3. If the errors are still not fixed after 30 days:

Handling of errors and HTTP status codes	
	◦ If the site is generally available to Google, Google will behave as if there is no robots.txt file (but still keep checking for a new version). ◦ If the site has general availability problems, Google will stop crawling the site, while still periodically requesting a robots.txt file.
Other errors	A robots.txt file which cannot be fetched due to DNS or networking issues, such as timeouts, invalid responses, reset or interrupted connections, and HTTP chunking errors, is treated as a server error .

Caching

Google generally caches the contents of robots.txt file for up to 24 hours, but may cache it longer in situations where refreshing the cached version isn't possible (for example, due to timeouts or 5xx errors). The cached response may be shared by different crawlers. Google may increase or decrease the cache lifetime based on [max-age Cache-Control](#) HTTP headers.

File format

The robots.txt file must be a [UTF-8](#) encoded plain text file and the lines must be separated by CR, CR/LF, or LF.

Google ignores invalid lines in robots.txt files, including the Unicode [Byte Order Mark](#) (BOM) at the beginning of the robots.txt file, and use only valid lines. For example, if the content downloaded is HTML instead of robots.txt rules, Google will try to parse the content and extract rules, and ignore everything else.

Similarly, if the character encoding of the robots.txt file isn't UTF-8, Google may ignore characters that are not part of the UTF-8 range, potentially rendering robots.txt rules invalid.

Google enforces a robots.txt file size limit of 500 [kibibytes](#) (KiB). Content which is after the maximum file size is ignored. You can reduce the size of the robots.txt file by consolidating rules that would result in an oversized robots.txt file. For example, place excluded material in a separate directory.

Syntax

A valid robots.txt line consists of a field, a colon, and a value. Field names are case-insensitive (for example, User-agent and user-agent are treated the same). Spaces are optional, but recommended to improve readability. Space at the beginning and at the end of the line is ignored. To include comments, precede your comment with the # character. Keep in mind that everything after the # character will be ignored. The general format is <field>: <value><#optional-comment>.

Google supports the following fields (other fields such as crawl-delay aren't supported):

- user-agent: identifies which crawler the rules apply to.
- allow: a URL path that may be crawled.
- disallow: a URL path that may not be crawled.
- sitemap: the complete URL of a sitemap.

The allow and disallow fields are also called rules (also known as directives). These rules are always specified in the form of rule: [path] where [path] is optional. By default, there are no restrictions for crawling for the

designated crawlers. Crawlers ignore rules without a [path].

The [path] value, if specified, is relative to the root of the website from where the robots.txt file was fetched (using the same protocol, port number, host and domain names). The path value must start with / to designate the root and the value is case-sensitive. Learn more about [URL matching based on path values](#).

user-agent

The user-agent line identifies which crawler rules apply to. See [Google's crawlers and user-agent strings](#) for a comprehensive list of user-agent strings you can use in your robots.txt file.

Both the user-agent field name and its value are case-insensitive.

disallow

The disallow rule specifies paths that must not be accessed by the crawlers identified by the user-agent line the disallow rule is grouped with. Crawlers ignore the rule without a path.

Google can't index the content of pages which are disallowed for crawling, but it may still index the URL and show it in search results without a snippet. Learn how to [block indexing](#).

The field name (disallow) is case-insensitive, but its value is case-sensitive.

Usage:

```
disallow: [path]
```

allow

The allow rule specifies paths that may be accessed by the designated crawlers. When no path is specified, the rule is ignored.

The field name (allow) is case-insensitive, but its value is case-sensitive.

Usage:

```
allow: [path]
```

sitemap

Google, Bing, and other major search engines support the sitemap field in robots.txt, as defined by [sitemaps.org](#).

The field name (sitemap) is case-insensitive, but its value is case-sensitive.

Usage:

```
sitemap: [absoluteURL]
```

The [absoluteURL] line points to the location of a sitemap or sitemap index file. It must be a fully qualified URL, including the protocol and host, and doesn't have to be URL-encoded. The URL doesn't have to be on the same host as the robots.txt file. You can specify multiple sitemap fields, with no limit to the number of sitemaps you can

include. The sitemap field isn't tied to any specific user agent and may be followed by all crawlers, provided it isn't disallowed for crawling.

For example:

```
user-agent: otherbot
disallow: /kale

sitemap: https://example.com/sitemap.xml
sitemap: https://cdn.example.org/other-sitemap.xml
sitemap: https://ja.example.org/テスト-サイトマップ.xml
```

Grouping of lines and rules

You can group together rules that apply to multiple user agents by repeating user-agent lines for each crawler.

For example:

```
user-agent: a
disallow: /c

user-agent: b
disallow: /d

user-agent: e
user-agent: f
disallow: /g

user-agent: h
```

In this example there are four distinct rule groups:

- One group for user agent "a".
- One group for user agent "b".
- One group for both "e" and "f" user agents.
- One group for user agent "h".

For the technical description of a group, see [section 2.1 of the REP](#).

Order of precedence for user agents

Only one group is valid for a particular crawler. Google's crawlers determine the correct group of rules by finding in the robots.txt file the group with the most specific user agent that matches the crawler's user agent. Other groups are ignored. All non-matching text is ignored (for example, both googlebot/1.2 and googlebot* are equivalent to googlebot). The order of the groups within the robots.txt file is irrelevant.

If there's more than one specific group declared for a user agent, all the rules from the groups applicable to the specific user agent are combined internally into a single group. User agent specific groups and global groups (*) are not combined.

Examples

Matching of user-agent fields

```
user-agent: googlebot-news
(group 1)

user-agent: *
(group 2)

user-agent: googlebot
(group 3)
```

This is how the crawlers would choose the relevant group:

Group followed per crawler	
Googlebot News	googlebot-news follows group 1, because group 1 is the most specific group.
Googlebot (web)	googlebot follows group 3.
Googlebot Storebot	Storebot-Google follows group 2, because there is no specific Storebot-Google group.
Googlebot News (when crawling images)	When crawling images, googlebot-news follows group 1. googlebot-news doesn't crawl the images for Google Images, so it only follows group 1.
Otherbot (web)	Other Google crawlers follow group 2.
Otherbot (news)	Other Google crawlers that crawl news content, but don't identify as googlebot-news follow group 2. Even if there is an entry for a related crawler, it is only valid if it's specifically matching.

Grouping of rules

If there are multiple groups in a robots.txt file that are relevant to a specific user agent, Google's crawlers internally merge the groups. For example:

```
user-agent: googlebot-news
disallow: /fish

user-agent: *
disallow: /carrots

user-agent: googlebot-news
disallow: /shrimp
```

The crawlers internally group the rules based on user agent, for example:

```
user-agent: googlebot-news
disallow: /fish
disallow: /shrimp

user-agent: *
disallow: /carrots
```

Rules other than allow, disallow, and user-agent are ignored by the robots.txt parser. This means that the following robots.txt snippet is treated as one group, and thus both user-agent a and b are affected by the disallow: / rule:

```
user-agent: a
sitemap: https://example.com/sitemap.xml

user-agent: b
disallow: /
```

When the crawlers process the robots.txt rules, they ignore the sitemap line. For example, this is how the crawlers would understand the previous robots.txt snippet:

```
user-agent: a
user-agent: b
disallow: /
```

URL matching based on path values

Google uses the path value in the allow and disallow rules as a basis to determine whether or not a rule applies to a specific URL on a site. This works by comparing the rule to the path component of the URL that the crawler is trying to fetch. The path in the rules can be either raw UTF-8 characters or percent-encoded UTF-8 strings (per [RFC 3986](#)). As defined in [RFC 9309 Section 2.2.2](#), Google's crawlers compare rules against URLs using their percent-encoded forms, and automatically canonicalize raw UTF-8 rule paths into percent-encoded forms. This means that, for example, Disallow: /foo/bar/👉 and Disallow: /foo/bar/%E3%83%84 are treated identically by the parser.

Google, Bing, and other major search engines support a limited form of *wildcards* for path values. These wildcard characters are:

- * designates 0 or more instances of any valid character.
- \$ designates the end of the URL.

The following table shows how the different wildcard characters affect parsing:

Example path matches	
/	Matches the root and any lower level URL.
/*	Equivalent to /. The trailing wildcard is ignored.
/\$	Matches only the root. Any lower level URL is allowed for crawling.

Example path matches

<code>/fish</code>	Matches any path that starts with <code>/fish</code>. Note that the matching is case-sensitive. Matches: • <code>/fish</code> • <code>/fish.html</code> • <code>/fish/salmon.html</code> • <code>/fishheads</code> • <code>/fishheads/yummy.html</code> • <code>/fish.php?id=anything</code> Doesn't match: • <code>/Fish.asp</code> • <code>/catfish</code> • <code>/?id=fish</code> • <code>/desert/fish</code>
<code>/fish*</code>	Equivalent to <code>/fish</code>. The trailing wildcard is ignored. Matches: • <code>/fish</code> • <code>/fish.html</code> • <code>/fish/salmon.html</code> • <code>/fishheads</code> • <code>/fishheads/yummy.html</code> • <code>/fish.php?id=anything</code> Doesn't match: • <code>/Fish.asp</code> • <code>/catfish</code> • <code>/?id=fish</code> • <code>/desert/fish</code>
<code>/fish/</code>	Matches anything in the <code>/fish/</code> folder. Matches: • <code>/fish/</code> • <code>/fish/?id=anything</code> • <code>/fish/salmon.htm</code> Doesn't match: • <code>/fish</code> • <code>/fish.html</code> • <code>/animals/fish/</code> • <code>/Fish/Salmon.asp</code>

Example path matches	
<code>/*.php</code>	Matches any path that contains <code>.php</code>. Matches: • <code>/index.php</code> • <code>/filename.php</code> • <code>/folder/filename.php</code> • <code>/folder/filename.php?parameters</code> • <code>/folder/any.php.file.html</code> • <code>/filename.php/</code> Doesn't match: • <code>/</code> (even if it maps to <code>/index.php</code>) • <code>/windows.PHP</code>
<code>/*.php\$</code>	Matches any path that ends with <code>.php</code>. Matches: • <code>/filename.php</code> • <code>/folder/filename.php</code> Doesn't match: • <code>/filename.php?parameters</code> • <code>/filename.php/</code> • <code>/filename.php5</code> • <code>/windows.PHP</code>
<code>/fish*.php</code>	Matches any path that contains <code>/fish</code> and <code>.php</code>, in that order. Matches: • <code>/fish.php</code> • <code>/fishheads/catfish.php?parameters</code> Doesn't match: <code>/Fish.PHP</code>

Order of precedence for rules

When matching robots.txt rules to URLs, crawlers use the most specific rule based on the length of the rule path. In case of conflicting rules, including those with wildcards, Google uses the least restrictive rule.

The following examples demonstrate which rule Google's crawlers will apply on a given URL.

Sample situations	
https://example.com/page	allow: /p disallow: / Applicable rule: allow: /p, because it's more specific.
https://example.com/folder/page	allow: /folder disallow: /folder Applicable rule: allow: /folder, because in case of conflicting rules, Google uses the least restrictive rule.
https://example.com/page.htm	allow: /page disallow: /*.htm Applicable rule: disallow: /*.htm, because the rule path is longer and it matches more characters in the URL, so it's more specific.
https://example.com/page.php5	allow: /page disallow: /*.ph Applicable rule: allow: /page, because in case of conflicting rules, Google uses the least restrictive rule.
https://example.com/	allow: /\$ disallow: / Applicable rule: allow: /\$, because it's more specific.
https://example.com/page.htm	allow: /\$ disallow: / Applicable rule: disallow: /, because the allow rule only applies on the root URL.

<https://developers.google.com/search/docs/crawling-indexing/canonicalization>

What is canonicalization

Canonicalization is the process of selecting the representative -canonical- URL of a piece of content. Consequently, a canonical URL is the URL of a page that Google chose as the most representative from a set of duplicate pages. Often called deduplication, this process helps Google show only one version of the otherwise duplicate content in its search results.

There are many reasons why a site may have duplicate content:

- **Region variants:** for example, separate USA and UK URLs with `hreflang`, but essentially the same content in the same language
- **Device variants:** for example, a page with both a mobile and a desktop version
- **Protocol variants:** for example, the HTTP and HTTPS versions of a site
- **Site functions:** for example, the results of sorting and filtering functions of a category page
- **Accidental variants:** for example, the demo version of the site is accidentally left accessible to crawlers

Some duplicate content on a site is normal and it's not a violation of [Google's spam policies](#). However, having the same content accessible through many different URLs can be a bad user experience (for example, people might wonder which is the right page, and whether there's a difference between the two) and it may make it harder for you to track how your *content* performs in search results.

How Google indexes and chooses the canonical URL

[Original embedded video]

When [Google indexes a page](#), it determines the primary content (or *centerpiece*) of each page. If Google finds multiple pages that seem to be the same or the primary content very similar, it clusters them together. Google then chooses the page that, based on the factors (or *signals*) the indexing process collected, is objectively the most complete and useful for search users, and marks it as the canonical. The canonical page will be crawled most regularly; duplicates are crawled less frequently in order to reduce the crawling load on sites.

There are a handful of factors that play a role in canonicalization: whether the page is served over HTTP or HTTPS, redirects, presence of the URL in a sitemap, and `rel="canonical"` link annotations. You can [indicate your preference to Google](#) using these techniques, but Google may choose a different page as canonical than you do, for various reasons. That is, indicating a canonical preference is a hint, not a rule.

Different language versions of a single page are considered duplicates only if the primary content is in the same language (that is, if only the header, footer, and other non-critical text is translated, but the body remains the same, then the pages are considered to be duplicates). For regional variants in the same language (for example, a piece of content for the USA and the UK that's accessible at different URLs but in the same language), use both canonicalization and `hreflang` to help Google better understand which regional URL to show in search results. To learn more, see our documentation on [handling duplicate pages on multi-regional sites](#).

Google uses the canonical page as the main source to evaluate content and quality. A Google Search result usually points to the canonical page, unless one of the duplicates is explicitly better suited for a search user. For example,

the search result will probably point to the mobile page if the user is on a mobile device, even if the desktop page is the canonical.

Read more about [how to indicate your preference for the canonical URL](#), and whether you need to.

<https://developers.google.com/search/docs/crawling-indexing/consolidate-duplicate-urls>

How to specify a canonical URL with rel="canonical" and other methods

To specify a [canonical URL](#) for duplicate or very similar pages to Google Search, you can indicate your preference using a number of methods. These are, in order of how strongly they can influence canonicalization:

- **Redirects:** A strong signal that the target of the redirect should become canonical.
- **rel="canonical" link annotations:** A strong signal that the specified URL should become canonical.
- **Sitemap inclusion:** A weak signal that helps the URLs that are included in a sitemap become canonical.

Keep in mind that these methods can stack and thus become more effective when combined. This means that when you use two or more of the methods, that will increase the chance of your preferred canonical URL appearing in search results.

While we encourage you to use these methods, none of them are required; your site will likely do just fine without specifying a canonical preference. That's because if you don't specify a canonical URL, [Google will identify which version of the URL is objectively the best version to show to users in Search](#).

If you use a CMS, such as WordPress, Wix, or Blogger, you might not be able to edit your HTML directly. Instead, your CMS might have a search engine settings page or some other mechanism to tell search engines about the canonical URL. Search for instructions about modifying the <head> of your page on your CMS (for example, search for "wordpress set the canonical element").

Reasons to specify a canonical URL

[Original embedded video]

While it's generally not critical to specify a canonical preference for your URLs, there are a number of reasons why you would want to explicitly tell Google about a canonical page in a set of duplicate or similar pages:

- **To specify which URL that you want people to see in search results.** You might prefer people to reach your green dress product page through <https://www.example.com/dresses/green/green-dress.html> rather than <https://example.com/dresses/cocktail?gclid=ABCD>.
- **To consolidate signals for similar or duplicate pages.** It helps search engines to be able to consolidate the signals they have for the individual URLs (such as links to them) into a single, preferred URL. This means that signals from other sites to <https://example.com/dresses/cocktail?gclid=ABCD> get consolidated with links to <https://www.example.com/dresses/green/green-dress.html> if the latter becomes canonical.
- **To simplify tracking metrics for a piece of content.** With a variety of URLs, it can be more challenging for you to get consolidated metrics for a specific piece of content.
- **To avoid spending crawling time on duplicate pages.** You may want Googlebot to get the most out of your site, so it's better for it to spend time crawling new (or updated) pages on your site, rather than crawling duplicate versions of the same content.

Best practices

For all canonicalization methods, follow these best practices:

- **Don't** use the robots.txt file for canonicalization purposes. Google may still index URLs that are disallowed in robots.txt without their content.
- **Don't** use the URL removal tool for canonicalization. It hides *all* versions of a URL from Search.
- **Don't** specify different URLs as canonical for the same page using different canonicalization techniques (for example, don't specify one URL in a sitemap, but a different URL for that same page using rel="canonical").
- **Don't** specify a URL fragment as canonical, as [Google generally doesn't support URL fragments](#).
- **Do** include a rel="canonical" link on the canonical page itself (also known as a self-referential canonical).
- **We don't recommend** using [noindex](#) to prevent selection of a canonical page within a single site, because it will completely block the page from Search. rel="canonical" link annotations are the preferred solution.
- If you're using [hreflang elements](#), make sure to specify a canonical page in the same language, or the best possible substitute language if a canonical page doesn't exist for the same language.
- When linking within your site, link to the canonical URL rather than a duplicate URL. Linking consistently to the URL that you consider to be canonical helps Google understand your preference.
- If you're using client-side rendering with JavaScript, it's important to make sure that the information about the canonical URL is as clear as possible. The best way to do this is to specify the canonical URL in the HTML source code and make sure that JavaScript doesn't change the canonical link element. If you can't set the canonical URL in the HTML source code, leave it out and only set it with JavaScript. This ensures that the information about the canonical URL is as clear as possible.

Comparison of canonicalization methods

The following table compares the different canonicalization methods, highlighting their strengths and weaknesses when it comes to maintenance and efficacy in different scenarios.

Method and description		
rel="canonical" link element	Add a <link> element in the code for all duplicate pages, pointing to the canonical page.	
	Pros: • Can map an infinite number of duplicate pages.	Cons: • Can be complex to maintain the mapping on larger sites, or sites where the URLs change often. • Only works for HTML pages, not for files such as PDF. In such cases, you can use the rel="canonical" HTTP header.
rel="canonical" HTTP header	Send a rel="canonical" header in your page response.	
	Pros:	Cons:

	- Doesn't increase page size. - Can map an infinite number of duplicate pages.	Can be complex to maintain the mapping on larger sites, or sites where the URLs change often.
Sitemap	Specify your canonical pages in a sitemap.	
	Pros: Simple to implement and maintain, especially on large sites.	Cons: - Google must still determine the associated duplicate for any canonicals that you declare in the sitemap. - Less powerful signal to Google than the <code>rel="canonical"</code> mapping technique.
Redirects	Use permanent redirects to tell Google that a redirected URL is a worse version than the URL it redirects to. Use this only when deprecating a duplicate page.	
AMP variant	If one of your variants is an AMP page, follow the AMP guidelines to indicate the canonical page and AMP variant.	

Use `rel="canonical"` link annotations

Google supports explicit `rel canonical` link annotations as described in [RFC 6596](#). `rel="canonical"` annotations that suggest alternate versions of a page are ignored; specifically, `rel="canonical"` annotations with `hreflang`, `lang`, `media`, and `type` attributes are not used for canonicalization. Instead, use the appropriate link annotations to specify alternate versions of a page; for example, link `rel="alternate" hreflang` for language and country annotations.

You can provide the `rel="canonical"` link annotations in two ways:

- The `rel="canonical"` link element in the HTML
- The `rel="canonical"` link HTTP header

We recommend that you choose one of these and go with that; while supported, using both methods at the same time is more error prone (for example, you might provide one URL in the HTTP header, and another URL in the `rel="canonical"` link element).

The `rel="canonical"` link element

A `rel="canonical"` link element (also known as a *canonical element*) is an element used in the head section of HTML to indicate that another page is representative of the content on the page.

Suppose you want `https://example.com/dresses/green-dresses` to be the canonical URL, even though a variety of URLs can access this content. Indicate this URL as canonical with these steps:

1. Add a `<link>` element with the attribute `rel="canonical"` to the `<head>` section of duplicate pages, pointing to the canonical page. For example:

```
<html>
<head>
<title>Explore the world of dresses</title>
<link rel="canonical" href="https://example.com/dresses/green-dresses" />
<!-- other elements -->
</head>
<!-- rest of the HTML -->
```

We recommend adding this same self-referential `rel="canonical"` link element to the canonical page itself as well.

2. If the canonical page has a mobile variant on a separate URL, add a `rel="alternate"` link element to it, pointing to the mobile version of the page:

```
<html>
<head>
<title>Explore the world of dresses</title>
<link rel="alternate" media="only screen and (max-width: 640px)"
href="https://m.example.com/dresses/green-dresses">
<link rel="canonical" href="https://example.com/dresses/green-dresses" />
<!-- other elements -->
</head>
<!-- rest of the HTML -->
```

3. Add any `hreflang` or other elements that are appropriate for the page.

Use absolute paths rather than relative paths with the `rel="canonical"` link element. Even though relative paths are supported by Google, they can cause problems in the long run (for example, if you unintentionally allow your testing site to be crawled) and thus we don't recommend them.

Good example: `https://www.example.com/dresses/green/green-dress.html`

Bad example: `/dresses/green/green-dress.html`

The `rel="canonical"` link element is only accepted if it appears in the `<head>` section of the HTML, so make sure at least the `<head>` section is valid HTML.

If you use JavaScript to add the `rel="canonical"` link element, make sure to [inject the canonical link element properly](#).

The `rel="canonical"` HTTP header

If you can change the configuration of your server, you can use a link [HTTP response header](#) with a `rel="canonical"` target attribute as defined by [RFC5988](#) rather than an HTML element to indicate the canonical URL for a document supported by Search, including non-HTML documents such as PDF files.

Google supports this method for web search results only.

If you publish content in many file formats, such as PDF or Microsoft Word, each on their own URL, you can return a `rel="canonical"` HTTP header to tell Googlebot what is the canonical URL for the non-HTML files. For example, to indicate that the PDF version of the `.docx` version should be canonical, you might add this HTTP header for the `.docx` version of the content:

```
HTTP/1.1 200 OK
Content-Length: 19
...
Link: <https://www.example.com/downloads/white-paper.pdf>; rel="canonical"
...
```

As with the `rel="canonical"` link element, use absolute URLs in the `rel="canonical"` HTTP header.

Use a sitemap

Pick a canonical URL for each of your pages and submit them in a [sitemap](#). All pages listed in a sitemap are suggested as canonicals; Google will decide which pages (if any) are duplicates, based on similarity of content.

Supplying the preferred canonical URLs in the sitemaps is a straightforward way of defining canonicals for a large site, and sitemaps are a useful way to tell Google which pages you consider most important on your site.

Use redirects

Use this method when you want to get rid of existing duplicate pages. All [permanent redirection methods](#) have the same effect on Google Search, however the time it takes for search engines to notice the different redirect methods may differ.

For the quickest effect, use HTTP (also known as *server-side*) redirects.

Suppose your page can be reached in multiple ways:

- <https://example.com/home>
- <https://home.example.com>
- <https://www.example.com>

Pick one of those URLs as your canonical URL, and use redirects to send traffic from the other URLs to your preferred URL.

Other signals

Apart from explicitly provided methods, Google also uses a set of canonicalization signals that are generally based on site setup: preferring HTTPS over HTTP, and URLs in hreflang clusters.

Prefer HTTPS over HTTP for canonical URLs

Google prefers HTTPS pages over equivalent HTTP pages as canonical, except when there are issues or conflicting signals such as the following:

- The HTTPS page has an invalid SSL certificate.
- The HTTPS page contains insecure dependencies (other than images).
- The HTTPS page redirects users to or through an HTTP page.

- The HTTPS page has a `rel="canonical"` link to the HTTP page.

Although our systems prefer HTTPS pages over HTTP pages by default, you can ensure this behavior by taking any of the following actions:

- Add redirects from the HTTP page to the HTTPS page.
- Add a `rel="canonical"` link from the HTTP page to the HTTPS page.
- Implement [HSTS](#).

To prevent Google from incorrectly making the HTTP page canonical, **avoid** the following practices:

- Avoid bad TLS/SSL certificates and HTTPS-to-HTTP redirects because they cause Google to prefer HTTP very strongly. Implementing HSTS cannot override this strong preference.
- Don't include the HTTP version of your pages in your sitemap or [hreflang annotations](#) rather than the HTTPS version.
- Avoid implementing your SSL/TLS certificate for the wrong host-variant. For example, `example.com` serving the certificate for `subdomain.example.com`. The certificate must match your complete site URL, or be a wildcard certificate that can be used for multiple subdomains on a domain.

Prefer URLs in hreflang clusters

To help with sites' localization efforts, for canonicalization purposes Google prefers URLs that are part of hreflang clusters. For example, if `https://example.com/de-de/cats` and `https://example.com/de-ch/cats` reciprocally point to each other with hreflang annotations, but not to `https://example.com/de-at/cats`, the pages for de-de and de-ch will be preferred as canonicals instead of the /de-at/ page that doesn't appear in the hreflang cluster.

Read more about [troubleshooting and fixing canonicalization issues](#).

<https://developers.google.com/search/docs/crawling-indexing/canonicalization-troubleshooting>

Fix canonicalization issues

Even if you explicitly designate a canonical page, Google might choose a different canonical for various reasons, such as the quality of the content or technical signals. To troubleshoot canonicalization issues, follow these steps:

1. **Check which page Google considers to be the canonical page:** Use the [URL Inspection tool](#) to check [which page Google considers canonical](#) and think about whether the Google-selected canonical makes more sense than your preferred canonical URL for your users coming from Google Search.

Important: If a canonical URL is in a Search Console property that you don't own, you won't be able to see any of the traffic for your duplicate page.

2. **Look for technical canonicalization issues:** Verify whether an unexpected canonical URL preference is being signaled through technical misconfigurations. Review the [common canonicalization issues table](#) to check for issues like incorrect canonical elements, server misconfigurations, or missing localization annotations.
3. **Ensure clustered pages are sufficiently different:** Fixing canonicalization issues technically boils down to ensuring that the pages that are [clustered together](#) are sufficiently different. Things to keep in mind:
 - **Re-evaluation takes time:** Even after fixing content issues, Google might hold pages in a duplicate cluster for **up to two weeks**.
 - **Content difference matters:** Pages will generally split out faster if the difference between the new content and the other clustered pages is clear and significant.
4. **Request re-indexing:** Once you fix the content issues, use the "Request Indexing" feature in the Search Console [URL Inspection tool](#) to ask Google to re-evaluate the pages that are clustered. However, since this feature is subject to quotas, reserve it for your most important URLs.

Common canonicalization issues

There are various reasons why the selected canonical URL differs from the canonical URL you'd prefer to see in Search. The most common issues are:

Common canonicalization issues

Language variants without localized annotations	If you have multiple websites that serve substantially the same content localized to different users around the world, be sure to follow our guidelines for localized sites. For example, if you have different sites for your English-speaking users in the United States, United Kingdom, and Australia respectively, but the content is the same, adding hreflang annotations to your pages can help the right pages surface for users in different regions.
Incorrect canonical elements	Some content management systems (CMS) or CMS plugins can make incorrect use of canonicalization techniques to point to undesired URLs. Check your HTML with your browser's developer tools to see if so. If your site is indicating an unexpected canonical URL preference, perhaps through incorrect use of rel="canonical" or a 3xx redirect, contact your CMS provider and report this error to them.
Misconfigured servers	Some hosting misconfigurations may cause unexpected cross-domain URL selection. For example: • A server may be misconfigured to return content from example.com in response to a request for a URL on other.example • Two unrelated web servers may return identical soft 404 pages that Google fails to identify as error pages. If you notice this is the case, get in touch with your hosting provider.
Malicious hacking	Some attacks on websites introduce code that returns an HTTP 3xx redirect or inserts a cross-domain rel="canonical" link annotation into the HTML <head> or HTTP header, usually pointing to a URL hosting malicious or spammy content. In these cases, our algorithms may choose the malicious or spammy URL instead of the URL on the compromised website.
Syndicated content	The canonical link element is not recommended for those who want to avoid duplication by syndication partners, because the pages are often very different. The most effective solution is for partners to block indexing of your content. For more, see Avoid article duplication in Google News, which also has advice about blocking syndicated content from Google Search.
A copycat website	In rare situations, our algorithm may select a URL from an external site that is hosting your content without your permission. If you believe that another site is duplicating your content in violation of copyright law, you may contact the site's host to request removal. In addition, you can request that Google remove the infringing page from our search results by filing a request under the Digital Millennium Copyright Act.

<https://developers.google.com/search/docs/crawling-indexing/mobile/mobile-sites-mobile-first-indexing>

Mobile site and mobile-first indexing best practices

Google uses the mobile version of a site's content, crawled with the [smartphone agent](#), for indexing and ranking. This is called **mobile-first indexing**.

[Original embedded video]

While it's not required to have a mobile version of your pages to have your content included in Google's Search results, it is very strongly recommended. These best practices apply to mobile sites in general, and by definition to mobile-first indexing.

To make sure that your users have the best experience, follow the best practices detailed in this guide.

Create a mobile-friendly site

If you haven't already, create a mobile-friendly website so your users visiting your site through a mobile phone can have a stellar experience. There are three configurations you can choose from to create a mobile-friendly site:

- **Responsive design:** Serves the same HTML code on the same URL regardless of the users' device (for example, desktop, tablet, mobile, non-visual browser), but can display the content differently based on the screen size. **Google recommends Responsive Web Design because it's the easiest design pattern to implement and maintain.**
- **Dynamic serving:** Uses the same URL regardless of device. This configuration relies on [user-agent sniffing](#) and the [Vary: user-agent HTTP response header](#) to serve a different version of the HTML to different devices.
- **Separate URLs:** Serves different HTML to each device, and on separate URLs. Like dynamic serving, this configuration relies on the user-agent and Vary HTTP headers to redirect users to the device-appropriate version of the site.

The contents of this guide only apply to dynamic serving and separate URL configurations. In case of responsive design, the content and the metadata are the same on the mobile and desktop version of the pages.

If you use a CMS, such as **Wix** or **Blogger**, you might need to find a new mobile-friendly theme for your site if you can't modify your existing theme. Try searching for the name of your CMS and "mobile friendly", for example, "wix mobile friendly".

Make sure that Google can access and render your content

Make sure that Google can access and render your mobile page content and resources.

- **Use the same robots meta tags on the mobile and desktop site.** If you use a different robots meta tag on the mobile site (especially the noindex or nofollow tags), Google may fail to crawl and index your page when your site is enabled for mobile-first indexing.

- **Don't lazy-load primary content upon user interaction.** Google won't load content that requires user interactions (for example, swiping, clicking, or typing) to load. [Make sure that Google can see lazy-loaded content.](#)
- **Let Google crawl your resources.** Some resources have different URLs on the mobile site from those on the desktop site. If you want Google to crawl your URLs, make sure that you're not blocking the URL with the [disallow rule](#).

Make sure that content is the same on desktop and mobile

Even with the equivalent content, differences in DOM or layout between desktop and mobile page can result in Google understanding the content differently. However having the same content on the desktop and mobile version ensures that the two versions can rank for the same keywords.

- **Make sure that your mobile site contains the same content as your desktop site.** If your mobile site has less content than your desktop site, consider updating your mobile site so that its primary content is equivalent to your desktop site. You can have a different design on mobile to maximize user experience (for example, moving content into accordions or tabs); just make sure that the content is equivalent to the desktop site, since indexing on your site comes from the mobile site.

If it's your intention that the mobile page should have less content than the desktop page, you can expect some traffic loss when your site is enabled mobile-first indexing, since Google can't get as much information from your page as before. Instead of removing content, consider moving content into accordions or tabs to save space.

- **Use the same clear and meaningful headings** on the mobile site as you do on the desktop site.

Check your structured data

If you have structured data on your site, make sure that it's present on both versions of your site. Here are some specific things to check:

- **Make sure that your mobile and desktop sites have the same structured data.** If you have to prioritize which types you add to your mobile site, start with [Breadcrumb](#), [Product](#), and [VideoObject](#) structured data.
- **Use correct URLs in structured data.** Make sure that URLs in the structured data on the mobile versions are updated to the mobile URLs.
- **If you use Data Highlighter, train it on your mobile site.** If you use [Data Highlighter](#) to provide structured data, regularly check the [Data Highlighter dashboard](#) for extraction errors.

Put the same metadata on both versions of your site

Make sure that the [title element](#) and the [meta description](#) are equivalent across both versions of your site.

Check the placement of your ads

Don't let ads harm your mobile page ranking. Follow the [Better Ads Standard](#) when displaying ads on mobile devices. For example, ads at the top of the page can take up too much room on a mobile device, which is a bad user experience

Check visual content

Check your images

Make sure that the images on your mobile site follow the [image best practices](#). In particular, we recommend that you:

- **Provide [high quality images](#).** Don't use images that are too small or have a low resolution on the mobile site.
- **Use a [supported format for images](#).** Don't use unsupported formats or tags. For example, Google supports SVG format images, but our systems can't index a .jpg image in the <image> tag inside an inline SVG.
- **Don't use URLs that change every time the page loads for images.** Google won't be able to process and index your resources properly if you use constantly-changing URLs for them.
- **Make sure that the mobile site has the same alt text for images as the desktop site.** Use [descriptive alt text](#) for images on your mobile site as you do on your desktop site.
- **Make sure that the mobile page content quality is as good as the desktop page.** Use the same [titles](#), [captions](#), [filenames](#), and [text](#) relevant to the images on the mobile site as you do for the desktop site.

If your site is using different image URLs for the desktop and mobile site, you may see a temporary image traffic loss while your site transitions to mobile-first indexing. This is because the image URLs on the mobile site are new to Google indexing, and it will take some time for the new image URLs to gain enough historical search results to get a better ranking. To avoid a temporary image traffic loss, use the same image URLs across both versions of your site. If you don't mind a temporary image traffic loss, you don't need to take any action.

Check your videos

Make sure that the videos on your mobile site follow the [video best practices](#). In particular, we recommend that you:

- **Don't use URLs that change every time the page loads for your videos.** Google won't be able to process and index your resources properly if you use constantly changing URLs for them.
- **Use a [supported format](#) for your videos and put videos in supported tags.** Videos are identified in the page by the presence of an HTML tag, for example: <video>, <embed>, or <object>.
- **Use the same video structured data on both your mobile site and desktop site.** For more information, [check your structured data](#).
- **Place the video in an easy to find position on the page when viewed on a mobile device.** For example, it might harm the video's ranking if users need to scroll down too much to find the video on mobile page.

Additional best practices for separate URLs

If your site has separate URLs for the desktop and mobile versions of a page (also known as m-dot), we recommend the following additional best practices:

- **Make sure that the error page status is the same on both the desktop and mobile sites.** If a page on your desktop site serves normal contents and your mobile site's version of that page serves an error page, this page will be missing from the index.
- **Make sure that your mobile version doesn't have URL fragments.** The fragment part of the URL is the end of the URL that starts with #. Most of the time, URL fragments are not indexable, these pages will be missing from the index after your domain is enabled for mobile-first indexing.

- **Ensure that the desktop versions that serve different contents have equivalent mobile versions.** If different URLs redirect to the same URL (for example, to the home page on mobile devices) after your domain is enabled for mobile-first indexing, all these pages will be missing from the index.
- **Verify both versions of your site in [Search Console](#)** to make sure that you have access to data and messages for both versions. Your site may experience a data shift when Google switches to mobile-first indexing for your site.
- **Check hreflang links on separate URLs.** When you use `rel=hreflang` [link elements](#) for [internationalization](#), link between mobile and desktop URLs separately. Your mobile URLs' hreflang must point to mobile URLs, and similarly desktop URL hreflang must point to desktop URLs.

Here's an example of hreflang for the home page of a site with separate URLs for mobile and desktop.

Mobile

In this example, the mobile site URL is `https://m.example.com/`.

```
<link rel="canonical" href="https://example.com/">
<link rel="alternate" hreflang="es" href="https://m.example.com/es/">
<link rel="alternate" hreflang="fr" href="https://m.example.com/fr/">
<link rel="alternate" hreflang="de" href="https://m.example.com/de/">
<link rel="alternate" hreflang="th" href="https://m.example.com/th/">
```

Desktop

In this example, the desktop site URL is `https://example.com/`.

```
<link rel="canonical" href="https://example.com/">
<link rel="alternate" media="only screen and (max-width: 640px)" href="https://m.example.com/">
<link rel="alternate" hreflang="es" href="https://example.com/es/">
<link rel="alternate" hreflang="fr" href="https://example.com/fr/">
<link rel="alternate" hreflang="de" href="https://example.com/de/">
<link rel="alternate" hreflang="th" href="https://example.com/th/">
```

- **Ensure that your mobile site has enough capacity** to handle a potential increase in [crawl rate](#) on the mobile version of your site.
- **Verify that your [robots.txt rules](#)** work as you intend for both versions of your site. The robots.txt file lets you specify which parts of a website may be crawled or not. In most cases, use the same robots.txt rules for both mobile and desktop versions of your site.
- **Use the correct `rel=canonical` and `rel=alternate` link elements** between your mobile and desktop versions. The desktop URL is always the canonical, and the mobile version is the alternate of that URL.

Here's an example of `rel=canonical` and `rel=alternate` for a separate URLs site setup.

Mobile

In this example, the mobile site URL is `https://m.example.com/` and includes a link element that points to the desktop URL as the canonical URL.

```
<link rel="canonical" href="https://example.com/">
```

Desktop

In this example, the desktop site URL is `https://example.com/` and includes a `link` element that points to itself as the canonical URL, followed by another `link` element that points to the mobile version as the alternate version of this URL.

```
<link rel="canonical" href="https://example.com/">
<link rel="alternate" media="only screen and (max-width: 640px)" href="https://m.example.com/">
```

Troubleshooting

Here's a list of the most common errors that can stop sites from being enabled for mobile-first indexing or could cause a drop in ranking after a site is enabled for mobile-first indexing. If your site isn't enabled for mobile-first indexing yet, you've seen a drop in ranking after your site is enabled for mobile-first indexing, or you've received a message in Search Console, check the following list of common errors and resolve possible errors you may have:

Errors	
Missing structured data	<i>error</i> What caused the issue: The mobile page doesn't have all the structured data markup that the desktop page has. <i>done</i> Fix the issue 1. Verify that the structured data is present on both versions of your site (desktop and mobile). 2. Make sure that your mobile and desktop sites have the same structured data. 3. Use correct URLs in structured data. Make sure that URLs in the structured data on the mobile versions are updated to the correct URLs. 4. Check extraction errors for your structured data. If you use Data Highlighter to provide structured data, regularly check the Data Highlighter dashboard for extraction errors. 5. Use the URL Inspection tool to check that the content is visible on the rendered page (the rendered page is how Google sees your page).
noindex tag on pages	<i>error</i> What caused the issue: A mobile page is blocked from indexing by a <code>noindex</code> tag. <i>done</i> Fix the issue: Use the same robots meta tags on the mobile site and the desktop site. Don't use the <code>noindex</code> tag on the mobile page (otherwise, Google won't index your page when your site is enabled for mobile-first indexing).
Missing image	<i>error</i> What caused the issue: The mobile page doesn't have all the important images that the desktop page has. <i>done</i> Fix the issue 1. Make sure that your mobile site contains the same content as your desktop site. If your mobile site has less content than your desktop site, consider updating your mobile site so that its primary content is equivalent with your desktop site. Only the content shown on the mobile site is used for indexing. 2. Use the same robots meta tags on the mobile site and the desktop site. Don't use the <code>nofollow</code> tag on the mobile page (otherwise, Google won't crawl and

index the images on your page when your site is enabled for mobile-first indexing).

3. Use a [supported format and tag for images](#). For example, Google supports SVG format images, but our systems can't index a .jpg image in the <image> tag inside an inline SVG.
4. Don't lazy-load primary content upon user interaction. Google won't load content that requires user interactions (for example, swiping, clicking, or typing) to load. [Make sure that Google can see lazy-loaded content](#).

Blocked image	<i>error</i> What caused the issue: An important image on the mobile page is blocked by robots.txt. <i>done</i> Fix the issue: Let Google crawl your resources. Some images have different URLs on the mobile site from those on the desktop site. If you want Google to crawl your URLs, don't block the URL with the disallow rule.
Low quality image	<i>error</i> What caused the issue: An important image on the mobile page is too small or low resolution. <i>done</i> Fix the issue: Provide high quality images. Don't use images that are too small or have a low resolution on the mobile site.
Missing alt text	<i>error</i> What caused the issue: An important image on the mobile page is missing alt text. <i>done</i> Fix the issue: Use the same alt text for images on your mobile site as you do on your desktop site.
Missing page title	<i>error</i> What caused the issue: A mobile page is missing a title. <i>done</i> Fix the issue: Make sure that the titles and meta descriptions are equivalent across both versions of your site.
Missing meta description	<i>error</i> What caused the issue: A mobile page is missing the meta description. <i>done</i> Fix the issue: Make sure that the titles and meta descriptions are equivalent across both versions of your site.
Mobile URL is an error page	<i>error</i> What caused the issue: The mobile page is an error page. <i>done</i> Fix the issue: Make sure error page status is the same across desktop and mobile site. If a page on your desktop site serves normal contents and your mobile site's version of that page serves an error page, this page will be missing from the index.
Mobile URL has anchor fragment	<i>error</i> What caused the issue: The mobile URL includes an anchor fragment; Google can't index URLs that include fragments. <i>done</i> Fix the issue: Make sure your mobile version doesn't have URL fragments. Most of the time, fragment URLs are not indexable, and these pages will be missing from the index after your domain is enabled for mobile-first indexing.
Mobile page blocked by robots.txt	<i>error</i> What caused the issue: The mobile page is blocked by a robots.txt rule. <i>done</i> Fix the issue: Verify that your robots.txt rules and robots meta tags work as you intended for both versions of your site. Use the same robots.txt rules for both mobile and desktop versions of your site.
Duplicate mobile page target	<i>error</i> What caused the issue: Multiple desktop pages redirect to the same mobile page. <i>done</i> Fix the issue: Ensure desktop versions which serve different contents have equivalent mobile versions. If different URLs redirects to the same URL, on mobile devices, after your domain is enabled for mobile-first indexing, all these pages will be missing from the index.

Desktop site redirects to the mobile home page	<i>error</i> What caused the issue: Most or all pages on your desktop site redirect to the mobile site's home page. <i>done</i> Fix the issue: Ensure the desktop version has an equivalent mobile version. If different URLs redirect to the home page on mobile devices, all these pages will be missing from the index after your domain is migrated for mobile-first indexing.
Page quality issues	<i>error</i> What caused the issue: The mobile page has issues with ads, missing content, titles, or descriptive elements for images on the page. <i>done</i> Fix the issue: 1. Don't let ads harm your mobile page ranking. Follow the Better Ads Standard when displaying ads on mobile devices. 2. Make sure your mobile site contains the same content as your desktop site. If your mobile site has less content than your desktop site, consider updating your mobile site so that its primary content is equivalent with your desktop site. Only the content shown on the mobile site is used for indexing. 3. Make sure you use the same clear and meaningful headings on the mobile site as you do on the desktop site. 4. Use the same titles, captions, filenames, and text relevant to the images on the mobile site you do for the desktop site.
Video issues	<i>error</i> What caused the issue: The mobile page has a video that is not in a supported format, is placed in a difficult to find location, is missing meta descriptions, or is very slow to load. <i>done</i> Fix the issue: 1. Use a supported format for your videos and put them in supported tags. Videos are identified in the page by the presence of an HTML tag, for example: <code><video></code>, <code><embed></code>, or <code><object></code>. 2. Don't lazy-load primary content upon user interaction. Google won't load content that requires user interactions (for example, swiping, clicking or typing) to load. Make sure Google can see lazy-loaded content. 3. Place the video in an easy to find location on your mobile site. For example, it might harm the video's ranking if users need to scroll down too much to find the video on mobile page.
Hostload issues	<i>error</i> What caused the issue: Some of the hosts don't have enough hostload. <i>done</i> Fix the issue: Ensure that your mobile site has enough capacity to handle a potential increase in crawl rate on the mobile version of your site.

<https://developers.google.com/search/docs/crawling-indexing/javascript/javascript-seo-basics>

Understand the JavaScript SEO basics

Do you suspect that JavaScript issues might be blocking your page or some of your content from showing up in Google Search? Learn how to fix JavaScript-related problems with our [troubleshooting guide](#).

JavaScript is an important part of the web platform because it provides many features that turn the web into a powerful application platform. Making your JavaScript-powered web applications discoverable via Google Search can help you find new users and re-engage existing users as they search for the content your web app provides. While Google Search runs JavaScript with an [evergreen version of Chromium](#), there are [a few things that you can optimize](#).

This guide describes how Google Search processes JavaScript and best practices for improving JavaScript web apps for Google Search.

[Original embedded video]

How Google processes JavaScript

Google processes JavaScript web apps in three main phases:

1. Crawling
2. Rendering
3. Indexing

[Original illustration: Googlebot takes a URL from the crawl queue, crawls it, then passes it into the processing stage. The processing stage extracts links that go back on the crawl queue and queues the page for rendering. The page goes from the render queue to the renderer which passes the rendered HTML back to processing which indexes the content and extracts links to put them into the crawl queue.]

Googlebot queues pages for both crawling and rendering. It is not immediately obvious when a page is waiting for crawling and when it is waiting for rendering. When Googlebot fetches a URL from the crawling queue by making an HTTP request, it first checks if you allow crawling. Googlebot reads the [robots.txt](#) file. If it marks the URL as disallowed, then Googlebot skips making an HTTP request to this URL and skips the URL. Google Search won't render JavaScript from blocked files or on blocked pages.

Googlebot then parses the response for other URLs in the href attribute of HTML links and adds the URLs to the crawl queue. To prevent link discovery, use the [nofollow mechanism](#).

It's fine to use JavaScript to inject links into the DOM, as long as such links follow the [best practices for crawlable links](#).

Crawling a URL and parsing the HTML response works well for classical websites or server-side rendered pages where the HTML in the HTTP response contains all content. Some JavaScript sites may use the [app shell model](#) where the

initial HTML does not contain the actual content and Google needs to execute JavaScript before being able to see the actual page content that JavaScript generates.

Googlebot queues all pages with a 200 HTTP status code for rendering, unless a [robots meta tag or header](#) tells Google not to index the page. The page may stay on this queue for a few seconds, but it can take longer than that. Once Google's resources allow, a headless Chromium renders the page and executes the JavaScript. Googlebot parses the rendered HTML for links again and queues the URLs it finds for crawling. Google also uses the rendered HTML to index the page.

All pages with a 200 HTTP status code are sent to the rendering queue, no matter whether JavaScript is present on the page. If the HTTP status code is non-200 (for example, on error pages with 404 status code), rendering might be skipped.

Keep in mind that [server-side or pre-rendering](#) is still a great idea because it makes your website faster for users and crawlers, and not all bots can run JavaScript.

Describe your page with unique titles and snippets

Unique, descriptive [<title> elements](#) and [meta descriptions](#) help users quickly identify the best result for their goal. You can use JavaScript to set or change the meta description as well as the [<title> element](#).

Set the canonical URL

The [rel="canonical" link tag](#) helps Google find the canonical version of a page. You can use JavaScript to set the canonical URL, but keep in mind that you shouldn't use JavaScript to change the canonical URL to something else than the URL you specified as the canonical URL in the original HTML. The best way to set the canonical URL is to use HTML, but if you have to use JavaScript, make sure that you always set the canonical URL to the same value as the original HTML. If you can't set the canonical URL in the HTML, then you can use JavaScript to set the canonical URL and leave it out of the original HTML.

Write compatible code

Browsers offer many APIs and JavaScript is a quickly-evolving language. Google has some limitations regarding which APIs and JavaScript features it supports. To make sure your code is compatible with Google, follow our [guidelines for troubleshooting JavaScript problems](#).

We recommend [using differential serving and polyfills](#) if you feature-detect a missing browser API that you need. Since some browser features cannot be polyfilled, we recommend that you check the polyfill documentation for potential limitations.

Use meaningful HTTP status codes

Googlebot uses [HTTP status codes](#) to find out if something went wrong when crawling the page.

To tell Googlebot if a page can't be crawled or indexed, use a meaningful status code, like a 404 for a page that could not be found or a 401 code for pages behind a login. You can use HTTP status codes to tell Googlebot if a page has moved to a new URL, so that the index can be updated accordingly.

Here's a [list of HTTP status codes](#) and how they effect Google Search.

Avoid soft 404 errors in single-page apps

In client-side rendered single-page apps, routing is often implemented as client-side routing. In this case, using meaningful HTTP status codes can be impossible or impractical. To avoid [soft 404 errors](#) when using client-side rendering and routing, use one of the following strategies:

- Use a [JavaScript redirect](#) to a URL for which the server responds with a 404 HTTP status code (for example `/not-found`).
- Add a `<meta name="robots" content="noindex">` to error pages using JavaScript.

Here is sample code for the redirect approach:

```
fetch(`/api/products/${productId}`)
  .then(response => response.json())
  .then(product => {
    if(product.exists) {
      showProductDetails(product); // shows the product information on the page
    } else {
      // this product does not exist, so this is an error page.
      window.location.href = '/not-found'; // redirect to 404 page on the server.
    }
  })
```

Here is sample code for the noindex tag approach:

```
fetch(`/api/products/${productId}`)
  .then(response => response.json())
  .then(product => {
    if(product.exists) {
      showProductDetails(product); // shows the product information on the page
    } else {
      // this product does not exist, so this is an error page.
      // Note: This example assumes there is no other robots meta tag present in the HTML.
      const metaRobots = document.createElement('meta');
      metaRobots.name = 'robots';
      metaRobots.content = 'noindex';
      document.head.appendChild(metaRobots);
    }
  })
```

Use the History API instead of fragments

Google can only discover your links if they are [<a> HTML elements with an href attribute](#).

For single-page applications with client-side routing, use the [History API](#) to implement routing between different views of your web app. To ensure that Googlebot can parse and extract your URLs, don't use fragments to load different page content. The following example is a bad practice, because Googlebot can't reliably resolve the URLs:


```

<nav>
  <ul>
    <li><a href="#/products">Our products</a></li>
    <li><a href="#/services">Our services</a></li>
  </ul>
</nav>

<h1>Welcome to example.com!</h1>
<div id="placeholder">
  <p>Learn more about <a href="#/products">our products</a> and <a href="#/services">our services</a></p>
</div>
<script>
window.addEventListener('hashchange', function goToPage() {
  // this function loads different content based on the current URL fragment
  const pageToLoad = window.location.hash.slice(1); // URL fragment
  document.getElementById('placeholder').innerHTML = load(pageToLoad);
});
</script>

```

Instead, you can make sure your URLs are accessible to Googlebot by implementing the History API:

```

<nav>
  <ul>
    <li><a href="/products">Our products</a></li>
    <li><a href="/services">Our services</a></li>
  </ul>
</nav>

<h1>Welcome to example.com!</h1>
<div id="placeholder">
  <p>Learn more about <a href="/products">our products</a> and <a href="/services">our services</a></p>
</div>
<script>
function goToPage(event) {
  event.preventDefault(); // stop the browser from navigating to the destination URL.
  const hrefUrl = event.target.getAttribute('href');
  const pageToLoad = hrefUrl.slice(1); // remove the leading slash
  document.getElementById('placeholder').innerHTML = load(pageToLoad);
  window.history.pushState({}, window.title, hrefUrl) // Update URL as well as browser history.
}

// Enable client-side routing for all links on the page
document.querySelectorAll('a').forEach(link => link.addEventListener('click', goToPage));

</script>

```

Properly inject the rel="canonical" link tag

While we don't recommend using JavaScript for this, it is possible to inject a `rel="canonical"` link tag with JavaScript. Google Search will pick up the injected canonical URL when rendering the page. Here is an example to inject a `rel="canonical"` link tag with JavaScript:

```

fetch('/api/cats/' + id)
  .then(function (response) { return response.json(); })
  .then(function (cat) {
    // creates a canonical link tag and dynamically builds the URL
    // e.g. https://example.com/cats/simba
    const linkTag = document.createElement('link');
    linkTag.setAttribute('rel', 'canonical');
    linkTag.href = 'https://example.com/cats/' + cat.urlFriendlyName;
    document.head.appendChild(linkTag);
  });

```

When using JavaScript to inject the `rel="canonical"` link tag, make sure that this is the only `rel="canonical"` link tag on the page. Incorrect implementations might create multiple `rel="canonical"` link tag or change an existing `rel="canonical"` link tag. Conflicting or multiple `rel="canonical"` link tags may lead to unexpected results.

Use robots meta tags carefully

You can prevent Google from indexing a page or following links through the robots meta tag. For example, adding the following meta tag to the top of your page blocks Google from indexing the page:

```

<!-- Google won't index this page or follow links on this page -->
<meta name="robots" content="noindex, nofollow">

```

You can use JavaScript to add a robots meta tag to a page or change its content. The following example code shows how to change the robots meta tag with JavaScript to prevent indexing of the current page if an API call doesn't return content.

```

fetch('/api/products/' + productId)
  .then(function (response) { return response.json(); })
  .then(function (apiResponse) {
    if (apiResponse.isError) {
      // get the robots meta tag
      var metaRobots = document.querySelector('meta[name="robots"]');
      // if there was no robots meta tag, add one
      if (!metaRobots) {
        metaRobots = document.createElement('meta');
        metaRobots.setAttribute('name', 'robots');
        document.head.appendChild(metaRobots);
      }
      // tell Google to exclude this page from the index
      metaRobots.setAttribute('content', 'noindex');
      // display an error message to the user
      errorMsg.textContent = 'This product is no longer available';
      return;
    }
    // display product information
    // ...
  });

```

When Google encounters the `noindex` tag, it may skip rendering and JavaScript execution, which means using JavaScript to change or remove the robots meta tag from `noindex` may not work as expected. If you *do* want the page indexed, don't use a `noindex` tag in the original page code.

Use long-lived caching

Googlebot caches aggressively in order to reduce network requests and resource usage. WRS may ignore caching headers. This may lead WRS to use outdated JavaScript or CSS resources. Content fingerprinting avoids this problem by making a fingerprint of the content part of the filename, like `main.2bb85551.js`. The fingerprint depends on the content of the file, so updates generate a different filename every time. Check out the [web.dev guide on long-lived caching strategies](#) to learn more.

Use structured data

When using [structured data](#) on your pages, you can use [JavaScript](#) to generate the required JSON-LD and inject it into the page. Make sure to [test your implementation](#) to avoid issues.

Follow best practices for web components

Google supports web components. When Google renders a page, it [flattens the shadow DOM and light DOM](#) content. This means Google can only see content that's visible in the rendered HTML. To make sure that Google can still see your content after it's rendered, use the [Rich Results Test](#) or the [URL Inspection Tool](#) and look at the rendered HTML.

If the content isn't visible in the rendered HTML, Google won't be able to index it.

The following example creates a web component that displays its light DOM content inside its shadow DOM. One way to make sure both light DOM and shadow DOM content is displayed in the rendered HTML is to use a [Slot](#) element.

```
<script>
class MyComponent extends HTMLElement {
  constructor() {
    super();
    this.attachShadow({ mode: 'open' });
  }

  connectedCallback() {
    let p = document.createElement('p');
    p.innerHTML = 'Hello World, this is shadow DOM content. Here comes the light DOM: <slot></slot>';
    this.shadowRoot.appendChild(p);
  }
}

window.customElements.define('my-component', MyComponent);
</script>

<my-component>
  <p>This is light DOM content. It's projected into the shadow DOM.</p>
  <p>WRS renders this content as well as the shadow DOM content.</p>
</my-component>
```

After rendering, Google can index this content:

```
<my-component>
  Hello World, this is shadow DOM content. Here comes the light DOM:
  <p>This is light DOM content. It's projected into the shadow DOM</p>
  <p>WRS renders this content as well as the shadow DOM content.</p>
</my-component>
```

Fix images and lazy-loaded content

Images can be quite costly on bandwidth and performance. A good strategy is to use lazy-loading to only load images when the user is about to see them. To make sure you're implementing lazy-loading in a search-friendly way, follow [our lazy-loading guidelines](#).

<https://developers.google.com/search/docs/crawling-indexing/javascript/fix-search-javascript>

Fix Search-related JavaScript problems

This guide helps you identify and fix JavaScript issues that may be blocking your page, or specific content on JavaScript powered pages, from showing up in Google Search. While Google Search does run JavaScript, there are some differences and limitations that you need to account for when designing your pages and applications to accommodate how crawlers access and render your content. Our [guide on JavaScript SEO basics](#) has more information on how you can optimize your JavaScript site for Google Search.

Googlebot is designed to be a good citizen of the web. Crawling is its [main priority](#), while making sure it doesn't degrade the experience of users visiting the site. Googlebot and its Web Rendering Service (WRS) component continuously analyze and identify resources that don't contribute to essential page content and may not fetch such resources. For example, reporting and error requests that don't contribute to essential page content, and other similar types of requests are unused or unnecessary to extract essential page content. Client-side analytics may not provide a full or accurate representation of Googlebot and WRS activity on your site. Use [the crawl stats report in Google Search Console](#) to monitor Googlebot and WRS activity and feedback on your site.

If you suspect that JavaScript issues might be blocking your page, or specific content on JavaScript powered pages, from showing up in Google Search, follow these steps. If you're not sure if JavaScript is the main cause, follow our [general debugging guide](#) to determine the specific issue.

1. **To test how Google crawls and renders a URL**, use the [Rich Results Test](#) or the [URL Inspection Tool](#) in Search Console. You can see loaded resources, JavaScript console output and exceptions, rendered DOM, and more information.

Optionally, we also recommend collecting and auditing JavaScript errors encountered by users, including Googlebot, on your site to identify potential issues that may affect how content is rendered. Here's an example that shows how to log JavaScript errors that are logged in the [global onerror handler](#). Note that some types of JavaScript errors, such as a parse error, cannot be logged with this method.

```

window.addEventListener('error', function(e) {
    var errorText = [
        e.message,
        'URL: ' + e.filename,
        'Line: ' + e.lineno + ', Column: ' + e.colno,
        'Stack: ' + (e.error && e.error.stack || '(no stack trace)')
    ].join('\n');

    // Example: log errors as visual output into the host page.
    // Note: you probably don't want to show such errors to users, or
    //       have the errors get indexed by Googlebot; however, it may
    //       be a useful feature while actively debugging the page.
    var DOM_ID = 'rendering-debug-pre';
    if (!document.getElementById(DOM_ID)) {
        var log = document.createElement('pre');
        log.id = DOM_ID;
        log.style.whiteSpace = 'pre-wrap';
        log.textContent = errorText;
        if (!document.body) document.body = document.createElement('body');
        document.body.insertBefore(log, document.body.firstChild);
    } else {
        document.getElementById(DOM_ID).textContent += '\n\n' + errorText;
    }

    // Example: log the error to remote service.
    // Note: you can log errors to a remote service, to understand
    //       and monitor the types of errors encountered by regular users,
    //       Googlebot, and other crawlers.
    var client = new XMLHttpRequest();
    client.open('POST', 'https://example.com/logError');
    client.setRequestHeader('Content-Type', 'text/plain; charset=UTF-8');
    client.send(errorText);

});

```

2. **Make sure to prevent soft 404 errors.** In a single-page application (SPA), this can be especially difficult. To prevent error pages from being indexed, you can use one or both of the following strategies:

- Redirect to a URL where the server responds with a 404 status code.

```

fetch(`https://api.kitten.club/cats/${id}`)
    .then(res => res.json())
    .then((cat) => {
        if (!cat.exists) {
            // redirect to page that gives a 404
            window.location.href = '/not-found';
        }
    });

```

- Add or change the robots meta tag to noindex.

```

fetch(`https://api.kitten.club/cats/${id}`)
  .then(res => res.json())
  .then((cat) => {
    if (!cat.exists) {
      const metaRobots = document.createElement('meta');
      metaRobots.name = 'robots';
      metaRobots.content = 'noindex';
      document.head.appendChild(metaRobots);
    }
  });

```

When a SPA is using client-side JavaScript to handle errors they often report a 200 HTTP status code instead of the [appropriate status code](#). This can lead to error pages being indexed and possibly shown in search results.

3. Expect Googlebot to decline [user permission requests](#).

Features that require user permission don't make sense for Googlebot, or for all users. For example, if you make the Camera API required, Googlebot can't provide a camera to you. Instead, provide a way for users to access your content without being forced to allow camera access.

4. Don't use URL fragments to load different content.

A SPA may use URL fragments (for example <https://example.com/#/products>) for loading different views. The [AJAX-crawling scheme has been deprecated](#) since 2015, so you can't rely on URL fragments to work with Googlebot. We recommend using the [History API](#) to load different content based on the URL in a SPA.

5. Don't rely on data persistence to serve content.

WRS loads each URL (refer to [How Google Search Works](#) for an overview of how Google discovers content), following server and client redirects, same as a regular browser. However, WRS does not retain state across page loads:

- Local Storage and Session Storage data are cleared across page loads.
- HTTP Cookies are cleared across page loads.

6. Use content fingerprinting to avoid caching issues with Googlebot.

Googlebot caches aggressively in order to reduce network requests and resource usage. WRS may ignore caching headers. This may lead WRS to use outdated JavaScript or CSS resources. Content fingerprinting avoids this problem by making a fingerprint of the content part of the filename, like `main.2bb85551.js`. The fingerprint depends on the content of the file, so updates generate a different filename every time. Check out the [web.dev guide on long-lived caching strategies](#) to learn more.

7. Ensure that your application uses [feature detection](#) for all critical APIs that it needs and provide a fallback behavior or polyfill where applicable.

Some web features may not yet be adopted by all user agents and some may intentionally disable certain features. For example, if you use WebGL to render photo effects in the browser, feature detection shows that Googlebot doesn't support WebGL. To fix this, you could skip the photo effect or decide to use server-side rendering to prerender the photo effects, which makes your content accessible to everyone, including Googlebot.

8. Make sure your content works with HTTP connections.

Googlebot uses HTTP requests to retrieve content from your server. It does not support other types of connections, such as [WebSockets](#) or [WebRTC](#) connections. To avoid problems with such connections, make sure to provide an HTTP fallback to retrieve content and use robust error handling and [feature detection](#).

9. Make sure your web components render as expected. Use the [Rich Results Test](#) or the [URL Inspection Tool](#) to check if the rendered HTML has all content you expect.

WRS flattens the [light DOM and shadow DOM](#). If the web components you use aren't using [<slot> mechanism](#)

for light DOM content, consult the documentation of the web component for further information or use another web component instead. For more information, see [best practices for web components](#).

10. If you're using a JavaScript-based paywall, consider the implementation.

Some JavaScript paywall solutions include the full content in the server response, then use JavaScript to hide it until subscription status is confirmed. This isn't a reliable way to limit access to the content. Make sure your paywall only provides the full content once the subscription status is confirmed.

11. After you fix the items in this checklist, test your page with the [Rich Results Test](#) or the [URL inspection tool](#) in Search Console again.

If you fixed the issue, a green check mark appears and no errors display. If you still see errors, post in the [Search Central help community](#).

<https://developers.google.com/search/docs/crawling-indexing/javascript/lazy-loading>

Fix lazy-loaded content

Deferring loading of non-critical or non-visible content, also commonly known as "lazy-loading", is a common performance and UX best practice. For more information, see [web.dev's resources on lazy-loading images and video](#). However, if not implemented correctly, this technique can inadvertently hide content from Google. This document explains how to make sure Google can crawl and index lazy-loaded content.

Load content when it's visible in the viewport

To ensure that Google sees all content on your page, make sure that your lazy-loading implementation loads all relevant content whenever it is visible in the viewport. Here are a few methods to implement lazy-loading:

- [Browser built-in lazy-loading](#) for images and iframes
- [IntersectionObserver API](#) and a [polyfill](#)
- A JavaScript library that supports loading data when it enters the viewport

The methods mentioned don't rely on user actions, such as scrolling or clicking, to load content, which is important as Google Search does not interact with your page.

Don't add lazy-loading to content that is likely to be immediately visible when a user opens a page. That might cause content to take longer to load and show up in the browser, which will be very noticeable to the user.

Make sure to [test your implementation](#).

Support paginated loading for infinite scroll

At a high level, infinite scroll is a technique that loads more content, more distinct pages, as the user scrolls down a long page. This could be one long article that's split into multiple chunks, or a collection of items that's similarly split into chunks. To implement infinite scroll in an indexable way, make sure your website supports paginated loading of these chunks by doing the following:

- Give each chunk its own persistent, unique URL.
- Ensure that the content shown on each URL remains the same every time it's loaded in a browser. One way this can be done is to use absolute page numbers in the URL, for example by using `?page=12` as a query parameter.
- Avoid using relative elements like `?date=yesterday` in these URLs. This allows search engines and users to consistently find the same content under a given URL, making it easier for search engines to properly index the content, and allowing users to share and reengage with that part of your content.
- Link sequentially to the individual URLs so that search engines can discover the URLs in a paginated set. Find out more about [best practices when implementing pagination](#).
- When a new page chunk is loaded in response to the user scrolling, and it becomes the primary visible element for the user, update the displayed URL using the [History API](#). This allows the user to refresh, share, and link to the current URL displayed in the browser.

Test

After you set up your implementation, make sure it works correctly. You can use the [URL Inspection Tool](#) in Search Console to see if all content was loaded. Check the rendered HTML to make sure your content is in the rendered HTML by looking for it in URL Inspection Tool. If your image or video URLs appear in the src attribute on the `<img>` or `<video>` elements in the rendered HTML, your setup works correctly.

<https://developers.google.com/search/docs/crawling-indexing/javascript/dynamic-rendering>

Dynamic rendering as a workaround

Dynamic rendering was a workaround and not a long-term solution for problems with JavaScript-generated content in search engines. Instead, we recommend that you use [server-side rendering](#), [static rendering](#), or [hydration](#) as a solution.

On some websites, JavaScript loads additional content when the page is open in a browser. This is called [client-side rendering](#). Google Search sees this content along with the content in the HTML of a website. Keep in mind that there are some [limitations for JavaScript in Google Search](#) and some pages may encounter problems with content not showing up in the rendered HTML. Other search engines may choose to ignore JavaScript and won't see JavaScript-generated content.

Dynamic rendering is a workaround for websites where JavaScript-generated content is not available to search engines. A dynamic rendering server detects bots that may have problems with JavaScript-generated content and serves a server-rendered version without JavaScript to these bots while showing the client-side rendered version of the content to users.

Dynamic rendering is a workaround and not a recommended solution, because it creates additional complexities and resource requirements.

[Original embedded video]

Sites that might use dynamic rendering

Dynamic rendering was a workaround for indexable, public JavaScript-generated content that changes rapidly, or content that uses [JavaScript features that aren't supported by the crawlers](#) you care about. Not all sites need to use dynamic rendering, and there are better solutions than dynamic rendering as explained in [an overview of rendering on the web](#).

Understand how dynamic rendering works

Dynamic rendering requires your web server to detect crawlers (for example, by checking the user agent). When your web server identifies a request from a crawler that does not support JavaScript or the JavaScript features required to render your content, this request is routed to a rendering server. Requests from users and crawlers without JavaScript issues are served normally. The rendering server responds to requests with a version of the content that's suitable to the crawler, for example, it may serve a static HTML version. You can choose to enable the dynamic renderer for all pages or on a per-page basis.

[Original illustration: A diagram that shows how dynamic rendering works. The diagram shows the server serving initial HTML and JavaScript content directly to the browser. In contrast, the diagram shows the server serving initial HTML and JavaScript to a renderer, which converts the initial HTML and JavaScript to static HTML. Once the content is converted, the renderer serves static HTML to the crawler.]

Dynamic rendering is not cloaking

Googlebot generally doesn't consider dynamic rendering as [cloaking](#). As long as your dynamic rendering produces similar content, Googlebot won't view dynamic rendering as cloaking.

When you're setting up dynamic rendering, your site may produce error pages. Googlebot doesn't consider these error pages as cloaking and [treats the error as any other error page](#).

Using dynamic rendering to serve completely different content to users and crawlers can be considered cloaking. For example, a website that serves a page about cats to users and a page about dogs to crawlers is cloaking.

<https://developers.google.com/search/docs/crawling-indexing/valid-page-metadata>

Use valid HTML to specify page metadata

Using valid HTML for page metadata ensures that Google can use the metadata as documented. Google tries to understand HTML even when it is invalid or inconsistent with the [HTML standard](#), but errors in the markup can cause problems with how your metadata is used in Google Search. The primary element for specifying metadata about a page is the `<head>` element of an HTML document. If you use an invalid element in the `<head>` element, Google ignores any elements that appear after the invalid element.

Use valid elements in the `<head>` element

The `<head>` element must only contain the following valid elements (and no other invalid elements), as per the HTML standard:

- `title`
- `meta`
- `link`
- `script`
- `style`
- `base`
- `noscript`
- `template`

Don't use invalid elements in the `<head>` element

No element other than the aforementioned is permitted by the HTML standard in the `<head>` element. Common elements that appear in the `<head>` element, rendering it invalid are:

- `iframe`
- `img`

We strongly recommend that you don't use these invalid elements in the `<head>` element, but if you must, place these invalid elements after the ones you want Google to see. Once Google detects one of these invalid elements, it assumes the end of the `<head>` element and stops reading any further elements in the `<head>` element.

<https://developers.google.com/search/docs/crawling-indexing/special-tags>

meta tags and attributes that Google supports

This page explains what meta tags are, which meta tags and HTML attributes Google supports to control indexing, and other important points to note when implementing meta tags on your site.

[Original embedded video]

meta tags

meta tags are HTML tags used to provide additional information about a page to search engines and other clients. Clients process the meta tags and ignore those they don't support. meta tags are added to the <head> section of your HTML page and generally look like this:

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <meta name="description" content="Author: A.N. Author, Illustrator: P. Picture, Category: Books, Price:
£9.24, Length: 784 pages">
  <meta name="google-site-verification" content="+nxGUDJ4QpAZ5l9Bsjudi102tLVC21AIh5d1Nl23908vVuFHs34=">
  <title>Example Books - high-quality used books for children</title>
  <meta name="robots" content="noindex,nofollow">
</head>
</html>
```

If you use a CMS, such as **Wix**, **WordPress**, or **Blogger**, you might not be able to edit your HTML directly, or you might prefer not to. Instead, your CMS might have a search engine settings page or some other mechanism to tell search engines about meta tags.

If you want to add a meta tag to your website, search for instructions about modifying the <head> of your page on your CMS (for example, search for "wix add meta tags").

Google supports the following meta tags:

List of meta tags that Google supports

description	<pre><meta name="description" content="A description of the page"></pre> Use this tag to provide a short description of the page. In some situations, this description is used in the snippet shown in search results.
robots and googlebot	<pre><meta name="robots" content="..., ..."></pre> <pre><meta name="googlebot" content="..., ..."></pre> These meta tags control the behavior of search engine crawling and indexing. The <code><meta name="robots" ...></code> tag applies to all search engines, while the <code><meta name="googlebot ...></code> tag is specific to Google. In the case of conflicting robots (or googlebot) meta tags, the more restrictive tag applies. For example, if a page has both the <code>max-snippet:50</code> and <code>nosnippet</code> tags, the <code>nosnippet</code> tag will apply. The default values are <code>index</code>, <code>follow</code> and don't need to be specified. For a full list of values that Google supports, see the list of valid rules. You can also specify this information in the header of your pages using the X-Robots-Tag HTTP header rule. This is particularly useful if you wish to limit indexing of non-HTML files like graphics or other kinds of documents. More information about robots meta tags.
notranslate	<pre><meta name="googlebot" content="notranslate"></pre> When Google recognizes that the contents of a page aren't in the language that the user likely wants to read, Google may provide a translated title link and snippet in search results. If the user clicks the translated title link, all further user interaction with the page is through Google Translate, which will automatically translate any links followed. In general, this gives you the chance to provide your unique and compelling content to a much larger group of users. However, there may be situations where this is not desired. This meta tag tells Google that you don't want us to provide a translation for this page.
nopageread aloud	<pre><meta name="google" content="nopageread aloud"></pre> Prevents various Google text-to-speech services from reading aloud web pages using text-to-speech (TTS).
google-site-	

List of meta tags that Google supports

verification	<pre><meta name="google-site-verification" content="..."></pre> You can use this tag on the top-level page of your site to verify ownership for Search Console. Note that while the values of the name and content attributes must match exactly what is provided to you (including upper and lower case), it doesn't matter if you change the tag from XHTML to HTML or if the format of the tag matches the format of your page.
Content-Type and charset	<pre><meta http-equiv="Content-Type" content="...; charset=..."></pre> <pre><meta charset="..."></pre> These tags define the page's content type and character set respectively. Make sure that you surround the value of the content attribute in the http-equiv meta tag with quotes—otherwise the charset attribute may be interpreted incorrectly. We recommend using Unicode/UTF-8 where possible.
refresh	<pre><meta http-equiv="refresh" content="...;url=..."></pre> This tag, commonly called meta-refresh, sends the user to a new URL after a certain amount of time, and is sometimes used as a simple form of redirection. However, it is not supported by all browsers and can be confusing to the user. We recommend using a server-side 301 redirect instead.
viewport	<pre><meta name="viewport" content="..."></pre> This tag tells the browser how to render a page on a mobile device. Presence of this tag indicates to Google that the page is mobile friendly. Read more about how to configure the viewport meta tag.
rating	<pre><meta name="rating" content="adult"></pre> <pre><meta name="rating" content="RTA-5042-1996-1400-1577-RTA"></pre> Labels a page as containing sexually-explicit adult content, to signal that it be filtered by SafeSearch results. Learn more about labeling SafeSearch pages.

HTML tag attributes

HTML tag attributes are additional values of HTML tags that configure the parent tag. For example, the href attribute of the <a> tag configures the resource the anchor tag points to: .

Google Search supports a limited number of HTML attributes for indexing purposes. Attributes like src and href are used for discovering resources such as images and URLs. Google also supports various rel attributes that allow site owners to qualify outbound links.

The data-nosnippet attribute of div, span, and section tags allow you to exclude parts of an HTML page from snippets.

Other points to note

- Google can read both HTML and XHTML-style meta tags, regardless of the code used on the page.
- To ensure machine readability, the head section must be valid HTML and in case of attributes, all parent tags closed accordingly.
- With the exception of google-site-verification, letter case is generally not important in meta tags.
- You can use other meta tags if they are important to your site, but Google will ignore meta tags that it doesn't support.
- If you're considering using JavaScript to inject or change meta tags, proceed with caution. We recommend that you avoid using JavaScript to inject or change meta tags whenever possible, and if you must use it, test your implementations thoroughly.
- To check the meta tags and attributes on your pages, use the URL Inspection Tool.

Unsupported tags and attributes

The following tags and attributes aren't supported by Google Search and are ignored. We're including them here because they're either very common in HTML or we used to support them.

Unsupported tags and attributes	
meta-keyword tag	<meta name="keywords" content="..."> The meta-keyword tag is not used by Google Search, and it has no effect on indexing and ranking at all.
HTML tag lang attributes	Google Search detects the language of a page based on the textual content of the page. It doesn't rely on code annotations such as the lang.
next and prev rel attribute values	<link rel="next" href="..."> <link rel="prev" href="..."> Google no longer uses these HTML <link> tags, and they have no effect on indexing.

nositelinkssearchbox

```
<meta name="google" content="nositelinkssearchbox">
```

The nositelinkssearchbox rule is no longer used by Google Search to control whether the sitelink search box is shown for a given page, as the feature no longer exists.

<https://developers.google.com/search/docs/crawling-indexing/robots-meta-tag>

Robots meta tag, data-nosnippet, and X-Robots-Tag specifications

This document details how the page- and text-level settings can be used to adjust how Google presents your content in search results. You can specify page-level settings by including a meta tag on HTML pages or in an HTTP header. You can specify text-level settings with the data-nosnippet attribute on HTML elements within a page.

Keep in mind that these settings can be read and followed only if crawlers are allowed to access the pages that include these settings.

The `<meta name="robots" content="noindex">` rule applies to search engine crawlers. To block non-search crawlers, such as AdsBot-Google, you might need to add rules targeted to the specific crawler (for example, `<meta name="AdsBot-Google" content="noindex">`).

Using the robots meta tag

The robots meta tag lets you use a granular, page-specific approach to controlling how an individual HTML page should be indexed and served to users in Google Search results. Place the robots meta tag in the `<head>` section of a given page, like this:

```
<!DOCTYPE html>
<html><head>
  <meta name="robots" content="noindex">
  (...)
</head>
<body>(…)</body>
</html>
```

If you use a CMS, such as Wix, WordPress, or Blogger, you might not be able to edit your HTML directly, or you might prefer not to. Instead, your CMS might have a search engine settings page or some other mechanism to tell search engines about meta tags.

If you want to add a meta tag to your website, search for instructions about modifying the `<head>` of your page on your CMS (for example, search for "wix add meta tags").

In this example, the robots meta tag instructs search engines not to show the page in search results. The value of the name attribute (robots) specifies that the rule applies to all crawlers. Both the name and the content attributes are case-insensitive. To address a specific crawler, replace the robots value of the name attribute with the user agent token of the crawler that you are addressing. Google supports two user agent tokens in the robots meta tag; other values are ignored:

1. googlebot: for all text results.

2. googlebot-news: for news results.

For example, to instruct Google specifically not to show a snippet in its search results, you can specify `googlebot` as the name of the meta tag:

```
<meta name="googlebot" content="nosnippet">
```

To show a full snippet in Google's web search results, but no snippet in Google News, specify `googlebot-news` as the name of the meta tag:

```
<meta name="googlebot-news" content="nosnippet">
```

To specify multiple crawlers individually, use multiple robots meta tags:

```
<meta name="googlebot" content="notranslate">
<meta name="googlebot-news" content="nosnippet">
```

Note: Google Search doesn't enforce placement of meta robots in the HTML head and will respect robots meta tags in the body section of an HTML document as well.

To block indexing of non-HTML resources, such as PDF files, video files, or image files, use the [X-Robots-Tag response header](#) instead.

Using the X-Robots-Tag HTTP header

The X-Robots-Tag can be used as an element of the HTTP header response for a given URL. Any rule that can be used in a robots meta tag can also be specified as an X-Robots-Tag. Here's an example of an HTTP response with an X-Robots-Tag instructing crawlers not to index a page:

```
HTTP/1.1 200 OK
Date: Tue, 25 May 2010 21:42:43 GMT
(...)
X-Robots-Tag: noindex
(...)
```

Multiple X-Robots-Tag headers can be combined within the HTTP response, or you can specify a comma-separated list of rules. Here's an example of an HTTP header response which has a `noimageindex` X-Robots-Tag combined with an `unavailable_after` X-Robots-Tag.

```
HTTP/1.1 200 OK
Date: Tue, 25 May 2010 21:42:43 GMT
(...)
X-Robots-Tag: noimageindex
X-Robots-Tag: unavailable_after: 25 Jun 2010 15:00:00 PST
(...)
```

The X-Robots-Tag may optionally specify a user agent before the rules. For instance, the following set of X-Robots-Tag HTTP headers can be used to conditionally allow showing of a page in search results for different search engines:

```
HTTP/1.1 200 OK
Date: Tue, 25 May 2010 21:42:43 GMT
(...)
X-Robots-Tag: googlebot: nofollow
X-Robots-Tag: otherbot: noindex, nofollow
(...)
```

Rules specified without a user agent are valid for all crawlers. The HTTP header, the user agent name, and the specified values are not case sensitive.

Conflicting robots rules: In the case of conflicting robots rules, the more restrictive rule applies. For example, if a page has both `max-snippet:50` and `nosnippet` rules, the `nosnippet` rule will apply.

Valid indexing and serving rules

The following rules, also available in [machine-readable format](#), can be used to control indexing and serving of a [snippet](#) with the robots meta tag and the X-Robots-Tag. Each value represents a specific rule. [Multiple rules may be combined](#) in a comma-separated list or in separate meta tags. These rules are case-insensitive.

It is possible that these rules may not be treated the same by all other search engines.

Rules	
all	There are no restrictions for indexing or serving. This rule is the default value and has no effect if explicitly listed.
noindex	Do not show this page, media, or resource in search results. If you don't specify this rule, the page, media, or resource may be indexed and shown in search results. To remove information from Google, follow our step-by-step guide .
nofollow	Do not follow the links on this page. If you don't specify this rule, Google may use the links on the page to discover those linked pages. Learn more about nofollow .
none	Equivalent to <code>noindex, nofollow</code> .
nosnippet	Do not show a text snippet or video preview in the search results for this page. A static image thumbnail (if available) may still be visible, when it results in a better user experience. This applies to all forms of search results (at Google: web search, Google Images, Discover, AI

	Overviews, AI Mode) and will also prevent the content from being used as a direct input for AI Overviews and AI Mode. If you don't specify this rule, Google may generate a text snippet and video preview based on information found on the page. To exclude certain sections of your content from appearing in search result snippets, use the data-nosnippet HTML attribute.
indexifembedded	Google is allowed to index the content of a page if it's embedded in another page through iframes or similar HTML tags, in spite of a <code>noindex</code> rule. <code>indexifembedded</code> only has an effect if it's accompanied by <code>noindex</code>.
max-snippet:[number]	Use a maximum of [number] characters as a textual snippet for this search result. (Note that a URL may appear as multiple search results within a search results page.) This does not affect image or video previews. This applies to all forms of search results (such as Google web search, Google Images, Discover, Assistant, AI Overviews, AI Mode) and will also limit how much of the content may be used as a direct input for AI Overviews and AI Mode. However, this limit does not apply in cases where a publisher has separately granted permission for use of content. For instance, if the publisher supplies content in the form of in-page structured data or has a license agreement with Google, this setting does not interrupt those more specific permitted uses. This rule is ignored if no parseable [number] is specified. If you don't specify this rule, Google will choose the length of the snippet. Special values: • 0: No snippet is to be shown. Equivalent to <code>nosnippet</code>. • -1: Google will choose the snippet length that it believes is most effective to help users discover your content and direct users to your site. Examples: To stop a snippet from displaying in search results: <pre><meta name="robots" content="max-snippet:0"></pre> To allow up to 20 characters to be shown in the snippet: <pre><meta name="robots" content="max-snippet:20"></pre> To specify that there's no limit on the number of characters that can be shown in the snippet: <pre><meta name="robots" content="max-snippet:-1"></pre>
max-image-preview:[setting]	Set the maximum size of an image preview for this page in search results. If you don't specify the <code>max-image-preview</code> rule, Google may show an image preview of the default size. Accepted [setting] values: • none: No image preview is to be shown.

	• standard: A default image preview may be shown. • large: A larger image preview, up to the width of the viewport, may be shown. This applies to all forms of search results (such as Google web search, Google Images, Discover, Assistant). However, this limit does not apply in cases where a publisher has separately granted permission for use of content. For instance, if the publisher supplies content in the form of in-page structured data (such as AMP and canonical versions of an article) or has a license agreement with Google, this setting will not interrupt those more specific permitted uses. If you don't want Google to use larger thumbnail images when their AMP pages and canonical version of an article are shown in Search or Discover, specify a <code>max-image-preview</code> value of <code>standard</code> or <code>none</code>. Example: <pre><meta name="robots" content="max-image-preview:standard"></pre>
max-video-preview: [number]	Use a maximum of [number] seconds as a video snippet for videos on this page in search results. If you don't specify the <code>max-video-preview</code> rule, Google may show a video snippet in search results, and you leave it up to Google to decide how long the preview may be. Special values: • <code>0</code>: At most, a static image may be used, in accordance to the <code>max-image-preview</code> setting. • <code>-1</code>: There is no limit. This applies to all forms of search results (at Google: web search, Google Images, Google Videos, Discover, Assistant). This rule is ignored if no parseable [number] is specified. Example: <pre><meta name="robots" content="max-video-preview:-1"></pre>
notranslate	Don't offer translation of this page in search results. If you don't specify this rule, Google may provide a translation of the title link and snippet of a search result for results that aren't in the language of the search query. If the user clicks the translated title link, all further user interaction with the page is through Google Translate, which will automatically translate any links followed.
noimageindex	Do not index images on this page. If you don't specify this value, images on the page may be indexed and shown in search results.
unavailable_after: [date/time]	Do not show this page in search results after the specified date/time. The date/time must be specified in a widely adopted format including, but not limited to RFC 822, RFC 850, and ISO 8601. The rule is ignored if no valid date/time is specified. By default there is no expiration date for content.

If you don't specify this rule, this page may be shown in search results indefinitely. Googlebot will decrease the crawl rate of the URL considerably after the specified date and time.

Example:

```
<meta name="robots" content="unavailable_after: 2020-09-21">
```

Reference of historical and other unused rules

The following rules aren't used by Google Search and are ignored. We're including these rules here because people have often asked about them or we used them in the past.

List of historical and other unused rules	
noarchive	The noarchive rule is no longer used by Google Search to control whether a cached link is shown in search results, as the cached link feature no longer exists.
nocache	The nocache rule isn't used by Google Search.
nositelinkssearchbox	The nositelinkssearchbox rule is no longer used by Google Search to control whether the sitelink search box is shown for a given page, as the feature no longer exists.

Handling combined indexing and serving rules

You can create a multi-rule instruction by combining robots meta tag rules with commas or by using multiple meta tags. Here is an example of a robots meta tag that instructs web crawlers to not index the page and to not crawl any of the links on the page:

Comma-separated list

```
<meta name="robots" content="noindex, nofollow">
```

Multiple meta tags

```
<meta name="robots" content="noindex">  
<meta name="robots" content="nofollow">
```

Here is an example that limits the text snippet to 20 characters, and allows a large image preview:

```
<meta name="robots" content="max-snippet:20, max-image-preview:large">
```


For situations where multiple crawlers are specified along with different rules, the search engine will use the sum of the negative rules. For example:

```
<meta name="robots" content="nofollow">
<meta name="googlebot" content="noindex">
```

The page containing these meta tags will be interpreted as having a noindex, nofollow rule when crawled by Googlebot.

Using the data-nosnippet HTML attribute

You can designate textual parts of an HTML page not to be used as a snippet. This can be done on an HTML-element level with the data-nosnippet HTML attribute on span, div, and section elements. The data-nosnippet is considered a [boolean attribute](#). As with all boolean attributes, any value specified is ignored. To ensure machine-readability, the HTML section must be valid HTML and all appropriate tags must be closed accordingly.

Examples:

```
<p>This text can be shown in a snippet
<span data-nosnippet>and this part would not be shown</span>.</p>

<div data-nosnippet>not in snippet</div>
<div data-nosnippet="true">also not in snippet</div>
<div data-nosnippet="false">also not in snippet</div>
<!-- all values are ignored -->

<div data-nosnippet>some text</html>
<!-- unclosed "div" will include all content afterwards -->

<mytag data-nosnippet>some text</mytag>
<!-- NOT VALID: not a span, div, or section -->

<p>This text can be shown in a snippet.</p>
<div data-nosnippet>
<p>However, this is not in snippet.</p>
<ul>
  <li>Stuff not in snippet</li>
  <li>More stuff not in snippet</li>
</ul>
</div>
```

Google typically renders pages in order to index them, however rendering is not guaranteed. Because of this, extraction of data-nosnippet may happen both before and after rendering. To avoid uncertainty from rendering, do not add or remove the data-nosnippet attribute of existing nodes through JavaScript. When adding DOM elements through JavaScript, include the data-nosnippet attribute as necessary when initially adding the element to the page's DOM. If custom elements are used, wrap or render them with div, span, or section elements if you need to use data-nosnippet.

Using structured data

Robots meta tags govern the amount of content that Google extracts automatically from web pages for display as search results. But many publishers also use schema.org structured data to make specific information available for

[search presentation](#). Robots meta tag limitations don't affect the use of that structured data, with the exception of `article.description` and the `description` values for structured data specified for other creative works. To specify the maximum length of a preview based on these `description` values, use the `max-snippet` rule. For example, recipe structured data on a page is eligible for inclusion in the recipe carousel, even if the text preview would otherwise be limited. You can limit the length of a text preview with `max-snippet`, but that robots meta tag doesn't apply when the information is provided using structured data for rich results.

To manage the use of structured data for your web pages, modify the structured data types and values themselves, adding or removing information in order to provide only the data you want to make available. Also note that structured data remains usable for search results when declared within a `data-nosnippet` element.

Practical implementation of X-Robots-Tag

You can add the X-Robots-Tag to a site's HTTP responses through the configuration files of your site's web server software. For example, on Apache-based web servers you can use `.htaccess` and `httpd.conf` files. The benefit of using an X-Robots-Tag with HTTP responses is that you can specify crawling rules that are applied globally across a site. The support of regular expressions allows a high level of flexibility.

For example, to add a `noindex`, `nofollow` X-Robots-Tag to the HTTP response for all `.PDF` files across an entire site, add the following snippet to the site's root `.htaccess` file or `httpd.conf` file on Apache, or the site's `.conf` file on NGINX.

Apache

```
<Files ~ "\.pdf$">
Header set X-Robots-Tag "noindex, nofollow"
</Files>
```

NGINX

```
location ~* \.pdf$ {
    add_header X-Robots-Tag "noindex, nofollow";
}
```

You can use the X-Robots-Tag for non-HTML files like image files where the usage of robots meta tags in HTML is not possible. Here's an example of adding a `noindex` X-Robots-Tag rule for images files (`.png`, `.jpeg`, `.jpg`, `.gif`) across an entire site:

Apache

```
<Files ~ "\.(png|jpe?g|gif)$">
Header set X-Robots-Tag "noindex"
</Files>
```

NGINX

```
location ~* \.(png|jpe?g|gif)$ {  
    add_header X-Robots-Tag "noindex";  
}
```

You can also set the X-Robots-Tag headers for individual static files:

Apache

```
# the htaccess file must be placed in the directory of the matched file.  
<Files "unicorn.pdf">  
Header set X-Robots-Tag "noindex, nofollow"  
</Files>
```

NGINX

```
location = /secrets/unicorn.pdf {  
    add_header X-Robots-Tag "noindex, nofollow";  
}
```

Combining robots.txt rules with indexing and serving rules

robots meta tags and X-Robots-Tag HTTP headers are discovered when a URL is crawled. If a page is disallowed from crawling through the robots.txt file, then any information about indexing or serving rules will not be found and will therefore be ignored. If indexing or serving rules must be followed, the URLs containing those rules cannot be disallowed from crawling.

<https://developers.google.com/search/docs/crawling-indexing/block-indexing>

Block Search indexing with noindex

`noindex` is a rule set with either a `<meta>` tag or HTTP response header and is used to prevent indexing content by search engines that support the `noindex` rule, such as Google. When Googlebot crawls that page and extracts the tag or header, Google will drop that page entirely from Google Search results, regardless of whether other sites link to it.

Important: For the `noindex` rule to be effective, the page or resource **must not** be blocked by a `robots.txt` file, and it has to be otherwise accessible to the crawler. If the page is blocked by a `robots.txt` file or the crawler can't access the page, the crawler will never see the `noindex` rule, and the page can still appear in search results, for example if other pages link to it.

Using `noindex` is useful if you don't have root access to your server, as it allows you to control access to your site on a page-by-page basis.

Implementing noindex

There are two ways to implement `noindex`: as a `<meta>` tag and as an HTTP response header. They have the same effect; choose the method that is more convenient for your site and appropriate for the content type. Specifying the `noindex` rule in the `robots.txt` file is not supported by Google.

You can also combine the `noindex` rule with other rules that control indexing. For example, you can join a `nofollow` hint with a `noindex` rule: `<meta name="robots" content="noindex, nofollow" />`.

`<meta>` tag

To prevent *all search engines* that support the `noindex` rule from indexing a page on your site, place the following `<meta>` tag into the `<head>` section of your page:

```
<meta name="robots" content="noindex">
```

To prevent *only Google web crawlers* from indexing a page:

```
<meta name="googlebot" content="noindex">
```

Be aware that some search engines might interpret the `noindex` rule differently. As a result, it is possible that your page might still appear in results from other search engines.

[Read more about the `noindex` `<meta>` tag.](#)

If you use a CMS, such as **Wix**, **WordPress**, or **Blogger**, you might not be able to edit your HTML directly, or you might prefer not to. Instead, your CMS might have a search engine settings page or some other mechanism to tell search engines about meta tags.

If you want to add a meta tag to your website, search for instructions about modifying the `<head>` of your page on your CMS (for example, search for "wix add meta tags").

HTTP response header

Instead of a `<meta>` tag, you can return an X-Robots-Tag HTTP header with a value of either `noindex` or `none` in your response. A response header can be used for non-HTML resources, such as PDFs, video files, and image files. Here's an example of an HTTP response with an X-Robots-Tag header instructing search engines not to index a page:

```
HTTP/1.1 200 OK
(...)
X-Robots-Tag: noindex
(...)
```

[Read more about the noindex response header.](#)

Debugging noindex issues

We have to crawl your page in order to see `<meta>` tags and HTTP headers. If a page is still appearing in results, it's probably because we haven't crawled the page since you added the `noindex` rule. Depending on the importance of the page on the internet, it may take months for Googlebot to revisit a page. You can request that Google recrawl a page using the [URL Inspection tool](#).

If you need to remove a page of your site quickly from Google's search results, see our [documentation about removals](#).

Another reason could also be that the `robots.txt` file is blocking the URL from Google web crawlers, so they can't see the tag. To unblock your page from Google, you must [edit your robots.txt file](#).

Finally, make sure that the `noindex` rule is visible to Googlebot. To test if your `noindex` implementation is correct, use the [URL Inspection tool](#) to see the HTML that Googlebot received while crawling the page. You can also use the [Page Indexing report](#) in Search Console to monitor the pages on your site from which Googlebot extracted a `noindex` rule.

<https://developers.google.com/search/docs/crawling-indexing/qualify-outbound-links>

Qualify your outbound links to Google

For certain links on your site, you might want to tell Google your relationship with the linked page. In order to do that, use one of the following `rel` attribute values in the `<a>` tag.

For regular links that you expect Google to fetch and parse without any qualifications, you don't need to add a `rel` attribute. For example:

```
<p>My favorite horse is the <a href="https://horses.example.com/Palomino">palomino</a>.</p>
```

For other links, use one or more of the following values:

rel values	
rel="sponsored"	Mark links that are advertisements or paid placements (commonly called <i>paid links</i>) with the sponsored value. Read more about Google's stance on paid links. <pre>Appenzeller</pre> Note: The nofollow attribute was previously recommended for these types of links and is still an acceptable way to flag them, though sponsored is preferred.
rel="ugc"	We recommend marking user-generated content (UGC) links, such as comments and forum posts, with the ugc value. <pre>Appenzeller</pre> If you want to recognize and reward trustworthy contributors, you might remove this attribute from links posted by members or users who have consistently made high-quality contributions over time. Read more about how to prevent user-generated spam on your site and platform.
rel="nofollow"	Use the nofollow value when other values don't apply, and you'd rather Google not associate your site with, or crawl the linked page from, your site. For links within your own site, use the robots.txt disallow rule. <pre>Appenzeller</pre>
Multiple values	You may specify multiple rel values as a space- or comma-separated list. Examples: <pre><p>I love Appenzeller cheese.</p></pre> <pre><p>I hate Blue cheese.</p></pre>

Links marked with these rel attributes will generally not be followed. Remember that the linked pages may be found through other means, such as sitemaps or links from other sites, and thus they may still be crawled. These rel attributes are used only in [<a> elements that Google can crawl](#), except nofollow, which is also available as [robots meta tag](#).

If you need to prevent Google from fetching a link to a page on your own site, use the [robots.txt disallow rule](#).

To prevent Google from indexing a page, allow crawling and use the [noindex robots rule](#).

<https://developers.google.com/search/docs/crawling-indexing/control-what-you-share>

Control what you share with Google

Google supports a variety of ways that allows site owners control what shows up in Google's search results. While most people focus on getting their pages indexed, sometimes it's important to do the opposite: prevent content from appearing in Search. There are a few reasons you might want to hide content from Google:

- **To restrict data:** You might have data hosted on your site that you want to show only to users who are already on your site. You can block Google from crawling such data so it doesn't show up in search results. Also keep in mind that certain files published on your site may have metadata that can show up in Search. [Learn more about keeping redacted information out of Search.](#)
- **To hide content of less value to your audience:** Your website might have low quality content that shouldn't show up in Search. For example, if your website allows users to create content, some of that content might be [low quality or even spam](#). Allowing indexing of such content may have a negative effect on your site's ranking in Google's search results.
- **To have Google focus on your important content:** If you have a very large site (over hundreds of thousands of URLs) and pages with less important content, or if you have a lot of duplicate content, you might want to prevent Google from crawling the duplicate or less important pages in order to focus on your more important content.

How to block content

Here are the main ways to block content from appearing in Google:

Methods	
Remove the content from your site	Applicable: all content types Removing content from your site is the best way to ensure that it won't appear in Google Search and anywhere else on the Internet.
Password-protect your files	Applicable: all content types If you have confidential or private content on your site, you need to password protect it to ensure only authorized users can access it. This will also prevent that content from appearing in Google Search, or if it already appears, it will eventually remove that content from our search results.
noindex rule	Applicable: all content types The noindex robots meta tag is a rule that tells Google not to index your content or let it appear in Google search results. Your content can still be linked to and visited through other web pages, or directly visited by users with a link, but the content will not appear in Google search results.

Disallow crawling with robots.txt	Applicable: images and video Google only indexes images and videos that Googlebot is allowed to crawl. To prevent Googlebot from accessing your media files, use robots.txt rules to block the files.
Opt out of specific Google properties	Applicable: web pages You can tell Google not to include content from your site in specific Google properties, such as Google Shopping, Google Hotels, and vacation rentals.

Remove existing content from Google

If the content hosted on your site is already appearing in Google, you can request the removal of those search results. Learn how to [remove a page hosted on your site from Google](#).

<https://developers.google.com/search/docs/crawling-indexing/remove-information>

Remove a page hosted on your site from Google

If you don't own the page, see [Remove your personal information from Google](#) instead.

For quick removals, use the [Removals tool](#) to remove a page hosted on your site from Google's search results within a day.

Protect or remove all variations of the URL for the content that you want to remove. In many cases, different URLs can point to the same page. For example: `example.com/puppies`, `example.com/PUPPIES`, and `example.com/petchooser?pet=puppies`. [Learn how to find the right URL to block.](#)

Make your removal permanent

Requests made in the Removals tool last for about 6 months. To permanently block a page from Google Search results, take one of the following actions:

- **Remove or update the content on your page.** This is the most secure way to prevent your information from appearing in other search engines that might not respect the `noindex` tag. It also ensures that other people can't access your page.
- **Password-protect your page.** Limiting access to your page enables the right users to view your page, while preventing Googlebot and other web crawlers from accessing it.
- **Add a `noindex` tag to your page.** A `noindex` tag only blocks your page from showing up in Google search results. Users and other search engines that don't support `noindex` can still access your page.

Don't use `robots.txt` as a way to block your page. [Learn more about the limitations of robots.txt.](#)

Remove an image from search results

Learn how to [remove images that are hosted on your site from search results.](#)

Remove information from other Google properties

To remove content from other Google properties, search the help documentation for your product to learn how to remove it. For example:

- **Google Shopping and some other properties:** [Opt out of search results for specific Google properties.](#)
- **Business information:** [Edit business information that you added to your Business Profile.](#)
- **Google knowledge panel:** [Update your Google knowledge panel.](#)

How do I remove content from a site that I don't own?

See the help article on how to [Remove your personal information from Google.](#)

<https://developers.google.com/search/docs/crawling-indexing/prevent-images-on-your-page>

Remove images hosted on your site from search results

Trying to remove images of yourself? See [Remove your personal information from Google](#) instead.

For emergency image removal

To quickly remove images hosted on your site from Google's search results, use the [Removals tool](#). Keep in mind that unless you also remove the images from your site or otherwise block the images as described in the [non-emergency image removal section](#), the images may resurface in Google's search results once the removal request expires.

For non-emergency image removal

There are two ways to remove images from your site from Google's search results:

- [robots.txt disallow rules](#)
- [The noindex X-Robots-Tag HTTP header](#)

The two methods have the same effect, choose the method that is more convenient for your site. Keep in mind that Googlebot has to crawl the URLs to extract the HTTP headers, so implementing both methods at the same time doesn't make sense.

If you don't have access to the site that's hosting your images (for example a CDN) or your CMS doesn't provide a way to block images with the noindex X-Robots-Tag HTTP header or robots.txt, you might need to delete the images altogether from your site.

Remove images using robots.txt rules

To prevent images from your site appearing in Google's search results, add a [robots.txt](#) file to the root of the site that hosts the image, for example <https://yoursite.example.com/robots.txt>. While it takes longer to remove an image from Google's search results using robots.txt rules than it does to use the Removals tool, it gives you more flexibility and control through the use of wildcards or subpath blocking. It also applies to all search engines, whereas the Removals tool only applies to Google.

For example, if you want Google to exclude the `dogs.jpg` image that appears on your site at `yoursite.example.com/images/dogs.jpg`, add the following to your robots.txt file:

```
User-agent: Googlebot-Image
Disallow: /images/dogs.jpg
```

The next time Google crawls the `dogs.jpg` image, we'll see this rule and drop your image from our search results.

Rules may include [special characters](#) for more flexibility and control. Specifically, the `*` character matches any sequence of characters which lets you to match multiple image paths with one rule.

To remove multiple images on your site from Google's index, add a disallow rule for each image, or if the images share a common pattern such as a suffix in the filename, use a the * character in the filename. For example:

```
User-agent: Googlebot-Image
# Repeated 'disallow' rules for each image:
Disallow: /images/dogs.jpg
Disallow: /images/cats.jpg
Disallow: /images/llamas.jpg

# Wildcard character in the filename for
# images that share a common suffix. For example,
# animal-picture-UNICORN.jpg and
# animal-picture-SQUIRREL.jpg
# in the "images" directory
# will be matched by this pattern.
Disallow: /images/animal-picture-*.jpg
```

To remove all the images on your site from our index, place the following rule in your robots.txt file:

```
User-agent: Googlebot-Image
Disallow: /
```

To remove all files of a specific file type (for example, to include .jpg but not .gif images), you'd use the following robots.txt entry:

```
User-agent: Googlebot-Image
Disallow: /*.gif$
```

By specifying Googlebot-Image as the User-agent, the images will be excluded from Google Images. If you would like to exclude the images from all Google searches (including Google Search and Google Images), specify the Googlebot user agent.

Remove images with the noindex X-Robots-Tag HTTP header

Alternatively, you can remove images hosted on your site from Google's search results by adding the noindex X-Robots-Tag to the HTTP response headers of the images you want to remove. In this case you must allow crawling the image URLs in order for Googlebot to be able to extract the noindex rule. To implement the noindex X-Robots-Tag HTTP response header, [follow our documentation about noindex](#).

Note that adding the noimageindex robots tag to a particular page will also prevent that images embedded in that page from getting indexed. However, if the same images also appear in other pages, they might get indexed through those pages. To make sure a particular image is blocked no matter where it appears, use the noindex X-Robots-Tag HTTP response header.

How do I remove images from properties that I don't own?

See the [Google Search help documentation on how to remove an image from search results](#).

<https://developers.google.com/search/docs/crawling-indexing/keep-redacted-information-out>

Keep redacted information out of Google Search

When publishing documents and images on the web, you may unintentionally publish information beyond what is immediately visible to the human eye. In particular, information that you might not see, or that was intended to be redacted, might be included in some document formats and visible to search engines.

Because search engines index public material on the web, including images, content that is not completely redacted can potentially be findable in search engines. Assistive technologies like screen readers can make this seemingly "hidden" content more easily accessible, and common image understanding techniques like optical character recognition (OCR) similarly make it possible to search for this content.

Even though putting text in a tiny font, using a font color that's the same as the background the text is on, or covering text with an image may make something invisible to the human eye, these methods don't actually redact material in a way that prevents search engines from indexing it and making it findable.

Similarly, some document types include information in various ways that aren't immediately visible. They might include the document's change history, allowing users to see text that has been redacted or altered. They might retain the full versions of images that contain cropped or redacted information. There might also be metadata that's included in a file, which is not immediately visible, that may list the names of people who accessed or edited the file.

All of this information can remain even when a document is exported or converted from one format to another. If you need to remove information from a file, it's critical that the information is removed completely from the file before that file is made public.

Here are some best practices for how to appropriately redact information from documents that you don't want to be indexed and made discoverable via Google Search.

Edit and export images before embedding them

Google Search lists images that it finds across the web, both those that are on web pages or those that are embedded into various document formats. Embedded images are sometimes edited using only the containing document's editing tools. This can cause this redaction to fail when an image is indexed apart from the document. That is why it's best to edit images before embedding them into a document, not after. In particular:

- Crop out unwanted information from images before embedding them into documents. Some document editing tools (such as word processors or slide creation tools) will maintain any uncropped images that you use in the public version of the document, so be sure to review the tool's documentation thoroughly.
- Completely remove or obscure any text or other non-public parts of the image, as OCR systems may turn any image text seen into searchable text.
- Remove any undesired metadata.

After following the suggestions in this document, export or save the updated images as non-vector or flattened image file formats such as PNG or WEBP. This prevents those parts of the images from being inadvertently included in a public document.

Edit or remove unwanted text before moving to a public file format

Before you generate the public document, remove any text that you don't want displayed in the final version of the file. Move to a public format that does not keep your previous change history. Here are more specific tips:

- Use proper document redacting tools if a file needs to have information redacted. For example, avoid placing black rectangles over text as a redaction method, as this can result in the text still being included in the public document.
- Double-check the document metadata in the public file.
- Follow the [document redaction best practices](#) for the format that you are using (PDF, image, etc).
- Consider information in the URL or file name itself. Even if a part of a website is [blocked by robots.txt](#), the URLs may be indexed in search (without their content). Use hashes in URL parameters instead of email addresses or names.
- Consider using authentication to limit access to the redacted content. Serve the resulting login page with a [noindex robots meta tag](#) to block indexing.
- When publishing, make sure that the website is [verified in Google Search Console](#). This allows quick removal action, if needed.

What to do if unredacted or improperly redacted documents are indexed in Search

1. Remove the live document from the website or location where you published it.
2. Use the [Removals tool](#) for the verified site to remove the documents in question from Search. Use a URL prefix if you need to remove many documents. For verified sites, a URL removal generally takes less than a day. This prevents the document in question from appearing for any searches for redacted content.
3. Host the properly redacted document under a different URL. This makes sure that any newly indexed version is of the new document, and not an older version of the document (since recrawling of URLs and updating them in a search index can take a bit of time). Update any links to those documents.
4. Contact any other site that may also be hosting the improperly redacted documents and ask them to take them down as well. Ask them to use the Removals tool in their Search Console account, or you can use the [Outdated Content tool](#) to ask Google's systems to update the search results.
5. Allow the URL removal requests to expire (this happens after the URLs were either updated in the Google Search index, or after about 6 months).

<https://developers.google.com/search/docs/crawling-indexing/301-redirects>

Redirects and Google Search

Redirecting URLs is the practice of resolving an existing URL to a different one, effectively telling your visitors and Google Search that a page has a new location. Redirects are particularly useful in the following circumstances:

- You've moved your site to a new domain, and you want to make the transition as seamless as possible.
- People access your site through several different URLs. If, for example, your home page can be reached in multiple ways (for instance, <https://example.com/home>, <http://home.example.com>, or <https://www.example.com>), it's a good idea to pick one of those URLs as your preferred ([canonical](#)) destination, and use redirects to send traffic from the other URLs to your preferred URL.
- You're merging two websites and want to make sure that links to outdated URLs are redirected to the correct pages.
- You removed a page and you want to send users to a new page.

If you're using a platform like Blogger or Shopify, the platform may already have built-in redirect solutions. Try searching for help articles (for example, search for "blogger redirects").

Overview of redirect types

While your users generally won't be able to tell the difference between the different types of redirects, Google Search uses certain types of redirects as a signal that the redirect target should be canonical. Choosing a redirect depends on how long you expect the redirect will be in place and what page you want Google Search to show in search results:

- **Permanent redirects:** Show the new redirect target in search results.
- **Temporary redirects:** Show the source page in search results.

The following table explains the various ways you can use to set up permanent and temporary redirects, ordered by how likely Google is able to interpret correctly (for example, a server side redirect has the highest chance of being interpreted correctly by Google). Choose the redirect type that works for your situation and site:

Redirect types	
Permanent	Googlebot follows the redirect, and the indexing pipeline uses the redirect as a signal that the redirect target should be canonical. Use permanent redirects when you're sure that the redirect won't be reverted. HTTP 301 (moved permanently) Set up server-side redirects.

	HTTP 308 (moved permanently)	
	meta refresh (0 seconds)	Set up meta refresh redirects .
	HTTP refresh (0 seconds)	
	JavaScript location	Set up JavaScript redirects . Only use JavaScript redirects if you can't do server-side or meta refresh redirects.
	Crypto redirect	Learn more about crypto redirects . Don't rely on crypto redirects for letting search engines know that your content has moved unless you have no other choice.
Temporary	Googlebot follows the redirect, but the indexing pipeline doesn't use the redirect as a signal that the redirect target should be canonical. The target page might still be indexed if other canonicalization signals are present.	
	HTTP 302 (found)	Set up server-side redirects .
	HTTP 303 (see other)	
	HTTP 307 (temporary redirect)	
	meta refresh (more than 0 seconds)	Set up meta refresh redirects .
	HTTP refresh (more than 0 seconds)	

Server-side redirects

Setting up server-side redirects requires access to the server configuration files (for example, the `.htaccess` file on Apache) or setting the redirect headers with server-side scripts (for example, PHP). You can create both permanent and temporary redirects on the server side.

Permanent server-side redirects

[\[Original embedded video\]](#)

If you need to change the URL of a page as it is shown in search engine results, we recommend that you use a permanent server-side redirect whenever possible. This is the best way to ensure that Google Search and people are directed to the correct page. The 301 and 308 status codes mean that a page has permanently moved to a new location.

Temporary server-side redirects

If you just want to send users to a different page temporarily, use a temporary redirect. This will also ensure that Google isn't influenced by the redirect which may help keep the old URL in its Search results. For example, if a service your site offers is temporarily unavailable, you can set up a temporary redirect to send users to a page that explains what's happening, without compromising the original URL in search results.

Implement server-side redirects

The implementation of server-side redirects depends on your hosting and server environment, or the scripting language of your site's backend.

To set up a permanent redirect with PHP, use the `header()` function. You must set the headers before sending anything to the screen:

```
header('HTTP/1.1 301 Moved Permanently');
header('Location: https://www.example.com/newurl');
exit();
```

Similarly, here's an example of how to set up a temporary redirect with PHP:

```
header('HTTP/1.1 302 Found');
header('Location: https://www.example.com/newurl');
exit();
```

If you have access to your web server configuration files, you may be able to write the redirect rules yourself. Follow your web server's guides:

- **Apache:** Consult the [Apache .htaccess Tutorial](#), the [Apache URL Rewriting Guide](#), and the [Apache mod_alias documentation](#). For example, you can use `mod_alias` to set up the simplest form of redirects:

```
# Permanent redirect:
Redirect permanent "/old" "https://example.com/new"

# Temporary redirect:
Redirect temp "/two-old" "https://example.com/two-new"
```

For more complex redirects, use `mod_rewrite`. For example:

```

RewriteEngine on
# redirect the service page to a new page with a permanent redirect
RewriteRule    "^/service$"    "/about/service"    [R=301]

# redirect the service page to a new page with a temporary redirect
RewriteRule    "^/service$"    "/about/service"    [R]

```

- **NGINX:** Read about [Creating NGINX Rewrite Rules](#) on the NGINX blog. As with Apache, you have multiple choices to create redirects. For example:

```

location = /service {
# for a permanent redirect
return 301 $scheme://example.com/about/service

# for a temporary redirect
return 302 $scheme://example.com/about/service
}

```

For more complex redirects, use the rewrite rule:

```

location = /service {
# for a permanent redirect
rewrite service?name=$1 ^service/offline/([a-z]+)/?$ permanent;

# for a temporary redirect
rewrite service?name=$1 ^service/offline/([a-z]+)/?$ redirect;
}

```

- For all other web servers, check with your server manager or hoster, or search for guides on your favorite search engine (for example, search for "LiteSpeed redirects").

meta refresh and its HTTP equivalent

If [server-side redirects](#) aren't possible to implement on your platform, meta refresh redirects may be a viable alternative. Google differentiates between two kinds of meta refresh redirects:

- **Instant meta refresh redirect:** Triggers as soon as the page is loaded in a browser. Google Search interprets instant meta refresh redirects as permanent redirects.
- **Delayed meta refresh redirect:** Triggers only after an arbitrary number of seconds set by the site owner. Google Search interprets delayed meta refresh redirects as temporary redirects.

Place the meta refresh redirect either in the <head> element in the HTML or in the HTTP header with server-side code. For example, here's an instant meta refresh redirect in the <head> element in the HTML:

```

<!doctype html>
<html>
<head>
<meta http-equiv="refresh" content="0; url=https://example.com/newlocation">
<title>Example title</title>
<!--...-->

```

Here's an example of the HTTP header equivalent, which you can inject with server-side scripts:

```
HTTP/1.1 200 OK
Refresh: 0; url=https://www.example.com/newlocation
...
```

To create a delayed redirect, which is interpreted as a temporary redirect by Google, set the content attribute to the number of seconds that the redirect should be delayed:

```
<!doctype html>
<html>
<head>
<meta http-equiv="refresh" content="5; url=https://example.com/newlocation">
<title>Example title</title>
<!--...-->
```

JavaScript location redirects

Google Search interprets and executes JavaScript using the Web Rendering Service once crawling of the URL has completed.

Only use JavaScript redirects if you can't do server-side or meta refresh redirects. While Google attempts to render every URL Googlebot crawled, rendering may fail for various reasons. This means that if you set a JavaScript redirect, Google might never see it if rendering of the content failed.

To set up a JavaScript redirect, set the location property to the redirect target URL in a script block in the HTML head. For example:

```
<!doctype html>
<html>
<head>
<script>
  window.location.href = "https://www.example.com/newlocation";
</script>
<title>Example title</title>
<!--...-->
```

Crypto redirects

If you can't implement any of the other redirect methods, you should still make an effort to let your users know that the page or its content has moved. The simplest way to do this is to add a link pointing to the new page accompanied by a short explanation. For example:

```
<a href="https://newsite.example.com/newpage.html">We moved! Find the content on our new site!</a>
```

This helps users find your new site and Google may understand this as a crypto redirect, (like the Loch Ness monster, its existence may be disputed; not all search engines may recognize this pseudo-redirect as an official

redirect).

Don't rely on crypto redirects for letting search engines know that your content has moved unless you have no other choice. Contact your hosting provider for help with other types of redirects before resorting to crypto redirects.

Alternate versions of a URL

When you redirect a URL, Google keeps track of both the redirect source (the old URL) and the redirect target (the new URL). One of the URLs will be the [canonical](#); which one, depends on signals such as whether the redirect was temporary or permanent. The other URL becomes an *alternate name* of the canonical URL. Alternate names are different versions of a canonical URL that users might recognize and trust more. Alternate names may appear in search results when a user's query hints that they might trust the old URL more.

For example, if you [moved to a new domain name](#), it's very likely that Google will continue to occasionally show the old URLs in the results, even though the new URLs are already indexed. This is normal and as users get used to the new domain name, the alternate names will fade away without you doing anything.

<https://developers.google.com/search/docs/crawling-indexing/site-move-no-url-changes>

Changing your hosting

Follow this guide to minimize the impact of changing your site's hosting infrastructure on the site's performance in Google Search. A change in hosting infrastructure means switching hosting providers or moving to a content distribution network (CDN). This guide is only for migrations that don't affect the user-visible URL.

Changing the URLs? If you're making visible URL changes, start with [Site moves with URL changes](#).

Overview

1. **Prepare the new hosting infrastructure.** Upload your content to the new servers or configure your CDN and your origin servers, and test it.
2. **Start the site move.** Change the DNS settings of your domain name to point to the new hosting infrastructure. This step is the actual site move step that starts the process of sending your traffic to the new infrastructure.
3. **Monitor traffic.** Keep tabs on the traffic served by the old and new hosting.
4. **Shut down.** Shut down the old hosting infrastructure when you're confident that all users, including Googlebot, are receiving content correctly from the new infrastructure and no one is using the old infrastructure.

Prepare the new hosting infrastructure

This section covers steps to take before you start the actual infrastructure move.

Copy and test your new site

First, upload a copy of your site to your new hosting provider. What a "copy of your website" means depends entirely on your old content management platform; it may be actual HTML files that you replicate on your new hosting platform, or a database export that you have to import in the new location. Once you do that, verify that it works as expected by thoroughly testing all aspects of how your users interact with your site. Here are a few suggestions:

- **Create a testing environment**, perhaps with IP-restricted access, through which you test all of the features before the website goes live.
- **Open your new site in a web browser** and review all elements of your site: pages, images, forms, and downloads (such as PDF files).
- **Allow for public testing** with a temporary hostname for your new infrastructure (like `beta.example.com`) so you can test accessibility by browsers. A temporary hostname can help you test whether Googlebot can reach your site or not. To prevent accidentally letting the test site get indexed, add the [noindex robots rule](#) to the HTML or the HTTP headers of your pages.

Check that Googlebot is able to access the new hosting infrastructure

If you don't already have a Search Console account, [create a new account](#) for your site to help you monitor Google access and traffic. If you created a temporary hostname for your new site, verify that property as well. Check that Googlebot can access your new infrastructure using the [URL Inspection Tool](#) in Search Console.

Caution: Check your firewall configuration or denial of service (DoS) protection. Make sure it does not block Googlebot's ability to reach the DNS or the hosting provider's servers. [Learn how to verify Googlebot.](#)

Lower the TTL value for your DNS records

You can help make your site move go faster if you lower your site DNS records' TTL value, which will allow the new settings to propagate to ISPs faster. DNS settings are usually cached by ISPs based on the specified [Time to Live \(TTL\) setting](#). Consider lowering the TTL to a conservative low value (for example, a few hours) at least a week in advance of the move to refresh DNS caches faster.

Review Search Console verification

Make sure your Search Console verification will continue to work after the hosting move.

If you're using the [HTML file method](#) to verify ownership of your site in Search Console, make sure you don't forget to include your current verification file in your new copy of the site.

Likewise, if you include in your content management system's (CMS) templates a [meta tag](#) or [Google Analytics](#) to verify ownership, ensure the new CMS copy includes these as well.

Start the move

The move process is as follows.

1. **Remove any temporary blocks to crawling.** While building the new copy of a site, some site owners use a robots.txt file to disallow all crawling by Googlebot and other crawlers, or use noindex meta tags or HTTP headers to block indexing of content. Be sure to remove any such blocks from the new copy of the site when you're ready to start the move.
2. **Update the DNS settings.** You start the move by updating the DNS records to point to the new hosting provider. Check with your DNS provider for how to do that.

Monitor traffic

To monitor that your infrastructure change is going smoothly:

- **Keep an eye on the server logs on both new and old servers.**
As DNS setting propagates and the site traffic moves, you'll notice a drop in traffic logged on the old servers and a corresponding increase in traffic on the new servers.
- **Use different public DNS checking tools.**
Check that different ISPs around the world are updating to your new DNS settings correctly.
- **Monitor crawling.**
Monitor the [Index coverage](#) graphs in Search Console.

A note about Googlebot's crawl rate

When you change hosting infrastructure, it's normal to see a temporary drop in Googlebot's crawl rate immediately after the launch, followed by a steady increase over the next few days, potentially to rates that may be higher than before the move.

This fluctuation occurs because we determine crawl rate for a site based on many signals, and these signals change when your hosting changes. As long as Googlebot doesn't encounter any serious problems or slowdowns when accessing your new serving infrastructure, it will try to crawl your site as fast as necessary and possible.

Shut down old hosting

Check the server logs on the old provider and, once the traffic to the old provider reaches zero, you can shut down your old hosting infrastructure. This completes the hosting change.

<https://developers.google.com/search/docs/crawling-indexing/site-move-with-url-changes>

How to move a site

This document describes how to change the URLs of existing pages on your site while minimizing negative impact on your Google Search results. Examples of this kind of site move include:

- URL changes from HTTP to HTTPS.
- Domain name changes such as `example.com` to `example.net` or merging multiple domains or hostnames.
- URL paths changes: `example.com/page.php?id=1` to `example.com/widget`, or `example.com/page.html` to `example.com/page.htm`

Not changing the URLs? If you are making infrastructure changes such as changing your hosting, [start here](#) instead.

Overview

1. **General best practices for site moves.** Know what to expect, and how it might affect your users and rankings. If moving from HTTP to HTTPS, review the [best practices for HTTPS](#).
2. **Prepare the new site** and test it thoroughly.
3. **Prepare a URL mapping** from the current URLs to their corresponding new format.
4. **Start the site move** by configuring the server to redirect from the old URLs to the new ones.
5. **Monitor the traffic** on both the old and new URLs.

General best practices for site moves

- **Split your move into smaller steps, if that makes sense for your site.**
If your site is large and it's technically possible, we recommend initially moving just a piece of the site to test any effects on traffic and search indexing. After that you can move the rest of your site all at once or in chunks. When choosing the initial test section of the site, pick a section that changes less frequently and isn't significantly affected by frequent or unpredictable events. Also keep in mind that while moving just one section is a great way to test your move, it's not necessarily representative of a whole site move when it comes to Search. The more pages that you move, the more likely you'll encounter additional problems to solve. Careful planning can minimize problems.
- **Change only one thing at a time**
Plan your changes to your site one after the other, not everything at the same time. For example, if you want to move your site to a new domain name, change your content management system (CMS), and update your site to use a new layout, do them one at a time: move to a new domain, then change your site's layout.
- **Time your move to coincide with lower traffic, if possible.**
If your traffic is seasonal or dips on certain weekdays, it makes sense to move your site during the recurring traffic dips. This means that fewer people will be affected by potential issues that can happen during the site move, and more of your server's resources can be dedicated to Googlebot crawling your site.
- **Expect temporary fluctuation in site ranking during the move.**
With any significant change to a site, you may experience ranking fluctuations while Google recrawls and reindexes your site. As a general rule: for medium-sized websites, it can take a few weeks or more for

Google to gradually start showing the new URLs instead of the old ones (and for larger sites, even longer). The speed at which Googlebot and our systems discover and process moved URLs largely depends on the number of URLs and your server speed. [Submitting a sitemap](#) can help make the discovery process quicker, and it's fine to move your site in sections.

- **Don't worry about link credit.**

301 and [other permanent redirects](#) don't cause a loss in [PageRank](#).

- **Make use of Search Console.**

Search Console is your friend, especially during a site move. Verify data for each property separately in Search Console. Use the [Index Status report](#) for a broad look. Use the [Sitemaps report](#) to view how many URLs submitted in a sitemap have been indexed.

- **Be patient, give it time**

To consider a site move complete, Googlebot will have to visit every URL on your old and new site at least once. There are no fixed crawl frequencies; how fast Googlebot crawls depends on the size of your site, and the speed of crawling that's possible. The move takes place on a per-URL basis.

Prepare the new site

The details of site preparation vary for each site move, but typically you'll do one or more of the following:

- **Set up the CMS** (preferably the same as the old site used) and import the content from the old site.
- **Transfer images and downloads** (such as PDF documents) that you host.
These might already be getting traffic from Google Search or links, and it's useful to tell users and Googlebot about their new location.
- **For a move to HTTPS**, get and configure the required TLS certificates on your server.
- **Set up a robots.txt for your new site** and make sure the rules in the new site's robots.txt file correctly reflect the parts you want blocked from crawling.

Note that some site owners block all crawling while in development. If you follow this strategy, make sure you prepare what the robots.txt file should look like once the site move starts. Likewise, if you use `noindex` rules during development, prepare a list of URLs from which you'll remove the `noindex` rules when you start the site move.

- **Provide errors for deleted or merged content** if you're not moving to the new site all your old content, make sure those URLs correctly return an HTTP 404 or 410 error response code on the new site.
- **Ensure correct Search Console settings** that may help with the site move.

If you haven't already, [verify](#) both the old and new sites in Search Console. Be sure to verify all variants of both the old and new sites. For example, verify `www.example.com` and `example.com`, and include both the HTTPS and HTTP site variants if you use HTTPS URLs. Do this for both old and new sites.

- **Review the Search Console verification**

Make sure your Search Console verification will continue to work after the site move. If you're using a different method of verification, keep in mind that verification tokens may be different when the URL changes.

If you're using the [HTML file method](#) to verify ownership of your site in Search Console, make sure you don't forget to include your current verification file in your new copy of the site.

Likewise, if you verify ownership with an include file that references the [meta tag](#) or [Google Analytics](#) to verify ownership, ensure the new CMS copy includes these as well.

- **Review any configured settings in Search Console** that you may have made for the old site, and make sure the new site's settings are updated to reflect those changes as well. For example:
 - **Crawl rate:** Set the crawl rate to "Let Googlebot determine" for both the old and new site.
 - **Disavowed backlinks:** If you've uploaded a file to disavow links on your old site, we recommend that you re-upload it again using the Search Console account of the new site.
- **Clean up your recently purchased domain;** you'll want to make sure it's clean of any outstanding issues from the previous owner. Check the following settings:
 - **Manual action** for previous spam. If there are any manual actions applied to the new site, address any problems listed there and file a [reconsideration request](#).
 - **Removed URLs.** Make sure that there aren't any URL removals left over from the previous owner, especially a site-wide URL removal.
- **Use web analytics** to analyze usage on both the old and new sites. Web analytics software can help with this. Typically, web analytics configuration consists of a piece of JavaScript embedded in your pages. The details for tracking different sites varies depending on your analytics software and its logging, processing, or filtering settings. Check with your analytics software provider for help. Additionally, if you have been planning to make any configuration changes to your analytics software, now is a good time. If you use Google Analytics, consider creating a new profile for your new site if you want clean separation in your content reports.
- **Ensure that your server has enough computing resources:** after a migration, Google will temporarily crawl your new site more heavily than usual. This is because your site redirects traffic from the old to the new site, and any crawls of the old site will be redirected to the new site, in addition to any other crawling. Ensure that your new site has sufficient capacity to handle the increased traffic from Google. If your site is particularly large, get in touch with your hosting providers and give them a heads-up about the site move you're planning.

Prepare URL mapping

It's important to map your old site's URLs to the URLs for the new site. This section describes a number of general approaches you can take to correctly assess the URLs on your two sites and facilitate mapping. The exact details of how you generate this mapping will vary depending on your current website infrastructure and the details of the site move.

Determine your old URLs

In the simplest of site moves, you may not need to generate a list of your old URLs. For example, you could use a wildcard server-side redirect if you're changing your site's domain (for example, moving from `example.com` to `example.net`).

In more complex site moves, you will need to generate a list of old URLs and map them to their new destinations. How you get a listing of old URLs depends on your current website's configuration, but here are some handy tips:

- **Start with your important URLs.** To find them:
 - Look in your [sitemaps](#) because it's likely your most important URLs have been submitted in Search Console that way

- Check your server logs or analytics software for the URLs that get the most traffic
- Check the [Links to your site feature](#) in Search Console for pages that have internal and external links
- **Use your content management system**, which can typically provide a straightforward way to get a listing of all URLs that host content.
- **Check your server logs** for URLs that were visited at least once recently. Pick a time period that makes sense for your site, keeping in mind seasonal variation of traffic.
- **Include images and videos**—Make sure that you include URLs of embedded content in your site move plans: videos, images, JavaScript, and CSS files. These URLs need to be moved in the same way as all other content on your website.

Create a mapping of old to new URLs

Once you have the listing of old URLs, decide where each one should redirect to. How you store this mapping depends on your servers and the site move. You might use a database, or configure some URL rewriting rules on your system for common redirect patterns.

Update all URL details on the new site

Once you have your URL mapping defined, you'll want to do three things to get the pages ready for receiving traffic.

1. **Update annotations** to point to the new URLs in the HTML or sitemaps entry for each page:
 1. Each new URL should have a self-referencing `rel="canonical"` `<link>` tag.
 2. If the site you moved has multilingual or multinational pages annotated using `rel-alternate-hreflang` annotations, be sure to update the annotations to use the new URLs.
2. **Update internal links.**

Change the internal links on the new site from the old URLs to the new URLs. You can use the mapping generated earlier to help find and update the links as needed.
3. **Save the following lists for your final move:**
 - A sitemap file containing the new URLs in the mapping. See our documentation about [building a sitemap](#).
 - A list of sites linking to your old URLs. You can find the [links to your site](#) in Search Console.

Plan your redirect strategy

Once you have a mapping and your new site is ready, the next step is to plan your redirect strategy. We recommend server side [permanent redirects](#) from the old URLs to the new URLs as you indicated in your mapping. Check with your server administrator (or hosting company) about what kind of server side redirects you can technically do. It might be redirect rules in your `.htaccess` files if your server is using [Apache HTTP server](#) or redirect functions in your CMS.

If none of the server side redirect setups are possible, you can fall back to client side redirects as a last resort.

Decide how you will move your site - all at once, or in sections:

- **Small or medium sites:** We recommend moving all URLs on your site simultaneously instead of moving one section at a time. This helps users interact with the site better in its new form, and helps our algorithms detect the site move and update our index faster.
- **Large sites:** You can choose to move larger sites one section at a time. This can make it easier to monitor, detect, and fix problems faster.

Keep in mind the following:

- **Use server side permanent redirects** if technically possible. Although Googlebot supports [several kinds of redirects](#), we recommend that you use HTTP permanent redirects if possible, such as 301 and 308.
- **Avoid chaining redirects.** While Googlebot can [follow up to 10 hops](#) in a "chain" of multiple redirects (for example, Page 1 > Page 2 > Page 3), we advise redirecting to the final destination directly. If this is not possible, keep the number of redirects in the chain low, ideally no more than 3 and fewer than 5. Chaining redirects adds latency for users, and not all user agents and browsers support long redirect chains.

Start the site move

Once the URL mapping is accurate and you finalized the plan for how you're going to redirect, you're ready to move.

1. **Implement or turn on the redirects:** depending on your redirect strategy, this might mean you push an update to your server configuration files or you update your CMS, likely with custom code.

Avoid irrelevant redirects

Don't redirect many old URLs to one irrelevant single URL destination, such as the home page of the new site. This can confuse users and might be treated as a [soft 404 error](#). However, if you have consolidated content previously hosted on multiple pages to a new single page, you can redirect the older URLs to that new, consolidated page.

2. **Check the rel="canonical" link annotations and robots meta rules:** Once the redirects are active, ensure that the rel="canonical" link annotations on the new site are using the new URLs. Similarly, if you added noindex robots meta rules to the new site to avoid prematurely indexing the new URLs, make sure to update them.
3. **Test the redirects.** You can use the [URL Inspection Tool](#) for testing individual URLs, or command line tools or scripts to test large numbers of URLs.
4. **If you're changing domain names or subdomains, submit a [Change of Address](#) in Search Console for the old site.**

You only need this tool when moving from one domain or subdomain to another (such as example.com to example.net, or a.example.com to b.example.com). You don't need it for HTTP to HTTPS moves, switching between www and non-www on the same domain, or moving paths within the same domain.

For domain migrations: If you're moving to a new domain name, make sure to submit **Change of Address** requests for all verified variants of the old domain (including subdomains and www/non-www versions) to the new domain (for example, from en.example.com, www.example.com, and example.com to new-example.net), even if you're not actively using these variants. Ensure that you have all of these variants verified in Search Console.

5. **Keep the redirects for as long as possible**, generally at least 1 year. This timeframe allows Google to transfer all signals to the new URLs, including recrawling and reassigning links on other sites that point to your old URLs.

From users' perspective, consider keeping redirects indefinitely. However, redirects are slow for users, so try to update your own links and any high-volume links from other websites to point to the new URLs.

6. **Submit the new sitemap in Search Console.** This will help Google learn about the new URLs. At this point you can remove your old sitemap, since Google will use the new sitemap going forward.

The time it takes Googlebot and our systems to discover and process all URLs in the site move depends on how fast your servers are and how many URLs are involved. As a general rule, a small to medium-sized website can take a few weeks for most pages to move, and larger sites take longer. The speed at which Googlebot and our systems discover and process moved URLs depends on the number of URLs and the server speed.

Note that the visibility of your content in Search may fluctuate temporarily during the move. This is normal and a site's rankings will settle down over time.

Update links

Immediately after the site move is started, try to update as many links as possible to improve the user experience and reduce your server load. These include:

- Internal links: based on the URL mapping you previously created, replace all URLs that pointed to your own pages.
- External links: Try to contact the sites in the saved list of sites linking to your current content, asking them to update their links to your new site. Consider prioritizing your efforts by the number of inbound visits for each link.
- Profile links such as from Facebook, Twitter, and LinkedIn.
- Ad campaigns to point to the new landing pages.

Monitor traffic

Once you've started the site move, monitor how the user and crawler traffic changes on the new site and also the old site. Ideally the traffic on the old site will go down, while on the new site the traffic goes up. You can monitor user and crawler activity on the sites with [Search Console](#) and other tools.

Use Search Console to monitor traffic

Many features of Search Console help you monitor a site move, including:

- **Sitemaps:** Submit the two sitemaps you saved earlier from the mapping. Initially, the sitemap containing the new URLs would have zero pages indexed, while the sitemap of the old URLs would have many pages indexed. Over time the number of pages indexed from the old URLs sitemap would drop to zero with a corresponding increase of indexing of the new URLs. Note that Search Console may show warnings for the sitemap that contains the old URLs about the URLs redirecting; this is normal and you can ignore these warnings: you are, in fact, moving to new URLs after all.
- **Index Coverage report:** The graphs would reflect the site move, showing a drop in indexed URL counts on the old site and an increase of indexing on the new site. Check regularly for any unexpected crawl errors.
- **Search queries:** As more pages of the new site get indexed and start ranking, the search queries reports would start showing the URLs on the new site getting search impressions and clicks.

Use other tools to monitor traffic

Keep an eye on your server access and error logs. In particular, check for crawling by Googlebot, any URLs that unexpectedly return HTTP error status codes, and normal user traffic.

If you installed any web analytics software on your site, or if your CMS provides analytics, it's also recommended that you review traffic this way so that you can see the progress of traffic from your old to new site. In particular,

Google Analytics offers real-time reporting, and this is a handy feature to use during the initial site move phase. You should expect to see traffic drop on the old site and rise on the new site.

More resources

Site migrations can be overwhelming and complicated, so there are lots of takes on how to proceed. We found [Aleyda Solis' site migration checklists](#) particularly useful, and also the [Screaming Frog tool guide](#) for site migrations.

If you're stuck at any point, look for help on Google Search Central.

There is plenty of good advice on [our help page](#) and specific cases answered in our [user forums](#). If you can't find an answer, you can ask a question to one of our Google Search specialists during our [SEO office hours](#).

Troubleshooting your site move

Here are some common mistakes when migrating a site with URL changes (including HTTP to HTTPS). These mistakes can prevent your new site from being indexed completely.

Common mistakes

noindex or robots.txt blocks	Don't forget to remove any noindex or robots.txt blocks that were only needed for the migration. It's fine if you don't have a robots.txt file on your site, but be sure to return a proper 404 HTTP status code if the robots.txt file doesn't exist. To test: • Examine your robots.txt file in your HTTPS site and see if anything needs to be changed. • Use the URL inspection tool for any pages that seem to be missing from Google in the new site.
Incorrect redirects	Check your redirects from the old site to the new one. We frequently see people redirecting to the wrong (non-existent) URLs on the new site. You can use Search Console to see if there are an unusually high number of "Not found" errors reported, or you can use other tools such as Screaming Frog to crawl your own site and see if the redirects work as expected.
Other crawl errors	Examine the Index Coverage report for a spike in other errors on your new site during migration events.
Insufficient server capacity	After a migration, Google will crawl your new site more heavily than usual. This is because your site redirects traffic from the old to the new site, and any crawls of the old site will be redirected to the new site, in addition to any other crawling. Ensure that your site has sufficient capacity to handle the increased traffic from Google.
Not updating sitemaps	Be sure that your sitemaps are all updated with the new URLs.

<https://developers.google.com/search/docs/crawling-indexing/website-testing>

Minimize A/B testing impact in Google Search

This page covers how to ensure that testing variations in page content or page URLs has minimal impact on your Google Search performance. It does not give instructions on how to build or design tests, but you can find more resources about testing at the end of this page.

Overview of testing

Website testing is when you try out different versions of your website (or a part of your website) and collect data about how users react to each version.

- **A/B testing** is where you test two (or more) variations of a change. For example, you may test different fonts on a button to see if you can increase button clicks.
- **Multivariate testing** is where you test more than one type of change at a time, looking for the impact of each change as well as potential synergies between the changes. For example, you might try several fonts for a button, but also try changing (and not changing) the font of the rest of the page at the same time. Is a new font easier to read and so should be used everywhere? Or is the benefit that the button font looks different to the rest of the page, helping it draw attention?

You can use software to compare behavior with different variations of your pages (parts of a page, entire pages, or entire multi-page flows), and track which version is most effective with your users.

You can run tests by creating multiple versions of a page, each with its own URL. When users try to access the original URL, you redirect some of them to each of the variation URLs and then compare users' behavior to see which page is most effective.

You can also run tests without changing the URL by inserting variations dynamically on the page. You can use JavaScript to decide which variation to display.

Depending on what types of content you're testing, it may not even matter much if Google crawls or indexes some of your content variations while you're testing. Small changes, such as the size, color, or placement of a button or image, or the text of your "call to action" ("Add to cart" vs. "Buy now!"), can have a surprising impact on users' interactions with your page, but often have little or no impact on that page's search result snippet or ranking.

In addition, if we crawl your site often enough to detect and index your experiment, we'll probably index the eventual updates you make to your site fairly quickly after you've concluded the experiment.

Best practices when testing

Here is a list of best practices to avoid any bad effects on your Google Search behavior while testing site variations:

Don't cloak your test pages

Don't show one set of URLs to Googlebot, and a different set to humans. This is called [cloaking](#), and is against our [spam policies](#), whether you're running a test or not. Remember that infringing our spam policies can get your site demoted or removed from Google search results—probably not the desired outcome of your test.

Cloaking counts whether you do it by server logic or by robots.txt, or any other method. Instead, use links or redirects as described next.

If you're using cookies to control the test, keep in mind that Googlebot generally doesn't support cookies. This means it will only see the content version that's accessible to users with browsers that don't accept cookies.

Use `rel="canonical"` links

If you're running a test with multiple URLs, you can use the `rel="canonical"` link attribute on all of your alternate URLs to indicate that the original URL is the preferred version. We recommend using `rel="canonical"` rather than a `noindex` meta tag because it more closely matches your intent in this situation. For instance, if you are testing variations of your home page, you don't want search engines not to index your home page; you just want them to understand that all the test URLs are close duplicates or variations on the original URL and should be grouped together, with the original URL as the canonical. Using `noindex` rather than `rel="canonical"` in such a situation can sometimes have unexpected bad effects.

Use 302 redirects, not 301 redirects

If you're running a test that redirects users from the original URL to a variation URL, use a [302 \(temporary\) redirect](#), not a [301 \(permanent\) redirect](#). This tells search engines that this redirect is temporary—it will only be in place as long as you're running the experiment— and that they should keep the original URL in their index rather than replacing it with the target of the redirect (the test page). [JavaScript-based redirects](#) are also fine.

Run the experiment only as long as necessary

The amount of time required for a reliable test will vary depending on factors like your conversion rates, and how much traffic your website gets; a good testing tool tells you when you've gathered enough data to draw a reliable conclusion. Once you've concluded the test, update your site with the desired content variation(s) and remove all elements of the test as soon as possible, such as alternate URLs or testing scripts and markup. If we discover a site running an experiment for an unnecessarily long time, we may interpret this as an attempt to deceive search engines and take action accordingly. This is especially true if you're serving one content variant to a large percentage of your users.

More information about testing

- [Google Analytics article](#) on content experiments
- [Google Analytics content testing tools](#)
- Ask questions about testing on the [Analytics Help Forum](#)
- Ask questions about impact on search results in the [Google Search Central Help Forum](#).

<https://developers.google.com/search/docs/crawling-indexing/pause-online-business>

Temporarily pause or disable a website

If you're unable to fulfill orders or many of your products out of stock, you may be considering temporarily closing your online business. If the situation is temporary, meaning you expect to be able to sell products in the coming weeks or months, we recommend that you take action that preserves as much of your site's standing in Search as possible. This guide explains how you can safely pause your online business.

Limit your site's functionality (recommended)

If your situation is temporary and you plan to reopen your online business, we recommend that you keep your site online and limit the functionality. This is the recommended approach since it minimizes any negative effects on your site's presence in Search. People can still find your products, read reviews, or add wishlists so they can purchase at a later time. We recommend doing the following:

- **Disable the cart functionality:** Disabling the cart functionality is the simplest approach, and doesn't change anything for your site's visibility in Search.
- **Display a banner or popup:** A banner or popup div on all pages including the landing page quickly makes the status clear to users. Mention any known and unusual delays, shipping times, pick-up or delivery options, so that users continue with the right expectations. To prevent the content in the banner or popup from being shown in a snippet in Search results, use the [data-nosnippet HTML attribute](#). Make sure to follow our [guidelines on popups and banners](#).
- **Update your structured data:** If your site uses structured data (for example, [Product](#), [Book](#), [Event](#)), make sure to adjust it appropriately (reflecting the current product availability, or changing events to cancelled). If your business has a physical storefront, update [Local Business structured data](#) to reflect current opening hours.
- **Check your Merchant Center feed:** If you use Merchant Center, follow the [best practices for the availability attribute](#).
- **Tell Google about your updates:** To ask Google to recrawl a limited number of pages (for example, the home page), use [Search Console](#). For a larger number of pages (for example, all of your product pages), use [sitemaps](#).

Not recommended: Disable the whole website

Warning: Google's systems are designed to be robust and to help websites recover from temporary issues. However, removing a site completely from Google's index is a significant change that can take quite some time to recover from. There's no fixed time for a recovery from a complete removal, and there's no mechanism to speed that up. This is why we strongly recommend limiting functionality instead of removing the site from Search.

You may decide to disable the whole website. This is an extreme measure that should only be taken for a very short period of time (a few days at most), as it will otherwise have **significant effects** on the website in Search, even when implemented properly.

Make sure that you consider the following side effects of disabling your entire site:

- Your customers won't know what's happening with your business if they can't find your business online at all.
- Your customers can't find or read first-hand information about your business and its example, reviews, specs, repair guides, or manuals won't be findable. Third-party information may not be as correct or comprehensive as what you can provide. This often also affects future purchase decisions.
- Knowledge Panels may lose information, like contact phone numbers and your site's logo.
- Search Console verification will fail, and you will lose all access to information about your business in Search. Aggregate reports in Search Console will lose data as pages are dropped from the index.
- Ramping back up after a prolonged period of time will be significantly harder if your website needs to be reindexed first. Additionally, it's uncertain how long this would take, and whether the site would appear similarly in Search afterwards.

If you decide that you need to do this (again, **not recommended**), here are some options:

- If you need to urgently disable the site for 1-2 days, then return an informational error page with a [503 HTTP response status code](#) instead of all content. Make sure to follow the [best practices for disabling a site](#).
- If you need to disable the site for a longer time, then provide an indexable home page as a placeholder for users to find in Search by using the 200 HTTP status code.
- If you quickly need to hide your site in Search while you consider the options, you can [temporarily remove a website from Search](#).

Best practices for disabling a site

Warning: Keep in mind that it's not possible for Google's systems to refresh titles, descriptions, metadata, or structured data included on a website if a page returns a 503 HTTP response status code.

While we don't recommend disabling your site, here are some best practices if you decide to do this:

- Continue to allow crawling through the robots.txt file. **Don't return a 503 HTTP response status code for the robots.txt file because this blocks all crawling.**
- Confirm a 503 HTTP response status code locally by using [curl](#) or a similar tool. For example:

```
curl -I -X GET "https://www.example.com/"
HTTP/1.1 503 Service Unavailable
Mime-Version: 1.0
Content-Type: text/html
(...)
```

- To minimize the server-side and client-side load of a 503 error page, follow these best practices:
 - Use the [retry-after HTTP header](#) with a best effort date or duration.
 - Use static HTML.
 - Minimize off-page resources; use inline CSS stylesheets and base-64-encoded images.
- Give your users clear guidance on future steps within the content of the error page. This could include:
 - Links to more information
 - The date when you expect the website to be online again, or when the information will be updated
 - How to contact customer service
- Don't disallow all crawling in the robots.txt file. Returning a valid robots.txt file that disallows all crawling may remove the website's content, and potentially its URLs, from Google Search.

- Don't block the website by returning 403, 404, 410 HTTP status codes, or with a `noindex` robots meta tag or `x-robots-tag` HTTP header. This will remove the website's URLs from Google Search.
- Don't use the temporary website removal tool in Search Console for closures. Doing so will make it impossible for users to find your website so that they can learn its status. Also, potential resellers or affiliates of your business's products may continue to be shown in Search.
- Don't block your robots.txt file with a 503 HTTP response status code.

FAQs

What if I only close the site for a few weeks?

Completely closing a site even for just a few weeks can have negative consequences on Google's indexing of your site. We recommend [limiting the site functionality](#) instead. Keep in mind that users may also want to find information about your products, your services, and your company, even if you're currently not selling anything.

What if I want to exclude all non-essential products?

That's fine. Make sure that people can't buy the non-essential products by [limiting the site functionality](#).

Can I ask Google to crawl less while my site is temporarily closed?

Yes, you can [reduce the Googlebot crawl rate](#), though it's not recommended for most cases. This may have some impact on the freshness of your results in Search. For example, it may take longer for Search to reflect that all of your products are currently not available. On the other hand, if Googlebot's crawling causes critical server resource issues, this is a valid approach. We recommend setting a reminder for yourself to reset the crawl rate once you're ready to go back in business.

How do I get a page indexed or updated quickly?

To ask Google to recrawl a limited number of pages (for example, the home page), use [Search Console](#). For a larger number of pages (for example, all of your product pages), [use sitemaps](#).

What if I block a specific region from accessing my site?

Google generally crawls from the US. If you block the US, Google Search won't be able to access your site at all. We don't recommend that you block an entire region from temporarily accessing your site; instead, we recommend [limiting the site functionality](#) for that region.

Should I use the [Removals Tool](#) to remove out-of-stock products?

No. If you do this, customers won't be able to find first-hand information about your products on Search, and there might still be third-party information for the product that may be incorrect or incomplete. It's better to still allow that page, and mark it out of stock. That way people can still understand what's going on, even if they can't purchase the item. If you remove the product from Search, people don't know why it's gone.

<https://developers.google.com/search/docs/appearance>

Overview of Search appearance topics

The topics in this section describe how you can influence how your website appears in Google Search.

• AI features • Business details • Byline dates • Favicons • Featured snippets • Flexible Sampling general guidance • Google Discover • Images • Preferred sources	• Regional differences in Search experience • Search profiles • Site names • Sitelinks • Snippets • Title links • Top Places List • Translated results • Videos • Visual Elements gallery • Web stories
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Using structured data

Google uses structured data to understand the content on the page. You can help us by providing specific information about your site, which can help your site display in richer features in search results.

• List of structured data markup that Google Search supports • Understand how structured data works • Generate structured data with JavaScript	• General structured data guidelines • Enriched search results
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Features that use structured data

Here's a list of the features that use structured data:

• Article • Breadcrumb • Carousel • Course list • Dataset • Discussion forum • Education Q&A • Employer aggregate rating • Event • Image metadata	• Local business • Math solver • Movie carousel • Organization • Practice problem • Product: ◦ Product snippet ◦ Merchant listing ◦ Variants	• Q&A • Recipe • Review snippet • Software app • Speakable • Subscription and paywalled content • Vacation rental • Video
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- [Job posting](#)

- [Merchant return policy](#)
- [Merchant shipping policy](#)
- [Loyalty program](#)
- [Profile page](#)

Early Adopters Program

To ensure the best experience for users, Google pilots some features with a limited number of organizations. Here are the features that are in the Early Adopters Program:

- [Package tracking](#)
- [Structured data carousels \(beta\)](#)

<https://developers.google.com/search/docs/appearance/ai-features>

AI features and your website

This guide covers how AI features like AI Overviews and AI Mode work in Google Search from a site owner's perspective and how to approach your content's inclusion in these experiences.

The best practices for SEO remain relevant for AI features in Google Search (such as AI Overviews and AI Mode). **There are no additional requirements to appear in AI Overviews or AI Mode, nor other special optimizations necessary.** That said, it's always good to review the [fundamental SEO best practices](#).

How AI features work in Search

As with Search overall, the AI features [AI Overviews](#) and [AI Mode](#) surface relevant links to help people find the information they're looking for quickly and reliably, as well as to help them explore content they may not have discovered before. These features offer unique opportunities for more types of sites to appear.

AI Overviews help people get to the gist of a complicated topic or question more quickly, and provide a jumping off point to explore links to learn more. They were designed to show up on queries where they can add additional benefits beyond what people might already get on Search. With AI Overviews, people have been visiting a greater diversity of websites for help with more complex questions.

AI Mode is particularly helpful for queries where further exploration, reasoning, or complex comparisons are needed. People can ask nuanced questions that might have previously taken multiple searches — from exploring a new concept, to comparing options, and beyond — and get a comprehensive AI-powered response with links to supporting websites.

[\[Original embedded video\]](#)

Both AI Overviews and AI Mode may use a "query fan-out" technique — issuing multiple related searches across subtopics and data sources — to develop a response. While responses are being generated, our advanced models identify more supporting web pages, allowing us to **display a wider and more diverse set of helpful links** associated with the response than with a classic web search, enabling new opportunities for exploration.

AI Mode and AI Overviews may use different models and techniques, so the set of responses and links they show will vary. AI Overviews are only shown when our systems determine that it is additive to classic Search, and as such, often don't trigger.

How to appear in AI features

You can apply the same [foundational SEO best practices](#) for AI features as you do for Google Search overall: making sure the page meets the [technical requirements for Google Search](#), following [Search policies](#), and focusing on the [key best practices](#), such as [creating helpful, reliable, people-first content](#).

Technical requirements for appearing in AI features

To be eligible to be shown as a supporting link in AI Overviews or AI Mode, a page must be indexed and eligible to be shown in Google Search with a snippet, fulfilling the [Search technical requirements](#). There are no additional technical requirements.

Just because a page meets all requirements, best practices, and complies with the policies, doesn't mean that Google will crawl, index, or serve its content. Indexing and serving isn't guaranteed. Learn more about [How Search Works](#).

SEO best practices

While specific optimization isn't required for AI Overviews and AI Mode, all existing [SEO fundamentals](#) continue to be worthwhile, for example:

- Ensuring that crawling is allowed in robots.txt, and by any CDN or hosting infrastructure
- Making your content easily findable through [internal links](#) on your website
- Providing a great [page experience](#) for users
- Making sure that important content is available in textual form
- Supporting your textual content with high-quality [images](#) and [videos](#), when applicable
- Making sure your [structured data](#) matches the visible text on the page
- Checking that your [Merchant Center](#) and [Business Profile](#) information is up-to-date

You don't need to create new machine readable files, AI text files, or markup to appear in these features. There's also no special schema.org structured data that you need to add.

To discover and diagnose potential technical issues quickly, [verify your site in Search Console](#).

Measuring the performance of your site

Just like the rest of the search results page, sites appearing in AI features (such as AI Overviews and AI Mode) are included in the overall search traffic in [Search Console](#). In particular, they're reported on in the [Performance report](#), within the "Web" [search type](#). Learn more about how [AI Overviews](#) and [AI Mode](#) are counted towards the overall data in Search Console, how to [analyze traffic changes](#) overall, and how to [combine Search Console and Analytics data](#).

In addition to Search Console, you could also track conversions and time spent on your site in other tools, such as Google Analytics. We've seen that when people click from search results pages with AI Overviews, these clicks are higher quality (meaning, users are more likely to spend more time on the site).

Controlling your content in AI features in Search

AI is built into Search and integral to how Search functions, which is why robots.txt directives for [Googlebot](#) is the control for site owners to manage access to how their sites are crawled for Search. To limit the information shown from your pages in Search, use [nosnippet](#), [data-nosnippet](#), [max-snippet](#), or [noindex](#) controls.

To limit AI training and grounding in some of Google's other systems, read more [about Google-Extended](#).

Troubleshooting preview controls

If you implemented [preview controls](#) and you're still seeing your content appear in AI features on Search, try the following steps:

1. Make sure that the preview control is correct and visible to Googlebot. To test if your implementation is correct, use the [URL Inspection tool](#) to see the HTML that Googlebot received while crawling the page.
2. Allow time for Google to recrawl and process the change in preview controls. Remember that crawling can take anywhere from several days to several months, depending on how often our systems determine a page needs to be refreshed. If you've made changes, you can [request that Google recrawl your pages](#).

If you tried the troubleshooting steps and still find issues, post in the [Google Search Central Help Community](#).

<https://developers.google.com/search/docs/appearance/publication-dates>

Influence your byline dates in Google Search

A *byline date* is the date that Google estimates that the web page was updated or published. When Google can determine the byline date of your page or video, it can expose this information in Search results, if this information is considered to be useful to the user. You can provide information to help Google determine the byline date.

Google doesn't depend on a single date factor because all factors can be prone to issues. That's why our systems look at several factors to determine our best estimate of when a page was published or significantly updated.

How to provide date information to Google

To provide date information to Google, follow these steps:

1. Follow the [best practices for influencing byline dates](#).
2. Add a user-visible date to the page and feature it prominently. Label your dates appropriately with text like "Publish" or "Last updated". Here are some examples of how you can highlight date information about a web page:
 - **Posted Feb 4, 2019**
 - **Published February 4, 2019**
 - **Last updated: Feb 14, 2018**
 - **Updated Feb 14, 2019 8pm ET**

You can provide a publication date and/or a last updated date.

```

<html>
  <head>
    <title>Analyzing Google Search traffic drops</title>
  </head>
  <body>
    <p>
      Posted Tuesday, July 20, 2021
    </p>
    <p>
      Suppose you open Search Console and find out that your Google Search traffic dropped. What should you do?
    </p>
  </body>
</html>

```

3. Specify dates with [structured data](#). We recommend that you add a subtype of [CreativeWork](#) (such as [Article](#), [BlogPosting](#), or [VideoObject](#)), and specify the `datePublished` and/or `dateModified` fields. Be sure to follow [Google's structured data guidelines](#) to help our crawlers understand your article dates.

```

<html>
  <head>
    <title>Analyzing Google Search traffic drops</title>
    <script type="application/ld+json">
      {
        "@context": "https://schema.org",
        "@type": "NewsArticle",
        "headline": "Analyzing Google Search traffic drops",
        "datePublished": "2021-07-20T08:00:00+08:00",
        "dateModified": "2021-07-20T09:20:00+08:00"
      }
    </script>
  </head>
  <body>
    <p>
      Posted Tuesday, July 20, 2021
    </p>
    <p>
      Suppose you open Search Console and find out that your Google Search traffic dropped. What should you do?
    </p>
  </body>
</html>

```

Best practices for influencing byline dates

While Google doesn't guarantee that a byline date, whether visible or in structured data, will be shown in search results, following these guidelines does help our algorithms find and process the information.

- **The date is required; the time is not:** However, we recommend you provide a time and timezone in markup for added precision.
- **If you choose to specify the timezone,** provide the [correct timezone](#), taking into account [daylight saving time](#) as appropriate.

- **Make your dates and times consistent.** Ensure that the date (and optional time and timezone) match between the equivalent user-visible and structured values. Time and timezone are optional in user-visible data even if provided in structured data.
- **Don't specify future dates, or the date of the action described on the page.** The dates must describe the publication or update date of the page, not the stories or events described therein. You can add [Event markup](#) to the page to describe the activities listed on the page, if you like.
- **Minimize the presence of other dates on the page:** If you've followed the best practices and find incorrect dates are being selected, consider removing some or all other dates that appear on the page.
- **If your page is intended to show up in Google News search,** follow [these additional guidelines](#).

<https://developers.google.com/search/docs/appearance/favicon-in-search>

Define a favicon to show in search results

If your site has a [favicon](#), it can be included in Google Search results for your site.

This documentation is for organic search results. For logos on Google Ads results, visit the [business logo specifications](#).

Implementation

Here's how to make your site eligible for a favicon in Google Search results:

- 1. Create a favicon that follows the [guidelines](#).
- 2. Add a <link> tag to the header of your [home page](#) with the following syntax:

```
<link rel="icon" href="/path/to/favicon.ico">
```

To extract the favicon information, Google relies on the following attributes of the link element:

Attributes		
rel	Google supports the following rel attribute values for specifying a favicon; use whichever one fits your use case:	
	icon	The icon that represents your site, as defined in the HTML standard. For historical reasons, we also support shortcut icon, which is an earlier, alternative version of icon.

	apple-touch-icon	An iOS-friendly icon that represents your site, per Apple's developer documentation .
	apple-touch-icon-precomposed	An alternative icon for earlier versions of iOS, per Apple's developer documentation .
href	The URL of the favicon. The URL can be a relative path (/smile.ico) or absolute path (https://example.com/smile.ico). The URL doesn't need to be hosted on your site (for example, your favicon could be hosted on a content delivery network (CDN)).	

3. Allow time for Google to recrawl and process the new information on your home page. Remember that crawling can take anywhere from several days to several weeks, depending on how often our systems determine content needs to be refreshed. You can [request indexing](#) of your site's home page by using the URL Inspection tool.

Guidelines

You must follow these guidelines to be eligible for a favicon in Google Search results.

A favicon isn't guaranteed to appear in Google Search results, even if all guidelines are met.

- Google Search only supports one favicon per site, where a *site* is defined by the hostname. For example, <https://www.example.com/> and <https://code.example.com/> are two different hostnames, and therefore can have two different favicons. However, <https://www.example.com/sub-site> is a subdirectory of a site, and you can only set one favicon for <https://www.example.com/>, which applies to the site and its subdirectories.
Supported: <https://example.com> (this is a domain-level home page)
Supported: <https://news.example.com> (this is a subdomain-level home page)
Not supported: <https://example.com/news> (this is a subdirectory-level home page)
- Googlebot-Image must be able to crawl the favicon file and Googlebot must be able to crawl the home page; they cannot be [blocked](#) for crawling.
- To help people quickly identify your site when they scan through search results, make sure your favicon is visually representative of your website's brand.
- Your favicon must be a square (1:1 aspect ratio) that's at least 8x8px. While the minimum size requirement is 8x8px, we recommend using a favicon that's larger than 48x48px so that it looks good on various surfaces. Google Search supports the following favicon file formats: BMP, GIF, ICO, PNG, JPEG, PPM, and TIFF.
- The favicon URL must be stable (don't change the URL frequently).
- Google won't show any favicon that it deems inappropriate, including pornography or hate symbols (for example, swastikas). If this type of imagery is discovered within a favicon, Google replaces it with a default icon.

Submit feedback about favicons in search results

If you have feedback about Google's handling of favicons in search results, [fill out our favicon feedback form](#). Note that feedback submitted here is designed to help our teams to improve the systems in Google Search overall and doesn't guarantee it will be acted upon individually.

<https://developers.google.com/search/docs/appearance/featured-snippets>

Featured snippets and your website

Featured snippets are special boxes where the format of a regular search result is reversed, showing the descriptive [snippet](#) first. They can also appear within a [related questions group](#) (also known as "People Also Ask"). [Read more about how Google's Featured Snippets work.](#)

How can I opt out of featured snippets?

There are two ways that you can opt out of featured snippets:

- [Block both featured and regular search snippets](#)
- [Block featured snippets only](#)

Block all snippets

To block all snippets (including featured snippets and regular snippets) from appearing for a given page, add the [nosnippet](#) rule to that page.

- Text marked by the [data-nosnippet HTML attribute](#) won't appear in featured snippets or regular snippets either.
- If both [nosnippet](#) and [data-nosnippet](#) rules appear in a page, [nosnippet](#) takes priority, and snippets won't be shown for the page.

Block featured snippets only

If you want to retain snippets in regularly-formatted search results, but you don't want to appear in featured snippets, experiment with setting the [max-snippet](#) rule to lower lengths. Featured snippets will only appear if enough text can be shown to generate a useful featured snippet.

Keep lowering the value if pages continue to show for featured snippets. In general, the shorter your [max-snippet](#) rule setting, the less likely the page will appear as a featured snippet.

Google does not provide an exact minimum length required to appear as a featured snippet. This is because the minimum length is variable based on a number of factors, including—but not limited to—the information in the snippet, the language, and the platform (mobile device, app, or desktop).

Using a low [max-snippet](#) setting doesn't guarantee that Google will stop showing featured snippets for your page. If you need a guaranteed solution, use the [nosnippet](#) rule.

How can I mark my page as a featured snippet?

You can't. Google systems determine whether a page would make a good featured snippet for a user's search request, and if so, elevates it.

What happens when a user clicks a featured snippet?

Clicking a featured snippet takes the user directly to the section of the page that appeared in the featured snippet. Scrolling to the position that appeared in the snippet happens automatically, without any additional annotation by the site. If a browser doesn't support the underlying technology needed, or if our systems can't confidently determine exactly where within a page to direct a click, clicking a featured snippet will take a user to the top of the source web page.

<https://developers.google.com/search/docs/appearance/flexible-sampling>

Flexible Sampling general guidance

In order to better understand the potential impact of sampling changes on Google users and publishers' subscription models, we developed a series of experiments in cooperation with our publishing partners. From these experiments we learned that even minor changes to the current sampling levels could degrade user experience and, as user access is restricted, unintentionally impact article ranking in Google Search.

There are two types of sampling we advise: **metering**, which provides users with a quota of articles to consume before requiring users to subscribe or log in, after which paywalls will start appearing; and **lead-in**, which offers a portion of an article's content without it being shown in full.

We encourage publishers to experiment cautiously with different amounts of sampling. Here is some general guidance for implementing flexible sampling.

Metering

In general, we think that monthly, rather than daily metering provides more flexibility and a safer environment for testing. The user impact of changing from one integer value to the next is less significant at, say, 10 monthly samples than at 3 daily samples. Monthly metering also has the advantage of focusing paywall views on your most engaged users, who are those most likely to subscribe, while allowing your newer and less engaged users to become acquainted with the value of your content before experiencing a paywall. ("Paywall," in this context, applies equally to barriers that require either subscription or merely registration for content access.)

How much content?

There is no single value for optimal sampling across different businesses. However, for most daily news publishers, we expect the value to fall between 6 and 10 articles per user per month. We think most publishers will find a number in that range that preserves a good user experience for new potential subscribers while driving conversion opportunities among the most engaged users.

As a starting point for your explorations, we encourage you to provide 10 articles per month to Google search users and iterate from there. We leave the exact number to the discretion of individual publishers, who are best positioned to understand the particular demands of their businesses. We encourage publishers to analyze the current percentage of search users who land on their paywalls, and select a monthly number that achieves a similar result. You can always lower the value later, after you have some confidence that you are on a stable footing.

Lead-in

In addition to metering, some publishers show the first few sentences of an article "above the fold" of their paywall after the meter has run out. We think this is a good practice. By exposing the article lead, publishers can let users experience the value of the content and so provide more value to the user than a page with completely blocked content. Lead-in also generates user curiosity about how article continues, which may assist in conversion.

Making Changes

Publishers will want to experiment with different sampling values to determine their effect on referral traffic and conversion.

Bear in mind that our user studies have shown that when users who have experienced only a small amount of content are required to subscribe, their interest in the product diminishes greatly. Our analysis shows that general user satisfaction starts to degrade significantly when paywalls are shown more than 10% of the time (which generally means that about 3% of the audience has been exposed to the paywall). We recommend caution in approaching that limit, because you may start to alienate users who have not yet become convinced of the value of your content.

Publishers with more advanced technical resources may want to focus their efforts more narrowly on those specific users in the engaged segment. By identifying users who consistently use up the monthly allotment, publishers could then target them by reducing the sample allowance for that audience specifically, and, by allowing more liberal consumption for other users, reduce the risk that overall user behavior and satisfaction is degraded.

How to indicate paywalled content

Enclose paywalled content with structured data in order to help Google differentiate paywalled content from the practice of [cloaking](#), where the content served to Googlebot is different from the content served to users. If you don't want the content to be accessible to the browser at the time of serving, choose a paywall implementation that doesn't supply the paywalled content to the browser.

Learn more about how to [indicate paywalled content with structured data](#) and refer to our [guidance on using JavaScript to implement paywalled content](#).

<https://developers.google.com/search/docs/appearance/google-discover>

Discover and your website

Discover is a part of Google Search that shows people content related to their interests, based on their [Web and App Activity](#). This page explains more about how content may appear in Discover and best practices for site owners to consider.

[Original illustration: How Discover may look on a phone]

How content appears in Discover

Content is automatically eligible to appear in Discover if it is [indexed by Google](#) and meets Discover's [content policies](#). No special tags or structured data are required. Note that being eligible to appear in Discover is not a guarantee of appearing.

Content that may appear in Discover includes a wide range of topics that aligns with a person's interests. Older content may appear if it's helpful and relevant to individuals based on their interests.

If your site violates one or more Discover content policies, Discover manual actions may appear in your Search Console under Security and Manual actions. Learn more about the [types of violations and how to fix them](#).

As part of Google Search, Discover makes use of many of the same signals and [systems](#) used by Search to determine what is helpful, people-first content. Given this, those looking for success with Discover should review our advice on [creating helpful, reliable, people-first content](#).

To increase the likelihood of your content appearing in Discover, we recommend the following:

- Avoid clickbait and similar tactics to artificially inflate engagement by using misleading or exaggerated details in preview content (title, snippets, or images) to increase appeal, or by withholding crucial information required to understand what the content is about.
- Use page titles and headlines that capture the essence of the content.
- Avoid sensationalism tactics that manipulate appeal by catering to morbid curiosity, titillation, or outrage.
- Provide content that's timely for current interests, tells a story well, or provides unique insights.
- Include compelling, high-quality images in your content that are relevant, especially large images that are more likely to generate visits from Discover. We recommend using images that meet the following specifications:
 - At least 1200 px wide
 - High resolution of more than 300,000 total pixels (for example, a 16:9 image that's 1280x720 pixels would have 921,600 total pixels and satisfies this specification)
 - 16x9 aspect ratio

Google tries to automatically crop the image for use in Discover. If you choose to crop your images yourself, be sure your images are well-cropped and positioned for landscape usage, and

avoid automatically applying an aspect ratio. For example, if you crop a vertical image into 16x9 aspect ratio, be sure the important details are included in the cropped version that you specify in the `og:image` meta tag).

- Enabled by the `max-image-preview:large` setting, or by using AMP
- Use either [schema.org markup](#) or the `og:image` meta tag to specify a large image that's relevant and representative of the web page, as this can influence which image is chosen as the thumbnail in Discover. Learn [more about how to specify your preferred image](#).
 - Avoid using generic images (for example, your site logo) in the schema.org markup or `og:image` meta tag.
 - For best results, avoid using text-heavy images in the schema.org markup or `og:image` meta tag.
- Provide an overall great page experience. For more advice, check out our help page on [Understanding page experience in Google Search results](#).

To deliver a good user experience, Discover seeks to present content that's suitable for interest-based feeds, such as articles and videos, as well as filter out content that's not wanted or that might confuse readers. For example, Discover might not recommend job applications, petitions, forms, code repositories, or satirical content without any context. Discover makes use of [SafeSearch](#) but beyond that filters out content that might be seen as shocking or unexpected.

Why Discover traffic might change over time

Traffic from Discover is less predictable or dependable when compared to keyword-driven search visits. Given its serendipitous nature, you should consider traffic from Discover as supplemental to your keyword-driven search traffic. Here are some reasons why Discover traffic may fluctuate:

- **Changing interests:** Discover is designed and always improving to show content aligned with what someone is interested in, which in part can be based on their search activity. If someone is no longer as interested in a particular topic—perhaps reflected by a decline in searching for it—their Discover feed might show other content they're more interested in. In turn, this may cause changes in traffic for publishers.
- **Content types:** Discover has and continues to adjust the types of content that might appear on the feed to better align with what people are looking for. Discover regularly shows content including but not limited to sports, health, entertainment, and lifestyle content from across the open web.
- **Updates to Google Search:** Periodically, we also make [updates to Search](#) designed to better provide people with links to helpful content. Because Discover is an extension of Search, updates can sometimes produce traffic changes. If you notice changes to your website's performance after an update, it may be useful to review our documentation on [Google Search's core updates and your website](#).

However, it may be that there's nothing you need to do after an update. Our ongoing work to improve Discover's user experience means sites may see changes in their traffic that are unrelated to the quality or publishing frequency of their content.

Monitor your performance on Discover

If you have content on Discover, you can monitor your performance using the [Performance report for Discover](#). This report shows impressions, clicks, and CTR for any of your content that has appeared on Discover in the last 16 months, as long as your data reaches a minimum threshold of impressions. The Discover performance report [includes traffic from Chrome](#), and fully tracks a site's Discover traffic across all surfaces where users interact with Discover.

<https://developers.google.com/search/docs/appearance/google-images>

Google image SEO best practices

Google provides several Search features and products that help users visually discover information on the web, such as the [text result images](#), Google Discover, and Google Images. While each feature and product looks different, the general recommendations for getting images to appear in them is the same.

[Original illustration: An illustration that shows images in Google search results, the images tab, and Discover]

You can optimize your images to appear in Google's search results by following these best practices:

1. [Help us discover and index your images](#)
2. [Optimize the image landing pages](#)

Help us discover and index your images

[Original embedded video]

The [technical requirements](#) for getting your content in Google's search results applies to images too. Since images are a substantially different format compared to HTML, it means there are additional requirements for getting images indexed; for example, finding the images on your site is different, and the presentation of the images also influences whether an image is indexed at all, and for the right keywords.

Use HTML image elements to embed images

Using standard HTML image elements helps crawlers find and process images. Google can find images in `src` attribute of `<img>` element (even when it's a child of other elements, such as the `<picture>` element). Google doesn't index CSS images.

Good:

```

```

Bad:

```
<div style="background-image:url(puppy.jpg)">A golden retriever puppy</div>
```

Use an image sitemap

You can provide the URL of images we might not have otherwise discovered by [submitting an image sitemap](#).

Unlike regular sitemaps, you can include URLs from other domains in the `<image:loc>` elements of the image sitemaps. This lets you use CDNs (content delivery networks) to host images. If you're using a CDN, we encourage

you to [verify ownership](#) of the CDN's domain name in Search Console so that we can inform you of any crawl errors that we may find.

Responsive images

Designing responsive web pages leads to better user experience, since people can access them across a plethora of device types. Refer to our [guide to responsive images](#) to learn about the best practices for handling images on your website.

Web pages use the `<picture>` element or the `srcset` attribute of an `img` element to specify responsive images. However, some browsers and crawlers don't understand these attributes. We recommend that you always specify a fallback URL using the `src` attribute.

The `srcset` attribute allows specifying different versions of the same image, specifically for different screen sizes. For example:

```

```

The `<picture>` element is a container that is used to group different `<source>` versions of the same image. It offers a fallback approach so the browser can choose the right image depending on device capabilities, like pixel density and screen size. The picture element also comes in handy for using new image formats with built-in graceful degradation for clients that may not yet support the new formats.

Per [section 4.8.1 of the HTML Standard](#), make sure that you provide an `img` element as a fallback with a `src` attribute when using the picture element using the following format:

```
<picture>
  <source type="image/svg+xml" srcset="pyramid.svg">
  <source type="image/webp" srcset="pyramid.webp">
  
</picture>
```

Use supported image formats

Google Search supports images referenced in the `src` attribute of `img` in the following file formats: BMP, GIF, JPEG, PNG, WebP, SVG, and AVIF. It's also a good idea to have the extension of your filename match with the file type.

You can also inline images as Data URIs. Data URIs provide a way to include a file, such as an image, inline by setting the `src` attribute of an `img` element as a Base64-encoded string using the following format:

```

```

While inlining images can reduce HTTP requests, carefully judge when to use them since it can considerably increase the size of the page. For more on this, refer to the [section on the advantage and disadvantages of inlining images on our web.dev page](#).

Optimize for speed and quality

[Original embedded video]

High-quality photos appeal to users more than blurry, unclear images. Also, sharp images are more appealing to users in the result thumbnail and can increase the likelihood of getting traffic from users. That said, images are often the largest contributor to overall page size, which can make pages slow and expensive to load. Make sure to apply the [latest image optimization](#) and [responsive image techniques](#) to provide a high quality and fast user experience.

Analyze your site speed with [PageSpeed Insights](#) and visit our [Why does speed matter?](#) to learn about best practices and techniques to improve website performance.

Optimize the image landing pages

While not immediately obvious, the content and metadata of the pages where an image is embedded can have a great influence on how and where the image may appear in Google's search results.

Specify a preferred image with metadata

Google's selection of an image preview is completely automated and takes into account a number of different sources to select which image on a given page is shown on Google (for example, a [text result image](#) or the preview image in Discover).

You can influence which image gets selected by providing your preferred image through one of the following metadata sources:

- Specify the schema.org [primaryImageOfPage](#) property with a URL or ImageObject.

```
<script type="application/ld+json">{
  "@context": "https://schema.org",
  "@type": "WebPage",
  "url": "https://example.com/url",
  "primaryImageOfPage": "https://example.com/images/cat.png"
}</script>
```

Or specify an image URL or ImageObject property and attach it to the main entity (using the schema.org [mainEntity](#) or [mainEntityOfPage](#) properties):

```
<script type="application/ld+json">{
  "@context": "https://schema.org",
  "@type": "BlogPosting",
  "mainEntityOfPage": "https://example.com/url",
  "image": "https://example.com/images/cat.png"
}</script>
```

- Specify the [og:image](#) meta tag.

```
<meta property="og:image" content="https://example.com/images/cat.png">
```

When choosing your preferred image for use in schema.org markup or the og:image meta tag, follow these best practices:

- Choose an image that's relevant and representative of the page.
- Avoid using a generic image (for example, your site logo) or an image with text in the schema.org markup or og:image meta tag.
- Avoid using an image with an extreme aspect ratio (such as images that are too narrow or overly wide).
- Use a high resolution, if possible.

Check your page title and description

Google Search automatically generates a title link and snippet to best explain each result and how it relates to the user query. This helps users decide whether or not to click a result. Here are two examples how the title links and snippet might look like on a Google Search results page:

[Original illustration: An illustration that shows titles and descriptions in image search results]

We use a number of different sources for this information, including information in the title and meta tags for each page.

You can help us improve the quality of the title link and snippet displayed for your pages by following Google's [title](#) and [snippet](#) guidelines.

Add structured data

If you include structured data, Google can display your images in certain [rich results](#), including a [prominent badge](#) in Google Images, which give users relevant information about your page and can drive better targeted traffic to your site.

Follow the [general structured data guidelines](#) as well as any guidelines specific to your structured data type; otherwise your structured data might be ineligible for rich result display in Google Images. In each of these structured data types, the image attribute is a required field to be eligible for a badge and rich result in Google Images. Here are two examples how rich results might look like on Google Images:

[Original illustration: An illustration that shows how rich results can appear in Google Images]

Use descriptive filenames, titles, and alt text

Google extracts information about the subject matter of the image from the content of the page, including captions and image titles. Wherever possible, make sure images are placed near relevant text and on pages that are relevant to the image subject matter.

Likewise, the filename can give Google very light clues about the subject matter of the image. When possible, use filenames that are short, but descriptive. For example, my-new-black-kitten.jpg is better than IMG00023.JPG. Avoid using generic filenames like image1.jpg, pic.gif, 1.jpg when possible. If your site has thousands of images, you might want to consider automating the naming of the images. If you localize your images, remember to also translate the filenames, keeping in mind the [URL encoding guidelines](#) if you're using non-latin or special characters.

[Original embedded video]

The most important attribute when it comes to providing more metadata for an image is the alt text (text that describes an image), which also improves accessibility for people who can't see images on web pages, including

users who use screen readers or have low-bandwidth connections.

Google uses alt text along with computer vision algorithms and the contents of the page to understand the subject matter of the image. Also, alt text in images is useful as anchor text if you decide to use an image as a link.

When writing alt text, focus on creating useful, information-rich content that uses keywords appropriately and is in context of the content of the page. Avoid filling alt attributes with keywords (also known as [keyword stuffing](#)) as it results in a negative user experience and may cause your site to be seen as spam.

Bad (missing alt text):

```

```

Bad (keyword stuffing):

```

```

Better:

```

```

Best:

```

```

Also consider the accessibility of your alt text, per [W3 guidelines](#). For the `<img>` element, you can add the alt attribute of the element, while for inline `<svg>` elements, you can use the `<title>` element. For example:

```
<svg aria-labelledby="svgtitle1">  
  <title id="svgtitle1">Googlebot wearing an apron and chef hat, struggling to make pancakes on the  
  stovetop</title>  
</svg>
```

We recommend testing your content by [auditing for accessibility](#) and [using a slow network connection emulator](#).

If an image is referenced on multiple pages within a larger website, consider the [site's overall crawl budget](#). In particular, consistently reference the image with the same URL, so that Google can cache and reuse the image without needing to request it multiple times.

Opt out of Google Images inline linking

If you choose, you can prevent the full-sized image from appearing in the Google Images search results page by opting out of inline linking in Google Images search results. **To opt out of inline linking:**

1. When your image is requested, examine the [HTTP referrer header](#) in the request.
2. If the request is coming from a Google domain, reply with a 200 HTTP status code, or a 204 HTTP status code and no content.

Google will still crawl your page and see the image, but will display a thumbnail image generated at crawl time in search results. This opt-out is possible at any time, and doesn't require re-processing of a website's images. This behavior isn't considered image [cloaking](#) and won't result in manual actions.

Alternatively, you can [prevent the image from appearing in search results entirely](#).

Optimize for SafeSearch

SafeSearch is a setting in Google user accounts that specifies whether to show, blur, or block explicit images, videos, and websites in Google Search results. Make sure Google understands the nature of your site so that Google can apply SafeSearch filters to your site if appropriate. [Learn more about labeling pages for SafeSearch](#).

<https://developers.google.com/search/docs/appearance/establish-business-details>

Establish your business details with Google

[Original illustration: Google knowledge panel in Search results]

You can enhance the appearance and coverage of your official website and presence in Search results by first establishing it with Google. This makes it easier for users to recognize your official site and reach information you provide more easily when they search.

Google offers a variety of ways to help you provide key business details so that they show up for users in Search results. This guide describes how to make your businesses location, official site, and content info available for results, Google knowledge panel, and Google Maps.

Claim your local business

Manage how your business appears on Google Maps and Google Search by [claiming your Business Profile](#). Once you verify yourself as the owner of a listing, you can provide or edit your address, contact info, business type, and photos. This enables your local business information to show up in Google Maps and Google knowledge panel.

Register your website with Search Console

The first step in establishing your website as an official presence is to [verify website ownership in Search Console](#). This process verifies the owners and operators of the site. Once you've verified your website, you can use Search Console to understand and monitor how Google displays information about your site.

Update your site's Google knowledge panel

Google algorithms find information, like your site's name, corporate contact information, and social profiles, that is publicly available on the web. You can update or provide more information for your site for broader reach and recognition in Search results. If you're verified as an official representative, you can [update your Google knowledge panel](#) to override the information Google finds automatically.

Add structured data

Google Search works hard to understand the content of a page. You can help us by providing explicit clues about the meaning of a page to Google by including structured data on the page. Structured data is a standardized format for providing information about a page and classifying the page content. Learn more about [structured data](#).

Here are some structured data features that all websites can benefit from:

- **Site logo:** You can specify which image that you want Google to use for your organization's logo in Search results and your Google knowledge panel. To do this, add [Organization structured data](#) to your official website that identifies the [location of your preferred logo](#).

- **Breadcrumbs:** Breadcrumb trails on a page indicate the page's position in the site hierarchy. A user can navigate all the way up in the site hierarchy, one level at a time, by starting from the last breadcrumb in the breadcrumb trail. To help Google understand your breadcrumbs, you can add [Breadcrumb structured data](#) to your site.

Highlight your customer support methods

Customers are often looking for ways to contact businesses, and Google works to show the best available information to help them in a [variety of ways](#). You can highlight your customer support methods in Google Search by [following several best practices](#) that help ensure we're showing the most accurate information for your business or service.

Troubleshooting

If you're experiencing issues with your site's information, try the following troubleshooting tips:

- **Make suggested changes to your Google knowledge panel.**

If you have waited at least one week since your marked-up page was last crawled by Google, report information that you believe is incorrect in a Google knowledge panel, by clicking the **Feedback** link at the bottom of the Google knowledge panel. If you are a recognized authority or the site or entity represented in the Google knowledge panel, you can view more options for providing specific edits.

Learn more about how to [update your Google knowledge panel](#).

- **Use the [Rich Results Test](#) to test your markup.** Give our system a week to find and update your site details in Search results.

<https://developers.google.com/search/docs/appearance/top-places-list>

Top Places List

If your business is mentioned in any "top places" lists online, Google might show a rich result that shows the lists that mention your business. For example, if your restaurant appears in a "Top 10 Chinese restaurants in NYC" list and a "Best restaurants in the city" list on [qualified websites](#), these lists would be shown in your business's search result. This feature is only for businesses with a physical location.

[Original illustration: Top places list for a restaurant search result]

Note: The actual appearance in search results might be different.

Requirements for the listing site

In order for your site's "top places" list to be eligible to appear in a Top Places List result, the list must meet these criteria:

- The list must be curated by the content provider, be genuine, independent, and not sponsored.
- The list must not consist of templated sentences built from data or automated metrics.
- The list must not contain vulgar or potentially offensive language.

Prevent your website's lists from appearing in a Top Places List feature

To prevent lists from your site from appearing in a Top Places List, [opt out of display in local search results and other Google properties](#).

<https://support.google.com/webmasters/answer/3035947>

Opt out of display on local search results and other Google properties

Complete-topic source brief · paraphrase. Open the original Google Support page for its full wording.

Domain-wide control hides Googlebot-fetched material from Shopping, Flights, Hotels/vacation rentals and specialized local results within 30 days. Subdomains are included; other domains need separate choices. Individual subdomains/paths cannot be chosen. Local scope is worldwide; other listed services are covered on google.com. Manage through Search Console.

<https://developers.google.com/search/docs/appearance/page-experience>

Understanding page experience in Google Search results

Google's core ranking systems look to reward content that provides a good page experience. Site owners seeking to be successful with our systems shouldn't focus on only one or two aspects of page experience. Instead, check if you're providing an overall great page experience across many aspects.

Self-assess your content's page experience

Answering yes to the following questions means you're probably on track in providing a good page experience:

- Do your pages have good Core Web Vitals?
- Are your pages served in a secure fashion?
- Does your content display well on mobile devices?
- Does your content avoid using an excessive amount of ads that distract from or interfere with the main content?
- Do your pages avoid using intrusive interstitials?
- Is your page designed so visitors can easily distinguish the main content from other content on your page?

These questions don't encompass all page experience aspects to consider. However, questions like these, and consulting the following resources, may help you align with providing an overall good page experience.

Page experience resources

Here are some resources that can help you measure, monitor, and optimize your page experience:

- [Understanding Core Web Vitals and Google Search results](#): Learn more about Core Web Vitals and how they work in Google Search results.
- [Search Console's HTTPS report](#): Check if you're serving secure HTTPS pages and what to fix, if you're not.
- [Check if a site's connection is secure](#): Learn how to check if your site's connection is secure, as reported by Chrome. If the page isn't served over HTTPS, learn how to [secure your site with HTTPS](#).
- [Avoid intrusive interstitials and dialog](#): Learn how to avoid interstitials that can make content less accessible.
- [Chrome Lighthouse](#): This toolset from Chrome can help you identify a range of improvements to make related to page experience, including mobile usability.
- [CrUX ad metrics](#): These experimental CrUX metrics from Chrome can help you understand your site's ad count, ad density, and ad weight metrics. They provide insight into real users' ad experiences on your site, as seen and reported by Chrome.

FAQ

Is there a single "page experience signal" that Google Search uses for ranking?

There is no single signal. Our core ranking systems look at a variety of signals that align with overall page experience.

What aspects of page experience are used in ranking?

Core Web Vitals are used by our ranking systems. We recommend site owners achieve good Core Web Vitals for success with Search and to ensure a great user experience generally. Keep in mind that getting good results in reports like Search Console's [Core Web Vitals report](#) or third-party tools doesn't guarantee that your pages will rank at the top of Google Search results; there's more to great page experience than Core Web Vitals scores alone. These scores are meant to help you to improve your site for your users overall, and trying to get a perfect score just for SEO reasons may not be the best use of your time.

Beyond Core Web Vitals, other page experience aspects don't directly help your website rank higher in search results. However, they can make your website more satisfying to use, which is generally aligned with what our ranking systems seek to reward. Therefore it's still worth working to improve page experience overall.

Is page experience evaluated on a site-wide or page-specific basis?

Our core ranking systems generally evaluate content on a page-specific basis, including when understanding aspects related to page experience. However, we do have some site-wide assessments.

How important is page experience to ranking success?

Google Search always seeks to show the most relevant content, even if the page experience is sub-par. But for many queries, there is lots of helpful content available. Having a great page experience can contribute to success in Search, in such cases.

Recent updates on our blog

Here's everything we've announced about page experience on the [Google Search Central blog](#):

<https://developers.google.com/search/docs/appearance/avoid-intrusive-interstitials>

Avoid intrusive interstitials and dialogs

[Original illustration: Example of an intrusive interstitial]

Intrusive [interstitials](#) and dialogs are page elements that obstruct users' view of the content, usually for promotional purposes. Interstitials are overlays on the whole page and dialogs are overlays only on a part of the page, sometimes also obfuscating the underlying content.

Websites often need to show dialogs for various reasons; however, interrupting users with intrusive interstitials may frustrate them and erode their trust in your website.

Intrusive dialogs and interstitials make it hard for Google and other search engines to understand your content, which may lead to poor search performance. Equally, if users find your site hard to use, they are unlikely to want to visit those websites again, including through search engines.

Create unintrusive dialogs

Creating unintrusive dialogs means that users can access your content without being interrupted by a dialog. This applies to all kinds of promotional dialogs, including app install prompts. Following these best practices will help your users have a good experience on your site, in addition to helping Google Search understand the content and structure of your site.

Use banners instead of interstitials

[Original illustration: Examples of banners that provide a better user experience]

Instead of full page interstitials, use banners that take up only a small fraction of the screen to grab your users' attention. Banners ensure that users and search engines can get to the content as soon as they land on the page.

These can be implemented in various ways. For example, for app install banners, you can use a browser-supported banner, such as [Smart App Banners](#) for Safari or [in-app install experience](#) for Chrome. Another option is to create an HTML banner, similar to a typical small advertisement, that links to the correct app store for download. You can reuse these small containers for other kinds of notifications, such as newsletter sign-up prompts.

Use common libraries

Many CMSes have plugins that create standard dialogs and interstitials for the most common use cases, such as newsletter sign-up prompts. For example, if you are using WordPress, search for "newsletter signup wordpress". Using such plugins can be helpful for Google, other search engines, and the internet in general, as the plugin developers can deploy improvements at scale.

Avoid common mistakes

Unless they're legally mandatory, avoid the following common mistakes when designing a dialog or interstitial to help Google Search crawl and understand your content:

- Don't obscure the entire page with interstitials.
- Don't redirect the user to a separate page for their consent or input.

Mandatory interstitials

Certain sites are required to show an interstitial due to the kind of content they are publishing. For example, a casino site may need to show an age gate, which is a type of interstitial where the user must provide their age before accessing the content.

Mandatory interstitials are exempted from the guidelines discussed in this document; however, if possible, we recommend that sites follow these best practices:

- Ensure that the content is overlaid with the interstitial. This will ensure that Google can still at least index some of the content and potentially show it in search results.
- Don't redirect the incoming HTTP requests to a different page for collecting consent or providing data. Redirecting all URLs to a single page will remove all but that one page from search results, as Googlebot can only fetch that page.

For content behind a mandatory age gate for adult users, we recommend allowing Googlebot to crawl your content without triggering the age gate. You can do this by [verifying Googlebot requests](#) and serving the content without age gate.

<https://developers.google.com/search/docs/appearance/core-web-vitals>

Understanding Core Web Vitals and Google search results

Core Web Vitals is a set of metrics that measure real-world user experience for loading performance, interactivity, and visual stability of the page. We highly recommend site owners achieve good Core Web Vitals for success with Search and to ensure a great user experience generally. This, along with other page experience aspects, aligns with what our core ranking systems seek to reward. Learn more in [Understanding page experience in Google Search results](#).

Core Web Vitals metrics

- **Largest Contentful Paint (LCP)**: Measures loading performance. To provide a good user experience, strive to have LCP occur within the first 2.5 seconds of the page starting to load.
- **Interaction To Next Paint (INP)**: Measures responsiveness. To provide a good user experience, strive to have an INP of less than 200 milliseconds.
- **Cumulative Layout Shift (CLS)**: Measures visual stability. To provide a good user experience, strive to have a CLS score of less than 0.1.

Optimizing your Core Web Vitals

Here are some resources that can help you measure, monitor, and optimize your Core Web Vitals:

- Check the [Core Web Vitals report in Search Console](#). This shows how your pages perform.
- Learn more about [Core Web Vitals](#), a guide about Core Web Vitals, including how to measure, debug, improve and best practices.
- Learn about the different tools that can help you [measure and report Core Web Vitals](#). These tools measure LCP, INP, and CLS.

Recent updates on our blog

Here's everything we've announced about Core Web Vitals on the [Google Search Central blog](#):

<https://developers.google.com/search/docs/appearance/preferred-sources>

Help your readers find your site through preferred sources in Google Search

If you're a website owner, you can help your audience find your publication as a preferred source in Google Search. When a user selects your site as a preferred source, your content is more likely to appear in "[Top Stories](#)", highlighted with a "preferred" badge. In AI Mode and AI Overviews, your content can be highlighted with a "preferred" badge for users who have selected your site as a preferred source.

[Original illustration: Preferred sources in AI Mode]

[Original illustration: Preferred sources in Top Stories]

Feature availability

The preferred sources feature is available globally for the "Top Stories" feature in all languages where Google Search is available.

Preferred sources can also appear in AI Mode and AI Overviews in all languages and locales where those features are available. To make your site eligible for display as a preferred source in these features, you must make sure your site is [included in Search generative AI features in Search Console](#).

Only domain-level and subdomain-level sites are eligible to appear in the [source preferences tool](#). For example, <https://www.example.com/> and <https://code.example.com/> are eligible for preferred sources, but the subdirectory <https://www.example.com/blog> isn't eligible.

How to help users find your site as a preferred source

If your site appears in the [source preferences tool](#) (to check, enter your site in the tool's search box), you can use the following methods to guide your readers to select your site as a preferred source:

- **Standard JavaScript implementation (recommended):** Embed an interactive button to your web pages, which lets readers seamlessly add your site as a preferred source and returns them to your page. By adding only two lines to your HTML, this option renders an automatically localized, Google-styled button that supports device and language customization.
- **Advanced JavaScript implementation with custom design assets:** If you want to use your own design assets and customize the standard JavaScript button, follow the advanced JavaScript implementation instructions. The end-user UI flow remains the same as the standard JavaScript implementation.
- **Deeplink implementation:** If you can't implement the interactive button (for example, your CMS doesn't support it), you can use a deeplink to direct people to adding your site as a preferred source. You can do this by adding a text link or a clickable image on your web pages. You can also use the deeplink in your social posts, email newsletters, or promotions.

These methods are examples of how you can build your audience and help people find your site as a preferred source. It's not required to do them in order to appear as a preferred source.

Standard JavaScript implementation (recommended)

By adding only two lines to your HTML, you can add an automatically translated, interactive button, which lets readers seamlessly add your site as a preferred source and return to where they left off on your page. You can also set the theme of the button (dark or light), plus override the button language. We recommend this implementation because it provides the best user experience for readers.

Here's how it'll look:

[Original illustration: Add to preferred sources button with JavaScript]

To embed the JavaScript button on your web page, do the following:

1. Add the following `<script>` tag to your web page (preferably in the `<head>` element), which loads the preferred sources library.

```
<script async src="https://news.google.com/swg/js/v1/publisher.js"></script>
```

2. Add the following `<div>` tag anywhere you'd like the button to appear in the body of your web page.

```
<div google-add-preferred-source-btn></div>
```

Set the button theme

By default, the "Add to preferred sources" button appears in a light theme. To adjust the theme, add the `data-theme` data attribute to your `<div>` element and set it to either `light` or `dark`:

```
<div google-add-preferred-source-btn data-theme="dark"></div>
```

Override the button language

By default, the "Add to preferred sources" button displays in the user's language based on their browser settings. To override this behavior, add the `data-lang` data attribute to your `<div>` element and set it to the language code you'd prefer (download the [list of supported language codes](#)):

```
<div google-add-preferred-source-btn data-lang="en"></div>
```

[Download supported language codes](#)

Advanced JavaScript implementation with custom design assets

By default, the standard JavaScript button automatically scans the DOM for elements with the `google-add-preferred-source-btn` attribute and renders the standard badge. However, for custom integrations where you want to trigger the flow from your own UI elements or application milestones, you can initialize the client programmatically and bind the flow to custom triggers.

When integrating the preferred sources library into custom user interfaces, application milestones, or modern framework applications, you can take full programmatic control over runtime settings and flow initiation using our JavaScript SDK. We provide two distinct integration approaches depending on your tooling and architecture: ES Module imports (ESM) and standard script callback queues (IIFE). Both distribution models expose identical capabilities and methods.

ES Module imports

For modern build setups and module-based environments, import the library directly into your code.

```
import { preferredSource } from
  "https://news.google.com/swg/js/v1/publisher.mjs";

// 1. Initialize directly using the imported module instance
preferredSource.init({
  theme: 'light', // Theme choice: "light" or "dark" (default "light")
  lang: 'en'      // Optional: override language (defaults to page language)
});

// 2. Programmatically bind flow invocation using a click handler
const button = document.querySelector('#myButton');
button.onclick = () => {
  preferredSource.addPreferredSource();
};
```

Standard script callback queues

For standard script tag integrations, load the following script in the `<head>` of your document with the `preferred-sources-control="manual"` attribute to prevent automatic button rendering:

```
<script async preferred-sources-control="manual" src="https://news.google.com/swg/js/v1/publisher.js">
</script>
```

Note: If the `preferred-sources-control="manual"` attribute is omitted, it'll search for and immediately initialize any elements with the `google-add-preferred-source-btn` attribute.

Then use the global `PREFERRED_SOURCE` callback queue to initialize options and bind custom triggers:

```

<script>
  (self.PREFERRED_SOURCE = self.PREFERRED_SOURCE || []).push(
    function(preferredSource) {
      // 1. Initialize with options
      preferredSource.init({
        theme: 'light',
        lang: 'en'
      });

      // 2. Programmatically bind trigger button
      const button = document.querySelector('#myButton');
      button.addEventListener('click', () => {
        preferredSource.addPreferredSource();
      });
    });
</script>

```

For interactive frontend and backend implementations showcasing live usage across buttons, declarative containers, IIFE scripts, and ES Module bundles, explore the [preferred sources demo](#).

Deeplink implementation

If you can't use JavaScript on your website, you can alternatively use a deeplink to direct your users to the [source preferences tool](#), where they can add your site as a preferred source and confirm their selection.

Use the following URL format and replace `example.com` with your publication's domain name, which takes users directly to your site in the [source preferences tool](#):

```
https://www.google.com/preferences/source?q=Your_Website's_URL
```

Example text link

```

<a
  href="https://www.google.com/preferences/source?q=example.com">
  Add as Preferred Source
</a>

```

Example clickable image link

```

<a
  href="https://www.google.com/preferences/source?q=example.com">
  
</a>

```

Button image assets

You can design your own custom promotion badge or download official translated graphic assets provided by Google:

[Download button assets](#)

<https://developers.google.com/search/docs/appearance/ranking-systems-guide>

A guide to Google Search ranking systems

Google uses automated ranking systems that [look at many factors and signals](#) about hundreds of billions of web pages and other content in our Search index to present the most relevant, useful results, all in a fraction of a second. This page is a guide to understanding some of our more notable ranking systems. It covers some systems that are part of our core ranking systems, which are the underlying technologies that produce search results in response to queries. It also covers some systems involved with specific ranking needs.

Our ranking systems are designed to work on the page level, using a variety of signals and systems to understand how to rank individual pages. Site-wide signals and classifiers are also used and contribute to our understanding of pages. Having some good site-wide signals does not mean that all content from a site will always rank highly, just as having some poor site-wide signals does not mean all the content from a site will rank poorly.

We regularly improve our ranking systems through [rigorous testing and evaluation](#) and provide notice of [updates to our ranking systems](#) when those might be useful to content creators and others.

You can also visit our [How Search Works](#) site to understand how our [ranking systems](#), combined with other processes, work together so that Google Search delivers on our mission to organize the world's information and make it universally accessible and useful.

BERT

Bidirectional Encoder Representations from Transformers ([BERT](#)) is an AI system Google uses that allows us to understand how combinations of words express different meanings and intent.

Crisis information systems

Google has developed systems to provide helpful and timely information during times of crisis, whether those involve personal crisis situations, natural disasters, or other wide-spread crisis situations:

- **Personal crisis:** Our systems work to understand when people are seeking information about personal crisis situations to display hotlines and content from trusted organizations for certain queries related to suicide, sexual assault, poison ingestion, gender-based violence, or drug addiction. Learn more about how [personal crisis information is displayed in Google Search](#).
- **SOS Alerts:** During times of natural disasters or wide-spread crisis situations, our SOS Alerts system works to show updates from local, national, or international authorities. These updates may include emergency phone numbers and websites, maps, translations of useful phrases, donation opportunities, and more. Learn more about [how SOS Alerts work](#) and how they're part of Google's [crisis alerts](#) that help in times of floods, wildfires, earthquakes, hurricanes, and other disasters.

Deduplication systems

Searches on Google may find thousands or even millions of matching web pages. Some of these may be very similar to each other. In such cases, our systems show only the most relevant results to avoid unhelpful duplication. Learn

more about [how deduplication works](#) and [how to see omitted results](#) if desired, when deduplication happens.

Deduplication also happens with [featured snippets](#). If a web page listing is elevated to become a featured snippet, we don't repeat the listing later on the first page of results. This declutters the results and helps people locate relevant information more easily.

Exact match domain system

Our ranking systems consider the words in domain names as one of many factors to determine if content is relevant to a search. However, our exact match domain system works to ensure we don't give too much credit for content hosted under domains designed to exactly match particular queries. For example, someone might create a domain name containing the words "best-places-to-eat-lunch" in hopes all those words in the domain name would propel content high in the rankings. Our system adjusts for this.

Freshness systems

We have various "query deserves freshness" systems designed to show fresher content for queries where it would be expected. For example, if someone is searching about a movie that's just been released, they probably want recent reviews rather than older articles from when production began. For another example, ordinarily a search for "earthquake" might bring back material about preparation and resources. However, if an earthquake happened recently, then news articles and fresher content might appear.

Link analysis systems and PageRank

We have various systems that understand how pages link to each other as a way to determine what pages are about and which might be most helpful in response to a query. Among these is PageRank, one of our core ranking systems used when Google first launched. Those curious can learn more by reading the original [PageRank research paper](#) and [patent](#). How PageRank works has evolved a lot since then, and it continues to be part of our core ranking systems.

Local news systems

We have systems that work to identify and surface local sources of news whenever relevant, [such as through](#) our "Top stories" and "Local news" features.

MUM

Multitask Unified Model ([MUM](#)) is an AI system capable of both understanding and generating language. It's not currently used for general ranking in Search but rather for some specific applications such as to [improve searches for COVID-19 vaccine information](#) and to [improve featured snippet callouts we display](#).

Neural matching

[Neural matching](#) is an AI system that Google uses to understand representations of concepts in queries and pages and match them to one another.

Original content systems

We have systems to help ensure we are showing original content prominently in search results, [including original reporting](#), ahead of those who merely cite it. This includes support of a special [canonical markup](#) creators can use to help us better understand what is the primary page if a page has been duplicated in several places.

Removal-based demotion systems

Google has policies that allow the removal of certain types of content. If we process a significant volume of such removals involving a particular site, we use that as a signal to improve our results. In particular:

- **Legal removals:** When we receive a significant volume of [valid copyright removal requests](#) involving a given site, [we are able to use that](#) to demote other content from the site in our results. This way, if there is other infringing content, people are less likely to encounter it versus the original content. We apply similar demotion signals to complaints involving defamation, counterfeit goods, and court-ordered removals. In the case of child sexual abuse material (CSAM), we always remove such content when it is identified and we demote all content from sites with a significant proportion of CSAM content.
- **Personal information removals:** If we process a significant volume of personal information removals involving a site with [exploitative removal practices](#), we demote other content from the site in our results. [We also look to see](#) if the same pattern of behavior is happening with other sites and, if so, apply demotions to content on those sites. We may apply similar demotion practices for sites that receive a significant volume of removals of content involving [doxing content](#), [explicit personal imagery created or shared without consent](#), or [explicit non-consensual fake content](#).

Passage ranking system

[Passage ranking](#) is an AI system we use to identify individual sections or "passages" of a web page to better understand how relevant a page is to a search.

RankBrain

[RankBrain](#) is an AI system that helps us understand how words are related to concepts. It means we can better return relevant content even if it doesn't contain all the exact words used in a search, by understanding the content is related to other words and concepts.

Reliable information systems

Multiple systems work in various ways to show the most reliable information possible, such as to [help surface more authoritative pages and demote low-quality content](#) and to [elevate quality journalism](#). In cases where reliable information might be lacking, our systems automatically display [content advisories](#) about rapidly-changing topics or when our systems don't have high confidence in the overall quality of the results available for the search. These provide tips on how to search in ways that might lead to more helpful results. Learn more about our [approach to delivering high-quality information in Search](#).

Reviews system

The [reviews system](#) aims to better reward high quality reviews, content that provides insightful analysis and original research, and is written by experts or enthusiasts who know the topic well.

Site diversity system

Our site diversity system works so that we generally won't show more than two web page listings from the same site in our top results, so that no single site tends to dominate all the top results. However, we may still show more than two listings in cases where our systems determine it's especially relevant to do so for a particular search. Site diversity generally treats subdomains as part of a root domain. IE: listings from a subdomain (subdomain.example.com) and the root domain (example.com) will all be considered from the same single site. However, sometimes subdomains are treated as separate sites for diversity purposes when deemed relevant to do so.

Spam detection systems

No one wants their email inbox filled with spam, which is why spam filters are so helpful. Search faces a similar challenge, because the internet includes huge amounts of spam that, if not dealt with, would prevent us from showing the most helpful and relevant results. We employ a range of [spam detection systems](#), including [SpamBrain](#), to deal with content and behaviors that violate our [spam policies](#). These systems are constantly [updated](#) to keep up with the latest ways that the spam threat evolves.

Retired systems

The following systems are noted for historical purposes. They've either been incorporated into successor systems or made part of our core ranking systems.

Helpful content system

[Announced in 2022](#) as the "Helpful Content Update", this was a system designed to better ensure people see original, helpful content written by people, for people, in search results, rather than content made primarily to gain search engine traffic. In March 2024, it evolved and [became part of](#) our core ranking systems, as our systems use a variety of signals and systems to present helpful results to users.

Hummingbird

This was a major improvement to our overall ranking systems made in August 2013. Our ranking systems have continued to evolve since then, just as they had been evolving before.

Panda system

This was a system designed to better ensure high-quality and original content was appearing in our search results. [Announced in 2011](#) and given the nickname of the "Panda," it evolved and became part of our core ranking systems in 2015.

Penguin system

This was a system designed to combat link spam. [Announced in 2012](#) and given the nickname of the "Penguin Update", it was [integrated](#) into our core ranking systems in 2016.

<https://developers.google.com/search/docs/appearance/reviews-system>

Google Search's reviews system and your website

The *reviews system* aims to better reward high quality reviews, which is content that provides insightful analysis and original research and is written by experts or enthusiasts who know the topic well. This page explains more about how the reviews system works, and what you can do to assess and improve your content.

How the reviews system works

The reviews system works to ensure that people see reviews that share in-depth research, rather than thin content that simply summarizes a bunch of products, services or other things. The reviews system is improved at a regular and ongoing pace.

The reviews system is designed to evaluate articles, blog posts, pages or similar first-party standalone content written with the purpose of providing a recommendation, giving an opinion, or providing analysis. It does not evaluate third-party reviews, such as those posted by users in the reviews section of a product or services page.

Reviews can be about a single thing, or head-to-head comparisons, or ranked-lists of recommendations. Reviews can be about any topic. There can be reviews of products such as laptops or winter jackets, pieces of media such as movies or video games, or services and businesses such as restaurants or fashion brands.

The reviews system primarily evaluates review content on a page-level basis. However, for sites that have a substantial amount of review content, any content within a site might be evaluated by the system. If you don't have a lot of reviews, a site-wide evaluation is not likely to happen.

Currently, this system applies to the following languages globally: English, Spanish, German, French, Italian, Vietnamese, Indonesian, Russian, Dutch, Portuguese, Polish.

In the case of products, [product structured data](#) might help us better identify if something is a product review, but we don't solely depend on it.

What does this system mean for my site?

To learn more about how to create content that's successful with the reviews system, see our [help page on how to write high quality reviews](#).

Content impacted by the reviews system may recover over time, if you've made improvements to your content. However, note that our automated assessment of review content is only one of many factors used in ranking content, so changes can happen at any time for various reasons.

<https://developers.google.com/search/docs/appearance/core-updates>

Google Search's core updates and your website

Several times a year, Google makes significant, broad changes to our [search algorithms and systems](#). We refer to these as core updates, and we give notice when they happen on our [list of Google Search ranking updates](#).

In general, most sites don't need to worry about core updates and may not even realize one has happened. However, if you have questions about a change in traffic that correlates with a core update, this page is for you. It explains more about how core updates work, and what you can do to assess and perhaps improve your content.

How core updates work

Core updates are designed to ensure that overall, we're delivering on our mission to present helpful and reliable results for searchers. These changes are broad in nature, and don't target specific sites or individual web pages. As content on the web changes, we assess and update our systems to keep pace, as a whole.

One way to think about a core update is to imagine that a friend asked you for your top food recommendations. While you do have a list of your 20 favorite restaurants, things have changed since you initially wrote it in 2019. Some new restaurants that didn't exist before are now candidates for your list. You might reassess some restaurants and realize they move to a higher spot on the list, given how many consistently positive experiences you had there, or to account for your friend's preference for dog-friendly restaurants. The list will change, and restaurants that move down aren't necessarily "bad"; there are just other restaurants that make your top 20.

Check if there's a traffic drop in Search Console

If you're seeing a drop in position and suspect it might correlate with the [timing of a core update](#), use Search Console to determine whether you need to make a change.

1. **Confirm that the core update has finished rolling out.** Check the [Search Status Dashboard](#) and take note of the start and end date of the core update.
2. **Compare the right dates:** We recommend waiting at least a full week after a core update completes before analyzing your site in Search Console. After a week, try [comparing this week with a week before](#) the core update started rolling out; this way you'll be able to better pinpoint what exactly has changed.
3. **Review your top pages and queries.** Assess how they were ranking before and after the core update: is the drop small or large?
 - **Small drop in position (dropping from position 2 to 4):** There's no need to take drastic action (in fact, we recommend avoiding making changes to content that's already performing well).
 - **Large drop in position (dropping from position 4 to 29):** Do a [deeper assessment](#).
4. **Analyze different search types separately:** This can help you understand whether the drop you've seen happened in Web Search, Google Images, Video mode, or the News tab.

Assessing a large drop in position

If you're seeing a sustained, large drop in position for your site as a whole, read through the [self-assessment](#) to check if your site overall (not just individual pages) is delivering content that's helpful, reliable, and people first.

Specifically, we recommend keeping the following in mind:

- **Take a close look at your site as a whole, and try to be objective.** You might also ask others you trust (that are unaffiliated with your site) to do an [assessment using the questions](#).
- **Evaluate the pages that were most impacted.** Look closely at these to understand how these pages may perform against the [self-assessment questions](#). For example, there may be other pages on the web that are doing a better job of helping the searcher.

Things to keep in mind when making changes

- **Avoid doing "quick fix" changes** (like removing some page element because you heard it was bad for SEO). Instead, focus on making changes that make sense for your users and are sustainable in the long term.
- **Consider how you can improve your content in meaningful ways.** For example, it could be that rewriting or restructuring your content makes it easier for your audience to read and navigate the page.
- **Deleting content is a last resort**, and only to be considered if you think the content can't be salvaged. In fact, if you're considering deleting entire sections of your site, that's likely a sign those sections were created for search engines first, and not people. If that's the case for your site, then deleting the unhelpful content can help the good content on your site perform better.

How long does it take to see an effect in Search results

If you've made improvements, it may take time to see an effect in Search results: some changes can take effect in a few days, but it could take several months for our systems to learn and confirm that the site as a whole is now producing helpful, reliable, people-first content in the long term. If it's been a few months and you still haven't seen any effect, that could mean waiting until the next core update.

However, you don't necessarily have to wait for a major core update to see the effect of your improvements. We're continually making updates to our search algorithms, including smaller core updates. These updates are not announced because they aren't widely noticeable, but they are another way that your content can see a rise in position (if you've made improvements).

Keep in mind that there's no guarantee that changes you make to your website will result in noticeable impact in search results, and that positions in Google Search results aren't static or fixed in place. Google's search results are dynamic in nature because user expectations evolve and the open web itself is constantly changing with new and updated content. This constant change can cause both gains and drops in organic Search traffic.

<https://developers.google.com/search/docs/appearance/spam-updates>

Google Search spam updates and your site

While Google's automated systems to [detect search spam](#) are constantly operating, we occasionally make notable improvements to how they work. When we do, we refer to this as a *spam update* and share when they happen on our [list of Google Search ranking updates](#).

For example, [SpamBrain](#) is our AI-based spam-prevention system. From time-to-time, we improve that system to make it better at spotting spam and to help ensure it catches new types of spam.

Sites that see a change after a spam update should review our [spam policies](#) to ensure they are complying with those. Sites that violate our policies may rank lower in results or not appear in results at all. Making changes may help a site improve if our automated systems learn over a period of months that the site complies with our spam policies.

In the case of a link spam update (an update that specifically deals with link spam), making changes might not generate an improvement. This is because when our systems remove the effects spammy links may have, any ranking benefit the links may have previously generated for your site is lost. Any potential ranking benefits generated by those links cannot be regained.

<https://developers.google.com/search/docs/appearance/aggregator-features>

Regional differences in Search experience

Google is evolving the search results page for certain types of queries to provide new opportunities for Vertical Search Services (VSS), Comparison Shopping Services (CSS), direct suppliers, and content providers.

To find the features relevant to your website, select where your company is based and the query types that are most relevant to your business:

EEA

▼

Car hire

▼

Clear all

Feature	Region	Queries
[Original illustration: Aggregator unit in Google Search] Aggregator unit A unit that provides visibility for ecosystem participants (aggregators) on the search results page. Learn about the aggregator unit	EEA	Hotels, flights, ground transportation, local businesses, and products
[Original illustration: Supplier unit in Google Search] Supplier unit A unit that provides visibility for ecosystem participants (direct suppliers) on the search results page. Learn about the supplier unit	EEA	Hotels, flights, ground transportation, local businesses, and products
[Original illustration: Ecosystem carousel in Google Search] Ecosystem carousel A carousel that may appear alongside answers that require data visualizations and interactivity with data (for example, stock charts, weather information, sports scores). Learn about the ecosystem carousel	EEA	Weather, sports, finance, and translate
[Original illustration: Job sites in Google Search] Job sites features A dedicated "Job sites" carousel and refinement chip that includes links to aggregator sites from across the web with the option to see . Learn about the job sites features	EEA	Jobs

Feature	Region	Queries
[Original illustration: Places sites in Google Search] Places sites features A dedicated "Places sites" carousel and refinement chip that includes links to aggregator sites from across the web with the option to see . Learn about the places sites features	Türkiye	Hotels and local businesses
[Original illustration: South African badge on a Google Search result] [Original illustration: South African refinement chip in Google Search] South Africa badge and refinement chip A badge and refinement chip are available for businesses based in South Africa for specific query types. Learn about the South Africa features	South Africa	Travel, products, car hire, food delivery, and ground transportation
[Original illustration: Structured data carousels in Google Search] Structured data carousels A list-like rich result that people can scroll horizontally to see entities from a given site (also known as a host carousel). To be eligible, your site must implement structured data markup. Learn about the carousel	EEA	Hotels, local businesses, things to do, products, ground transportation, flights, and vacation rentals
	South Africa	Hotels, things to do, flights, products, food delivery, car hire, vacation rentals, and ground transportation
	Türkiye	Hotels, local businesses, and vacation rentals

<https://developers.google.com/search/docs/appearance/aggregator-unit>

Aggregator unit in Google Search

People in [European Economic Area \(EEA\)](#) countries who are searching for things like "hotels near me" may notice that the experience has changed. These experiences include the aggregator units and [supplier units](#). This page explains how the aggregator unit works and how businesses that serve users in EEA can express interest in the aggregator unit.

[Original illustration: Aggregator unit in Google Search]

How the aggregator unit works

The aggregator unit is a multi-provider feature designed for Vertical Search Services (VSSs), including Online Travel Agencies (OTAs), Comparison Shopping Services, metasearch engines, and directories. Aggregator units allow users to more easily see results from aggregators for queries related to hotels, flights, long-distance trains, products, and local businesses.

In this unit, eligible aggregators populate their unit with results relevant to the user's query. The top-ranked provider's results are expanded by default. For example, if a user searched for "hotels in paris", they may see a list of top-ranked hotels with photos, prices, or star ratings from an eligible provider. User clicks within the unit lead directly to that aggregator's website.

Users have the ability to select a different provider from a list of alternative aggregators, if available, expanding a unit populated with results from the new aggregator. Only one aggregator unit will show at a time.

Note: To provide visibility to direct providers, Google Search shows another unit alongside the aggregator unit. [Learn more about the supplier unit](#).

Feature availability

The aggregator unit is available to users in the [EEA](#) for queries related to hotels, flights, long distance trains or buses, products, and local businesses.

Eligibility criteria

Businesses are eligible to appear in the aggregator unit when they are approved as a Vertical Search Service (VSS), they provide the necessary data, and they meet the established quality standards to participate:

- **Express interest:** If your business serves users in EEA, and you would like to express interest in participating in the aggregator units, you can start by filling out the applicable form:
 - For ground transportation, flight, hotel, and local business queries, fill out the [Google Search Aggregator Features form](#).
 - For product queries, contact us through the [comparison shopping services contact form](#).

- **Have relevant content:** Given the aggregator unit will only appear in response to certain queries (for example, for hotels, flights, ground transportation, and local businesses), businesses must have the relevant content for the respective query to be eligible for the aggregator unit.
- **Provide data:** Eligible aggregators that wish to participate must provide data to populate a unit by direct data feed integrations or real-time APIs for queries such as flights or long distance trains and buses. For more information on the data required for the aggregator unit, see the applicable guides:
 - For hotel queries, see the [Lodging Point of Interest Feed](#) documentation.
 - For ground transportation queries, see the [Transport features API](#) documentation.
 - For flight queries, see the [Partner standard Live API](#) documentation.
 - For product queries, learn [how to start showing product pages across Google](#).
 - For local business queries, see the [Local Point of Interest Feed](#) documentation.
- **Comply with Google Search content policies:** All content in the aggregator unit must comply with Google Search's [content policies](#).

Aggregator unit best practices

The unit is designed to help users connect with aggregators when it is helpful for the query. For flight and transport queries, the unit is populated by real-time APIs. For lodging and local business queries, the unit is populated by direct data feed integrations. Just like all of our search features, we show information in the unit based on what is most helpful to users for their queries. To learn more about how we think about Search quality, you can view the Search Quality Raters guidelines, as outlined in [our blog](#).

To help deliver the best possible experience for users, we encourage you to follow these best practices for this feature:

- **Include rich entity details:** Where available, providing more comprehensive attributes helps users better evaluate options. Depending on your business type, consider including:
 - Images of the entity
 - Detailed descriptions and specifications
 - Verified user ratings and review counts
 - Specific categories (for example, "Boutique hotel" or "Eco-resort" instead of just a generic "Hotel")
 - Key amenities, features, and operating hours
- **Use accurate titles:** Factual, descriptive titles formatted in title case are easier for users to read. We recommend avoiding all caps, excessive punctuation, emoji, or promotional text (such as "BEST DEALS" or "Free shipping") in entity names.
- **Provide high-quality imagery:** Clear, original photography gives users a better sense of the place or service. For the best presentation, use images with clean backgrounds without watermarks or promotional overlay badges. Where possible, provide high-resolution photos.
- **Keep pricing and availability up to date:** Accurate data helps prevent frustrating experiences for users. Ensure prices and availability in feeds match the landing page as closely as possible, and refresh feeds regularly to remove expired or out-of-stock listings.
- **Monitor your overall performance:** Keep up with what's happening across Search with [Search Console](#).

<https://developers.google.com/search/docs/appearance/supplier-unit>

Supplier unit in Google Search

People in [European Economic Area \(EEA\)](#) countries who are searching for things like "hotels near me" may notice that the experience has changed. These experiences include the [aggregator units](#) and supplier units. This page explains how the supplier unit works and how businesses that serve users in the EEA can participate in the supplier unit.

[Original illustration: Supplier unit in Google Search]

How the supplier unit works

The supplier unit is a dedicated feature for direct providers (for example, individual hotels or airlines, brick-and-mortar business owners, or providers of services like plumbing) that appears alongside the [aggregator unit](#). The supplier unit only appears if the aggregator unit appears.

Feature availability

The supplier unit is available to users in the [EEA](#) when the [aggregator unit](#) appears for queries related to hotels, flights, long distance trains or buses, products, and local businesses.

How to participate in the supplier unit

To be eligible to appear in the supplier unit, your website must meet the following requirements:

- Your website must serve users in the EEA.
- Your business is a direct provider (for example, individual hotels or airlines, brick-and-mortar business owners, or providers of services like plumbing) that could appear for queries related to hotels, flights, long-distance trains or buses, products, and local businesses.

Note: You don't need to provide additional data beyond what's accessible through web crawling to appear in this feature. However, supplier results can be enhanced with data feeds, if available. [Learn more about supplier feeds.](#)

<https://developers.google.com/search/docs/appearance/ecosystem-carousel>

Ecosystem carousel in Google Search

To provide additional, relevant sources of information, the ecosystem carousel may appear when people search for specific information that benefit from data visualizations or interactivity with data (for example, stock charts, weather information, sports scores, translation).

[Original illustration: Ecosystem carousel in Google Search]

After clicking a tile in the carousel, the searcher is taken directly to that website. The order of websites in the carousel is determined by Search ranking algorithms based on non-discriminatory ranking principles designed to ensure users see the most relevant results for their location and language. To be eligible to appear in the carousel, check to see if you [meet the requirements](#), and if you do, [express interest in the feature](#).

Feature availability

This feature is available in [European Economic Area](#) (EEA) countries for queries related to weather, sports, finance, and translate on both desktop and mobile devices.

Requirements

To be eligible to appear in the ecosystem carousel, your website must meet the following requirements:

- Your website must serve users in the EEA.
- Your website publishes relevant finance, weather, sports information, or provides translation.
- Your website must be an authoritative source or a specialized provider in your respective domains (for example, major media outlets, or commercial weather services).

Note: No new structured data markup or special feeds are required for this specific feature.

Express interest

If your business serves users in the EEA, and you believe you are an eligible provider of finance, weather, or sports information, or provide translation, you can express interest by filling out our [interest form](#).

<https://developers.google.com/search/docs/appearance/job-sites>

Job sites: aggregator carousel and refinement chip

This page describes features for job sites in Google Search, including the aggregator carousel and refinement chips for job-related queries in [European Economic Area \(EEA\)](#) countries. Publishers don't need to add any markup to be eligible for the aggregator carousel or refinement chips.

[Original illustration: Job sites aggregator carousel and refinement chip in Google Search]

The "Job sites" aggregator carousel lets users easily see the top aggregator results for their query. Users can also click **More sites** to see additional relevant aggregator sites for their query.

The "Job sites" refinement chip can help users focus the Search page on aggregator text results. Searchers can activate the refinement chips at the top of the Search results page or by clicking **More sites** on the aggregator carousel.

Feature availability

These experiences are available to users in the [EEA](#) who are searching for job-related queries.

Learn more and express interest

If your business serves users in the EEA and you want to learn more and express interest in these experiences, you can start by filling out the [interest form](#).

<https://developers.google.com/search/docs/appearance/places-sites>

Places sites: aggregator carousel and refinement chip

This page describes features for Places sites in Google Search, including the aggregator carousel and refinement chips for local business and hotel queries in Türkiye. Publishers don't need to add any markup to be eligible for the aggregator carousel or refinement chips.

[Original illustration: Places sites aggregator carousel and refinement chip in Google Search]

The "Places sites" aggregator carousel allows users to easily see the top aggregator results for their respective query. Users can also click **More sites** to see additional relevant aggregator sites for their query.

The "Places sites" refinement chip can help users focus the Search page on aggregator text results. Searchers can activate the refinement chips at the top of the Search results page or by clicking **More sites** on the aggregator carousel.

Feature availability

These new experiences are available to users in Türkiye who are searching for queries related to local businesses and hotels.

Learn more and express interest

If your business serves users in Türkiye and you would like to learn more and express interest in these new experiences, you can start by filling out the [interest form](#).

<https://developers.google.com/search/docs/appearance/south-africa-features>

Search experiences in South Africa: badge and refinement chip

This page describes features for South Africa in Google Search, including the South African badge and South African refinement chips, for travel, products, car hire, bus booking, and food delivery searches. These experiences aim to:

- Improve the visibility of South African platforms on the search results page.
- Make it easier for users to filter and explore South African results.

"South African" badge

A "South African" badge is shown on the search results page, for applicable web results and advertisements from eligible South African platforms. This badge doesn't change the ranking of results. This badge is available for web results for travel, car hire, bus booking, products, and food delivery queries in South Africa. It's also available for advertisements in travel and shopping for eligible South African platforms.

[Original illustration: South African badge on a Google Search result]

"South African" refinement chip

Refinement chips can help users focus the Search page on specific types of web results. Searchers in South Africa can activate the "South African" refinement chip at the top of the Search results page to refine their view to eligible South African travel and shopping platforms. Eligible South African platforms don't need to add any markup to appear using the refinement chip.

[Original illustration: South African refinement chip in Google Search]

Learn more and express interest

These experiences are launched to eligible sites in South Africa only. If your business is based in South Africa, you think you're eligible for these experiences, and you'd like to express interest, you can start by filling out this [interest form](#).

<https://developers.google.com/search/docs/appearance/structured-data/carousels-beta>

Structured data carousels (beta)

Google uses [structured data](#) to understand the content on the page and show that content in a richer appearance in search results, which is called a *rich result*. This guide focuses on a [new carousel rich result that's in beta](#), which is a list-like rich result that people can scroll horizontally to see more entities from a given site (also known as a host carousel). Each tile in the carousel may have information from your site about the price, rating, and images for entities on the page.

To be eligible for this beta rich result, add `ItemList` structured data in combination with at least one one of the following supported structured data items:

- [LocalBusiness](#) and its subtypes, for example:
 - [Restaurant](#)
 - [Hotel](#)
 - [VacationRental](#)
- [Product](#)
- [Event](#)

Here's how carousels can look in Google Search when you add `ItemList` markup in combination with a supported content type:

[Original illustration: [New carousel rich result](#)]

Feature availability

This feature is in beta and you may see changes in requirements or guidelines, as we develop this feature. This feature is also only available in [European Economic Area](#) (EEA) countries, Türkiye, and South Africa, on both desktop and mobile devices. In EEA countries, this experience is available for queries related to hotels, vacation rentals, ground transportation, flights, local businesses, things to do (events, tours, and activities), and products. In Türkiye, this experience is only available for queries related to hotels, vacation rentals, and local businesses. In South Africa, this experience is available for queries related to hotels, vacation rentals, things to do (events, tours, and activities), flights, products, food delivery, car hire, and bus booking.

If your business serves users in the EEA or Türkiye, fill out the applicable form:

- For ground transportation, hotel, vacation rental, local business, and things to do (for example, events, tours, and activities) queries, use this [Google Search aggregator features interest form](#).
- For flight features, use this [flight queries interest form](#).
- For products queries in [CSS Program](#) countries, get started with the [Comparison Shopping Services \(CSS\) program](#).

If your business is based in South Africa, fill out the [Google Search South African Badging and Refinement Chips Interest Form](#).

Add structured data

Structured data is a standardized format for providing information about a page and classifying the page content. If you're new to structured data, you can learn more about [how structured data works](#).

Here's an overview of how to add structured data to your site.

1. Pick a single summary page that contains some information about every entity in the list. For example, a category page that lists the "Top hotels in Paris", with links out to specific detail pages on your site for more information about each hotel. You can mix and match different types of entities (for example, hotels, restaurants), if needed for your scenario. For example, if you have a "Things to do in Switzerland" article that lists both local events and local businesses.
2. Add the [required properties](#) to that summary page. You don't need to add markup to the detail pages in order to be eligible for this beta feature. Based on the format you're using, learn where to [insert structured data on the page](#).

Using a CMS? It may be easier to use a plugin that's integrated into your CMS.
Using JavaScript? Learn how to [generate structured data with JavaScript](#).

3. Add the required and recommended properties for the specific content type that the carousel is about:
 - [LocalBusiness](#) and its subtypes, for example:
 - [Restaurant](#)
 - [Hotel](#)
 - [VacationRental](#)
 - [Product](#)
 - [Event](#)

Depending on your scenario, you may choose the best type to use. For example, if you have a list of hotels and vacation rentals on your page, use both `Hotel` and `VacationRental` types. While it's ideal to use the type that's closest to your scenario, you can choose to use a more generic type (for example, `LocalBusiness`).

4. Follow the [guidelines](#).
5. Validate your code using the [Rich Results Test](#).
6. Deploy a few pages that include your structured data and use the [URL Inspection tool](#) to test how Google sees the page. Be sure that your page is accessible to Google and not blocked by a robots.txt file, the noindex tag, or login requirements. If the page looks okay, you can [ask Google to recrawl your URLs](#).

Note: Allow time for re-crawling and re-indexing. Remember that it may take several days after publishing a page for Google to find and crawl it.

7. To keep Google informed of future changes, we recommend that you [submit a sitemap](#). You can automate this with the [Search Console Sitemap API](#).

Guidelines

For your page to be eligible for carousel rich results (beta), you must follow the [Search Essentials](#) and [general structured data guidelines](#). In addition, the following guidelines apply to carousel rich results (beta):

- Use of generic types is allowed. However, to use recommended properties, you must use the respective types. For example, to use `amenityFeature`, use the `LodgingBusiness` type.
- Use of additional or extra fields is allowed, but may not appear in the rich result.
- Your site must have a summary page and multiple detail pages. Currently, this feature isn't designed to support other scenarios, such as an all-in-one page where the "details" are anchor points within the same page.
- The markup must be on a summary or category page, which is a list-like page that contains information about at least three entities and then links out to other pages on your site for more information on those entities. While you don't need to add markup to the detail pages, you must include the detail page URLs in your summary page's markup.
- Mark up all items that are on the summary or category page. For paginated categories, add an `ItemList` to each subsequent page and include the entities that are listed on that page. For infinite scroll, focus on marking up the entities that are initially loaded in the viewport.

Examples

The following is a high level structure of the carousel. The order specified in the markup is the order that will be used to order the tiles in the carousel rich result.


```

<html>
<head>
  <title>Top 5 Restaurants in Italy</title>
  <script type="application/ld+json">
    {
      "@context": "https://schema.org",
      "@type": "ItemList",
      "itemListElement": [
        {
          "@type": "ListItem",
          "position": 1,
          "item": {
            "@type": "Restaurant",
            "name": "Trattoria Luigi",
            "image": [
              "https://example.com/photos/1x1/photo.jpg",
              "https://example.com/photos/4x3/photo.jpg",
              "https://example.com/photos/16x9/photo.jpg"
            ],
            "priceRange": "$$$",
            "servesCuisine": "Italian",
            "aggregateRating": {
              "@type": "AggregateRating",
              "ratingValue": 4.5,
              "reviewCount": 250
            },
            "url": "https://www.example.com/trattoria-luigi"
          }
        },
        {
          "@type": "ListItem",
          "position": 2,
          "item": {
            "@type": "Restaurant",
            "name": "La Pergola",
            "image": [
              "https://example.com/photos/1x1/photo.jpg",
              "https://example.com/photos/4x3/photo.jpg",
              "https://example.com/photos/16x9/photo.jpg"
            ],
            "priceRange": "$$$",
            "servesCuisine": "Italian",
            "aggregateRating": {
              "@type": "AggregateRating",
              "ratingValue": 4.9,
              "reviewCount": 1150
            },
            "url": "https://www.example.com/la-pergola"
          }
        },
        {
          "@type": "ListItem",
          "position": 3,
          "item": {
            "@type": "Restaurant",
            "name": "Pasta e Basta",
            "image": [
              "https://example.com/photos/1x1/photo.jpg",
              "https://example.com/photos/4x3/photo.jpg",
              "https://example.com/photos/16x9/photo.jpg"
            ]
          }
        }
      ]
    }
  </script>

```

```
    ],
    "priceRange": "$$$",
    "servesCuisine": "Italian",
    "aggregateRating": {
      "@type": "AggregateRating",
      "ratingValue": 4.2,
      "reviewCount": 690
    },
    "url": "https://www.example.com/pasta-e-basta"
  }
]
}
</script>
</head>
<body>
</body>
</html>
```

```

<html>
<head>
  <title>Top 5 Restaurants in Italy</title>
  <script type="application/ld+json">
    {
      "@context": "https://schema.org",
      "@type": "ItemList",
      "itemListElement": [
        {
          "@type": "ListItem",
          "position": 1,
          "item": {
            "@type": "Restaurant",
            "name": "Trattoria Luigi",
            "image": [
              "https://example.com/photos/1x1/photo.jpg",
              "https://example.com/photos/4x3/photo.jpg",
              "https://example.com/photos/16x9/photo.jpg"
            ],
            "priceRange": "$$$",
            "servesCuisine": "Italian",
            "aggregateRating": {
              "@type": "AggregateRating",
              "ratingValue": 4.5,
              "reviewCount": 250
            },
            "url": "https://www.example.com/trattoria-luigi"
          }
        },
        {
          "@type": "ListItem",
          "position": 2,
          "item": {
            "@type": "Restaurant",
            "name": "La Pergola",
            "image": [
              "https://example.com/photos/1x1/photo.jpg",
              "https://example.com/photos/4x3/photo.jpg",
              "https://example.com/photos/16x9/photo.jpg"
            ],
            "priceRange": "$$$",
            "servesCuisine": "Italian",
            "aggregateRating": {
              "@type": "AggregateRating",
              "ratingValue": 4.9,
              "reviewCount": 1150
            },
            "url": "https://www.example.com/la-pergola"
          }
        },
        {
          "@type": "ListItem",
          "position": 3,
          "item": {
            "@type": "Restaurant",
            "name": "Pasta e Basta",
            "image": [
              "https://example.com/photos/1x1/photo.jpg",
              "https://example.com/photos/4x3/photo.jpg",
              "https://example.com/photos/16x9/photo.jpg"
            ]
          }
        }
      ]
    }
  </script>

```

```

    ],
    "priceRange": "$$$",
    "servesCuisine": "Italian",
    "aggregateRating": {
      "@type": "AggregateRating",
      "ratingValue": 4.2,
      "reviewCount": 690
    },
    "url": "https://www.example.com/pasta-e-basta"
  }
]
}
</script>
</head>
<body>
</body>
</html>

```

Structured data type definitions

You must include the required properties for your content to be eligible for display as a rich result. You can also include the recommended properties to add more information about your content, which could provide a better user experience.

ItemList

ItemList is the container item that holds all elements in the list. All URLs of the elements in the list must point to different pages on the same domain.

The full definition of **ItemList** is available at schema.org/ItemList.

Required properties	
<code>itemListElement</code>	ListItem List of items. To specify a list, define an ItemList that contains at least three <code>itemListElement.item</code> elements.
<code>itemListElement.item</code>	Subtype of LocalBusiness, Product, or Event An individual item in a list. Populate this object with: - The general properties that all carousels must have (image, url, name) - Any other properties required for this data type, as described for your content type: LocalBusiness and its subtypes Product Event Example: For a hotel, provide <code>priceRange</code> and <code>amenityFeature</code> properties.

Required properties	
itemListElement.position	Integer The item's position in the carousel. This is a 1-based number.

Common list item properties (LocalBusiness, Product, or Event)

All of the carousel item types have the following properties in common.

Required properties	
image	Repeated URL or ImageObject One or more images of the entity or item (for example, an image of the hotel). Don't include logos in this image property. Additional image guidelines: - Image URLs must be crawlable and indexable. To check if Google can access your URLs, use the URL Inspection tool. - Images must represent the marked up content. - Images must be in a file format that's supported by Google Images. - For best results, we recommend providing multiple high-resolution images (minimum of 50K pixels when multiplying width and height) with the following aspect ratios: 16x9, 4x3, and 1x1. For example: <pre>"image": ["https://example.com/photos/1x1/photo.jpg", "https://example.com/photos/4x3/photo.jpg", "https://example.com/photos/16x9/photo.jpg"]</pre>
name	Text The string name of the entity or item. For example, the name of a hotel or a vacation listing. The item.name is displayed as the title of an individual item in the carousel. HTML formatting is ignored.
url	URL The canonical URL of the item detail page (for example, the standalone page for a single hotel or vacation listing that was referenced in the summary page). All URLs in the list must be unique, but live on the same domain (the same domain, or sub or super domain as the summary page). Anchor links within a summary or category page aren't supported; your site must have standalone detail pages for each item in the list.

Recommended properties	
aggregateRating.bestRating	Number The highest value allowed in this rating system (for example, 5 / 10). If bestRating is omitted, 5 is assumed.
aggregateRating.ratingCount	Number The total number of ratings for the item on your site.
aggregateRating.ratingValue	Number or Text A numerical quality rating for the item, either a number, fraction, or percentage (for example, 4, 60%, or 6 / 10). Google understands the scale for fractions and percentages, since the scale is implied in the fraction itself or the percentage. The default scale for numbers is a 5-point scale, where 1 is the lowest value and 5 is the highest value. If another scale is intended, use bestRating and worstRating. For decimal numbers, use a dot instead of a comma to specify the value (for example 4.4 instead of 4,4). In Microdata and RDFa, you can use content attributes to override the visible content. That way, you can show the user whatever style convention you want, while also satisfying the dot requirement for structured data. For example: <pre>4,4 stars</pre>

Additional type-specific properties definitions

LocalBusiness (and subtypes)

In addition to the [ListItem properties](#), Google supports the following LocalBusiness properties (including its subtypes) for carousel rich results. Nest these properties under itemListElement.item.

Recommended properties	
amenityFeature	LocationFeatureSpecification For LodgingBusiness only: An amenity feature (for example, a characteristic or service) of the accommodation. <pre>"amenityFeature": { "@type": "LocationFeatureSpecification", "name" : "beachAccess", "value": true }</pre>
priceRange	Text

Recommended properties	
	The relative price range of a business, commonly specified by a normalized number of currency signs. Provide the price range in either of the following formats: • Price level: for example, "\$", "\$\$", "\$\$\$" • Range: for example, "\$-\$" This field must be shorter than 12 characters. If it's longer than 12 characters, Google won't show a price range for the business.
servesCuisine	Text For restaurants only: The type of cuisine the restaurant serves.

Product

In addition to the [ListItem properties](#), Google supports the following Product properties for carousel rich results. Nest these properties under `itemListElement.item`.

Recommended properties	
offers	Offer or AggregateOffer A nested Offer or AggregateOffer to sell the product. Include the recommended properties for either Offer or AggregateOffer (whichever is applicable to your content). If you're using Offer, including the following properties: • <code>offers.price</code> • <code>offers.priceCurrency</code> If you're using AggregateOffer, including the following properties: • <code>offers.highPrice</code> • <code>offers.lowPrice</code> • <code>offers.priceCurrency</code>
offers.highPrice	Number The highest price of all offers available. If you're specifying a single price with <code>price</code>, don't need to include the <code>highPrice</code> and <code>lowPrice</code> properties.
offers.lowPrice	Number The lowest price of all offers available. If you're specifying a single price with <code>price</code>, don't need to include the <code>highPrice</code> and <code>lowPrice</code> properties.
offers.price	Number The offer price of a product, or of a price component when attached to <code>PriceSpecification</code> and its subtypes. If you're specifying a price range with <code>lowPrice</code> and <code>highPrice</code>, don't include the <code>price</code> property.

Recommended properties	
offers.priceCurrency	Text The currency used to describe the product price, in three-letter ISO 4217 format. If a currency is not provided, Google defaults to USD.

Event

In addition to the [ListItem properties](#), Google supports the following Event properties for carousel rich results. Nest these properties under `itemListElement.item`.

Recommended properties	
offers	Offer or AggregateOffer A nested Offer or AggregateOffer to sell the event. Include the recommended properties for either Offer or AggregateOffer (whichever is applicable to your content). If you're using Offer, including the following properties: • offers.price • offers.priceCurrency If you're using AggregateOffer, including the following properties: • offers.highPrice • offers.lowPrice • offers.priceCurrency
offers.highPrice	Number The highest price of all offers available. If you're specifying a single price with price, don't need to include the highPrice and lowPrice properties.
offers.lowPrice	Number The lowest price of all offers available. If you're specifying a single price with price, don't need to include the highPrice and lowPrice properties.

Recommended properties	
offers.price	Number The price for your tickets, including service charges and fees. Don't forget to update it as prices change or tickets sell out. If you're specifying a price range with lowPrice and highPrice, don't include the price property. If the event is available without payment, fees, or service charges, set the price to 0. <pre>"offers": { "@type": "Offer", "price": 0 }</pre>
offers.priceCurrency	Text The currency used to describe the event price, in three-letter ISO 4217 format. If a currency is not provided, Google defaults to USD.

Examples for common scenarios

Restaurant example

Here is an example of a restaurant carousel in JSON-LD.

```

<html>
  <head>
    <title>Top 5 Restaurants in Paris</title>
    <script type="application/ld+json">
      {
        "@context": "https://schema.org",
        "@type": "ItemList",
        "itemListElement": [
          {
            "@type": "ListItem",
            "position": 1,
            "item": {
              "@type": "Restaurant",
              "name": "Trattoria Luigi",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "priceRange": "$$$",
              "servesCuisine": "Italian",
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.5,
                "reviewCount": 250
              },
              "url": "https://www.example.com/restaurant-location-1"
            }
          },
          {
            "@type": "ListItem",
            "position": 2,
            "item": {
              "@type": "Restaurant",
              "name": "La Pergola",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "priceRange": "$$$",
              "servesCuisine": "Italian",
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.9,
                "reviewCount": 1150
              },
              "url": "https://www.example.com/restaurant-location-2"
            }
          },
          {
            "@type": "ListItem",
            "position": 3,
            "item": {
              "@type": "Restaurant",
              "name": "Pasta e Basta",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ]
            }
          }
        ]
      }
    </script>
  </head>
</html>

```

```
    ],
    "priceRange": "$$$",
    "servesCuisine": "Italian",
    "aggregateRating": {
      "@type": "AggregateRating",
      "ratingValue": 4.2,
      "reviewCount": 690
    },
    "url": "https://www.example.com/restaurant-location-3"
  }
}
]
}
</script>
</head>
<body>
</body>
</html>
```

```

<html>
  <head>
    <title>Top 5 Restaurants in Paris</title>
    <script type="application/ld+json">
      {
        "@context": "https://schema.org",
        "@type": "ItemList",
        "itemListElement": [
          {
            "@type": "ListItem",
            "position": 1,
            "item": {
              "@type": "Restaurant",
              "name": "Trattoria Luigi",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "priceRange": "$$$",
              "servesCuisine": "Italian",
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.5,
                "reviewCount": 250
              },
              "url": "https://www.example.com/restaurant-location-1"
            }
          },
          {
            "@type": "ListItem",
            "position": 2,
            "item": {
              "@type": "Restaurant",
              "name": "La Pergola",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "priceRange": "$$$",
              "servesCuisine": "Italian",
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.9,
                "reviewCount": 1150
              },
              "url": "https://www.example.com/restaurant-location-2"
            }
          },
          {
            "@type": "ListItem",
            "position": 3,
            "item": {
              "@type": "Restaurant",
              "name": "Pasta e Basta",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ]
            }
          }
        ]
      }
    </script>
  </head>
</html>

```

```
    ],
    "priceRange": "$$$",
    "servesCuisine": "Italian",
    "aggregateRating": {
      "@type": "AggregateRating",
      "ratingValue": 4.2,
      "reviewCount": 690
    },
    "url": "https://www.example.com/restaurant-location-3"
  }
}
]
}
</script>
</head>
<body>
</body>
</html>
```

Lodging (Hotels and VacationRental) example

Here is an example of a lodging carousel in JSON-LD.

```

<html>
  <head>
    <title>Top 5 Hotels in Paris</title>
    <script type="application/ld+json">
      {
        "@context": "https://schema.org",
        "@type": "ItemList",
        "itemListElement": [
          {
            "@type": "ListItem",
            "position": 1,
            "item": {
              "@type": "Hotel",
              "name": "Four Seasons Hotel George V, Paris",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "priceRange": "$$$$ ",
              "amenityFeature": {
                "@type": "LocationFeatureSpecification",
                "name": "internetType",
                "value": "Free"
              },
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.9,
                "reviewCount": 50
              },
              "url": "https://www.example.com/four-seasons"
            }
          },
          {
            "@type": "ListItem",
            "position": 2,
            "item": {
              "@type": "VacationRental",
              "name": "Downtown Condo",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "priceRange": "$$",
              "amenityFeature": {
                "@type": "LocationFeatureSpecification",
                "name": "instantBookable",
                "value": true
              },
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.7,
                "reviewCount": 827
              },
              "url": "https://www.example.com/downtown-condo"
            }
          },
          {
            "@type": "ListItem",

```

```

        "position": 3,
        "item": {
            "@type": "Hotel",
            "name": "Ritz Paris",
            "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
            ],
            "priceRange": "$$$$ ",
            "amenityFeature": {
                "@type": "LocationFeatureSpecification",
                "name" : "freeBreakfast",
                "value": true
            },
            "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.9,
                "reviewCount": 1290
            },
            "url": "https://www.example.com/ritz-paris"
        }
    ]
}
</script>
</head>
<body>
</body>
</html>

```

```

<html>
  <head>
    <title>Top 5 Hotels in Paris</title>
    <script type="application/ld+json">
      {
        "@context": "https://schema.org",
        "@type": "ItemList",
        "itemListElement": [
          {
            "@type": "ListItem",
            "position": 1,
            "item": {
              "@type": "Hotel",
              "name": "Four Seasons Hotel George V, Paris",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "priceRange": "$$$$ ",
              "amenityFeature": {
                "@type": "LocationFeatureSpecification",
                "name": "internetType",
                "value": "Free"
              },
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.9,
                "reviewCount": 50
              },
              "url": "https://www.example.com/four-seasons"
            }
          },
          {
            "@type": "ListItem",
            "position": 2,
            "item": {
              "@type": "VacationRental",
              "name": "Downtown Condo",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "priceRange": "$$",
              "amenityFeature": {
                "@type": "LocationFeatureSpecification",
                "name": "instantBookable",
                "value": true
              },
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.7,
                "reviewCount": 827
              },
              "url": "https://www.example.com/downtown-condo"
            }
          },
          {
            "@type": "ListItem",

```



```

    "position": 3,
    "item": {
      "@type": "Hotel",
      "name": "Ritz Paris",
      "image": [
        "https://example.com/photos/1x1/photo.jpg",
        "https://example.com/photos/4x3/photo.jpg",
        "https://example.com/photos/16x9/photo.jpg"
      ],
      "priceRange": "$$$$ ",
      "amenityFeature": {
        "@type": "LocationFeatureSpecification",
        "name" : "freeBreakfast",
        "value": true
      },
      "aggregateRating": {
        "@type": "AggregateRating",
        "ratingValue": 4.9,
        "reviewCount": 1290
      },
      "url": "https://www.example.com/ritz-paris"
    }
  ]
}
</script>
</head>
<body>
</body>
</html>

```

Things to do example example

Here is an example of a things to do carousel in JSON-LD.

```

<html>
  <head>
    <title>Top 5 Things To Do in Paris</title>
    <script type="application/ld+json">
      {
        "@context": "https://schema.org",
        "@type": "ItemList",
        "itemListElement": [
          {
            "@type": "ListItem",
            "position": 1,
            "item": {
              "@type": "Event",
              "name": "Paris Seine River Dinner Cruise",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "offers": {
                "@type": "Offer",
                "price": 45.00,
                "priceCurrency": "EUR"
              },
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.2,
                "reviewCount": 690
              },
              "url": "https://www.example.com/event-location1"
            }
          },
          {
            "@type": "ListItem",
            "position": 2,
            "item": {
              "@type": "LocalBusiness",
              "name": "Notre-Dame Cathedral",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "priceRange": "$",
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.8,
                "reviewCount": 4220
              },
              "url": "https://www.example.com/localbusiness-location"
            }
          },
          {
            "@type": "ListItem",
            "position": 3,
            "item": {
              "@type": "Event",
              "name": "Eiffel Tower With Host Summit Tour",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",

```

```
        "https://example.com/photos/4x3/photo.jpg",
        "https://example.com/photos/16x9/photo.jpg"
    ],
    "offers": {
        "@type": "Offer",
        "price": 59.00,
        "priceCurrency": "EUR"
    },
    "aggregateRating": {
        "@type": "AggregateRating",
        "ratingValue": 4.9,
        "reviewCount": 652
    },
    "url": "https://www.example.com/event-location2"
}
}
]
}
</script>
</head>
<body>
</body>
</html>
```

```

<html>
  <head>
    <title>Top 5 Things To Do in Paris</title>
    <script type="application/ld+json">
      {
        "@context": "https://schema.org",
        "@type": "ItemList",
        "itemListElement": [
          {
            "@type": "ListItem",
            "position": 1,
            "item": {
              "@type": "Event",
              "name": "Paris Seine River Dinner Cruise",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "offers": {
                "@type": "Offer",
                "price": 45.00,
                "priceCurrency": "EUR"
              },
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.2,
                "reviewCount": 690
              },
              "url": "https://www.example.com/event-location1"
            }
          },
          {
            "@type": "ListItem",
            "position": 2,
            "item": {
              "@type": "LocalBusiness",
              "name": "Notre-Dame Cathedral",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "priceRange": "$",
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.8,
                "reviewCount": 4220
              },
              "url": "https://www.example.com/localbusiness-location"
            }
          },
          {
            "@type": "ListItem",
            "position": 3,
            "item": {
              "@type": "Event",
              "name": "Eiffel Tower With Host Summit Tour",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",

```

```
        "https://example.com/photos/4x3/photo.jpg",
        "https://example.com/photos/16x9/photo.jpg"
    ],
    "offers": {
        "@type": "Offer",
        "price": 59.00,
        "priceCurrency": "EUR"
    },
    "aggregateRating": {
        "@type": "AggregateRating",
        "ratingValue": 4.9,
        "reviewCount": 652
    },
    "url": "https://www.example.com/event-location2"
}
}
]
}
</script>
</head>
<body>
</body>
</html>
```

Product example

Here is an example of a product carousel in JSON-LD.

```

<html>
  <head>
    <title>Top coats of the season</title>
    <script type="application/ld+json">
      {
        "@context": "https://schema.org",
        "@type": "ItemList",
        "itemListElement": [
          {
            "@type": "ListItem",
            "position": 1,
            "item": {
              "@type": "Product",
              "name": "Puffy Coat Series by Goat Coat",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "offers": {
                "@type": "AggregateOffer",
                "lowPrice": 45.00,
                "highPrice": 60.00,
                "priceCurrency": "EUR"
              },
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.9,
                "reviewCount": 50
              },
              "url": "https://www.example.com/puffy-coats"
            }
          },
          {
            "@type": "ListItem",
            "position": 2,
            "item": {
              "@type": "Product",
              "name": "Wool Coat Series by Best Coats Around",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "offers": {
                "@type": "AggregateOffer",
                "lowPrice": 189.00,
                "highPrice": 200.00,
                "priceCurrency": "EUR"
              },
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.7,
                "reviewCount": 827
              },
              "url": "https://www.example.com/wool-coats"
            }
          },
          {
            "@type": "ListItem",

```

```
    "position": 3,
    "item": {
      "@type": "Product",
      "name": "Antarctic Coat by Cold Coats",
      "image": [
        "https://example.com/photos/1x1/photo.jpg",
        "https://example.com/photos/4x3/photo.jpg",
        "https://example.com/photos/16x9/photo.jpg"
      ],
      "offers": {
        "@type": "Offer",
        "price": 45.00,
        "priceCurrency": "EUR"
      },
      "aggregateRating": {
        "@type": "AggregateRating",
        "ratingValue": 4.9,
        "reviewCount": 1290
      },
      "url": "https://www.example.com/antarctic-coat"
    }
  ]
}
</script>
</head>
<body>
</body>
</html>
```

```

<html>
  <head>
    <title>Top coats of the season</title>
    <script type="application/ld+json">
      {
        "@context": "https://schema.org",
        "@type": "ItemList",
        "itemListElement": [
          {
            "@type": "ListItem",
            "position": 1,
            "item": {
              "@type": "Product",
              "name": "Puffy Coat Series by Goat Coat",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "offers": {
                "@type": "AggregateOffer",
                "lowPrice": 45.00,
                "highPrice": 60.00,
                "priceCurrency": "EUR"
              },
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.9,
                "reviewCount": 50
              },
              "url": "https://www.example.com/puffy-coats"
            }
          },
          {
            "@type": "ListItem",
            "position": 2,
            "item": {
              "@type": "Product",
              "name": "Wool Coat Series by Best Coats Around",
              "image": [
                "https://example.com/photos/1x1/photo.jpg",
                "https://example.com/photos/4x3/photo.jpg",
                "https://example.com/photos/16x9/photo.jpg"
              ],
              "offers": {
                "@type": "AggregateOffer",
                "lowPrice": 189.00,
                "highPrice": 200.00,
                "priceCurrency": "EUR"
              },
              "aggregateRating": {
                "@type": "AggregateRating",
                "ratingValue": 4.7,
                "reviewCount": 827
              },
              "url": "https://www.example.com/wool-coats"
            }
          },
          {
            "@type": "ListItem",

```



```

    "position": 3,
    "item": {
      "@type": "Product",
      "name": "Antarctic Coat by Cold Coats",
      "image": [
        "https://example.com/photos/1x1/photo.jpg",
        "https://example.com/photos/4x3/photo.jpg",
        "https://example.com/photos/16x9/photo.jpg"
      ],
      "offers": {
        "@type": "Offer",
        "price": 45.00,
        "priceCurrency": "EUR"
      },
      "aggregateRating": {
        "@type": "AggregateRating",
        "ratingValue": 4.9,
        "reviewCount": 1290
      },
      "url": "https://www.example.com/antarctic-coat"
    }
  ]
}
</script>
</head>
<body>
</body>
</html>

```

Troubleshooting

If you're having trouble implementing or debugging structured data, here are some resources that may help you.

- If you're using a content management system (CMS) or someone else is taking care of your site, ask them to help you. Make sure to forward any Search Console message that details the issue to them.
- Google does not guarantee that features that consume structured data will show up in search results. For a list of common reasons why Google may not show your content in a rich result, see the [General Structured Data Guidelines](#).
- You might have an error in your structured data. Check the [list of structured data errors](#) and the [Unparsable structured data report](#).
- If you received a structured data manual action against your page, the structured data on the page will be ignored (although the page can still appear in Google Search results). To fix [structured data issues](#), use the [Manual Actions report](#).
- Review the [guidelines](#) again to identify if your content isn't compliant with the guidelines. The problem can be caused by either spammy content or spammy markup usage. However, the issue may not be a syntax issue, and so the Rich Results Test won't be able to identify these issues.
- Structured data issues can affect how your site's content appears in search results. Use this [Troubleshoot missing rich results / drop in total rich results](#) guide to review a step-by-step approach to identify, fix, and validate these issues in Search Console.
- Allow time for re-crawling and re-indexing. Remember that it may take several days after publishing a page for Google to find and crawl it. For general questions about crawling and indexing, check the [Google Search crawling and indexing FAQ](#).

- Post a question in the [Google Search Central forum](#).

<https://developers.google.com/search/docs/appearance/search-profiles>

Add a Search profile badge to your website

If you're a website owner, publisher, or creator with a Search profile, you can add a badge to your website to help your audience find your Search profile.

[Original illustration: Example of a Search profile badge on a website and a Search profile on Google]

How Search profiles work on Google

A [Search profile](#) brings together your content from across the web and social platforms into a single destination on Google. When readers follow your Search profile, it makes your content that's linked on your Search profile (such as your content on Instagram, TikTok, YouTube, X, Facebook, and your website) more likely to appear for your audience on [Google Discover](#).

Add a Search profile badge to your website

After you've [claimed your Search profile](#), you can encourage your audience to find your content from across the web by adding a Search profile badge to your website:

1. Find your Search profile URL. Use the following URL pattern, replacing *handle* with your profile handle:

```
https://profile.google.com/@handle
```

2. [Download the Search profile badge assets](#) and follow [brand guidelines](#) when deciding where to place the badge on your site.
3. Add the following HTML snippet to your site:

```
<a href="https://profile.google.com/@example"
  aria-label="Find us on Google Search">
  
</a>
```

Alternative option: If you prefer a text link in article bylines, newsletters, or bio links, use the following snippet:

```
<a href="https://profile.google.com/@example">Find us on Google Search</a>
```

Follow brand guidelines and best practices

When adding a Search profile badge to your site, follow these brand guidelines and best practices:

- **Ensure accessible touch targets:** Keep the clickable target at least 48 × 48 dp on Android and 44 × 44 px on iOS and web.
- **Preserve logo integrity:** If you're using the official assets from Google, don't stretch, distort, rotate, or alter the colors of the "Super G" icon and Search profile button.
- **Avoid mixing and matching monochrome badges with colorful badges:** For example, if you're using monochrome for all icons, use the monochrome "G" icon.
- **Maintain badge clarity:** If you want to use both the Search profile badge and [Preferred Sources badge](#) together, use the higher emphasis Search profile button with the label for clarity and distinction from the Preferred Sources badge. Don't use the "Super G" icon for the Search profile paired with the Preferred Sources badge.

Troubleshooting

If you're having trouble adding the badge to your website, post a question in the [Google Search Central Help Community](#).

<https://developers.google.com/search/docs/appearance/site-names>

Provide a site name to Google Search

When Google lists a page in search results, it shows the name of the site the page comes from. This is called the site name. Note that the site name is different from the per-page [title links](#) (title links are specific to each web page, whereas the site name is for the entire site).

Feature availability

Site names are available in all languages where Google Search is available, on both mobile and desktop. Site names can appear for domain-level and subdomain-level sites (learn more in the [technical guidelines](#)).

How site names in Google Search are created

Google's generation of site names on the Google Search results page is completely automated and takes into account content from a site's home page and references to it that appear on the web. The goal of the site name in Google Search is to best represent and describe the source of each result.

To indicate your site name preference, add [WebSite structured data](#) to your home page. Our site name system will also consider content in `og:site_name`, `<title>`, heading elements, and other text on a home page. However, WebSite structured data is most important, if you want to specify a preference.

While we can't manually change automatically selected site names, you can [indicate alternatives](#) for our automated system to consider if your primary preference isn't selected.

Choosing your site name

- **Choose a unique name** that accurately reflects the identity of your site and isn't misleading for users. The name you choose must follow [Search content policies](#).
- **Use a concise, commonly-recognized name** for your site (for example, "Google" instead of "Google, Inc"). While there's no limit to how long a site name can be, long site names may be truncated on some devices.
- **Avoid using a generic name.** A generic name like "Best Dentists In Iowa" is unlikely to be selected by our system as a site name, unless that's an extremely well-recognized brand name.
- **Use your site name consistently across your home page.** Make sure whatever you use as the site name in structured data is consistent with how you refer to your site in [other sources](#) on your home page that our system considers.

- **Provide an alternative name.** While our site name system tries to use your preferred site name, sometimes the name isn't available for use. For example, our system generally won't use the same site name for two different sites that are global in nature. In other cases, our system might determine that a site is more commonly recognized by an acronym than by a full name. Providing an alternative name using the `alternateName` property allows Google to consider other options if your preferred choice isn't selected.

How to add a site name with structured data

[Original embedded video]

Structured data is a standardized format for providing information about a page and classifying the page content. If you're new to structured data, you can learn more about [how structured data works](#).

This section goes over the technical guidelines, required properties, and how to add and test site name structured data.

Using a CMS? It may be easier to use a plugin that's integrated into your CMS.

Using JavaScript? Learn how to [generate structured data with JavaScript](#).

Follow the guidelines

To help Google better understand your site name, make sure you follow the [Search Essentials](#), [General structured data guidelines](#), guidelines for [choosing your site name](#), and the following technical guidelines:

Technical guidelines

- **Only one name per site:** Currently, Google Search only supports one site name per site, where a *site* is defined by the domain or subdomain. Google Search does not support site names at the subdirectory level. Note that subdomain names starting with `www` or `m` are generally considered as being equivalent.
Supported: `https://example.com` (this is a domain-level home page)
Supported: `https://www.example.com` (this is also considered to be a domain-level home page)
Supported: `https://m.example.com` (this is also considered to be a domain-level home page)
Supported: `https://news.example.com` (this is a subdomain-level home page)
Not supported: `https://example.com/news` (this is a subdirectory-level home page)
- **Structured data must be on the home page of a site:** The [WebSite structured data](#) must be on the home page of the site. By home page, we mean the domain or subdomain level root URI. For example, `https://example.com` is the home page of the domain, while `https://example.com/de/index.html` isn't the home page.

Note: If there's no structured data on the subdomain's home page, then the domain-level site name may be used for the subdomain as a fallback.

- **The home page must be crawlable by Google:** If we don't have access to the content on your home page because it is [blocked](#), we may not be able to generate a site name.
- **For sites with duplicate home pages:** If you have duplicate home pages for the same content (for example, HTTP and HTTPS versions of your home page, or `www` and non-`www`), make sure that you're using the same structured data on all page duplicates, not just on the canonical page.

- If you already have **WebSite** structured data on your site, make sure that you nest the site name properties in the same node. In other words, avoid creating an additional **WebSite** structured data block on your home page if you can help it.

Add required site name properties

Add the required properties to the home page of your website, in either JSON-LD, RDFa, or microdata format. You don't need to include this markup on every page of your site; you only need to add this markup to the home page of your site.

Required properties	
name	Text The name of the website. Make sure the name meets the guidelines for choosing your site name.
url	URL The URL of the home page of the site. Set this to the canonical home page of your site's domain or subdomain. For example, <code>https://example.com/</code> or <code>https://news.example.com/</code>.

Here's an example of **WebSite** structured data that includes the required fields:

JSON-LD

```
<html>
  <head>
    <title>Example: A Site about Examples</title>
    <script type="application/ld+json">
      {
        "@context" : "https://schema.org",
        "@type" : "WebSite",
        "name" : "Example",
        "url" : "https://example.com/"
      }
    </script>
  </head>
  <body>
  </body>
</html>
```

Microdata

```
<html>
  <head>
    <title>Example: A Site about Examples</title>
  </head>
  <body>
    <div itemscope itemtype="https://schema.org/WebSite">
      <link itemprop="url" href="https://example.com" />
      <meta itemprop="name" content="Example"/>
    </div>
  </body>
</html>
```

Add an alternative site name

If you want to provide an alternate version of your site name (for example, an acronym or shorter name), you can do this by adding the `alternateName` property. This is optional.

Recommended properties	
alternateName	Text The alternate name of the website (for example, if there's a commonly recognized acronym or shorter name for your site), if applicable. Make sure the name meets the guidelines for choosing your site name. You can list more than one alternative name. Specify them in order of your preference, with the most important one listed first. For example: <pre><script type="application/ld+json"> { "@context": "https://schema.org", "@type": "WebSite", "name": "Burnt Toast", "alternateName": ["BT", "B-T", "Burnt Toast Shop"], "url": "https://www.example.com/" } </script></pre>

Here's an example of `WebSite` structured data that includes all required and recommended fields:

JSON-LD


```
<html>
  <head>
    <title>Example: A Site about Examples</title>
    <script type="application/ld+json">
      {
        "@context" : "https://schema.org",
        "@type" : "WebSite",
        "name" : "Example Company",
        "alternateName" : "EC",
        "url" : "https://example.com/"
      }
    </script>
  </head>
  <body>
  </body>
</html>
```

Microdata

```
<html>
  <head>
    <title>Example: A Site about Examples</title>
  </head>
  <body>
    <div itemscope itemtype="https://schema.org/WebSite">
      <link itemprop="url" href="https://example.com" />
      <meta itemprop="name" content="Example Company"/>
      <meta itemprop="alternateName" content="EC"/>
    </div>
  </body>
</html>
```

Test structured data

1. Validate your markup with a schema testing tool (for example, [Schema Markup Validator](#)) to make sure there are no syntax errors. Site names aren't supported in the Rich Results Test.
2. Use the [URL Inspection tool](#) to test how Google sees the page. Be sure that your home page is accessible to Google and not blocked by a robots.txt file, noindex, or login requirements.
3. If the page looks okay, you can [ask Google to recrawl your URL](#).

Allow time for re-crawling and re-indexing. Remember that it can take anywhere from a few days to a few weeks after publishing a page for Google to find and crawl it.

What to do if your preferred site name isn't selected

Our system generally tries to use a preferred site name from WebSite structured data, when indicated. However, if our system is less confident in a name you provided, it may sometimes generate site names using [other sources](#) or show a domain or subdomain name.

If your preferred site name is not selected by our automated system, try these steps:

1. Verify the following:

- The site name that's in the [WebSite structured data](#) on your home page is the preferred name for your site.
 - Your WebSite structured data doesn't have [structured data errors](#). Use a schema testing tool (for example, [Schema Markup Validator](#)) to make sure that there are no syntax errors (the Rich Results Test doesn't support site names).
 - Your structured data [follows our guidelines](#).
 - Confirm that [other sources](#) on your home page also use the preferred name for your site.
 - Confirm that you're not trying to set up a site name for a subdirectory. Site names aren't supported for subdirectories (for example, <https://example.com/news> is a subdirectory-level home page and cannot have its own site name). See our [technical guidance](#) for more information.
2. Make sure your redirects are working as intended and that Googlebot can access the redirect target. Then [request a recrawl of that page](#). If your page redirects to a page that's visible to Googlebot, the site name will reflect the redirect target.
3. If you have multiple versions of your site (for example, HTTP and HTTPS), make sure that you're using the same site name consistently.
4. If you've updated your site name structured data, allow time for Google to recrawl and process the new information. Remember that crawling can take anywhere from several days to several weeks, depending on how often our systems determine content needs to be refreshed. You can request a recrawl of a page by using the [URL Inspection tool](#).

Not seeing your preferred site name for internal pages? If your home page is already showing your preferred site name, remember to also allow time for Google to recrawl and process your internal pages.

If you've followed the guidance and your preferred site name still isn't selected, consider one of these options:

1. **First, try providing an alternative name using the `alternateName` property.** If our site name system isn't confident enough to use your preferred name, it strongly considers this option.
2. **Provide your domain or subdomain name as a backup option.** To provide your domain or subdomain as a backup option, add your domain or subdomain name as your [alternative name](#). Your domain or subdomain needs to be in all lowercase (for example, [example.com](#) not [Example.com](#)) for our system to detect this as a site name preference. Our system will strongly consider using it if your preferred name isn't selected. In this example, [Burnt Toast](#) is the most preferred option, followed by [BT](#), and ending with the domain [example.com](#) as the final name preference:

```
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "WebSite",
  "name": "Burnt Toast",
  "alternateName": ["BT", "B-T", "Burnt Toast Shop", "example.com"],
  "url": "https://www.example.com/"
}
</script>
```

3. **If that's still not working, then try providing your domain or subdomain name (in all lowercase) as your [preferred name](#), as a last-resort workaround option.** If you provide your domain or subdomain name as your preferred name, our system will generally select that (but we recommend only doing this as a last resort). In this example, the only preference is the domain [example.com](#):

```
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "WebSite",
  "name": "example.com",
  "url": "https://www.example.com/"
}
</script>
```

If you tried the [troubleshooting steps](#) and still find issues, post in the [Google Search Central Help Community](#). This will help us look for potential improvements to make in our system.

<https://developers.google.com/search/docs/appearance/sitelinks>

Sitelinks

Sitelinks are links from the same domain that are clustered together under a text result. Our systems analyze the link structure of your site to find shortcuts that will save users time and allow them to quickly find the information they're looking for.

We only show sitelinks for results when we think they'll be useful to the user. If the structure of your site doesn't allow our algorithms to find good sitelinks, or we don't think that the sitelinks for your site are relevant for the user's query, we won't show them.

Sitelinks best practices

At the moment, sitelinks are automated. We're always working to improve our sitelinks algorithms, and we may incorporate site owner input in the future. There are best practices you can follow, however, to improve the quality of your sitelinks.

- Make sure that the text you use as your page titles and in your headings is informative, relevant, and compact.
- Create a logical site structure that is easy for users to navigate, and make sure you link to your important pages from other relevant pages.
- Ensure that your internal links' anchor text is concise and relevant to the page they're pointing to.
- Avoid repetitions in your content.

If you need to remove a sitelink, consider removing the page from your site or using [noindex](#).

<https://developers.google.com/search/docs/appearance/snippet>

Control your snippets in search results

A *snippet* is the description or summary part of search result on Google Search and other properties (for example, Google News). Google primarily uses the content on the page to automatically determine the appropriate snippet. We may also use descriptive information in the [meta description](#) element when it describes the page better than other parts of the content.

While we can't manually change snippets for individual sites, we're always working to make them as relevant as possible. You can help improve the quality of the snippet displayed for your pages by following the [best practices for creating quality meta descriptions](#).

How snippets are created

Snippets are automatically created from page content. Snippets are designed to emphasize and preview the page content that best relates to a user's specific search. This means that Google Search might show different snippets for different searches.

Snippets are primarily created from the page content itself. However, Google sometimes uses the [meta description](#) HTML element if it might give users a more accurate description of the page than content taken directly from the page.

How to prevent snippets or adjust snippet length

You can prevent snippets from being created and shown for your site in search results, or let Google know about the maximum lengths that you want your snippets to be. To prevent Google from displaying a snippet for your page in search results, use the [nosnippet meta tag](#). To specify the maximum length for your snippets, use the [max-snippet:\[number\]](#) meta tag. You can also prevent certain parts of the page from being shown in a snippet by using the [data-nosnippet](#) attribute.

Best practices for creating quality meta descriptions

Google will sometimes use the `<meta name="description">` tag from a page to generate a snippet in search results, if we think it gives users a more accurate description than would be possible purely from the on-page content. A meta description tag generally informs and interests users with a short, relevant summary of what a particular page is about. They are like a pitch that convince the user that the page is exactly what they're looking

for. There's no limit on how long a meta description can be, but the snippet is truncated in Google Search results as needed, typically to fit the device width.

If you use a CMS, such as Wix, WordPress, or Blogger, you might not be able to edit your HTML directly, or you might prefer not to. Instead, your CMS might have a search engine settings page or some other mechanism to tell search engines about meta tags.

If you want to add a meta tag to your website, search for instructions about modifying the <head> of your page on your CMS (for example, search for "wix add meta tags").

Create unique descriptions for each page on your site

Identical or similar descriptions on every page of a site aren't helpful when individual pages appear in search results. Wherever possible, create descriptions that accurately describe the specific page. Use site-level descriptions on the main home page or other aggregation pages, and use page-level descriptions everywhere else. If you don't have time to create a description for every single page, try to prioritize your content; at the very least, create a description for the critical URLs like your home page and popular pages.

Include relevant information about the content in the description

The meta description doesn't just have to be in sentence format; it's also a great place to include information about the page. For example, news or blog postings can list the author, date of publication, or byline information. This can give potential visitors very relevant information that might not be displayed in the snippet otherwise. Similarly, product pages might have the key bits of information—price, age, manufacturer—scattered throughout a page. A good meta description can bring all this data together.

For example, the following meta description provides detailed information about a book, and information is clearly tagged and separated:

```
<meta name="description" content="Written by A.N. Author, Illustrated by V. Gogh, Price: $17.99, Length: 784 pages">
```

Programmatically generate descriptions

For some sites, like news media sources, generating an accurate and unique description for each page is easy: since each article is hand-written, it takes minimal effort to also add a one-sentence description. For larger database-driven sites, like product aggregators, hand-written descriptions can be impossible. In the latter case, however, programmatic generation of the descriptions can be appropriate and are encouraged. Good descriptions are human-readable and diverse. Page-specific data is a good candidate for programmatic generation.

Keep in mind that meta descriptions comprised of long strings of keywords don't give users a clear idea of the page's content, and are less likely to be displayed as a snippet.

Use quality descriptions

Make sure your descriptions are truly descriptive. Because meta descriptions aren't displayed in the pages the user sees, it's easy to let this content slide. But high-quality descriptions can be displayed in Google's search results, and can go a long way to improving the quality and quantity of your search traffic.

Here are some examples of how a meta description can be improved:

Bad (list of keywords):

```
<meta name="description" content="Sewing supplies, yarn, colored pencils, sewing machines, threads, bobbins, needles">
```

Better (explains what the shop sells and details like opening hours and location):

```
<meta name="description" content="Get everything you need to sew your next garment. Open Monday-Friday 8-5pm, located in the Fashion District.">
```

Bad (same description used for every news article):

```
<meta name="description" content="Local news in Whoville, delivered to your doorstep. Find out what happened today.">
```

Better (uses a snippet from the specific news article):

```
<meta name="description" content="Upsetting the small town of Whoville, a local elderly man steals everyone's presents the night before an important event. Stay tuned for live updates on the matter.">
```

Bad (doesn't summarize the page):

```
<meta name="description" content="Eggs are a source of joy in everyone's life. When I was a small child, I remember picking eggs from the hen house and bringing them to the kitchen. Those were the days.">
```

Better (summarizes the whole page):

```
<meta name="description" content="Learn how to cook eggs with this complete guide in 1 hour or less. We cover all the methods, including: over-easy, sunny side up, boiled, and poached.">
```

Bad (too short):

```
<meta name="description" content="Mechanical pencil">
```

Better (specific and detailed):

```
<meta name="description" content="Self-sharpening mechanical pencil that autocorrects your penmanship. Includes 2B auto-replenishing lead. Available in both Vintage Pink and Schoolbus Yellow. Order 50+ pencils, get free shipping.">
```

Best practices for "Read more" deep links in Google Search

A "Read more" deep link is a link within a snippet that leads users to a specific section on that page.

[Original illustration: A read more deep link in Google Search]

To increase the likelihood that "read more" deep links appear for your site in Google Search, follow these best practices:

- Make sure content is immediately visible on the page to a human (and not hidden behind an expandable section or tabbed interface, for example).
- Avoid using JavaScript to control the user's scroll position on page load (for example, don't force the user's scroll position to the top of the page).
- If you make [history API](#) calls or [window.location.hash](#) modifications on page load, make sure you don't remove the hash fragment from the URL, as this breaks deep linking behavior.

<https://developers.google.com/search/docs/appearance/structured-data/intro-structured-data>

Introduction to structured data markup in Google Search

[Original embedded video]

Google Search works hard to understand the content of a page. You can help us by providing explicit clues about the meaning of a page to Google by including structured data on the page. Structured data is a standardized format for providing information about a page and classifying the page content; for example, on a recipe page, what are the ingredients, the cooking time and temperature, the calories, and so on.

Why add structured data to a page?

Adding structured data can enable search results that are more engaging to users and might encourage them to interact more with your website, which are called *rich results*. Here are some case studies of websites that have implemented structured data for their site:

- Rotten Tomatoes added structured data to 100,000 unique pages and measured a 25% higher click-through rate for pages enhanced with structured data, compared to pages without structured data.
- The Food Network has converted 80% of their pages to enable search features, and has seen a 35% increase in visits.
- Rakuten has found that users spend 1.5x more time on pages that implemented structured data than on non-structured data pages, and have a 3.6x higher interaction rate on AMP pages with search features vs non-feature AMP pages.
- Nestlé has measured pages that show as rich results in search have an 82% higher click through rate than non-rich result pages.

Read more [case studies from sites that have implemented structured data](#).

How structured data works in Google Search

Google uses structured data that it finds on the web to understand the content of the page, as well as to gather information about the web and the world in general, such as information about the people, books, or companies that are included in the markup. For example, when a recipe page has [JSON-LD](#) structured data (describing the title of the recipe, the author of the recipe, and other details), Google Search can use that information to display a rich result for the recipe:

[Original illustration: How a recipe web page's structured data can influence a rich result in Google Search]

Because the structured data labels each individual element of the recipe, users can search for your recipe by ingredient, calorie count, cook time, and so on.

If you use a CMS, such as Wix, WordPress, or Shopify, you might not be able to edit your HTML directly. Instead, your CMS may have a search engine settings page, or you may be able to install a plugin that allows

you to specify structured data. Search for instructions about adding structured data to your CMS (for example, search for "wix structured data" or "wordpress structured data plugin").

[Original embedded video]

Structured data is coded using in-page markup on the page that the information applies to. The structured data on the page describes the content of that page. Don't create blank or empty pages just to hold structured data, and don't add structured data about information that is not visible to the user, even if the information is accurate. For more technical and quality guidelines, see the [Structured data general guidelines](#).

The [Rich Results Test](#) is an easy and useful tool for validating your structured data, and in some cases, previewing a feature in Google Search. Try it out:

```

<html>
  <head>
    <title>Non-Alcoholic Piña Colada</title>
    <script type="application/ld+json">
      {
        "@context": "https://schema.org/",
        "@type": "Recipe",
        "name": "Non-Alcoholic Piña Colada",
        "image": [
          "https://example.com/photos/1x1/photo.jpg",
          "https://example.com/photos/4x3/photo.jpg",
          "https://example.com/photos/16x9/photo.jpg"
        ],
        "author": {
          "@type": "Person",
          "name": "Mary Stone"
        },
        "datePublished": "2024-03-10",
        "description": "This non-alcoholic pina colada is everyone's favorite!",
        "recipeCuisine": "American",
        "prepTime": "PT1M",
        "cookTime": "PT2M",
        "totalTime": "PT3M",
        "keywords": "non-alcoholic",
        "recipeYield": "4 servings",
        "recipeCategory": "Drink",
        "nutrition": {
          "@type": "NutritionInformation",
          "calories": "120 calories"
        },
        "aggregateRating": {
          "@type": "AggregateRating",
          "ratingValue": 5,
          "ratingCount": 18
        },
        "recipeIngredient": [
          "400ml of pineapple juice",
          "100ml cream of coconut",
          "ice"
        ],
        "recipeInstructions": [
          {
            "@type": "HowToStep",
            "name": "Blend",
            "text": "Blend 400ml of pineapple juice and 100ml cream of coconut until smooth.",
            "url": "https://example.com/non-alcoholic-pina-colada#step1",
            "image": "https://example.com/photos/non-alcoholic-pina-colada/step1.jpg"
          },
          {
            "@type": "HowToStep",
            "name": "Fill",
            "text": "Fill a glass with ice.",
            "url": "https://example.com/non-alcoholic-pina-colada#step2",
            "image": "https://example.com/photos/non-alcoholic-pina-colada/step2.jpg"
          },
          {
            "@type": "HowToStep",
            "name": "Pour",
            "text": "Pour the pineapple juice and coconut mixture over ice.",
            "url": "https://example.com/non-alcoholic-pina-colada#step3",

```

```

        "image": "https://example.com/photos/non-alcoholic-pina-colada/step3.jpg"
      }
    ],
    "video": {
      "@type": "VideoObject",
      "name": "How to Make a Non-Alcoholic Piña Colada",
      "description": "This is how you make a non-alcoholic piña colada.",
      "thumbnailUrl": [
        "https://example.com/photos/1x1/photo.jpg",
        "https://example.com/photos/4x3/photo.jpg",
        "https://example.com/photos/16x9/photo.jpg"
      ],
      "contentUrl": "https://www.example.com/video123.mp4",
      "embedUrl": "https://www.example.com/videoplayer?video=123",
      "uploadDate": "2024-02-05T08:00:00+08:00",
      "duration": "PT1M33S",
      "interactionStatistic": {
        "@type": "InteractionCounter",
        "interactionType": { "@type": "WatchAction" },
        "userInteractionCount": 2347
      },
      "expires": "2024-02-05T08:00:00+08:00"
    }
  }
</script>
</head>
<body>
</body>
</html>

```

Structured data vocabulary and format

This documentation describes which properties are required, recommended, or optional for structured data with special meaning to Google Search. Most Search structured data uses schema.org vocabulary, but you should rely on the Google Search Central documentation as definitive for Google Search behavior, rather than the schema.org documentation. There are more attributes and objects on schema.org that aren't required by Google Search; they may be useful for other search engines, services, tools, and platforms.

Data-vocabulary.org markup is no longer eligible for Google rich result features. Learn more about [sunsetting support for data-vocabulary](#).

Be sure to check your structured data using the [Rich Results Test](#) during development, and the [Rich result status reports](#) after deployment, to monitor the validity of your pages, which might break after deployment due to templating or serving issues.

You must include all the required properties for an object to be eligible for appearance in Google Search with enhanced display. In general, defining more recommended features can make it more likely that your information can appear in Search results with enhanced display. **However**, it is more important to supply fewer but complete and accurate recommended properties rather than trying to provide every possible recommended property with less complete, badly-formed, or inaccurate data.

In addition to the properties and objects documented here, Google can make general use of the [sameAs](#) property and other schema.org structured data. Some of these elements may be used to enable future Search features, if

they are deemed useful.

Supported formats

Google Search supports structured data in the following formats, unless documented otherwise. In general, we recommend using a format that's easiest for you to implement and maintain (in most cases, that's JSON-LD); all 3 formats are equally fine for Google, as long as the markup is valid and properly implemented per the feature's documentation.

Formats	
JSON-LD* (Recommended)	A JavaScript notation embedded in a <script> tag in the <head> and <body> elements of an HTML page. The markup is not interleaved with the user-visible text, which makes nested data items easier to express, such as the Country of a PostalAddress of a MusicVenue of an Event. Also, Google can read JSON-LD data when it is dynamically injected into the page's contents, such as by JavaScript code or embedded widgets in your content management system.
Microdata	An open-community HTML specification used to nest structured data within HTML content. Like RDFa, it uses HTML tag attributes to name the properties you want to expose as structured data. It is typically used in the <body> element, but can be used in the <head> element.
RDFa	An HTML5 extension that supports linked data by introducing HTML tag attributes that correspond to the user-visible content that you want to describe for search engines. RDFa is commonly used in both the <head> and <body> sections of the HTML page.

In general, Google recommends using JSON-LD for structured data if your site's setup allows it, as it's the easiest solution for website owners to implement and maintain at scale (in other words, less prone to user errors).

Structured data guidelines

Be sure to follow the general structured data guidelines, as well as any guidelines specific to your structured data type; otherwise your structured data might be ineligible for rich result display in Google Search.

Get started with structured data

If you're new to structured data, check out schema.org beginner's guide to structured data. While the guide focuses on Microdata, the basic ideas are relevant for JSON-LD and RDFa.

Once you're comfortable with the basics of structured data, explore the list of structured data features in Google Search and pick a feature to implement. Each guide goes into detail on how to implement the structured data in a way that makes your site eligible for a rich result appearance on Google Search.

Choose a feature

Measuring the effect of structured data

You probably want to compare performance of your pages with structured data with those pages that don't have structured data, in order to decide if it's worth your effort. The best way to do that is to run a [before and after test on a few pages on your site](#). This can be a little tricky, since page views can vary for a single page for various reasons.

1. Take some pages on your site that are not using any structured data, and have several months of data in Search Console. Be sure to choose pages that won't be affected by the time of year or timeliness of the page content; use pages that won't change much, but are still popular enough to be read often enough to generate meaningful data.
2. Add structured data or other features to your pages. Confirm that your markup is valid, and that Google has found your structured data using the [URL Inspection tool](#) on your page.
3. Record the performance for a few months in the [Performance report](#), and filter by URL to compare performance of your page.

<https://developers.google.com/search/docs/appearance/structured-data/sd-policies>

General structured data guidelines

To be eligible for rich result appearance in Google Search results, structured data shouldn't violate the [Content policies for Google Search](#) (which include our [spam policies](#)). In addition, this page details the general guidelines that apply to all structured data: they must be followed in order to be eligible for appearance as a rich result in Google Search.

If your page contains a [structured data issue](#), it can result in a manual action. A structured data manual action means that a page loses eligibility for appearance as a rich result; it doesn't affect how the page ranks in Google web search. To check if you have a manual action, open the [Manual Actions report in Search Console](#).

Important: Google does not guarantee that your structured data will show up in search results, even if your page is marked up correctly according to the [Rich Results Test](#). Here are some common reasons why:

- Using structured data *enables* a feature to be present, it does not *guarantee* that it will be present. The Google algorithm tailors search results to create what it thinks is the best search experience for a user, depending on many variables, including search history, location, and device type. In some cases it may determine that one feature is more appropriate than another, or even that a [text result](#) is best.
- The structured data is not [representative of the main content of the page](#), or is potentially misleading.
- The structured data is incorrect in a way that the Rich Results Test was not able to catch.
- The content referred to by the structured data is [hidden from the user](#).
- The page doesn't meet the guidelines for structured data described on this page, the guidelines for a specific structured data feature, the [Search Essentials](#), or the [Content policies for Google Search](#).

Technical guidelines

You can test compliance with technical guidelines using the [Rich Results Test](#) and the [URL Inspection tool](#), which catch most technical errors.

Format

In order to be eligible for rich results, mark up your site's pages using one of [three supported formats](#):

- JSON-LD (recommended)
- Microdata
- RDFa

Access

Don't block your structured data pages to Googlebot using robots.txt, [noindex](#), or any other access control methods.

Quality guidelines

These quality guidelines are not easily testable using an automated tool. Violating a quality guideline can prevent syntactically correct structured data from being displayed as a rich result in Google Search, or possibly cause it to be [marked as spam](#).

Content

- Follow the [spam policies for Google web search](#).
- Provide up-to-date information. We won't show a rich result for time-sensitive content that is no longer relevant.
- Provide original content that you or your users have generated.
- **Don't** mark up content that is not visible to readers of the page. For example, if the JSON-LD markup describes a performer, the HTML body must describe that same performer.
- **Don't** mark up irrelevant or misleading content, such as fake reviews or content unrelated to the focus of a page.
- **Don't** use structured data to deceive or mislead users. Don't impersonate any person or organization, or misrepresent your ownership, affiliation, or primary purpose.
- Content in structured data must also follow the additional content guidelines or policies, as documented in the specific feature guide. For example, content in JobPosting structured data must follow the [job posting content policies](#).

Relevance

Your structured data must be a true representation of the page content. Here are some examples of irrelevant data:

- A sports live streaming site labeling broadcasts as local events.
- A woodworking site labeling instructions as recipes.

Completeness

- Specify all required properties listed in the [documentation for your specific rich result type](#). Items that are missing required properties are not eligible for rich results.
- The more recommended properties that you provide, the higher quality the result is to users. For example: users prefer job postings with explicitly stated salaries than those without; users prefer recipes with actual user reviews and genuine star ratings (note that reviews or ratings not by actual users may result in [manual action](#)). Rich result ranking takes extra information into consideration.

Location

- Put the structured data on the page that it describes, unless specified otherwise by the documentation.
- If you have duplicate pages for the same content, we recommend placing the same structured data on all page duplicates, not just on the canonical page.

Specificity

- Try to use the most specific applicable type and property names defined by schema.org for your markup.
- Follow all additional guidelines given in the [documentation for your specific rich result type](#).

Images

- When specifying an image as a structured data property, make sure that the image is relevant to the page that it's on. For example, if you define the `image` property of `NewsArticle`, the image must be relevant to that news article.
- All image URLs specified in structured data must be crawlable and indexable. Otherwise, Google Search can't find and display them on the search results page. To check if Google can access your URLs, use the [URL Inspection tool](#).

Multiple items on a page

Multiple items on a page means that there is more than one kind of thing on a page. For example, a page could contain a recipe, a video that shows how to make that recipe, and breadcrumb information for how people can discover that recipe. All of this user-visible information can also be marked up with structured data, which makes it easier for search engines like Google Search to understand the information on a page. When you add more items that apply to a page, Google Search has a fuller picture of what the page is about and can display that page in different search features.

[Original illustration: A recipe rich result that shows both a video and reviews]

Google Search understands multiple items on a page, whether you nest the items or specify each item individually:

- **Nesting:** When there is one main item, and additional items are grouped under the main item. This is particularly helpful when grouping related items (for example, a recipe with a video and reviews).
- **Individual items:** When each item is a separate block on the same page.

If there are items that are more helpful when they are linked together (for example, a recipe and a video), use `@id` in both the recipe and the video items to specify that the video is about the recipe on the page. If you didn't link the items together, Google Search may not know that it can show the video as a Recipe rich result.

These examples are trimmed for brevity, and they don't include all the required and recommended properties for the features. For a full example, refer to the [specific structured data type documentation](#).

Nesting

Here's an example of nested structured data, where `Recipe` is the main item, and `aggregateRating` and `video` are nested in the `Recipe`.

```

<html>
  <head>
    <title>How To Make Banana Bread</title>
    <script type="application/ld+json">
      {
        "@context": "https://schema.org/",
        "@type": "Recipe",
        "name": "Banana Bread Recipe",
        "image": [
          "https://example.com/photos/1x1/photo.jpg",
          "https://example.com/photos/4x3/photo.jpg",
          "https://example.com/photos/16x9/photo.jpg"
        ],
        "description": "The best banana bread recipe you'll ever find! Learn how to use up all those extra bananas.",
        "aggregateRating": {
          "@type": "AggregateRating",
          "ratingValue": 4.7,
          "ratingCount": 123
        },
        "video": {
          "@type": "VideoObject",
          "name": "How To Make Banana Bread",
          "description": "This is how you make banana bread, in 5 easy steps.",
          "contentUrl": "https://www.example.com/video123.mp4"
        }
      }
    </script>
  </head>
  <body>
  </body>
</html>

```

Individual items

Here's an example of individual items of structured data. There are two, distinct items: Recipe and BreadcrumbList.

```

<html>
  <head>
    <title>How To Make Banana Bread</title>
    <script type="application/ld+json">
      [{
        "@context": "https://schema.org/",
        "@type": "Recipe",
        "name": "Banana Bread Recipe",
        "description": "The best banana bread recipe you'll ever find! Learn how to use up all those extra
bananas.",
        "image": [
          "https://example.com/photos/1x1/photo.jpg",
          "https://example.com/photos/4x3/photo.jpg",
          "https://example.com/photos/16x9/photo.jpg"
        ]
      },
      {
        "@context": "https://schema.org",
        "@type": "BreadcrumbList",
        "itemListElement": [{
          "@type": "ListItem",
          "position": 1,
          "name": "Recipes",
          "item": "https://example.com/recipes"
        },{
          "@type": "ListItem",
          "position": 2,
          "name": "Bread recipes",
          "item": "https://example.com/recipes/bread-recipes"
        },{
          "@type": "ListItem",
          "position": 3,
          "name": "How To Make Banana Bread"
        }
      ]
    }
  ]
</script>
</head>
<body>
</body>
</html>

```

Additional tips

- To make sure that Google Search understands what the main purpose of the page is, include the main type of structured data that reflects the main focus of the page. For example, if a page is mainly about a recipe, make sure to include [Recipe structured data](#) in addition to [Video](#) and [Review](#) structured data. This is an opportunity for that page to be eligible for multiple search appearances (recipe rich results, video search, and review snippets). If the page only contained Video structured data, Google Search wouldn't know enough about the page to also display it as a recipe rich result.
- To ensure that the page fully represents the content that's visible to users, make sure all structured data items are complete. For example, if you include multiple reviews, make sure that you include all of the reviews that are visible to people on the page. If a page didn't mark up all of the reviews on a page, this would be misleading for people who expect to see all of those reviews, based on the appearance of the page in Search results.

<https://developers.google.com/search/docs/appearance/enriched-search-results>

Enriched search results

In addition to standard rich results, Google Search supports a more interactive and enhanced class of rich result called *enriched search results*. Enriched search results often include an immersive experience or other advanced interaction feature. For example, here is a Jobs enriched result that might appear if a user searched for "jobs in United States":

[Original illustration: Jobs enriched search result]

Enriched search enables the user to search across the various properties of a structured data item; for example, a user might search for chicken soup recipes under 200 calories, or recipes that take less than 1 hour of preparation time.

Implementing enriched search

Enriched search is a subset of rich results, and is implemented using [structured data](#). Some rich result types are only available as enriched search types (for example, recipes, jobs, and events); other rich result types can be extended to be an enriched search type with the addition of a few properties. The documentation for a rich result type explains whether and how it can be extended from a basic rich result to an enriched result.

[Technical information and a gallery of results is available here.](#)

Enriched search is driven by the Google Search ranking algorithm; in addition to adding the correct structured data on your pages, you must follow the following quality guidelines so that Google can properly index and rank your pages.

- The [Structured data quality guidelines](#)
- The [Search Essentials](#)
- The [enriched search quality guidelines](#)

A note about duplicate structured data content: Structured data is typically duplicated across many pages within a site, and for good reason. For example, you might post several Job listings for the same opening in multiple locations. These listings would have identical description values but different location values. The enriched search algorithm takes this into account, and these objects are not considered to be duplicates.

Enriched search types

The following search types support an enriched search experience:

- [Job Posting](#)
- [Recipe](#)
- [Event](#)

Enriched search quality guidelines

You must follow these spam policies to be eligible for enriched search. If the enriched search ranking algorithm decides that a large part of a site is not meeting the quality bar, it can exclude the entire site from enriched search results.

- **Required properties:** Each enriched search type defines a required set of properties. Items missing the required properties are ineligible.
- **Completeness:** The more additional (recommended) properties you provide, the higher quality the item is to our users. For a job posting, users prefer jobs with explicitly stated salaries than those without, and enriched search ranking also takes this into consideration. If your recipes have actual user reviews and genuine star ratings, that is also valuable to users of your site and enriched search. Completeness is one of the most important ranking signals for enriched search.
- **Relevance:** Your marked up data must be relevant to the enriched search you are participating in. Here are some examples of irrelevant data:
 - A sports live streaming site labeling broadcasts as local events.
 - A woodworking site labeling instructions as recipes.
- **Leaf content:** Enriched search is only available for leaf pages, not for listing pages. A leaf page is a page that describes the detailed properties of an item. A listing page, on the other hand, is a category page that links to multiple leaf pages. The following are examples of listing pages:
 - A page that describes "10 best ideas for cooking turkey" with links out to each recipe.
 - A page listing all the jobs at Mountain View, CA, with links to individual jobs.
- **Content policies:** Individual enriched search has additional content-type-specific policies for each data type, as described in its documentation. Documents or sites that violate these content policies may receive less favorable ranking or ineligible for the feature.

<https://developers.google.com/search/docs/appearance/structured-data/generate-structured-data-with-javascript>

Generate structured data with JavaScript

Modern websites use JavaScript to display lots of dynamic content. There are a few things you need to look out for when using JavaScript to generate structured data on your websites, and this guide covers best practices and implementation strategies. If you're new to structured data, you can learn more about [how structured data works](#).

There are different ways to generate structured data with JavaScript, but the most common are:

- [Google Tag Manager](#)
- [Custom JavaScript](#)

Using Product markup? Be aware that dynamically-generated markup can make Shopping crawls less frequent and less reliable, which can be an issue for fast-changing content like product availability and price. If you're a merchant optimizing for all types of shopping results, make sure your server has enough computing resources to handle increased traffic from Google.

Use Google Tag Manager to generate JSON-LD dynamically

[Google Tag Manager](#) is a platform that lets you manage tags on your website without editing the code. To generate structured data with Google Tag Manager, follow these steps:

1. [Set up and install Google Tag Manager](#) on your site.
2. Add a new **Custom HTML** tag to the container.
3. Paste a [supported structured data](#) block into the tag content.
4. Install the container as shown in the **Install Google Tag Manager** section of your container's admin menu.
5. To add the tag to your website, publish your container in the Google Tag Manager interface.
6. [Test your implementation](#).

Using variables in Google Tag Manager

Google Tag Manager (GTM) supports [variables](#) to use information on the page as part of your structured data. Use variables to extract the structured data from the page instead of duplicating the information in GTM. Duplicating the information in GTM increases the risk of having a mismatch between page content and the structured data inserted using GTM.

For example, you can dynamically create a [Recipe](#) JSON-LD block that uses the page title as the recipe name by creating the following custom variable named `recipe_name`:

```
function() { return document.title; }
```

You can then use `{{recipe_name}}` in your custom tag HTML.

We recommend to create variables to collect all the necessary information from the page using variables.

Here is an example for the custom HTML tag content:

```
<script type="application/ld+json">
{
  "@context": "https://schema.org/",
  "@type": "Recipe",
  "name": "{{recipe_name}}",
  "image": [ "{{recipe_image}}" ],
  "author": {
    "@type": "Person",
    "name": "{{recipe_author}}"
  }
}
</script>
```

Note: The previous sample assumes that you defined the variables `recipe_name`, `recipe_image` and `recipe_author` in GTM.

Generate structured data with custom JavaScript

Another way you can generate structured data is by using JavaScript to either generate all of your structured data or add more information to the server-side rendered structured data. Either way, Google Search can understand and process structured data that's available in the DOM when it renders the page. To learn more about how Google Search processes JavaScript, check out the [JavaScript basics guide](#).

Here is an example of JavaScript-generated structured data:

1. [Find the type of structured data](#) you are interested in.
2. Edit your website's HTML to include a JavaScript snippet like the following example (refer to the documentation from your CMS or hosting provider, or ask your developers).

```
fetch('https://api.example.com/recipes/123')
.then(response => response.text())
.then(structuredDataText => {
  const script = document.createElement('script');
  script.setAttribute('type', 'application/ld+json');
  script.textContent = structuredDataText;
  document.head.appendChild(script);
});
```

3. [Test your implementation with the Rich Results Test](#).

Using server-side rendering

If you are using [server-side rendering](#), you can also include the structured data in the rendered output. Check the documentation of your framework to find out how to generate the JSON-LD for [the type of structured data](#) you're interested in.

Test your implementation

To make sure Google Search can crawl and index your structured data, test your implementation:

1. Open the [Rich Results Test](#).
2. Enter the URL that you want to test.

We recommend that you use the URL input instead of the code input because there are JavaScript limitations when using the code input (for example, CORS restrictions).

3. Click **Test URL**.

Success: If you did everything correctly and your [structured data type is supported in the tool](#), you will see the message "Page is eligible for rich results".

If you are testing a structured data type that is not supported by the Rich Results test, check the rendered HTML. If the rendered HTML contains the structured data, Google Search will be able to process it.

Try again: If you see errors or warnings, it is most likely a syntax error or a missing property. Read the [documentation for your type of structured data](#) and make sure you've added all the properties. If your problem persists, make sure to also check the guide on [fixing search-related JavaScript problems](#).

<https://developers.google.com/search/docs/appearance/structured-data/search-gallery>

Structured data markup that Google Search supports

Google uses [structured data](#) to understand the content on the page and show that content in a richer appearance in search results, which is called a *rich result*. To make your site eligible for appearance as one of these rich results, follow the guide to learn how to implement structured data on your site. If you're just getting started, visit [Understand how structured data works](#).

Note: The actual appearance in search results might be different. You can preview most features with the [Rich Results Test](#).

Choose a category that describes your website

Structured data features		Filter column
Article A news, sports, or blog article displayed in various rich result features, such as the title of the article and larger-than-thumbnail images. Get started	[Original illustration: article example in search results]	News, Sports
Breadcrumb Navigation that indicates the page's position in the site hierarchy. Get started	[Original illustration: breadcrumb example in search results]	Generic
Carousel Rich results that display in a sequential list or gallery from a single site. This feature must be combined with one of the following features: Recipe , Course list , Restaurant , Movie . Get started	[Original illustration: An illustration of how a recipe host carousel can appear in Google Search. It shows 3 different recipes from the same website in a carousel format that users can explore and select a specific recipe]	Food and Drink, Education and Science, Entertainment
Course list A list of educational course from the same course provider. Courses can include the course title, provider, and a short description.	[Original illustration: course list rich result in search results]	Education and Science

Structured data features		Filter column
Get started		
Dataset Large data sets that appear in Google Dataset Search. Get started	[Original illustration: dataset example in search results]	Education and Science
Discussion forum User-generated content (traditionally short-form compared to Article), followed by a threaded or non-threaded discussion about that topic. Get started	[Original illustration: An illustration of the discussions and forums rich result]	
Education Q&A Education-related questions and answers that help students discover flashcards on Google Search. Get started	[Original illustration: Education Q&A carousel in search results]	Education and Science
Employer aggregate rating An evaluation of a hiring organization compiled from many users that's displayed in the job search experience on Google. Get started	[Original illustration: employer aggregate rating example in search results]	Jobs
Event An interactive rich result that shows a list of organized events, such as concerts or art festivals, that people may attend at a particular time and place. Get started	[Original illustration: How the event experience looks on Google Search]	Entertainment
Image metadata When you specify image metadata, Google Images can show more details about the image, such as who the creator is, how people can use an image, and credit information. Get started	[Original illustration: An example of Image Metadata in Google Images]	
Job posting	[Original illustration: job posting	Jobs

Structured data features		Filter column
An interactive rich result that allows job seekers to find a job. The job search experience on Google can feature your logo, reviews, ratings, and job details. Get started	example in search results	
Local business Business details displayed in the Google knowledge panel, including open hours, ratings, directions, and actions to book appointments or order items. Get started	[Original illustration: local business example in search results]	Organizations
Math solver Help students, teachers, and others with math problems by adding structured data to indicate the type of math problems and step-by-step walkthroughs for specific math problems. Get started	[Original illustration: math solvers in search results]	Education and Science
Movie The movie carousel helps users explore lists of movies on Google Search (for example, "best movies of 2023"). You can provide details about the movies, such as the title of each movie, director information, and images. Get started	[Original illustration: An illustration of how a movie rich result can appear in Google Search. It shows 3 different movies from the same website in a carousel format that users can explore and select a specific movie]	Entertainment
Organization Information about your organization, such as your logo, legal name of the organization, address, contact information, and company identifiers. This information can show up in knowledge panels and other visual elements (such as attribution). Get started	[Original illustration: organization example in search results]	Organizations

Structured data features		Filter column
Product Information about a product, including price, availability, and review ratings. Get started	[Original illustration: product example in search results]	Ecommerce
Profile page A page that primarily focuses on information about a single person or organization that is somehow affiliated with the overall website. Get started	[Original illustration: An illustration of the Perspectives filter in search results]	
Q&A Q&A Pages are web pages that contain data in a question and answer format, which is one question followed by its answers. Get started	[Original illustration: question answer page example in search results]	
Recipe Recipes that display as an individual rich result or part of a host carousel. Get started	[Original illustration: An illustration of how a recipe host carousel can appear in Google Search. It shows 3 different recipes from the same website in a carousel format that users can explore and select a specific recipe]	Food and Drink
Review snippet A short excerpt of a review or a rating from a review website, usually an average of the combined rating scores from reviewers. A review snippet can be about Book , Recipe , Movie , Product , Software App , and Local business . Get started	[Original illustration: review snippet example in search results]	Organizations, Ecommerce, Food and Drink, Entertainment
Software app Information about a software app, including rating information, a description of the app, and a link to the app. Get started	[Original illustration: software app example in search results]	

Structured data features		Filter column
Speakable Allow search engines and other applications to identify news content to read aloud on Google Assistant-enabled devices using text-to-speech (TTS). Get started	[Original illustration: speakable example that shows a conversation with the Google Home. A person asks Google Home what's the latest news with Nasa. Google Home responds with a list of three news articles.]	News
Subscription and paywalled content Indicate paywalled content on your site to help Google differentiate paywalled content from the practice of cloaking , which violates our spam policies . Get started	[Original illustration: A New York Times paywall example that shows a reader has reached the limit of articles]	News
Vacation rental Information about a vacation property, such as the name, description, images, location, rating, and reviews. Get started	[Original illustration: illustration of a vacation rental rich result]	
Video Video information in search results, with the option to play the video, specify video segments, and live-stream content. Get started	[Original illustration: video example in search results]	Food and Drink, News, Education and Science, Sports

<https://developers.google.com/search/docs/appearance/title-link>

Influencing your title links in search results

A *title link* is the title of a search result on Google Search and other properties (for example, Google News) that links to the web page. Google uses a number of different sources to automatically determine the title link, but you can indicate your preferences by following our [best practices for influencing title links](#).

Best practices for influencing title links

Title links are critical to giving users a quick insight into the content of a result and why it's relevant to their query. It's often the primary piece of information people use to decide which result to click, so it's important to use high-quality title text on your web pages.

- Make sure **every page on your site has a title specified in the `<title>` element**.
- Write **descriptive and concise** text for your `<title>` elements. Avoid vague descriptors like "Home" for your home page, or "Profile" for a specific person's profile.
Also avoid unnecessarily long or verbose text in your `<title>` elements. While there's no limit on how long a `<title>` element can be, the title link is truncated in Google Search results as needed, typically to fit the device width.
- Avoid **keyword stuffing**. It's sometimes helpful to have a few descriptive terms in the `<title>` element, but there's no reason to have the same words or phrases appear multiple times. Title text like "Foobar, foo bar, foobars, foo bars" doesn't help the user, and this kind of [keyword stuffing](#) can make your results look spammy to Google and to users.
- Avoid **repeated or boilerplate text in `<title>` elements**. It's important to have distinct text that describes the content of the page in the `<title>` element for each page on your site. Titling every page on a commerce site "Cheap products for sale", for example, makes it impossible for users to distinguish between two pages. Long text in the `<title>` element that varies by only a single piece of information ("boilerplate" titles) is also bad; for example, a common `<title>` element for all pages with text like "Band Name - See videos, lyrics, posters, albums, reviews and concerts" contains a lot of uninformative text.

One solution is to dynamically update the `<title>` element to better reflect the actual content of the page. For example, include the words "video" and "lyrics" only if that particular page contains video or lyrics.

- **Brand your titles** concisely. The `<title>` element on your site's home page is a reasonable place to include some additional information about your site. For example:

```
<title>ExampleSocialSite, a place for people to meet and mingle</title>
```

But displaying that text in the `<title>` element of every single page on your site will look repetitive if several pages from your site are returned for the same query. In this case, consider including just your [site name](#) at the beginning or end of each `<title>` element, separated from the rest of the text with a delimiter such as a hyphen, colon, or pipe, like this:

```
<title>ExampleSocialSite: Sign up for a new account.</title>
```

- **Make it clear which text is the main title** for the page. Google looks at various [sources when creating title links](#), including the main visual title, heading elements, and other large and prominent text, and it can be confusing if multiple headings carry the same visual weight and prominence. Consider ensuring that your main title is distinctive from other text on a page and stands out as being the most prominent on the page (for example, using a larger font, putting the title text in the first visible `<h1>` element on the page, etc).
- **Be careful about disallowing search engines** from crawling your pages. Using the [robots.txt](#) protocol on your site can stop Google from crawling your pages, but it may not always prevent them from being indexed. For example, Google may index your page if we discover it by following a link from someone else's site. If we don't have access to the content on your page, we will rely on off-page content to generate the title link, such as anchor text from other sites. To prevent a URL from being indexed, you can use the [noindex rule](#).
- **Use the same language and writing system (meaning, the script or alphabet for a given language) as the primary content** on your pages. For example, if a page is written in Hindi, make sure to also write the `<title>` element in Hindi (don't write title text in English or transliterate the title into Latin characters). Google tries to show a title link that matches the primary language and writing system of a page. If Google determines that a `<title>` element does not match the writing system or language of the page's primary content, we may choose a different text as the title link.
- **Avoid including flight price information in `<title>` elements.** Our systems will likely not show price information when generating title links for flight pages. This is because pricing for flights can change so fast (sometimes every few minutes), that what's shown in title links may not correspond to the actual price on the landing page.

How title links in Google Search are created

Google's generation of title links on the Google Search results page is completely automated and takes into account both the content of a page and references to it that appear on the web. The goal of the title link is to best represent and describe each result.

Google Search uses the following sources to automatically determine title links:

- Content in `<title>` elements
- Main visual title shown on the page
- Heading elements, such as `<h1>` elements
- Content in `og:title` meta tags
- Other content that's large and prominent through the use of style treatments
- Other text contained in the page
- Anchor text on the page
- Text within links that point to the page
- [WebSite structured data](#)

Keep in mind that Google has to recrawl and reprocess the page to notice updates to these sources, which may take a few days to a few weeks. If you've made changes, you can [request that Google recrawl your pages](#).

While we can't manually change title links for individual sites, we're always working to make them as relevant as possible. You can help improve the quality of the title link that's displayed for your page by following the [best practices](#).

Common issues and how Google manages them

Why the title link in search results might differ from the page's `<title>` element or main heading: If we've detected an [issue on the page](#), we may try to generate an improved title link from anchors, on-page text, or other [sources](#).

Here are the most common issues we see with title links in search results. To avoid these issues, follow the [best practices for influencing title links](#).

Common issues	
Half-empty <code><title></code> elements	When part of the title text is missing. For example: <code><title> Site Name</title></code> Google Search looks at information in header elements or other large and prominent text on the page to produce a title link: Product Name Site Name
Obsolete <code><title></code> elements	When the same page is used year-after-year for recurring information, but the <code><title></code> element didn't get updated to reflect the latest date. For example: <code><title>2020 admissions criteria - University of Awesome</title></code> In this example, the page has a large, visible title that says "2021 admissions criteria", and the <code><title></code> element wasn't updated to the current date. Google Search may detect this inconsistency and uses the right date from the visible title on the page in the title link: 2021 admissions criteria - University of Awesome
Inaccurate <code><title></code> elements	When the <code><title></code> elements don't accurately reflect what the page is about. For example, the page could have dynamic content with the following <code><title></code> element:


```
<title>Giant stuffed animals, teddy bears, polar bears - Site Name</title>
```

Google Search tries to determine if the `<title>` element isn't accurately showing what a page is about. Google Search might modify the title link to better help users if it determines that the page title doesn't reflect the page content. For example:

```
Stuffed animals - Site Name
```

Micro-boilerplate text in <title> elements	When there are repeated boilerplate text in <title> elements for a subset of pages within a site. For example, a television website has multiple pages that share the same <title> element that omits the season numbers, and it's not clear which page is for what season. That produces duplicate <title> elements like this: <pre><title>My so-called amazing TV show</title></pre> <pre><title>My so-called amazing TV show</title></pre> <pre><title>My so-called amazing TV show</title></pre> Google Search can detect the season number used in large, prominent title text and insert the season number in the title link: <pre>Season 1 - My so-called amazing TV show</pre> <pre>Season 2 - My so-called amazing TV show</pre> <pre>Season 3 - My so-called amazing TV show</pre>
No clear main title	When there's more than one large, prominent heading, and it isn't clear which text is the main title of the page. For example, a page has two or more headings that use the same styling or heading level. If Google Search detects that there are multiple large, prominent headings, it may use the first heading as the text for the title link. Consider ensuring that your main heading is distinctive from other text on a page and stands out as being the most prominent on the page (for example, using a larger font, putting the title text in the first visible <h1> element on the page, etc).
Mismatch of writing system or language used in <title> elements	When the writing system or language of the text in <title> elements doesn't match the writing system or language of the primary text on a page. For example, when a page is in written in Hindi, but the title includes text in English or is transliterated into Latin characters. If Google detects a mismatch, it may generate a title link that better matches the primary content. Consider ensuring that the script and language matches what is most prominent on the page.
Duplication of the site name in the <title> element	In the case of domain-level site names, Google may omit the site name from the title link, if it's repetitive with the site name that's already shown in the search result.

Submitting feedback about title links

If you're seeing your pages appear in the search results with modified title links, check whether your page has [one of the issues](#) that Google adjusts for. If not, consider whether the title link in search results is a better fit for the query. To discuss your pages' title links and get feedback about your pages from other site owners, join our [Google Search Central Help Community](#).

<https://developers.google.com/search/docs/appearance/translated-results>

Translated results in Google Search

Google Search strives to make information accessible and useful to all users. To help address content and perspective gaps when a user searches in their local language, sometimes Google may translate the title link and snippet of a search result for results that aren't in the language of the search query, when available. Translated results enable people to view results from other languages in their language, and can help publishers reach a larger audience.

[Original illustration: How a translated result could look in Google Search]

[Original illustration: How translated news results could look in Google Search]

Feature availability

Currently, Google may translate results into the following languages: Arabic, Bengali, English, French, German, Gujarati, Hindi, Indonesian, Kannada, Korean, Malayalam, Marathi, Persian, Portuguese, Spanish, Tamil, Telugu, Thai, Turkish, Urdu, Vietnamese. It's available on mobile and desktop.

How translated results work

If the user clicks the translated title link, they're presented with a page that's been machine translated. Users also have an option to view the original search result, and access the entire page in the original language.

Google doesn't host any translated pages. Opening a page through a translated result is no different than opening the original search result through [Google Translate](#) or using [Chrome in-browser translation](#). This means that Javascript on the page is usually supported, as well as embedded images and other page features.

If you run an ad network, you may need to take additional action to ensure that your ad network displays correctly after a user clicks a translated result. Learn more about [enabling your ad network to work with translation-related Google Search features](#).

Monitor performance in Search Console

To monitor clicks and impressions for translated results, you can use the [Search Appearance filter](#) in the [Performance report](#).

Opting in or out of translated results

This feature is applicable across all pages and results based on the user's language. You don't need to do anything to opt in.

Translated results are like other translation-related features in Google Search. To opt out of all translation features in Google Search, use the [notranslate rule](#), which can be implemented as a meta tag or an HTTP header:

```
<!-- opt out of translation features on all search engines that support this rule -->
<meta name="robots" content="notranslate">
```

```
<!-- opt out of translation features on Google -->
<meta name="googlebot" content="notranslate">
```

Alternatively, you can specify the rule as an HTTP response header:

```
HTTP/1.1 200 OK
Date: Tue, 25 May 2010 21:42:43 GMT
(...)
X-Robots-Tag: notranslate
(...)
```

<https://developers.google.com/search/docs/appearance/ad-network-and-translation>

Enabling your ad network to work with translation-related Google Search features

Google Search offers several [translation-related features](#) that enable users to access translated content. If you run an ad network and your ads aren't working properly on translated web pages, you'll need to follow the steps in this guide to make sure your ads render or attribute correctly.

Our approach

When users access translated content provided by [Google Translate](#) from within search results, Google retrieves the page from the publisher, rewrites the source URL, and translates the web page after the user clicks the translated result.

Convert the Google Translate URL to the original URL

If you run an ad network that relies on the publisher's source URL, you'll need to convert the Google Translate URL to make sure your ads are working properly. Follow these steps to decode the publisher's hostname:

1. Extract the domain prefix from the hostname, by removing the `.translate.goog` suffix.
2. Split the `_x_tr_enc` parameter by the `,` (comma) character and save it as `encoding_list`.
3. Prepend the value of the `_x_tr_hp` parameter to the domain prefix, if it exists.
4. If `encoding_list` contains `1` and the output begins with `1-`, remove the `1-` prefix from the output of step 2.
5. If `encoding_list` contains `0` and the output begins with `0-`, remove the `0-` prefix from the output of step 3.
If you removed the prefix, set `is_idn` to `true`. Otherwise, set `is_idn` to `false`.
6. Replace `/\b-\b/` (regex) with the `.` (dot) character.
7. Replace the `--` (double hyphen) character with the `-` (hyphen) character.
8. If `is_idn` is set to `true`, add the punycode prefix `xn--`.
9. **Optional:** Convert to Unicode.

Sample JavaScript code for decoding the hostname from a Google Translate URL

```

function decodeHostname(proxyUrl) {
  const parsedProxyUrl = new URL(proxyUrl);
  const fullHost = parsedProxyUrl.hostname;
  // 1. Extract the domain prefix from the hostname, by removing the
  //     ".translate.google" suffix
  let domainPrefix = fullHost.substring(0, fullHost.indexOf('.'));

  // 2. Split _x_tr_enc parameter by "," (comma), save as encodingList
  const encodingList = parsedProxyUrl.searchParams.has('_x_tr_enc') ?
    parsedProxyUrl.searchParams.get('_x_tr_enc').split(',') :
    [];

  // 3. Prepend value of _x_tr_hp parameter to the domain prefix, if it exists
  if (parsedProxyUrl.searchParams.has('_x_tr_hp')) {
    domainPrefix = parsedProxyUrl.searchParams.get('_x_tr_hp') + domainPrefix;
  }

  // 4. Remove '1-' prefix from the output of step 2 if encodingList contains
  //     '1' and the output begins with '1-'.
  if (encodingList.includes('1') && domainPrefix.startsWith('1-')) {
    domainPrefix = domainPrefix.substring(2);
  }

  // 5. Remove '0-' prefix from the output of step 3 if encodingList contains
  //     '0' and the output begins with '0-'.
  //     Set isIdn to true if removed, false otherwise.
  let isIdn = false;
  if (encodingList.includes('0') && domainPrefix.startsWith('0-')) {
    isIdn = true;
    domainPrefix = domainPrefix.substring(2);
  }

  // 6. Replace /\b-\b/ (regex) with '.' (dot) character.
  // 7. Replace '--' (double hyphen) with '-' (hyphen).
  let decodedSegment =
    domainPrefix.replaceAll(/\b-\b/g, '.').replaceAll('--', '-');

  // 8. If isIdn equals true, add the punycode prefix 'xn--'.
  if (isIdn) {
    decodedSegment = 'xn--' + decodedSegment;
  }
  return decodedSegment;
}

```

Reconstruct the URL

1. Using the original page URL, replace the hostname with the decoded hostname.
2. Remove all `_x_tr_*` parameters.

Test your code

You can create unit tests for your code using the following table. Given a `proxyUrl`, the `decodeHostname` must match the expected value.

The following table can only be used to test the hostname decoding. You'll need to ensure that the path, fragment, and original parameters of the URL are preserved as is.

proxyUrl	decodeHostname
https://example-com.translate.goog	example.com
https://foo-example-com.translate.goog	foo.example.com
https://foo--example-com.translate.goog	foo-example.com
https://0-57hw060o-com.translate.goog/?_x_tr_enc=0	xn--57hw060o.com (⚡ 😊 .com)
https://1-en--us-example-com/?_x_tr_enc=1	en-us.example.com
https://0-en---w45as309w-com.translate.goog/?_x_tr_enc=0	xn--en--w45as309w.com (en- ⚡ 😊 .com)
https://1-0-----16pw588q-com.translate.goog/?_x_tr_enc=0,1	xn----16pw588q.com (⚡ - 😊 .com)
https://lanfairpwillgwyngyllgogerychwyrndrobwlilllantysiliogogoch-co-uk.translate.goog/?_x_tr_hp=1	llanfairpwillgwyngyllgogerychwyrndrobwlilllantysiliogogoch.co.uk
https://lanfairpwillgwyngyllgogerychwyrndrobwlilllantysiliogogoch-co-uk.translate.goog/?_x_tr_hp=www-l	www.llanfairpwillgwyngyllgogerychwyrndrobwlilllantysiliogogoch.co.uk
https://a--aaaaaaaaaaaaaaaaaaaaaaaaaaaaa-com.translate.goog/?_x_tr_hp=a--xn--xn--xn--xn-xn-----aa-aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa.com	a-xn-xn-xn-xn-xn-----aa-aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa.com
https://g5h3969ntadg44juhyah3c9aza87iiar4i410avdl1d3f1fuq3nz05dg5b-com.translate.goog/?_x_tr_enc=0&_x_tr_hp=0-	xn--g5h3969ntadg44juhyah3c9aza87iiar4i410avdl1d3f1fuq3nz05dg5b.com (❤️ 🌲 😊💕🧸💖🌹😍🌸🌺😂😭😏😞😟💕👩🐱🐻🐰🐾🐶🐯.com)

<https://developers.google.com/search/docs/appearance/video>

Video SEO best practices

If you have videos on your site, following these video SEO best practices can help more people find your site through video results on Google. Videos can appear in several different places on Google, including the main search results page, Video mode, Google Images, and [Discover](#):

[Original illustration: Video content in Google search results, the video tab, and Discover]

Optimize your videos to appear on Google by following these best practices:

[Original embedded video]

1. [Help Google find your videos](#)
2. [Ensure your videos can be indexed](#)
3. [Enable specific video features](#)
4. [Remove, restrict, or update your videos as needed](#)
5. [Monitor your videos with Search Console](#)
6. [Troubleshoot video issues](#)

Help Google find your videos

The [technical requirements](#) for getting your content in Google's search results applies to videos too. There are some additional requirements to making your videos eligible to be discovered, crawled, and indexed by Google Search:

- Use HTML elements commonly used for embedding videos. Google can find videos referenced by a `<video>`, `<embed>`, `<iframe>`, or `<object>` element.
- Don't use [fragment identifiers](#) to load the video, as Google Search doesn't generally support URL fragments.
- If you're using JavaScript to inject the video, make sure that it [appears in the rendered HTML in the URL inspection tool](#).
- If you're using a media API (for example, the [Media Source API](#)), make sure the HTML video container element is still injected even if the media API call fails (in addition to providing metadata about the video). That way Google can still locate the position of the video container even if there's an issue with calling the media API.
- Don't rely on user actions (such as swiping, clicking, or typing) to load the video.

To make it easier for Google to find your videos, we recommend providing metadata about the video. We support [structured data](#), [video sitemaps](#), and the [Open Graph protocol \(OGP\)](#).

Ensure your videos can be indexed

To be eligible for video features, a video must meet the following indexing requirements:

- The [watch page](#) must be indexed.
- The indexed watch page must be performing well in Search before its video can be considered for indexing.

Just because the watch page is indexed doesn't mean that the video will also be indexed.

- The video must be embedded on a watch page.
- The video can't be hidden behind other elements.

If you're using a paywall (for example, users need to sign in or subscribe before seeing the video), add [paywall structured data](#) so that Google can still find and index the video.

- The video must have a [valid thumbnail](#) that's available at a [stable URL](#).

Use a supported video file type

To be eligible for video features, use a supported video file type. Google can process the following video file types: 3GP, 3G2, ASF, AVI, DivX, M2V, M3U, M3U8, M4V, MKV, MOV, MP4, MPEG, OGV, QVT, RAM, RM, VOB, WebM, WMV, and XAP.

[Data URLs](#) aren't supported.

Use stable URLs

Some CDNs use quickly expiring URLs. If the video's thumbnail URL changes too often, Google may not be able to successfully index your videos. To make sure your videos can be indexed, use a single unique and stable thumbnail URL for each video.

To make your videos [eligible for specific features](#) like key moments and video previews, make sure your video files are available at stable URLs too. This also helps Google discover and process the videos consistently, confirm they are still available, and collect signals on the videos.

If you are concerned about bad actors (for example, hackers or spammers) accessing your content, you can [verify Googlebot](#) before displaying a stable version of your media URLs. For example, you can choose to serve the [contentUrl](#) property only to trusted clients like Googlebot, whereas other clients accessing your page wouldn't see that field. With this setup, only trusted clients will be able to access the location of your video file.

Create a dedicated watch page for each video

To be eligible for video features (including video results on the main search results page, Video mode, [Key Moments](#), the [Live Badge](#), and other rich formats), create a dedicated watch page for each video, if it makes sense for your business.

[Original illustration: a web page where the video is the main content on the page]

A *watch page*'s main purpose is to show users a single video. The following pages are watch pages because watching an individual video is the main reason the user is visiting the page:

- A video landing page
- A TV episode video player page
- A news video watch page
- A sports highlight page
- An events clip page

These pages aren't watch pages because the video is complementary to the rest of the content on the page:

- A blog post that reviews an embedded video
- A product page with a 360 video of a product
- A video category page that lists multiple videos of equal prominence
- A movie review page with an embedded movie trailer

Make sure each watch page has a page title and description that are unique to that video. For tips, check our best practices for writing good [titles](#) and [descriptions](#).

It's fine to include the same video on both a watch page and another page alongside other information, like a news article or a product detail page; a non-watch page can still be eligible to appear as a [text result](#) and [Google Image result with a video badge](#).

Using third-party embedded players

If your website embeds videos from third-party platforms like YouTube, Vimeo, or Facebook, Google may index the video both on your web page and on the third-party platform's equivalent page. Both versions may appear in video features on Google, as long as the pages meet the [video indexing criteria](#).

For your own watch page where you've embedded the third-party player, we still recommend that you [provide structured data](#), and you may also include these pages in your [video sitemaps](#). To be eligible for more video features, check with your video host to ensure they allow Google to [fetch your video file](#).

Which URL is which?

There are several URLs associated with a video. Here's a summary of the main ones:

[Original illustration: [URLs on a video page](#)]

URLs related to the video	
1. Watch page	The URL of the watch page that's embedding the video. If you're using a video sitemap, this URL is the value for the <code><loc></code> video sitemap tag. <pre><?xml version="1.0" encoding="UTF-8"?> <urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:video="http://www.google.com/schemas/sitemap-video/1.1"> <url> <loc>https://example.com/videos/some_video_landing_page.html</loc> <video:video> ...</pre>
2. Video player	The URL of a specific player for the video. This is often the <code>src</code> value for an <code><iframe></code> element in the HTML of the watch page: <pre><iframe src="https://example.com/videooplayer.php?video=123" width="640" height="360" frameborder="0" allow="autoplay; fullscreen; picture-in-picture" allowfullscreen></iframe></pre>

URLs related to the video	
	How to provide the URL If you're using structured data, provide the video player URL as the value for the <code>VideoObject.embedUrl</code> property. <pre>"embedUrl": "https://example.com/videoplayer.php?video=123"</pre> If you're using a video sitemap, provide the video player URL as the value for the <code><video:player_loc></code> tag. <pre><video:player_loc>https://example.com/videoplayer.php?video=123</video:player_loc></pre>
3. Video file	The URL of the video file's actual content bytes, which could be hosted on the embedding site, a CDN, or on a streaming service. In the <code><object></code> element, the video file's URL is the value of the <code>data</code> attribute: <pre><object data="https://streamserver.example.com/video/123/file.mp4" width="400" height="300"></object></pre> In the <code><video></code> element, the video file's URL is the value of the <code>src</code> attribute of the <code><source></code> element: <pre><video controls width="250"> <source src="https://streamserver.example.com/video/123/file.webm" type="video/webm" /> <source src="https://streamserver.example.com/video/123/file.mp4" type="video/mp4" /> </video></pre> In the <code><embed></code> element, the video file's URL is the value of the <code>src</code> attribute: <pre><embed type="video/webm" src="https://streamserver.example.com/video/123/file.mp4" width="400" height="300"></embed></pre> How to provide the URL If you're using structured data, provide the video file's URL as the value for the <code>VideoObject.contentUrl</code> property. <pre>"contentUrl": "https://streamserver.example.com/video/123/file.mp4"</pre> If you're using a video sitemap, provide the video file's URL as the value for the <code><video:content_loc></code> tag.

URLs related to the video	
	<pre><video:content_loc>https://streamserver.example.com/video/123/file.mp4</video:content_loc></pre>

Provide a high-quality video thumbnail

To be eligible to appear in video features, a video must have a valid thumbnail image. If you [allow Google to fetch your video files](#), Google will try to automatically generate a thumbnail for you.

However, you can influence which thumbnail is shown in video features by providing your preferred thumbnail through one of the following metadata sources:

- If you're using the <video> HTML element, specify the poster attribute.
- In a video sitemap (including mRSS), specify the <video:thumbnail_loc> tag (or <media:thumbnail> respectively).
- For structured data, specify the [thumbnailUrl](#) property.
- For [OGP](#), specify the og:video:image property.

If you choose to specify multiple metadata sources (for example, specifying a thumbnail in both your sitemap and structured data), make sure you're using the same thumbnail URL per video across all metadata.

Video thumbnail specifications	
Supported thumbnail formats	BMP, GIF, JPEG, PNG, WebP, SVG, and AVIF
Size	Minimum 60x30 pixels, larger preferred.
Location	The thumbnail file must be accessible by Googlebot and Googlebot Images (don't block the file with robots.txt or a login requirement). Make sure that the file is consistently available at a stable URL .
Transparency	At least 80% of the thumbnail's pixels must have an alpha (transparency) value greater than 250.

Provide consistent and unique information in your structured data

To influence how your videos appear on Google, describe your video with [structured data](#). Ensure that any information that you provide in structured data is consistent with the actual video content and other metadata you provide. Make sure to provide unique information in the thumbnailUrl, name, and description properties for each video on your site.

Enable specific video features

Not all features are eligible to trigger on all queries. Google doesn't guarantee that adding markup will result in a specific video feature in Search results; learn why in our [general structured data guidelines](#).

Video previews

[Original illustration: Video preview in search results]

Google selects a few seconds from your video to display a moving preview, which can help users better understand what they'll find in your video. To make your videos eligible for this feature, [allow Google to fetch your video files](#). You can set the maximum duration for these video previews using the `max-video-preview` robots meta tag.

Key moments

[Original illustration: Video with key moments in search results]

The key moments feature is a way for users to navigate video segments like chapters in a book, which can help users engage more deeply with your content. Google Search tries to automatically detect the segments in your video and show key moments to users, without any effort on your part. Alternatively, you can tell Google about the important points of your video. We will prioritize key moments set by you, either through structured data or the YouTube description.

- **If your video is embedded on your web page or you're running a video platform**, there are two ways that you can enable key moments:
 - **Clip structured data**: Specify the exact start and end time to each segment, and what label to display for each segment. This is supported in all languages where Google Search is available.
 - **SeekToAction structured data**: Tell Google where timestamps typically go in your URL structure, so that Google can automatically identify key moments and link users to those points within the video. This is supported for the following languages: English, Spanish, Portuguese, Italian, Chinese, French, Japanese, German, Turkish, Korean, Dutch, and Russian.
- **If your video is hosted on YouTube**, you can specify the exact timestamps and labels in the video description on YouTube. Check out the [best practices for marking timestamps in YouTube descriptions](#). This is supported in all languages where Google Search is available. If you want to enable Video Chapters on YouTube, follow these [additional guidelines](#).

To opt out of the key moments feature completely (including any efforts Google may make to show key moments automatically for your video), use the `nosnippet` meta tag.

Live Badge

[Original illustration: Video with a LIVE badge in search results]

For livestreaming videos, you can enable a red "LIVE" badge to appear in search results by using `BroadcastEvent` structured data.

Allow Google to fetch your video files

Google needs to successfully fetch the actual bytes of a video file to enable features like [video previews](#) and [key moments](#).

Google won't expose the video file directly in search results, and users clicking a video result will land on your site to watch the video. You can also verify if a web crawler accessing your server [really is Googlebot](#).

Allow Google to find and fetch your video files by following these best practices:

- Allow Google to fetch the video's streaming file URL (such as M3U8). Don't block the URL of the actual video bytes with the [noindex](#) rule or [robots.txt](#) file.
- The video file must be available at a [stable URL](#).
- Use structured data to provide the [contentURL](#) value of a [supported file type](#).
- The host of the video watch page and the server that's streaming the actual video must have enough server resources to support crawling. So if your landing page at [example.com/puppies.html](#) has an embedded puppies video served by [streamserver.example.com](#), both [example.com](#) and [streamserver.example.com](#) must meet the [technical requirements for Google Search](#) and have [available server capacity](#).

If you're using a CDN, you might be able to get Search Console notifications about crawl errors. You'll need to be able to [verify ownership](#) of your allocated URL space on the CDN (either a subdomain or subdirectory that's reserved for your content).

Remove or restrict your videos

Remove a video

If you need your video removed from search results as quickly as possible, [file a removal request](#) for the watch page that's embedding the video. For the removal to remain permanent, the watch page must not be present or accessible to Google (that is, either return 404, use a [noindex robots meta tag](#), require server-side authentication). If the video is embedded on other pages or sites, those pages won't be removed unless you file additional removal requests for each page embedding that video.

To remove a video from your site, do one of the following:

- **Return a 404 (Not found)** for any watch page that's embedding the removed or expired video. In addition to the 404 response code, you can still return the HTML of the page to make the change transparent to most users.
- **Include a [noindex robots meta tag](#)** on any watch page that's embedding a removed or expired video. This prevents the watch page from being indexable.
- **Indicate an expiration date** in your structured data (the [expires](#) property) or video sitemap (use the `<video:expiration_date>` element). Here is an example of a video sitemap with a video that expired in November 2009:

```

<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"
        xmlns:video="http://www.google.com/schemas/sitemap-video/1.1">
<url>
<loc>https://example.com/videos/some_video_landing_page.html</loc>
<video:video>
  <video:thumbnail_loc>
    https://example.com/thumbs/123.jpg
  </video:thumbnail_loc>
  <video:title>
    Grilling steaks for summer
  </video:title>
  <video:description>
    Bob shows you how to grill steaks perfectly every time
  </video:description>
  <video:player_loc>
    https://example.com/videoplayer?video=123
  </video:player_loc>
  <video:expiration_date>2009-11-05T19:20:30+08:00</video:expiration_date>
</video:video>
</url>
</urlset>

```

When a video has an expiration date in the past, the video won't appear in video results. The watch page may still be shown as a text result, without a video thumbnail. This includes expiration dates from sitemaps, structured data, and [meta tags](#). Make sure that your expiration dates are correct for each video. While this is useful if your video is no longer available after the expiration date, it's easy to accidentally set the date to the past for an available video. If a video doesn't expire, don't include expiration information.

Restrict a video based on the user's location

You can restrict search results for your video based on the user's location. If your video doesn't have any country restrictions, omit the country restriction tags.

Restrict using structured data

If you use [VideoObject structured data](#) to describe a video, set the [regionsAllowed](#) property to specify which regions can get the video result. If you omit this property, all regions can see the video in search results.

Alternatively, you can use the [ineligibleRegion](#) property to specify which regions can't get the video result.

Restrict using a video sitemap

In a video sitemap, the `<video:restriction>` tag can be used to allow or deny the video from appearing in specific countries. Only one `<video:restriction>` tag is allowed per video entry.

The `<video:restriction>` tag must contain one or more space-delimited [ISO 3166-1 two- or three-letter country codes](#). The required relationship attribute specifies the type of restriction.

- `relationship="allow"`: The video can appear only for the specified countries. If no countries are specified, the video won't appear anywhere.
- `relationship="deny"`: The video can appear everywhere except for the specified countries. If no countries are specified, the video will appear everywhere.

In this video sitemap example, the video will only appear in search results in Canada and Mexico.

```
<url>
  <loc>https://example.com/videos/some_video_landing_page.html</loc>
  <video:video>
    <video:thumbnail_loc>
      https://example.com/thumbs/123.jpg
    </video:thumbnail_loc>
    <video:title>Grilling steaks for summer</video:title>
    <video:description>
      Bob shows you how to get perfectly done steaks every time
    </video:description>
    <video:player_loc>
      https://example.com/player?video=123
    </video:player_loc>
    <video:restriction relationship="allow">ca mx</video:restriction>
  </video:video>
</url>
```

Optimize for SafeSearch

SafeSearch is a setting in Google user accounts that specifies whether to show or block explicit images, videos, and websites in Google Search results. Make sure Google understands the nature of your site so that Google can apply SafeSearch filters to your site if appropriate. [Learn more about labeling SafeSearch pages.](#)

Monitor video watch pages with Search Console

The following Search Console reports and tools can help you monitor and optimize how your video content is performing on Google Search:

- **Video indexing report:** See how many of your indexed watch pages contain an indexed video, and see reasons for why other videos didn't get indexed.
- **Video rich result report:** Review and fix issues with your VideoObject structured data implementation.
- **Performance report:** Use the Videos search appearance filter to monitor how your videos are performing in Google Search.

Troubleshoot video issues

You can troubleshoot video issues with Search Console. See the [video troubleshooting guide](#) for help.

<https://developers.google.com/search/docs/appearance/visual-elements-gallery>

Visual Elements gallery of Google Search

Visual elements are the building blocks of the Google Search results page that a user can perceive or interact with. The Visual Elements gallery is an illustrated guide to the most common UI elements of Google web search: it explains what the elements look like, what they're called, and whether you can optimize your website for each element.

Anatomy of a Google Search results page

[Original embedded video]

The Google Search results page contains a set of different types of search result visual elements, and each search result has its own set of possible child visual elements. For example, a *text result* is a visual element in its own right, and it has various child visual elements, such as attribution, title link, and snippet.

How the visual elements look can change over time, and a given result can be displayed differently depending on whether you're using a desktop computer or a phone, what country you're in, the language of your search query, and many other factors. Here are the most common types of search result visual elements that you might see in Google Search:

Common types of search result visual elements	
Text result	A result in Google Search that's based on the textual content of the page. Learn more about the text result visual elements .
Rich result	A result that typically relies on structured data in the markup of your page to display graphical elements or interactive experiences. Explore the list of structured data features .
Image result	A result that's based on an image that's embedded on that web page. An image result is more likely to show for image-seeking queries. Learn more about the image result visual elements .

Common types of search result visual elements	
Video result	A result that's based on a video that's embedded on that web page. A video result is more likely to show for video-seeking queries. Learn more about the video result visual elements .
Exploration features	A feature that helps searchers expand and refine their initial search. Learn more about exploration features .

Attribution

Attribution describes the source of a search result, and can appear for a variety of search result types, including text, image, and video results. Attribution can include various aspects of the source, such as the name of the site, favicon, and URL to the web page.

Attribution visual elements		
Favicon	The small icon that's associated with the site. Learn how to provide a favicon .	
Site name	The name of the site. Learn how to provide a site name with structured data .	
Visible URL	The URL of the page that's shown in a readable format. A visible URL has two parts: domain and breadcrumb.	
	Domain	The site address as defined by the domain name. This is the name you chose when setting up the website (for example, example.com).
	Breadcrumb	The trail that shows the page's position within the site's hierarchy. Learn how to specify the trail with Breadcrumb structured data .

Text result

A *text result* (formerly known as a "web result" or "plain blue link") is a result in Google Search that's based on the textual content of the page. It includes visual elements such as attribution, title link, and snippet.

A text result may also include additional visual elements like rich attributes or a sitelinks group; keep in mind that a given text result may display differently depending on a variety of factors, like what device you're using, what you searched for, or what language you're using. You won't see a text result that includes all of the possible visual elements.

Text result visual elements			
Attribution	The source information for the web page. Learn how to control attribution .		
Title link	The title of a search result on Google Search and other properties (for example, Google News) that links to the web page. Learn how to influence title links .		
Snippet	The description or summary part of the search result on Google Search and other properties (for example, Google News). Learn how to control snippets .		
Byline date	The date that Google estimates the web page was updated or published. Learn how to provide a byline date .		
Sitelinks group	Two or more links from the same domain or its localized variations that are clustered together under a text result. For example, the links could be other pages on that domain, headings, or anchors within that page. A sitelinks group contains two or more sitelinks: <table border="1"> <tr> <td>Sitelink</td><td>A single link within a sitelinks group. While sitelinks are automated, there are some best practices you can follow for improving the quality.</td></tr> </table>	Sitelink	A single link within a sitelinks group. While sitelinks are automated, there are some best practices you can follow for improving the quality .
Sitelink	A single link within a sitelinks group. While sitelinks are automated, there are some best practices you can follow for improving the quality .		

Text result image

A *text result image* is the most relevant image from that particular web page for the given query. Tapping it takes the user to the web page that's embedding the image. A text result image is more likely to appear for image-seeking queries.

To optimize for a text result image, follow the [Image SEO best practices](#).

Rich attributes

Rich attributes are one or more lines of additional information about the web page, such as review stars and recipe information. This information is typically powered by [structured data](#), provided by you.

Image result

An *image result* is a result that's based on an image that's embedded on a web page. It's more likely to appear for image-seeking queries. To optimize your image for image results, follow the [image SEO best practices](#).

Image result visual elements	
Image thumbnail	An image thumbnail for the indexed image that's embedded on a web page. Tapping or clicking it takes the user to the image. To optimize your image for image results, follow the image SEO best practices .
Attribution	The source information for the web page that's embedding the image. Learn how to control attribution .

Video result

A *video result* is a result that's based on a video that's embedded on a web page. It's more likely to appear for video-seeking queries. To optimize your video for video results, follow the [video best practices](#).

Video result visual elements	
Video thumbnail	A video thumbnail for the indexed video that's embedded on a web page. Tapping or clicking it takes the user to the web page that's embedding the video. Learn how to specify a video thumbnail .
Title link	The title link for the video landing page. Learn how to influence title links .
Attribution	The source information for the video landing page. Learn how to control attribution .
Upload date	The date that the video was published as provided in its metadata. Learn optimize your videos .

Exploration features

Exploration features help searchers explore more questions or searches that are related to their original search query (also known as "People also ask"). While you can't control what shows up here, it can be helpful to pay attention to the related search queries when you're thinking about topics you could write about for your site.

Related searches group

[Original illustration: An illustration of how a related searches group could look in Google Search, which shows a series of related things that other people have searched for]

A *related searches group* is a cluster of related searches that other people have done. Tapping or clicking a related search takes the user to another search results page. These searches are automatically generated based on the initial query and other things people have searched for.

Related questions group

[Original illustration: An illustration of how a related questions group could look in Google Search, which shows a series of questions that are related to what the user initially searched for]

A *related questions group* is a cluster of questions that are related to what the user initially searched for (also known as "People Also Ask"). When a user expands the question, they're shown a featured snippet. Learn how to [manage featured snippets](#).

<https://developers.google.com/search/docs/appearance/enable-web-stories>

Enable Web Stories on Google

[Original illustration: An illustration of a Web Story in Google Search results]

Web Stories are a web-based version of the popular "Stories" format that blend video, audio, images, animation and text to create a dynamic consumption experience. This visual format lets you explore content at your own pace by tapping through it, or swiping from one piece of content to the next.

This guide explains how to make your Web Stories eligible to appear on Google Search (including [Discover](#)).

If you're a creator, check out [these resources on creating stories](#), without any coding involved.

Here's an overview of how to enable Web Stories on Google:

1. [Create the Web Story](#).
2. [Make sure the Web Story is valid AMP](#).
3. [Verify the metadata](#).
4. [Check if the Web Story is indexed](#).
5. Follow the [Web Story Content Policies](#).

Feature availability

Web Stories can appear as a single result on Google Search, which is available in all regions and languages where Google Search is available.

In the Discover feed, Web Stories can appear as a single card where you can tap through the story. While this appearance is available in all regions and languages where Google Discover is available, it's most likely to appear in the United States, India, and Brazil.

Create the Web Story

Web Stories are web pages under the hood and must follow the same guidelines and best practices that apply to publishing regular web pages. There are two ways to get started:

- Pick one of several [Story editor tools](#) to start creating stories without any coding involved.
- If you have engineering resources, you can [get started with AMP](#). To ensure your Web Story renders appropriately, we suggest using [Chrome Developer Tools](#) to simulate different device sizes and formats.

To ensure a smooth process, review the [Best practices for creating Web Stories](#).

[Original embedded video]

Make sure the Web Story is valid AMP

After you've developed the story, make sure the Web Story is valid AMP. A valid AMP story adheres to various [AMP specifications](#). This provides optimal performance and the best experience. You can use the following tools to check that your Web Story is valid AMP:

- [Web Stories Google Test Tool](#): Check that the Web Story is valid.
- [URL Inspection Tool](#): Check that the Web Story is valid AMP and the Google indexing status of a URL.
- [AMP Linter](#): Validate Web Stories during development using the command line.

Verify metadata

To make your Web Stories eligible to appear on Google Search or Google Discover, supply the necessary metadata so your Web Story appears correctly in the preview.

1. Refer to the [full list of metadata](#).
2. Verify that your Web Story preview appears correctly in the [Web Stories Google Test Tool](#).

Remember that the following fields are required on every Web Story: publisher-logo-src, poster-portrait-src, title, and publisher.

[Original illustration: Remember that the following fields are required on every Web Story: publisher-logo-src, poster-portrait-src, title, and publisher.]

Check if the Web Story is indexed

Check to see if Google Search has indexed your Web Story. Use the [URL Inspection Tool](#) to submit individual URLs or review status using [Page Indexing report](#) or [Sitemaps report](#). If your Web Story isn't indexed:

1. To make it easier for Google to discover your Web Story, link to your Web Stories from your site or add your Web Story URL to your sitemap.
2. All Web Stories must be canonical. Make sure that each Web Story has a `link rel="canonical"` to itself. For example: `<link rel="canonical" href="https://www.example.com/url/to/webstory.html">`

If there are multiple versions of the same story in different languages, make sure to [tell Google about localized versions](#).

3. Check that the Web Story URL isn't [blocked from Googlebot](#) in your robots.txt file or by a noindex tag.

<https://developers.google.com/search/docs/appearance/web-stories-creation-best-practices>

Best practices for creating Web Stories

To keep your readers engaged, follow our best practices for creating [Web Stories](#). We recommend focusing on the critical tasks first. If you have more time, follow the recommended best practices too.

Storytelling

Critical storytelling best practices	
Video first	Video is more engaging than text or images. Use as much video as possible, and supplement with images and text.

More storytelling best practices

Recommended storytelling best practices	
Bring your perspective	Go beyond the facts. Share your opinions. Be the protagonist of your own story. Make it relatable.
Have a narrative arc	Create suspense in your story from one page to another. Bring the user along in the journey by providing context and narrative. Deliver payoff for sticking with you to the end.

Design

Critical design best practices	
Reduce your character count	Avoid including multiple pages with walls of text. Consider reducing text to approximately 280 characters per page (the length of a tweet).
Don't block text	Make sure text is not blocked by other content on the page. Avoid burned in text; by not using burned in text, you prevent text from being blocked when it gets resized to fit various device sizes.
Keep text within bounds	Ensure that all text in your Web Story is visible to the reader. Avoid burned in text; by not using burned in text, you prevent text from overflowing when it gets resized to fit various device sizes.
Use animations mindfully	Bring your stories to life with animations. Avoid distracting or repetitive animations which can cause fatigue.

More design best practices

Recommended design best practices	
Use Web Stories-specific call to action	When re-creating stories that were originally created for a social platform like Instagram, Snapchat or YouTube, be sure to remove any reader call-to-action specific to a certain platform. Make sure that users are able to follow any actions suggested in your Web Story.
Use full bleed videos and images	Include full bleed assets in your stories to create a more immersive experience for readers.
Avoid low resolution or distorted images and videos	Use high-quality images, and take care when resizing images to portrait.
Add a logo to your cover page	Include a high-resolution logo that represents your brand.
Shorten video length	We recommend videos that are less than 15 seconds per page, or 60 seconds maximum.
Include audio	Use high-quality audio clips that are at least 5 seconds long with balanced volume, and ensure speech is audible.
Consider auto advance for video-only stories	Auto-advanced experience for video-based Web Stories could work well for a laid back experience.

SEO

Note: The same [SEO best practices](#) for web pages also apply to [Web Stories](#). A Web Story is still a web page.

Critical SEO best practices	
Provide high-quality content	Like any web page, providing high-quality content that is useful and interesting to your readers the most important thing you can do. Include a complete narrative and follow the storytelling best practices to keep your readers engaged.
Keep the title short	Keep titles shorter than 90 characters. We recommend using a descriptive title that is shorter than 70 characters.
Make sure Google Search can find your story	Don't include a noindex attribute in your story; this attribute blocks Google from indexing the page and prevents it from appearing on Google. Additionally, add your Web Stories to your sitemap. You can check to see if Google can find your Web Stories with the Index Coverage Report and Sitemaps Report in Search Console.
Make the story self-canonical	All Web Stories must be canonical. Make sure that each Web Story has a link rel="canonical" to itself. For example: <code><link rel="canonical" href="https://www.example.com/url/to/webstory.html"></code>

	Note: If there are multiple versions of the same story in different languages, make sure to tell Google about localized versions.
Attach metadata	Make sure that your Web Stories follow the AMP story metadata guidelines. Include markup that you would normally include on a web page, such as: • title and description meta tags • Structured data • OGP • Twitter card

More SEO best practices

Recommended SEO best practices	
Include structured data	We recommend including structured data in your Web Story to help Google Search understand the structure and content of your Web Story.
Include alt text on images	We recommend including alt text on your images to improve your story's discoverability.
Integrate stories into your website	We recommend integrating Web Stories into your website, such as linking them from your home page or category pages where applicable. For example, if your Web Story is about a travel destination and you have a page that lists all your travel articles, then also link the Web Stories on that category page. An additional special landing page like www.example.com/stories (which would then be linked from key pages like your home page) might also make sense. There is no need to indicate in the URL of a Web Story that it is using the Web Stories format or AMP Stories technology. Ideally, your Web Stories are integrated into a wider URL strategy. For example, if your "New York Travel" articles are using a format like <code>"/new-york/travel/title-of-article.html"</code>, then consider using the exact same directory structure and URL format for your Web Stories.
Use AMP story page attachments	AMP story page attachments can be used to present additional information alongside your Web Story. This can be useful to provide extra detail, deep dives, or onward journeys for the content presented in your Web Story.
Include subtitles on video	Add captions to your video to help readers better understand your story. Avoid captions that are burned into the video to ensure that they don't overlap with other content or flow off the screen.
Optimize video-only stories	We recommend that you use semantic HTML to build your Web Story. However, some Web Story editor tools may instead export a story that formats each slide as a video file that bakes in all the text into the video. In this case, we recommend that you add

	the precise text displayed inside of the video as a title attribute on the amp-video element. Again, only do this if you can't use semantic markup in your Web Stories.
Add support for landscape displays	To enable Web Stories to appear in desktop Google Search results, add support for landscape displays .

Technical

Critical technical best practices	
Make the story valid	Web Stories must be valid AMP pages. To avoid invalid AMP issues, test your Story using the AMP Validator tool and fix any detected errors.
Don't include text in the poster image	Avoid using images that contain burned in text, as this could obstruct the title of your story when users preview your story in Search results. If users are unable to clearly read the title, they may be less likely to continue reading.
Include the right poster image size and aspect ratio	Make sure that the image linked to your <code><amp-story> poster-portrait-src</code> attribute is at least 640x853px and use an aspect ratio of 3:4.
Include the right aspect ratio for the logo	Make sure that the logo image linked to your <code><amp-story> publisher-logo-src</code> attribute is at least 96x96 px and aspect ratio of 1:1.

More technical best practices

Recommended technical best practices	
Include <code>og:image</code>	We recommend including <code>og:image</code> in your <code><meta></code> tags to improve your story's discoverability.

Other resources

- [Web Stories](#): Resources on making Web Stories from the AMP Project.
- [Enable Web Stories on Google Search](#): Developer-oriented guide on building Web Stories that meet the technical guidelines required to appear on Google Search.
- [AMP stories website](#): Developer-focused Web Stories format capabilities.
- [Web accessibility](#): Tips to help ensure your Web Story is accessible to all users.
- [Structured data guidelines](#): Details on adding structured data to help Google Search understand your content.

<https://developers.google.com/search/docs/appearance/web-stories-content-policy>

Web Story Content Policies on Google

For your [Web Stories](#) to be eligible to appear on Google Discover and Search as single results, they must comply with [Discover policies](#), [Google Search Essentials](#), and [Google Search's overall policies](#). In addition, for Web Stories to be eligible to appear on richer experiences on Google Search (for example, the carousel view), Web Stories must comply with the [Search Feature content policies](#) and the following additional content policies. In cases of egregious violations of the Web Stories Content Policies, a site may stop permanently appearing on the richer experiences on Google Search.

Copyrighted content

We don't allow Web Stories that infringe anyone's copyright. Web Stories are meant to reflect original works, so we don't allow Web Stories that include someone else's copyrighted work unless you have received permission. Google does not assume any obligation or responsibility with respect to obtaining rights for your Web Stories to appear across Google. If your Web Story infringes on someone else's copyright, we may block it from appearing. For more information, review [our copyright procedures](#).

Text-heavy Web Stories

We don't allow Web Stories that are text heavy. Web Stories may not be eligible if the majority of [pages](#) have more than 180 characters of text. Usage of bite-sized video (less than 60 seconds per page) wherever possible is encouraged.

Low quality assets

We don't allow Web Stories that contain images and video assets that are stretched out or pixelated to the point that the viewer's experience is negatively impacted.

Lack of narrative

We don't allow Web Stories that are missing a binding theme or narrative structure from page to page.

Incomplete stories

We don't allow Web Stories that are incomplete or that require users to click links to other websites or apps to get essential information.

Overly commercial

We don't allow Web Stories in which the sole goal is to advertise a service or a product, and especially if you may directly benefit from users consuming your Web Story. Affiliate marketing links are permissible as long as they are restricted to a minor part of the Web Story. Display ads may be placed following the [Story Ad Guidelines](#). [Good affiliate programs](#) are supported based on [spam policies for Google web search](#).

<https://developers.google.com/search/docs/appearance/package-tracking>

Package tracking Early Adopters Program

The package tracking early adopters program is no longer accepting new partners.

Package tracking is a feature that displays package tracking related information on Google. When people come to Search looking to track a package shipped with your company, they'll be able to enter a package ID directly. The feature uses your API to retrieve the package tracking information and then displays it to the user. Here's how package tracking information may appear on Google Search when a user seeks to track a package.

[Original illustration: package tracking in search results]

Note: The actual appearance in search results might be different.

Feature availability

Package tracking is available in all languages and countries where Google Search is available. The program is no longer accepting new partners.

Availability and responsiveness

We expect almost no downtime from your API and require that your API respond within 700ms on average with the 95th percentile not exceeding 1,000ms. If your API doesn't meet these requirements, we may stop displaying your package tracking information.

Content

For the integration to work, your API must return the following information:

Required field	
CurrentStatus	The current status of the package. This includes the date and time this status became valid and any error states.

We strongly recommend that your API also return the following information:

Recommended fields	
DeliveredDate	The day and time the package was delivered (if it has been delivered).
PromisedDate	The date the package is expected to be delivered.

TrackingNumber	The tracking number for the package.
TrackingURL	The website URL that a user can visit to view package tracking information and possible additional details.
SupportPhoneNumbers	A list of support phone numbers available per region.
TransitEvents	The set of events that denote when a package makes interim progress on its journey to the recipient, including the Day, Time, City, State, and Country (where applicable).
CreateDate	The day and time the tracking number was created.
PickupDate	The date the package was picked up by the carrier.
TimestampEvent	The timestamp of an event associated with a given package.
LocationEvent	The location of an event associated with a given package.
CanReschedule	Whether this package can be re-scheduled.

We don't accept the following information:

- Any personal data about the recipient or sender of the package.
- Any geographical information about the recipient or sender of the package.

Upcoming deprecation of Crawl Rate Limiter Tool in Search Console

Google last updated / published: 2023-11-24 (publication) · Reviewed 6 October 2026

<https://developers.google.com/search/blog/2023/11/sc-crawl-limiter-byebye>

Friday, November 24, 2023

The crawl rate limiter tool in Search Console is being deprecated on Jan 8th, 2024. This tool has been [available for over a decade](#), but with the improvements we've made to our crawling logic and other tools available to publishers, its usefulness has dissipated.

Googlebot reacts to how the site—or more specifically the server handling the site—responds to Googlebot's HTTP requests. For example, if the server persistently returns HTTP 500 status codes for a range of URLs, Googlebot will automatically, and almost immediately slow down crawling. Similarly, Googlebot slows down automatically if the response time for requests gets significantly longer. If you do experience unusually heavy crawling that your site can't manage on its own, refer to this [help article](#).

In contrast, the rate limiter tool had a much slower effect; in fact it may have taken over a day for the new limits to be applied on crawling. Fortunately though, site owners rarely had to resort to using the tool, and those who have, in many cases set the crawling speed to the bare minimum. With the deprecation of the crawl limiter tool, we're also setting the minimum crawling speed to a lower rate, comparable to the old crawl rate limits. This means that we effectively continue honoring the settings that some site owners have set in the past if the Search interest is low, and our crawlers don't waste the site's bandwidth.

Because of the advances in the automated crawl rate handling, and in the spirit of keeping things simple for users, we'll be deprecating this tool in Search Console. We are keeping the Googlebot [report form](#) for reporting unusual Googlebot activities and for emergency cases, but keep in mind that the fastest way to reduce crawl rate is to instruct Googlebot through server responses as detailed in [our documentation](#).

If you have questions or comments, write in our [Google Search Central Community](#).

Posted by [Gary Illyes](#) and [Nir Kalush](#)

Introducing Search Generative AI performance reports in Search Console

Google last updated / published: not stated · Reviewed 6 October 2026

<https://developers.google.com/search/blog/2026/06/gen-ai-performance-reports>

Wednesday, June 3, 2026

Note: As of August 31, 2026, we've rolled out these insights to all websites worldwide.

Today, we're excited to announce the launch of new Search Generative AI performance reports in Search Console, including dedicated reports for Search and Discover, to help you understand your site's visibility within generative AI features on Search.

The new Search Console reports are designed to give you dedicated views of your impressions within generative AI features on Search, such as AI Overviews and AI Mode, as well as generative AI features in Discover. This data is included in the overall performance report, where it will continue to be tracked to give site owners an overview of the overall visibility of their site in Google Search. Today, we are launching a separate view dedicated to visibility from generative AI features.

We are rolling these reports out to a subset of websites, allowing us to thoroughly test them and receive feedback before making them widely available.

[Original illustration: [Generative AI performance report \(Search\)](#)]

To help you understand how pages from your site are shown, our new reports show the following information:

- **Impressions:** How often URLs from your site appeared in generative AI features in Search and Discover.
- **Pages:** Check which URLs appeared within AI features.
- **Countries:** Understand your visibility on a country basis.
- **Devices:** Identify the devices people are using when seeing your website (available for Search results).
- **Dates:** Monitor your performance over time with hourly, daily, weekly, and monthly granularity.

Learn more about the data available in our [help center documentation on the generative AI performance reports](#).

As always, we value your feedback: please let us know what you think using the "Submit feedback" link in Search Console, with the [dedicated feedback form for these reports](#), or in the [Google Search Central Community](#). We're continuing to work with website owners to understand what insights and data would be most helpful to inform their strategies, such as adding additional metrics over time.

Posted by [Hillel Maoz](#), Search Ecosystem Engineering Manager and [Moshe Samet](#), Product Manager Lead, Search Console

<https://support.google.com/webmasters/answer/16908024>

Search generative AI control

Complete-topic source brief · paraphrase. Open the original Google Support page for its full wording.

Available worldwide since August 31, 2026, this Search Console Settings control governs links/content/grounding for AI Overviews, AI Mode and listed Discover AI features. Include, exclude or inherit the closest configured parent; owners can override inheritance and parent owners are notified. Default inclusion applies at the highest level. Exclusion leaves other Search ranking/inclusion signals and Merchant Center/Ads choices unaffected. Training is separate: Google-Extended limits training; noindex blocks overall Search. Changes generally take days; exclusion normally takes one to two days after activation, with caching delays possible. Check the generative-AI performance report. The original provides parent/child examples and feedback.

<https://web.dev/articles/vitals>

[Original illustration: Philip Walton]

Philip Walton

Published: May 4, 2020, Last updated: October 31, 2024

Optimizing for quality of user experience is key to the long-term success of any site on the web. Whether you're a business owner, marketer, or developer, Web Vitals can help you quantify the experience of your site and identify opportunities to improve.

Overview

Web Vitals is an initiative by Google to provide unified guidance for quality signals that are essential to delivering a great user experience on the web.

Google has provided a number of tools over the years to measure and report on performance. Some developers are experts at using these tools, while others have found the abundance of both tools and metrics challenging to keep up with.

Site owners shouldn't have to be performance experts to understand the quality of experience they are delivering to their users. The Web Vitals initiative aims to simplify the landscape, and help sites focus on the metrics that matter most, the **Core Web Vitals**.

[Original illustration: Sign up for the web.dev newsletter.]

Core Web Vitals

Core Web Vitals are the subset of Web Vitals that apply to all web pages, should be measured by all site owners, and will be surfaced across all Google tools. Each of the Core Web Vitals represents a distinct facet of the user experience, is measurable **in the field**, and reflects the real-world experience of a critical **user-centric** outcome.

The metrics that make up Core Web Vitals will **evolve** over time. The current set focuses on three aspects of the user experience—*loading*, *interactivity*, and *visual stability*—and includes the following metrics (and their respective thresholds):

[Original illustration: Largest Contentful Paint threshold recommendations]

[Original illustration: Interaction to Next Paint threshold recommendations]

[Original illustration: Cumulative Layout Shift threshold recommendations]

- **Largest Contentful Paint (LCP)**: measures *loading* performance. To provide a good user experience, LCP should occur within **2.5 seconds** of when the page first starts loading.
- **Interaction to Next Paint (INP)**: measures *interactivity*. To provide a good user experience, pages should have a INP of **200 milliseconds** or less.
- **Cumulative Layout Shift (CLS)**: measures *visual stability*. To provide a good user experience, pages should maintain a CLS of **0.1** or less.

To ensure you're hitting the recommended target for these metrics for most of your users, a good threshold to measure is the **75th percentile** of page loads, segmented across mobile and desktop devices.

Tools that assess Core Web Vitals compliance should consider a page passing if it meets the recommended targets at the 75th percentile for all three of the Core Web Vitals metrics.

Note: To learn more about the research and methodology behind these recommendations, see: [Defining the Core Web Vitals metrics thresholds](#).

Lifecycle

Metrics on the Core Web Vitals track go through a lifecycle consisting of three phases: experimental, pending, and stable.

[Original illustration: The three lifecycle phases of Core Web Vitals metrics, visualized as a series of three chevrons. From left to right, the phases are Experimental, Pending, and Stable.]

The stages of the Core Web Vitals lifecycle.

Each phase is designed to signal to developers how they should think about each metric:

- **Experimental metrics** are prospective Core Web Vitals that may still be undergoing significant changes depending on testing and community feedback.
- **Pending metrics** are future Core Web Vitals that have passed the testing and feedback stage and have a well-defined timeline to becoming stable.
- **Stable metrics** are the current set of Core Web Vitals that Chrome considers essential for great user experiences.

The Core Web Vitals are at the following lifecycle stages:

- **LCP**: Stable
- **CLS**: Stable
- **INP**: Stable

Experimental

When a metric is initially developed and enters the ecosystem, it is considered an *experimental metric*.

The purpose of the experimental phase is to assess a metric's fitness, first by exploring the problem to be solved, and possibly iterate on what previous metrics may have failed to address. For example, **Interaction to Next Paint (INP)** was initially developed as an experimental metric to address the runtime performance issues present on the web more comprehensively than **First Input Delay (FID)**.

The experimental phase of Core Web Vitals lifecycle is also intended to give flexibility in a metric's development by identifying bugs and even exploring changes to its initial definition. It's also the phase in which community feedback is most important.

Pending

When the Chrome team determines that an experimental metric has received sufficient feedback and proven its efficacy, it becomes a *pending metric*. For example, INP was promoted in 2023 from experimental to pending status with the intent to eventually retire FID.

Pending metrics are held in this phase for a minimum of six months to give the ecosystem time to adapt. Community feedback remains an important aspect of this phase, as more developers begin to use the metric.

Stable

When a Core Web Vital candidate metric is finalized, it becomes a *stable metric*. This is when the metric can become a Core Web Vital.

Stable metrics are actively supported, and can be subject to bug fixes and definition changes. Stable Core Web Vitals metrics won't change more than once per year. Any change to a Core Web Vital will be clearly communicated in the metric's official documentation, as well as in the metric's changelog. Core Web Vitals are also included in any assessments.

Key point: Stable metrics aren't necessarily permanent. A stable metric can be retired and replaced by another metric that addresses the problem area more effectively. This is exactly what happened to FID as INP became a stable Core Web Vital metric in 2024.

Tools to measure and report Core Web Vitals

Google believes that the Core Web Vitals are critical to all web experiences. As a result, it is committed to surfacing these metrics in all of its popular tools. The following sections details which tools support the Core Web Vitals.

Field tools to measure Core Web Vitals

The [Chrome User Experience Report](#) collects anonymized, real user measurement data for each Core Web Vital. This data enables site owners to quickly assess their performance without requiring them to manually instrument analytics on their pages, and powers tools like [Chrome DevTools](#), [PageSpeed Insights](#), and Search Console's [Core Web Vitals report](#).

	LCP	INP	CLS
Chrome User Experience Report	check	check	check
Chrome DevTools	check	check	check
PageSpeed Insights	check	check	check

Search Console (Core Web Vitals report)	check	check	check
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Note: For guidance on how to use these tools, and which tool is right for your use case, see: [Getting started with measuring Web Vitals](#).

The data provided by Chrome User Experience Report offers a quick way to assess the performance of sites, but it does not provide the detailed, per-pageview telemetry that is often necessary to accurately diagnose, monitor, and quickly react to regressions. As a result, we strongly recommend that sites set up their own real-user monitoring.

Measure Core Web Vitals in JavaScript

Each of the Core Web Vitals can be measured in JavaScript using standard web APIs.

Note: Note that the Core Web Vitals measured in JavaScript using public APIs may differ from the Core Web Vitals reported by CrUX. Read about [the differences between these data sources](#) for more info.

The easiest way to measure all the Core Web Vitals is to use the [web-vitals](#) JavaScript library, a small, production-ready wrapper around the underlying web APIs that measures each metric in a way that accurately matches how they're reported by all the Google tools listed earlier.

With the [web-vitals](#) library, measuring each metric can be done by calling a single function. See the documentation for complete [usage](#) and [API](#) details.

```
import {onCLS, onINP, onLCP} from 'web-vitals';

function sendToAnalytics(metric) {
  const body = JSON.stringify(metric);
  // Use `navigator.sendBeacon()` if available, falling back to `fetch()`.
  (navigator.sendBeacon && navigator.sendBeacon('/analytics', body)) ||
    fetch('/analytics', {body, method: 'POST', keepalive: true});
}

onCLS(sendToAnalytics);
onINP(sendToAnalytics);
onLCP(sendToAnalytics);
```

Once you've configured your site to use the [web-vitals](#) library to measure and send your Core Web Vitals data to an analytics endpoint, the next step is to aggregate and report on that data to see if your pages are meeting the recommended thresholds for at least 75% of page visits.

While some analytics providers have built-in support for Core Web Vitals metrics, even those that don't should include basic custom metric features that allow you to measure Core Web Vitals in their tool.

Developers who prefer to measure these metrics directly using the underlying web APIs can instead use these metric guides for implementation details:

- [Measure LCP in JavaScript](#)
- [Measure INP in JavaScript](#)

- [Measure CLS in JavaScript](#)

For additional guidance on measuring these metrics using popular analytics services or your own in-house analytics tools, see [Best practices for measuring Web Vitals in the field](#).

Lab tools to measure Core Web Vitals

While all of the Core Web Vitals are, first and foremost, field metrics, many of them are also measurable in the lab.

Lab measurement is the best way to test the performance of features during development—before they've been released to users. It's also the best way to catch performance regressions before they happen.

The following tools can be used to measure the Core Web Vitals in a lab environment:

	LCP	INP	CLS
Chrome DevTools	check	check	check
Lighthouse	check	block (use TBT instead)	check

Note: Tools like Lighthouse that load pages in a simulated environment without a user cannot measure INP, as there is no user input. However, the Total Blocking Time (TBT) metric is lab-measurable and is a proxy for INP. Performance optimizations that improve TBT in the lab may improve INP in the field (see performance recommendations below).

While lab measurement is an essential part of delivering great experiences, it is not a substitute for field measurement.

The performance of a site can substantially vary based on a user's device capabilities, their network conditions, what other processes may be running on the device, and how they're interacting with the page. In fact, each of the Core Web Vitals metrics can have its score affected by user interaction. Only field measurement can accurately capture the complete picture.

Recommendations for improving your scores

The following guides offer specific recommendations for how to optimize your pages for each of the Core Web Vitals:

- [Optimize LCP](#)
- [Optimize INP](#)
- [Optimize CLS](#)

There are many ways to improve Core Web Vitals, and each approach comes with varying levels of impact, relevance, and ease of use for every situation. Refer to [The most effective ways to improve Core Web Vitals](#) for a short list of the Chrome team's top recommendations.

Other Web Vitals

While the Core Web Vitals are the critical metrics for understanding and delivering a great user experience, there are other supporting metrics.

These other metrics can serve as proxy—or as supplemental metrics for the three Core Web Vitals—to help capture a larger part of the experience or to aid in diagnosing a specific issue.

For example, the metrics [Time to First Byte \(TTFB\)](#) and [First Contentful Paint \(FCP\)](#) are both vital aspects of the *loading* experience, and are both useful in diagnosing issues with LCP (slow [server response times](#) or [render-blocking resources](#), respectively).

Similarly, a metric like [Total Blocking Time \(TBT\)](#) is a lab metrics is vital in catching and diagnosing potential *interactivity* issues that can impact INP. However, it is not part of the Core Web Vitals set because they are not field-measurable, nor do they reflect a [user-centric](#) outcome.

Changes to Web Vitals

Web Vitals and Core Web Vitals represent the best available signals developers have today to measure quality of experience across the web, but these signals are not perfect and future improvements or additions should be expected.

The **Core Web Vitals** are relevant to all web pages and featured across relevant Google tools. Changes to these metrics will have wide-reaching impact; as such, developers should expect the definitions and thresholds of the Core Web Vitals to be stable, and updates to have prior notice and a predictable, annual cadence.

The other Web Vitals are often context or tool specific, and may be more experimental than the Core Web Vitals. As such, their definitions and thresholds may change with greater frequency.

For all Web Vitals, changes will be clearly documented in this public [CHANGELOG](#).

Generative AI performance report (Search)

Complete-topic source brief · paraphrase. Open the original Google Support page for its full wording.

The Search generative AI report covers AI Overviews and AI Mode impressions; Discover has a separate report. It groups by pages, countries, dates and devices, with text/multimodal Web filters. Pages generally use canonical URLs; dates use Pacific Time. Chart and table aggregation can differ; standard performance-report limits apply. Export is available, with unavailable values exported as zero. The article explains inclusion settings, insufficient impressions and access/rollout wording when the report is absent. It documents impressions rather than an AI click, CTR or query report.

<https://support.google.com/merchants/answer/14743464>

AI-generated content

Complete-topic source brief · paraphrase. Open the original Google Support page for its full wording.

Merchant Center requires AI-generated images to retain provenance metadata, including the IPTC DigitalSourceType value TrainedAlgorithmicMedia. Preserve embedded source tags. Generated images can appear in primary, additional and lifestyle image attributes. AI-generated titles use structured_title; AI-generated descriptions use structured_description. Both supply content and digital_source_type=trained_algorithmic_media. The article also explains synthetic/composite source classifications and links to attribute specifications.

<https://developers.google.com/crawling/docs/crawl-budget>

Optimize your crawl budget

This guide describes how to optimize Google's crawling of very large and frequently updated sites.

If your site doesn't have a large number of pages that change rapidly, or if your pages seem to be crawled the same day that they are published, you don't need to read this guide. For Google Search specifically, [keeping your sitemap up to date](#) and [checking the Page Indexing report](#) regularly is adequate.

Who this guide is for

While the recommendations in this guide are generally good practices, this is an advanced guide intended primarily for the following types of sites:

- Large sites (1 million+ unique pages) with content that changes moderately often (once a week)
- Medium or larger sites (10,000+ unique pages) with very rapidly changing content (daily)
- Sites with a large portion of their total URLs classified by Search Console as [Discovered - currently not indexed](#)

The numbers given here are a **rough estimate** to help you classify your site. These are not exact thresholds.

General theory of crawling

[Original embedded video]

The web is a nearly infinite space, exceeding Google's ability to explore every publicly accessible URL. As a result, there are limits to how much time and resources Google can devote to crawling any single site. The allocation of these resources is commonly called a site's *crawl budget*. In this context, Google's crawling infrastructure defines a *site* as a unique hostname. For example, <https://www.example.com/> and <https://code.example.com/> are treated as separate sites and have separate crawl budgets. A site's crawl budget is determined by two main elements: *crawl capacity limit* and *crawl demand*.

For Google Search, not every page that is crawled will necessarily be indexed. After crawling, each page must be evaluated, [consolidated](#), and assessed to determine its suitability for the index.

Crawl capacity limit

Google wants to crawl your site without overwhelming your servers. To prevent this, Google's crawlers calculate a *crawl capacity limit* (also known as *hostload*). This limits the total amount of time your server spends holding connections open for Google, factoring in both the number of parallel connections and their duration. This ensures Google can cover all your important content without overloading your servers.

Every site starts with the same default, conservative crawl capacity limit. If there is demand to crawl more and the site remains healthy, Google's systems will automatically adjust this limit over time.

The crawl capacity limit can go up and down based on a few factors:

- **Crawl health:** If the site responds consistently and its response times (including latency and Time-to-First Byte) remain stable or improve, the limit goes up, meaning more connections can be used to crawl. If the site slows down (latency increases or response times become longer), or responds with server errors (5xx HTTP status codes) or rate-limiting signals (such as HTTP 429), the limit goes down and Google crawls less.
- **Google's crawling limits:** While Google's resources are extensive, they are finite, and we must prioritize resource allocation across the web.

Crawl demand

Each crawler has its own "demand" when it comes to crawling the web, determined by factors unique to that crawler. For example, AdsBot generally has a higher demand when a site is running dynamic ad targets, and Google Shopping has a higher demand for products you have in your merchant feeds.

For [Googlebot](#), demand varies based on a site's size, update frequency, page quality, and relevance, compared to other sites. The primary factors you can influence are:

- **Perceived inventory:** Without guidance from you, Google tries to crawl all or most of the URLs that it knows about on your site. If many of these URLs are duplicates, or you don't want them crawled for some other reason (removed, unimportant, and so on), this wastes a lot of Google crawling time on your site. This is the factor that you can positively control the most.
- **Popularity:** URLs that are more popular on the Internet tend to be crawled more often to keep them fresher in our systems.
- **Staleness:** Our systems want to recrawl documents frequently enough to pick up any changes.

Additionally, site-wide events like site moves may trigger an increase in crawl demand in order to reprocess the content under the new URLs.

While each crawler has a different *crawl demand*, the *crawl capacity limit* is shared across all crawlers. This means that high demand from one crawler can reduce the capacity available for others.

Summary

Taking crawl capacity and crawl demand together, Google defines a site's crawl budget as the set of URLs that Google can and wants to crawl. Even if the crawl capacity limit isn't reached, if crawl demand is low, Google will crawl your site less.

Best practices

To maximize your crawling efficiency, follow these best practices:

- **Manage your URL inventory:** Use the appropriate tools to tell Google which pages to crawl and which not to crawl. If Google spends too much time crawling URLs that it shouldn't, Google's crawlers might not explore the rest of your site, or might not increase your crawl budget.
 - **Consolidate duplicate content.** Eliminate duplicate content to focus crawling on unique content rather than unique URLs.

- **Block crawling of URLs using robots.txt.** Some pages might be important to users, but you don't necessarily want them to appear on Google surfaces or get reprocessed by Google's systems. Examples include infinite scrolling pages that duplicate information on linked pages, or differently sorted versions of the same page. If you can't consolidate them as described in the first bullet, block these unimportant pages using [robots.txt](#). Blocking URLs with robots.txt prevents Google from crawling them, and significantly decreases the chance the URLs will be processed by other Google systems (such as getting indexed by Google Search).

Don't use noindex, as Google will still request, but then drop the page when it sees a noindex meta tag or header in the HTTP response, wasting crawling time. **Don't use robots.txt to temporarily reallocate crawl budget** for other pages; use robots.txt to block pages or resources that you don't want Google to crawl at all. Google won't shift this newly available crawl budget to other pages unless Google is already hitting your site's crawl capacity limit.

- **Return a 404 or 410 status code for permanently removed pages.** Google won't forget a URL that it knows about, but a 404 status code is a strong signal not to crawl that URL again. Blocked URLs, however, will stay part of your crawl queue much longer, and will be recrawled when the block is removed.
- **Eliminate soft 404 errors.** soft 404 pages will continue to be crawled, and waste your budget. Check the [Page Indexing report](#) for soft 404 errors.
- **Keep your sitemaps up to date.** Google reads your sitemap regularly, so be sure to include all the content that you want Google to crawl. If your site includes updated content, we recommend including the <lastmod> tag.
- **Avoid long redirect chains**, which have a negative effect on crawling.
- **Make your pages efficient to load.** If Google can load and render your pages faster, we might be able to read more content from your site.
 - **Improve loading speed:** Optimize your server response times and resources to make pages load faster.
 - **Use HTTP caching:** Support [304 \(Not Modified\) HTTP status codes](#). If a page hasn't changed since Google last crawled it, returning a 304 code tells Google to reuse the cached version, saving your server bandwidth and resources.
- **Debug issues with crawl budget.** Check whether your site had any availability issues during crawling, and look for ways to make your crawling more efficient.

How do I get more crawl budget?

There are two ways to increase crawl budget:

- **Add more server resources:** If your site can't be crawled because of server capacity on your end (for example, you're getting [Hostload exceeded](#) in the URL inspection tool), add more server resources if that makes sense for your business.
- **Optimize your content's quality for the Google product you're targeting:** Google determines the crawling resources allocated to each site by factoring in elements that are relevant to the specific Google product. For example, for Google Search, this includes things like popularity, overall user value, content uniqueness, and serving capacity.

interview coaching is editorial application. Exact excerpts preserve wording with whitespace normalized. Google Support articles use concise complete-topic briefs and short excerpts with original links. Source dates are retained; illustrations and videos are linked. Reviewed 6 October 2026. This pack is not endorsed by Google.